

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

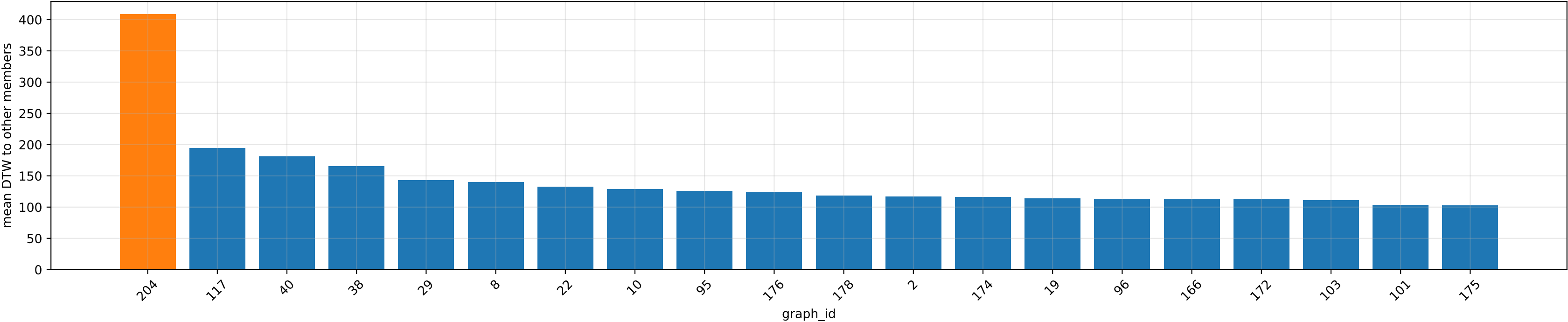
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	13	0	1 vs 19	204	204	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
408.604	nan	113.843

Per-member mean DTW



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

First applied split pairwise DTW metadata | seed=0

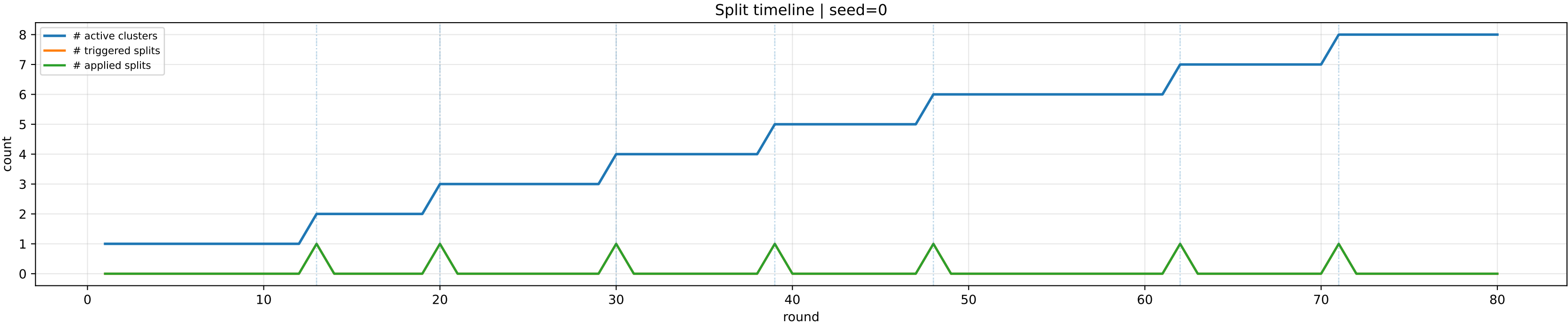
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	13	0	20	204.0	0.049	408.604	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 517.3 at (8, 204)

graph_id	178	96	176	19	174	175	172	95	101	103	22	29	8	166	10	2	40	38	204	117
178	0.0	107.8	76.2	102.5	132.4	57.0	122.2	121.5	91.6	72.1	122.8	74.9	106.0	56.4	133.1	75.3	144.2	93.1	373.6	183.4
96	107.8	0.0	113.0	49.3	58.1	60.2	37.8	77.3	33.3	51.2	66.0	127.2	121.3	103.3	66.3	109.9	182.5	153.1	445.4	190.1
176	76.2	113.0	0.0	100.5	124.0	85.7	122.0	119.3	92.7	114.3	139.6	98.7	96.9	53.2	118.8	88.3	138.0	143.6	370.8	164.3
19	102.5	49.3	100.5	0.0	70.1	50.4	53.8	68.0	33.1	50.2	56.8	154.6	82.0	97.5	48.2	85.2	207.5	181.5	459.1	216.4
174	132.4	58.1	124.0	70.1	0.0	73.9	36.5	67.3	48.0	93.6	96.3	121.3	100.1	124.3	63.5	91.0	166.4	154.8	432.4	154.0
175	57.0	60.2	85.7	50.4	73.9	0.0	66.6	66.8	28.9	26.5	71.2	104.2	91.8	73.1	72.9	53.1	183.2	157.2	434.8	194.1
172	122.2	37.8	122.0	53.8	36.5	66.6	0.0	66.8	35.2	63.7	74.0	121.4	111.0	123.4	61.9	95.3	171.6	159.1	443.0	175.1
95	121.5	77.3	119.3	68.0	67.3	66.8	66.8	0.0	53.4	71.1	45.9	190.1	41.0	91.4	28.9	45.1	251.2	225.3	502.9	262.2
101	91.6	33.3	92.7	33.1	48.0	28.9	35.2	53.4	0.0	34.5	61.4	132.7	105.0	88.5	58.1	69.8	190.8	165.6	443.2	194.3
103	72.1	51.2	114.3	50.2	93.6	26.5	63.7	71.1	34.5	0.0	54.4	124.2	112.1	76.2	65.8	76.1	192.5	165.6	443.2	218.3
22	122.8	66.0	139.6	56.8	96.3	71.2	74.0	45.9	61.4	54.4	0.0	185.7	71.8	101.3	52.7	73.1	252.6	226.6	504.2	267.0
29	74.9	127.2	98.7	154.6	121.3	104.2	121.4	190.1	132.7	124.2	185.7	0.0	182.8	112.4	195.7	142.1	112.6	81.8	298.5	161.3
8	106.0	121.3	96.9	82.0	100.1	91.8	111.0	41.0	105.0	112.1	71.8	182.8	0.0	78.9	49.1	21.5	263.2	239.7	517.3	266.7
166	56.4	103.3	53.2	97.5	124.3	73.1	123.4	91.4	88.5	76.2	101.3	112.4	78.9	0.0	109.0	62.6	117.5	141.6	389.7	142.4
10	133.1	66.3	118.8	48.2	63.5	72.9	61.9	28.9	58.1	65.8	52.7	195.7	49.1	109.0	0.0	62.1	256.0	230.1	507.7	267.0
2	75.3	109.9	88.3	85.2	91.0	53.1	95.3	45.1	69.8	76.1	73.1	142.1	21.5	62.6	62.1	0.0	209.4	184.6	461.5	220.8
40	144.2	182.5	138.0	207.5	166.4	183.2	171.6	251.2	190.8	192.5	252.6	112.6	263.2	117.5	256.0	209.4	0.0	85.7	239.7	79.3
38	93.1	153.1	143.6	181.5	154.8	157.2	159.1	225.3	165.6	165.6	226.6	81.8	239.7	141.6	230.1	184.6	85.7	0.0	254.5	102.5
204	373.6	445.4	370.8	459.1	432.4	434.8	443.0	502.9	443.2	443.2	504.2	298.5	517.3	389.7	507.7	461.5	239.7	254.5	0.0	242.0
117	183.4	190.1	164.3	216.4	154.0	194.1	175.1	262.2	194.3	218.3	267.0	161.3	266.7	142.4	267.0	220.8	79.3	102.5	242.0	0.0

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split timeline

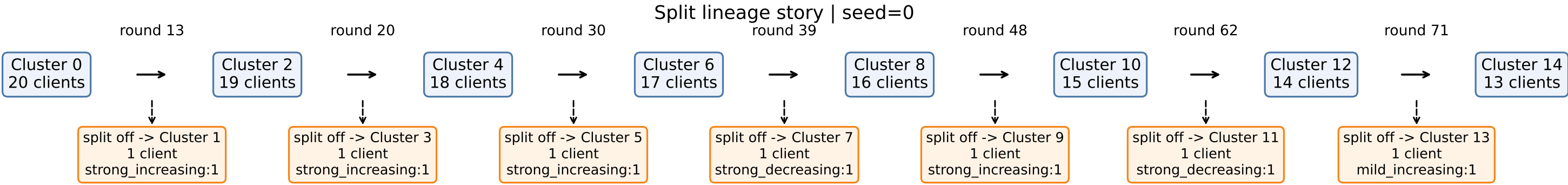
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050



Seed summary seed=0										
Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	13, 20, 30, 39, 48, 62, 71	13, 20, 30, 39, 48, 62, 71	1, 3, 5, 7, 9, 11, 13, 14	1:1, 3:1, 5:1, 7:1, 9:1, 11:1, 13:1, 14:13

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050



Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
13	0	20	159.763 < 160.736 , 258.657 > 252.512	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
20	2	19	157.940 < 158.292 , 297.851 > 280.172	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
30	4	18	156.506 < 156.915 , 322.936 > 268.856	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1
39	6	17	156.605 < 157.331 , 255.828 > 239.781	C7 (1 client)	strong_decreasing:1	C8 (16 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:3 , strong_increasing:1
48	8	16	155.579 < 157.260 , 244.690 > 214.442	C9 (1 client)	strong_increasing:1	C10 (15 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:3
62	10	15	157.011 < 157.305 , 215.090 > 209.820	C11 (1 client)	strong_decreasing:1	C12 (14 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:2
71	12	14	155.941 < 156.083 , 205.817 > 188.795	C13 (1 client)	mild_increasing:1	C14 (13 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:3 , strong_decreasing:2

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

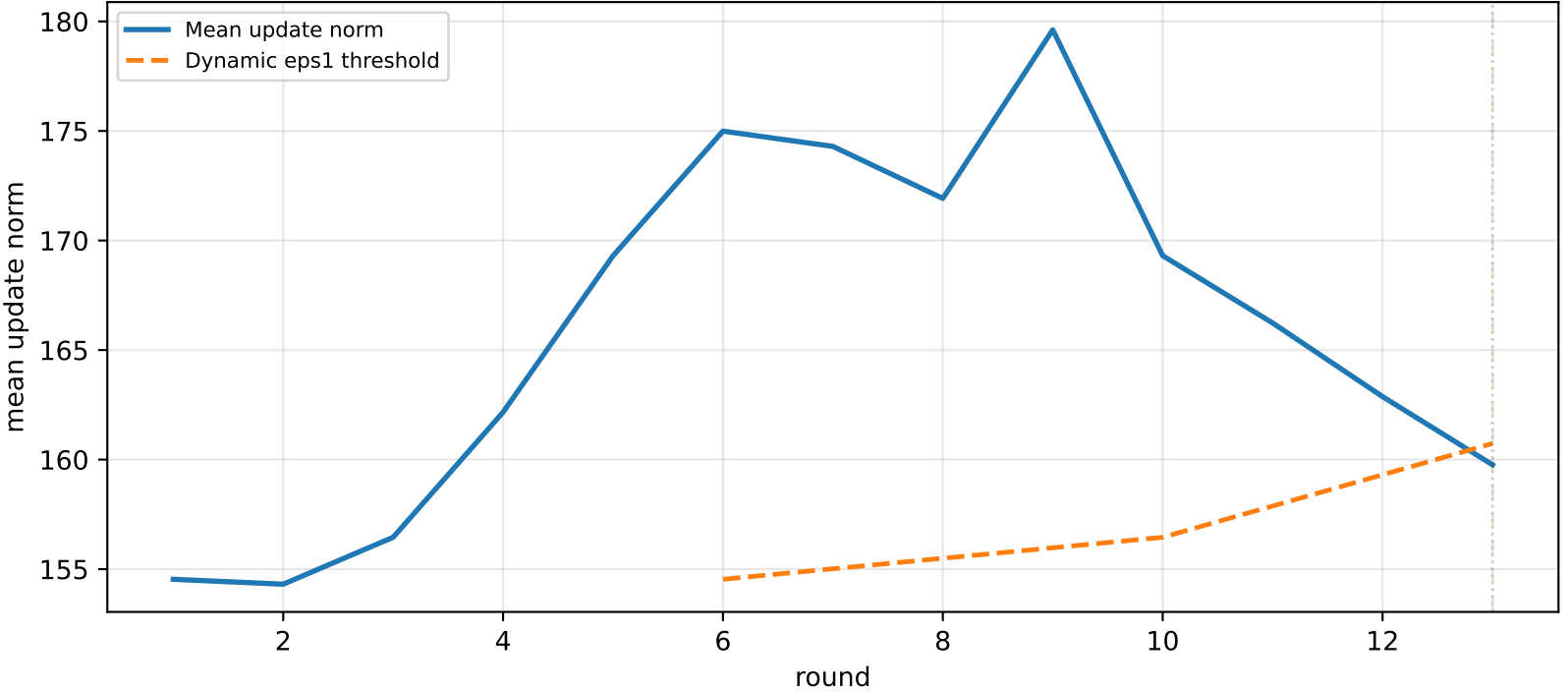
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	204	18	strong_increasing:1
3	1	117	19	strong_increasing:1
5	1	40	16	strong_increasing:1
7	1	2	15	strong_decreasing:1
9	1	38	17	strong_increasing:1
11	1	166	13	strong_decreasing:1
13	1	22	10	mild_increasing:1
14	13	178, 96, 176, 19, 174, 175, 172, 95, 101, 103, 29, 8, 10	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 11, 12, 14	flat_balanced:4 , mild_decreasing:4 , mild_increasing:3 , strong_decreasing:2

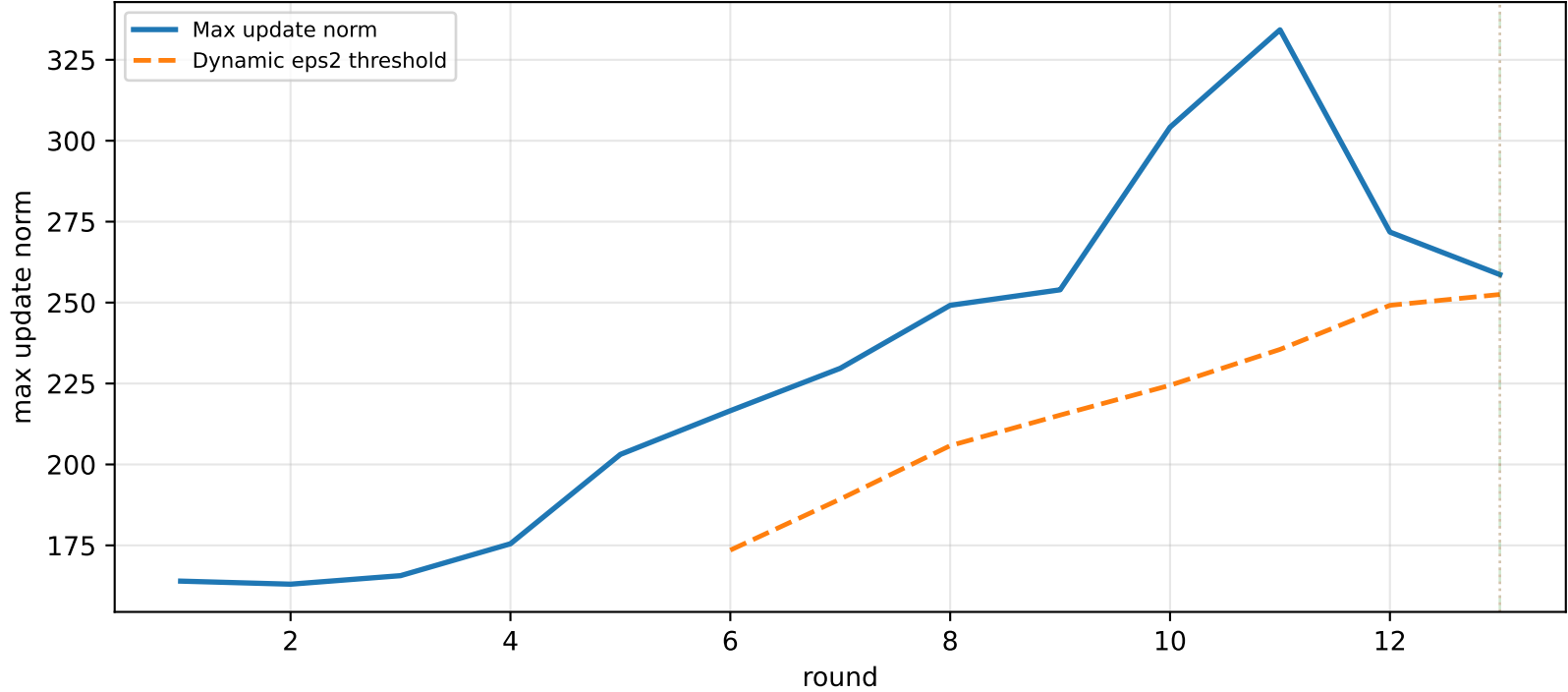
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

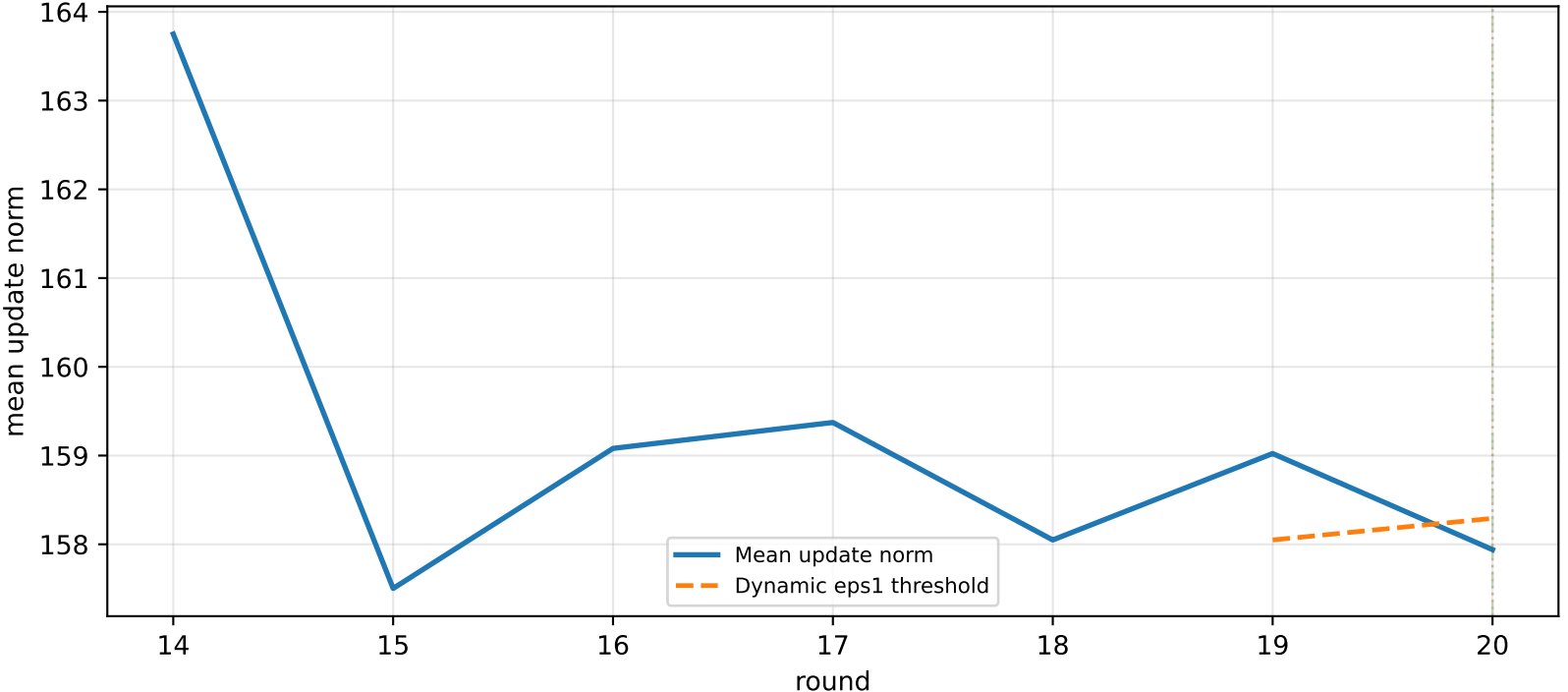
Cluster 0 | Mean update norm



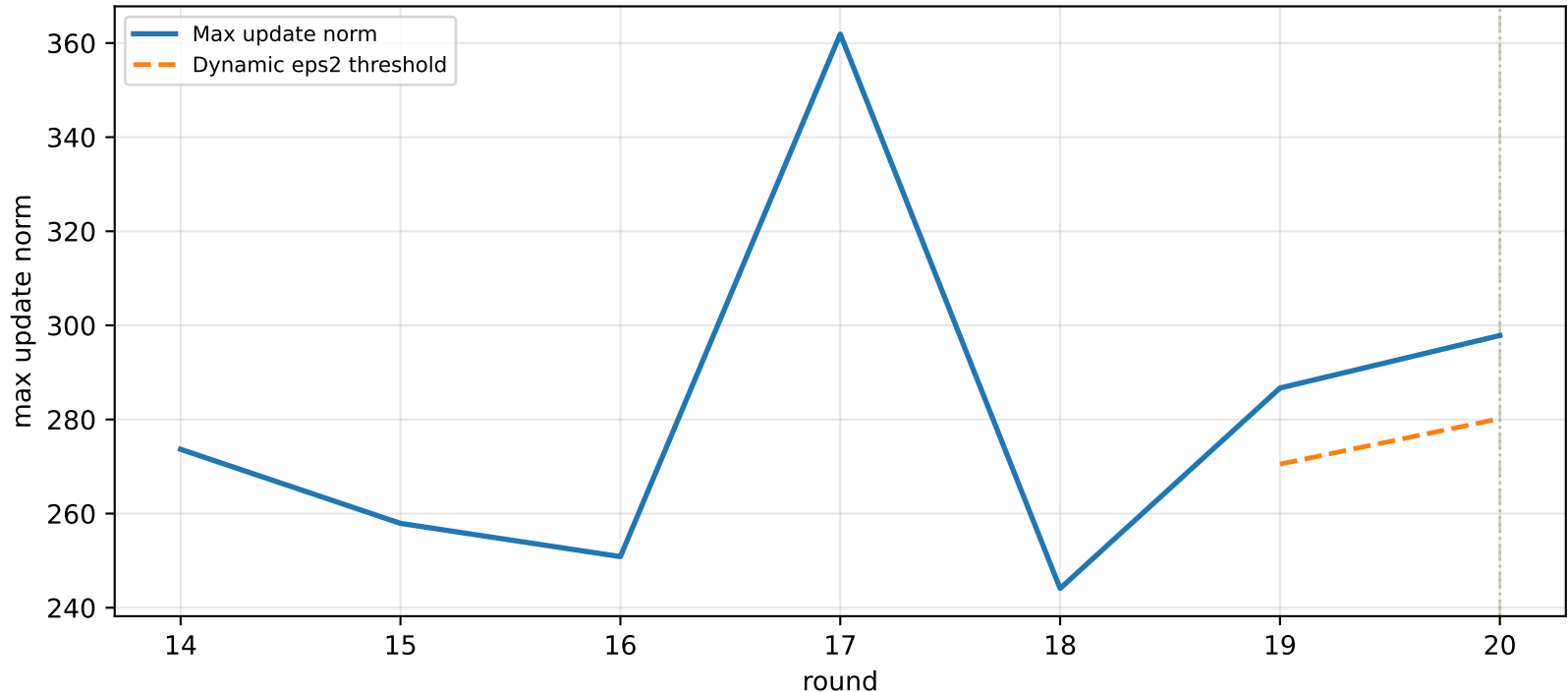
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



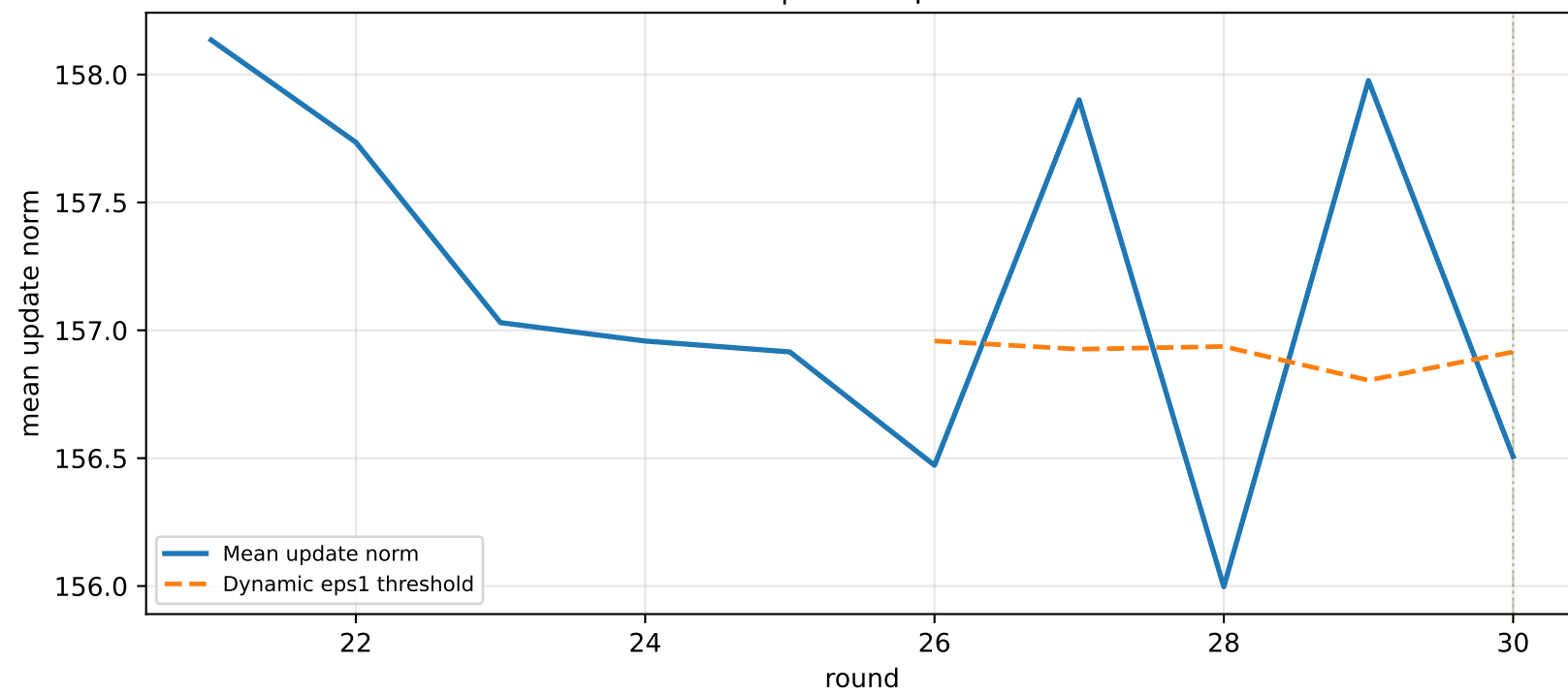
Cluster 2 | Max update norm



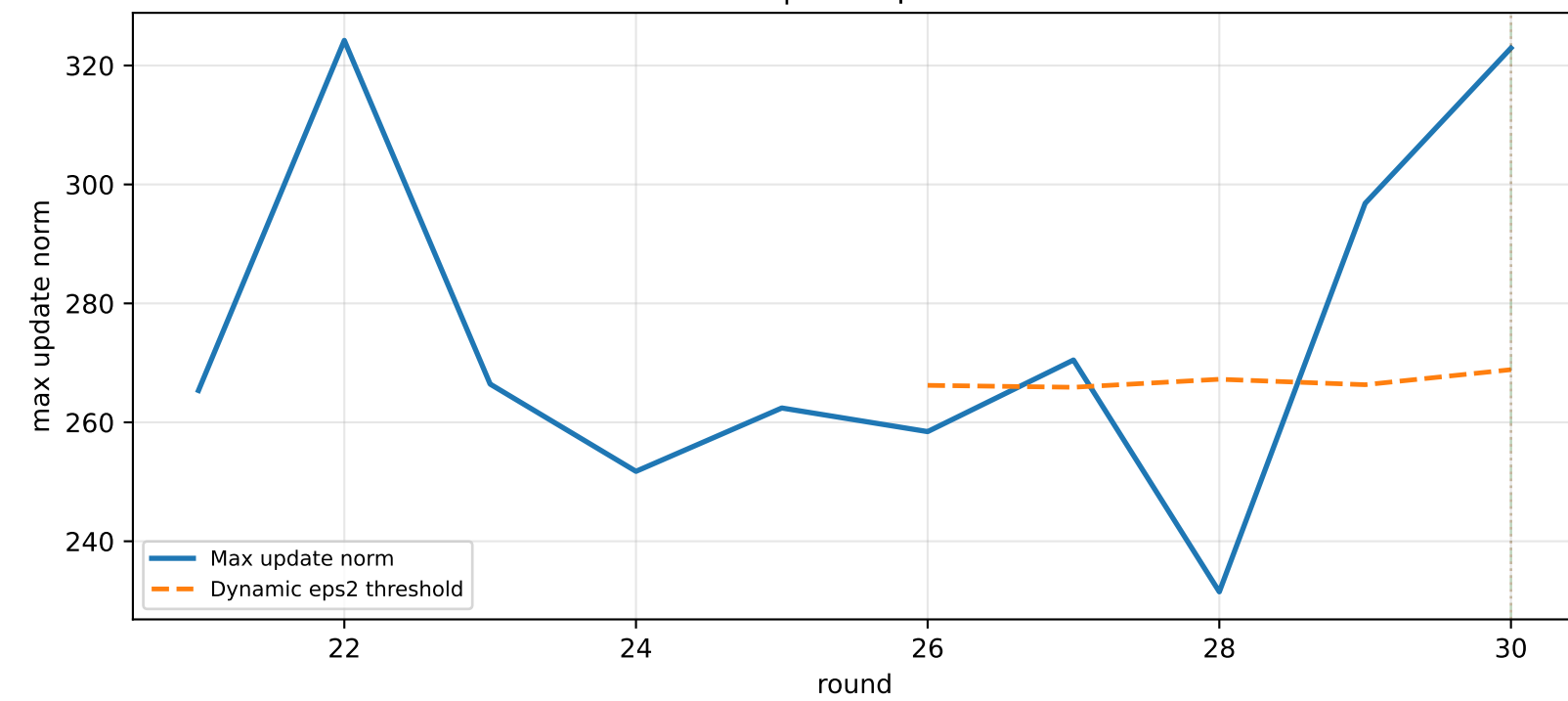
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

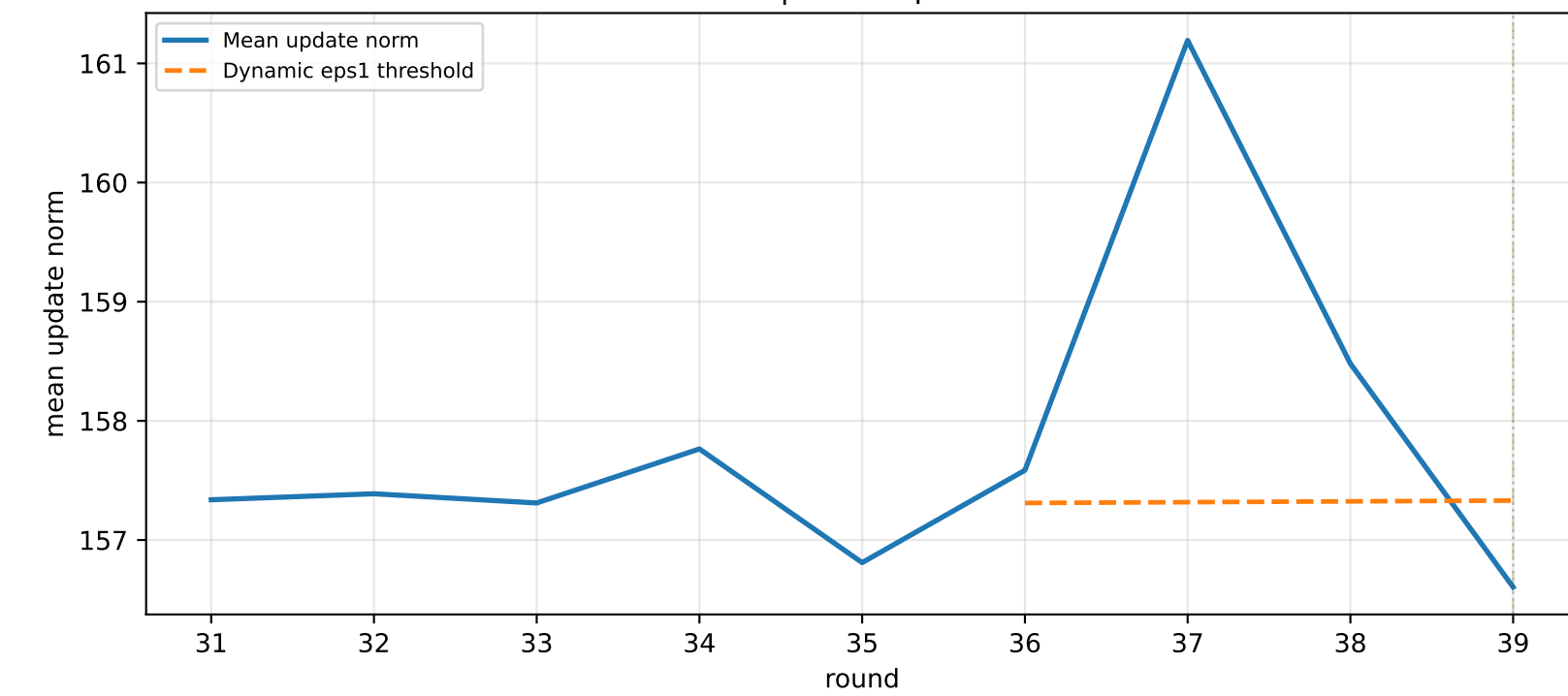
Cluster 4 | Mean update norm



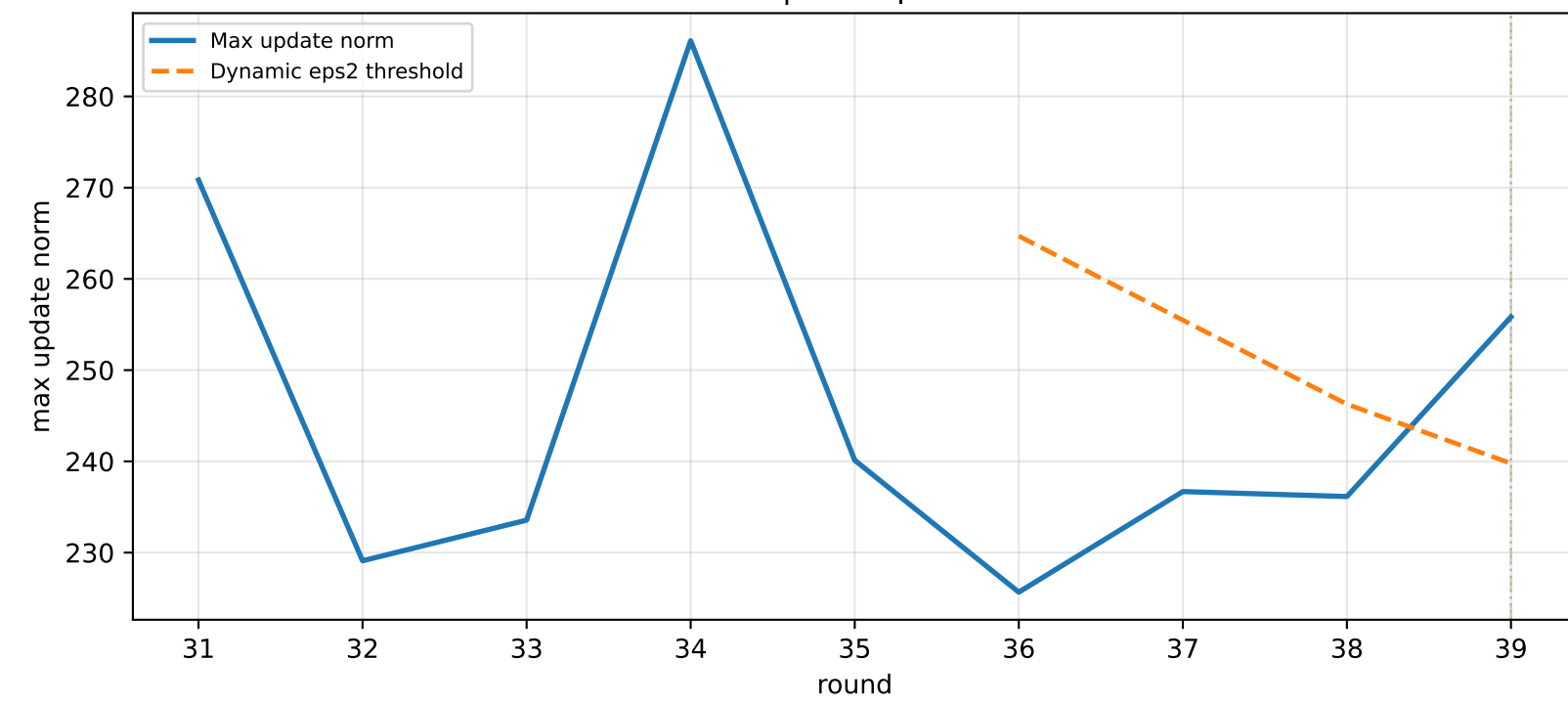
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



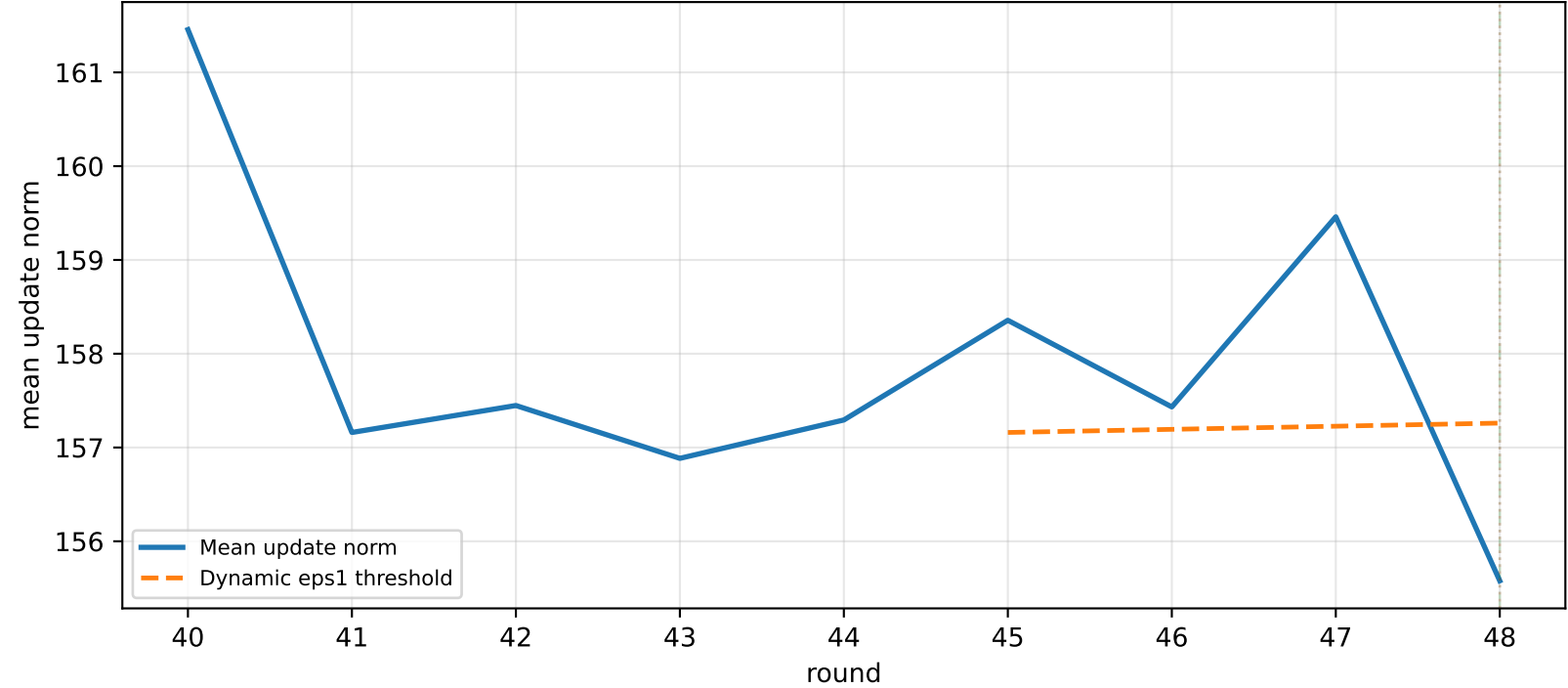
Cluster 6 | Max update norm



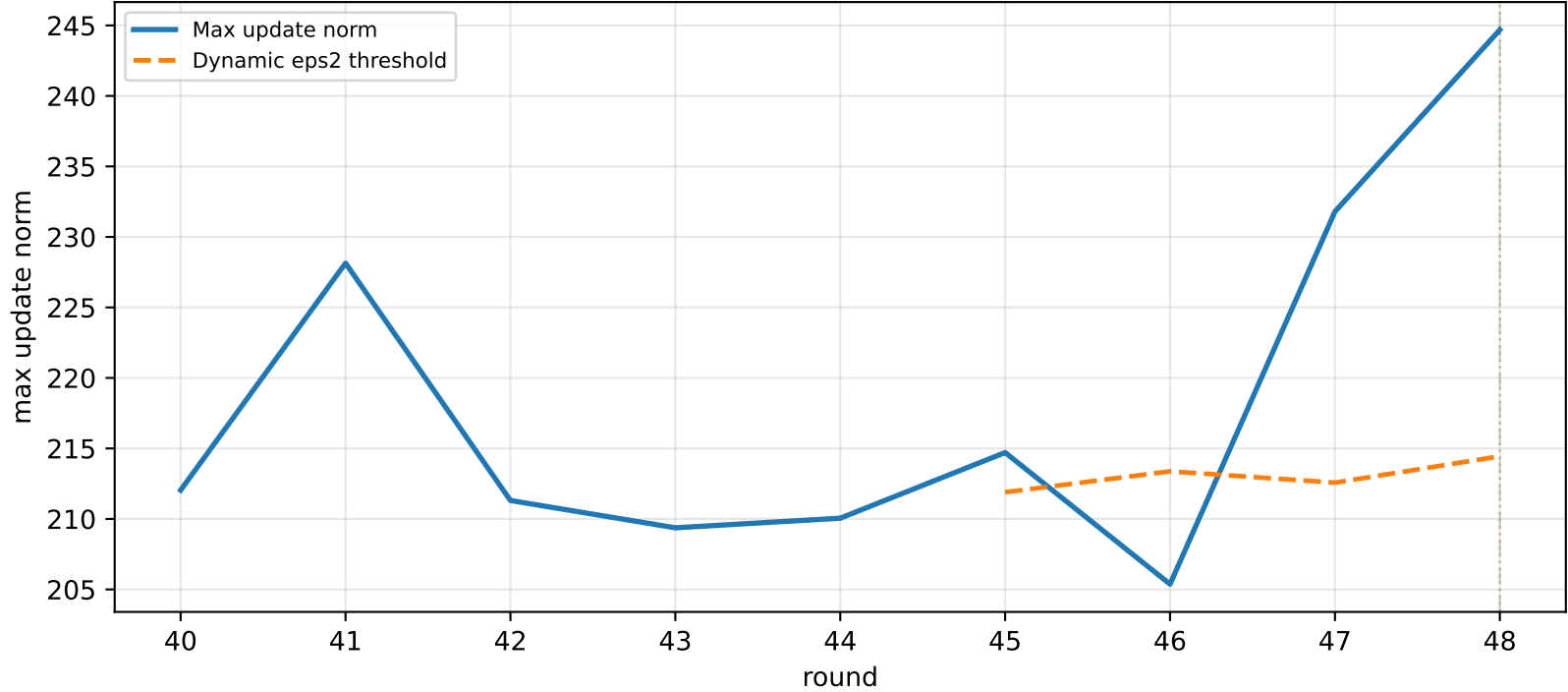
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (8-10)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

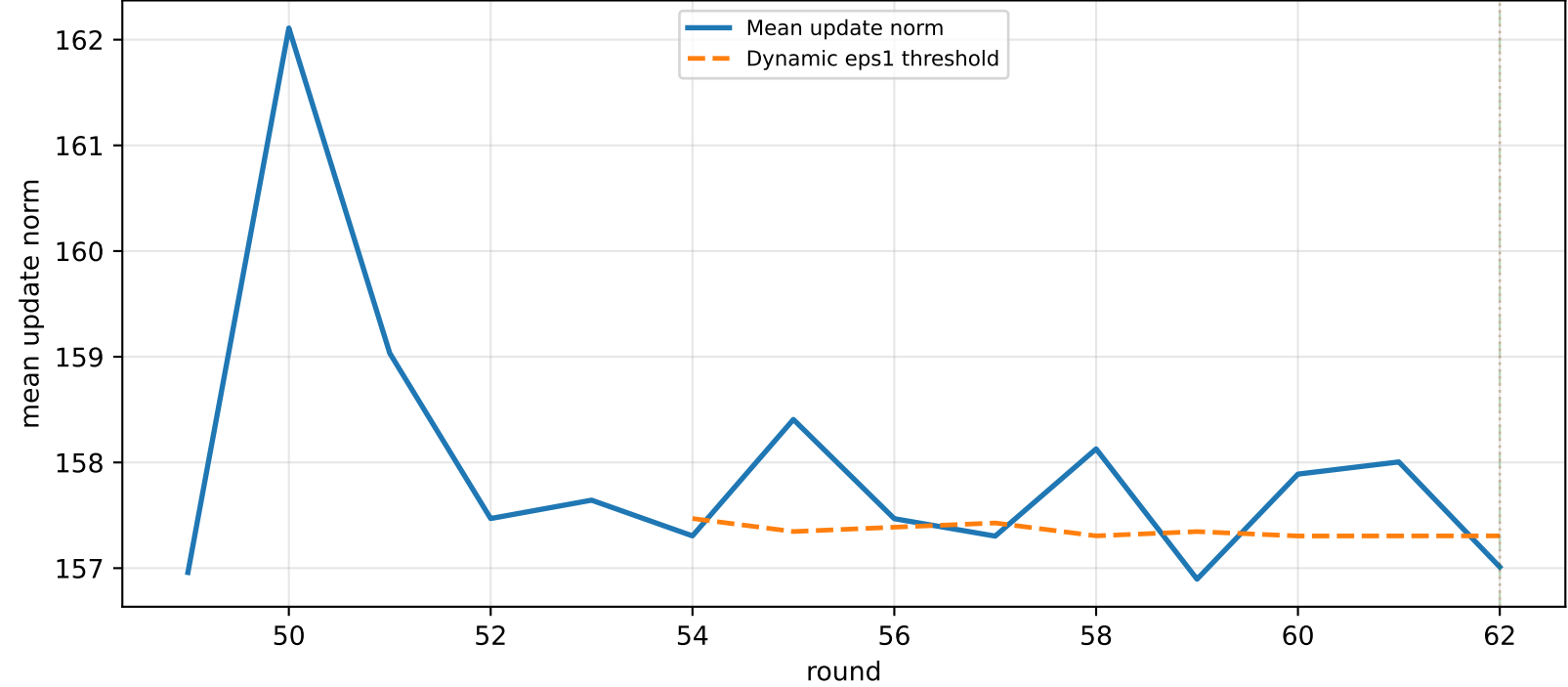
Cluster 8 | Mean update norm



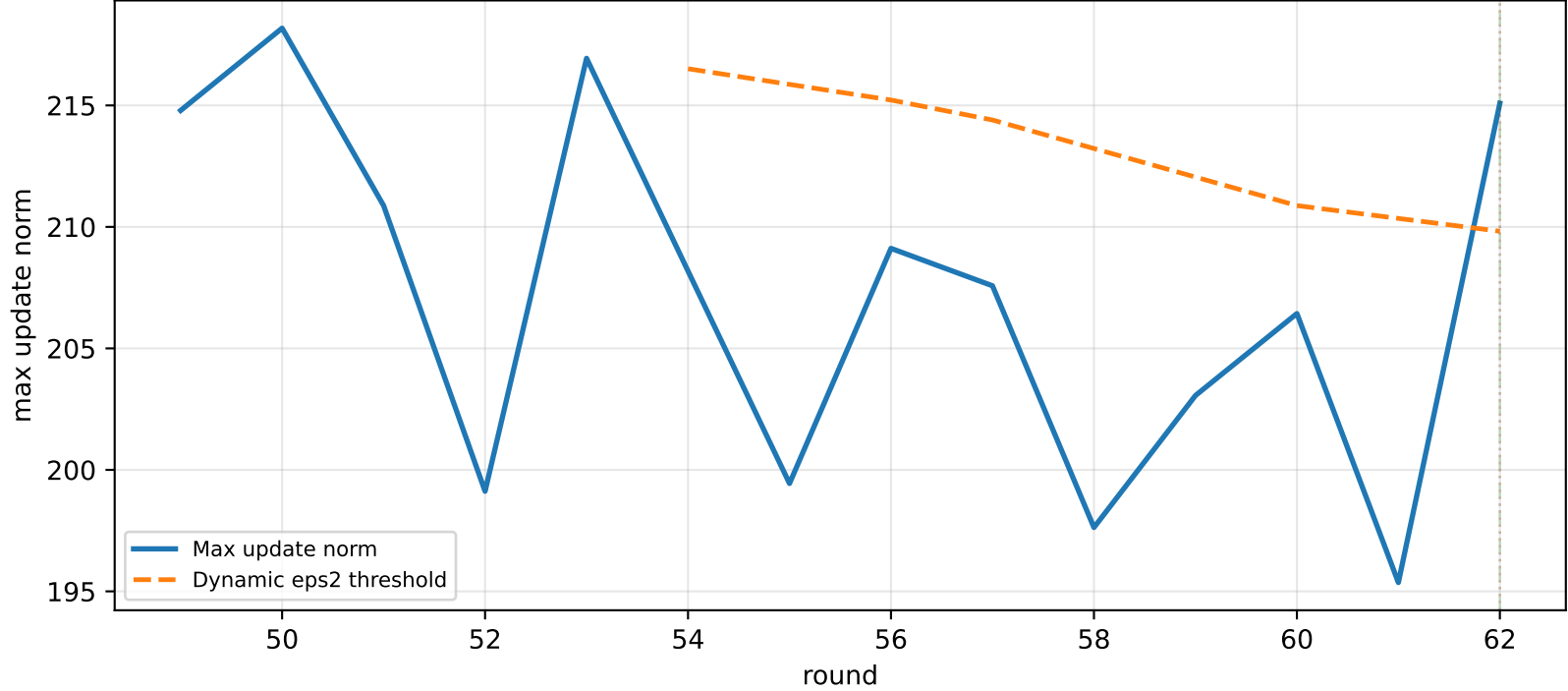
Cluster 8 | Max update norm



Cluster 10 | Mean update norm



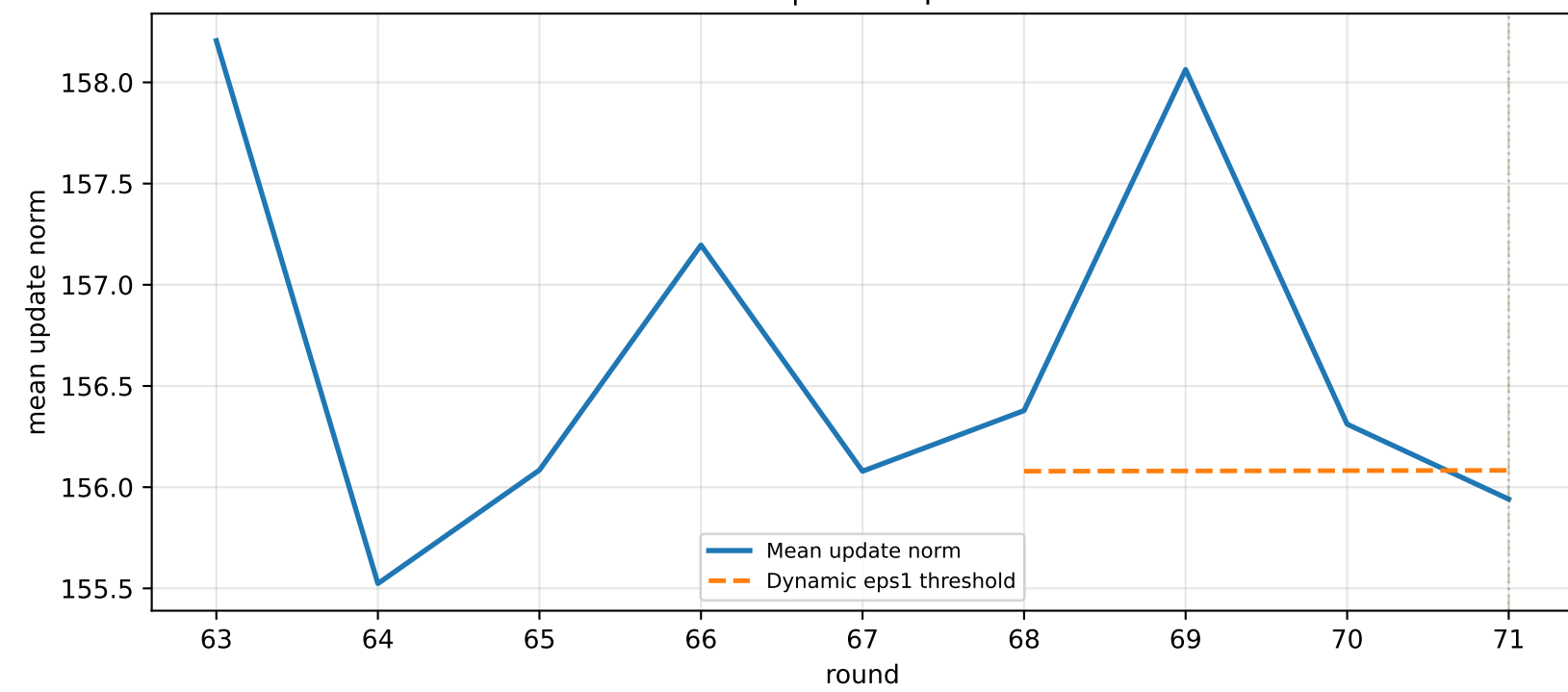
Cluster 10 | Max update norm



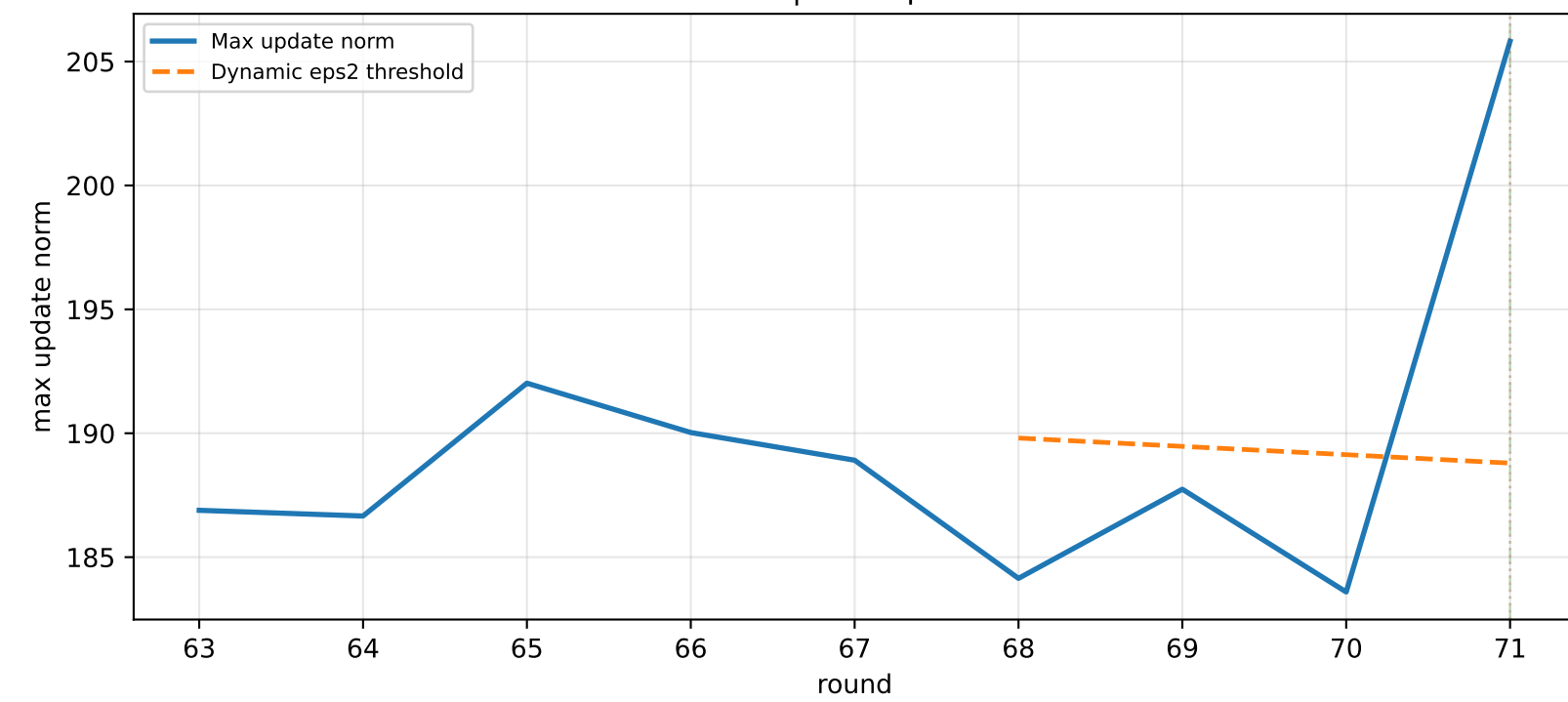
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (12-14)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

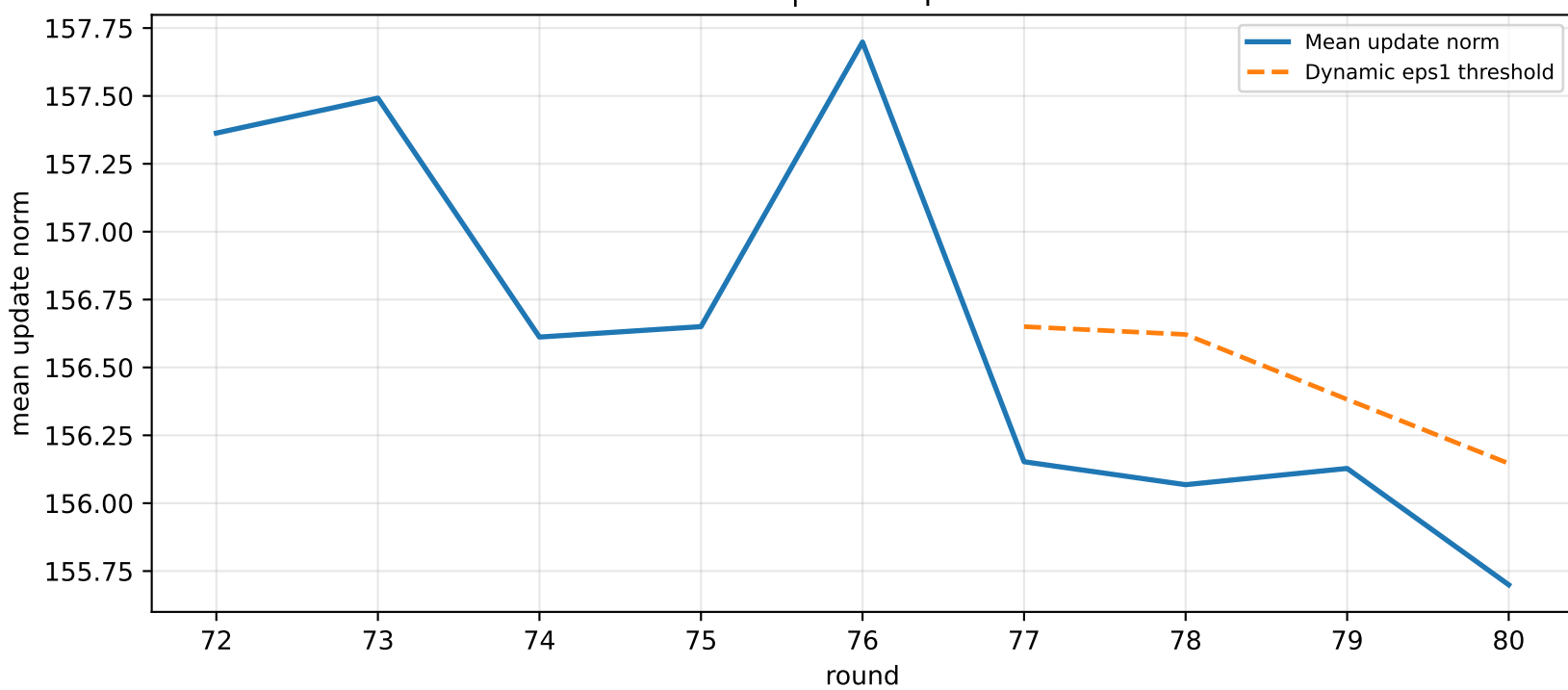
Cluster 12 | Mean update norm



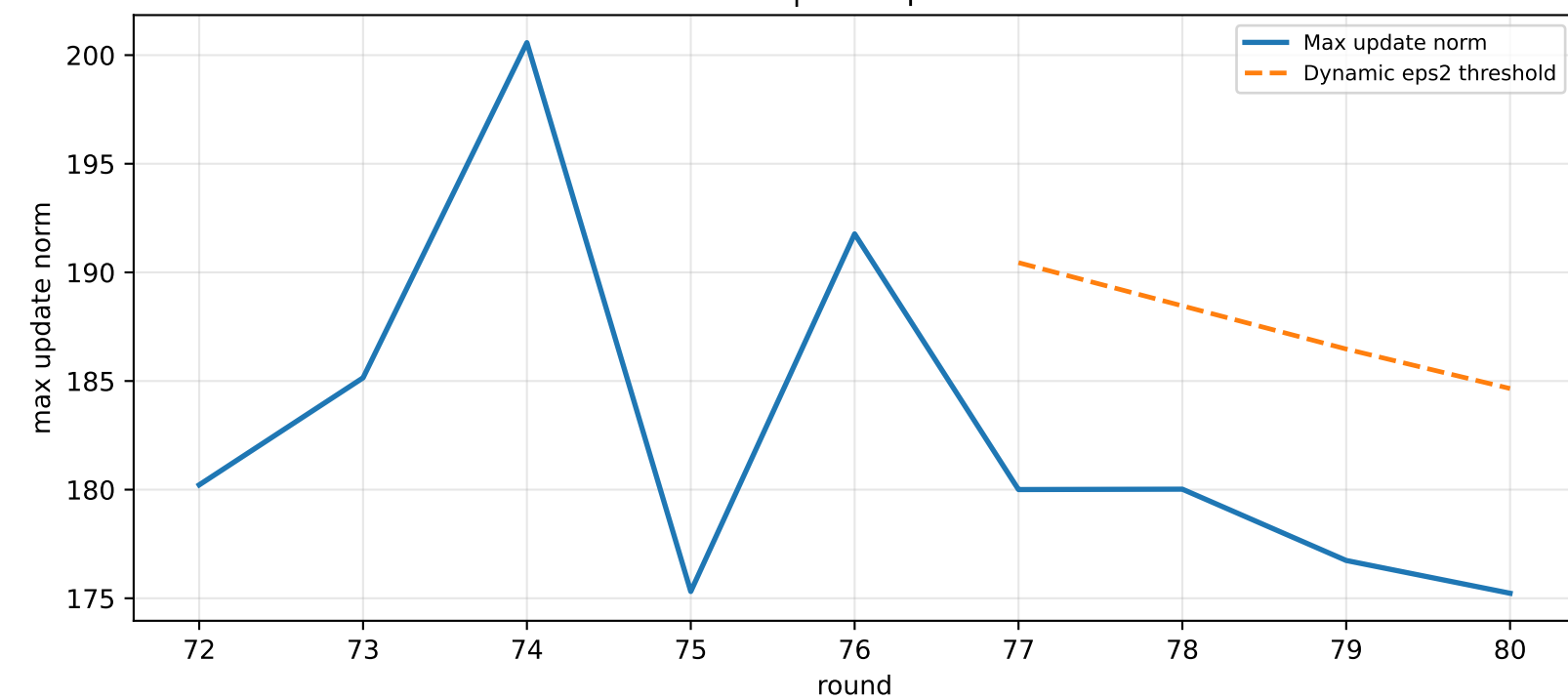
Cluster 12 | Max update norm



Cluster 14 | Mean update norm



Cluster 14 | Max update norm



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

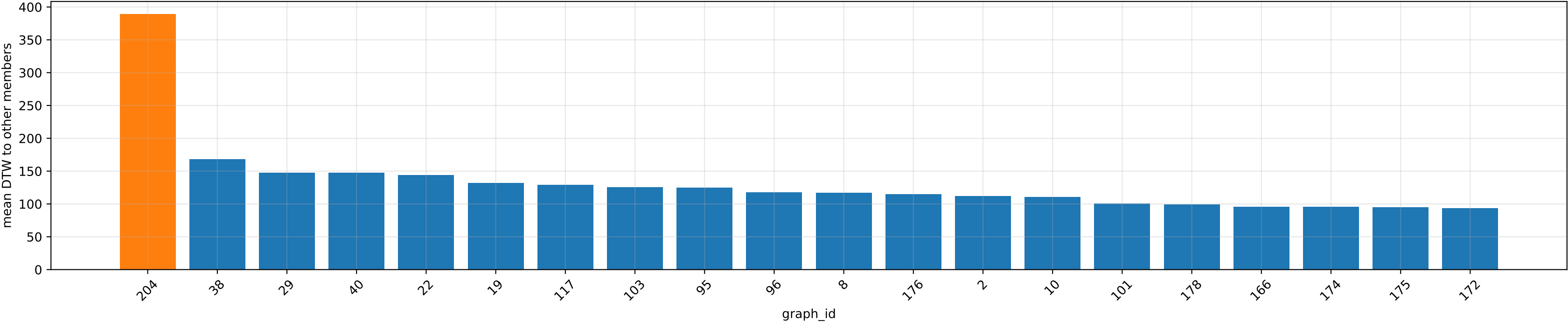
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	16	0	1 vs 19	204	204	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
388.979	nan	104.549

Per-member mean DTW



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

First applied split pairwise DTW metadata | seed=1

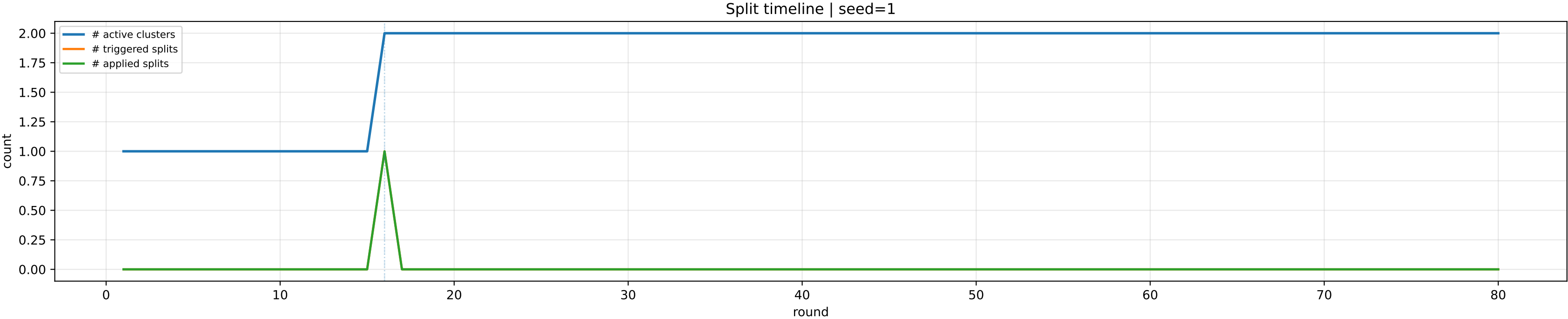
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	16	0	20	204.0	0.051	388.979	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 491.1 at (19, 204)

graph_id	178	96	176	19	174	175	172	95	101	103	22	29	8	166	10	2	40	38	204	117
178	0.0	34.4	92.6	50.8	32.8	28.2	36.9	42.8	57.2	120.4	125.7	160.1	30.9	45.3	41.8	86.8	142.7	186.1	437.2	125.5
96	34.4	0.0	117.4	37.1	62.4	63.6	67.2	14.3	57.3	157.7	126.6	179.8	27.4	83.4	41.6	123.7	185.8	223.4	474.5	162.8
176	92.6	117.4	0.0	134.3	97.4	63.9	56.9	121.6	88.4	84.1	128.5	87.0	111.9	93.3	126.6	70.9	174.6	107.4	357.1	72.7
19	50.8	37.1	134.3	0.0	81.3	81.3	84.2	29.0	78.9	174.3	133.5	202.6	21.2	101.2	48.8	140.4	202.4	240.1	491.1	179.4
174	32.8	62.4	97.4	81.3	0.0	30.0	26.9	72.5	55.1	100.3	112.3	152.4	61.6	29.2	43.5	77.2	113.0	158.6	409.7	100.1
175	28.2	63.6	63.9	81.3	30.0	0.0	11.1	78.8	40.4	92.3	144.4	122.6	64.9	39.4	63.4	85.4	126.5	159.9	410.9	102.5
172	36.9	67.2	56.9	84.2	26.9	11.1	0.0	81.6	37.9	90.5	136.7	109.4	68.5	32.6	63.3	81.5	124.9	155.6	407.3	102.0
95	42.8	14.3	121.6	29.0	72.5	78.8	81.6	0.0	79.1	165.4	125.0	203.2	20.5	92.2	33.9	130.7	193.3	231.1	482.2	170.4
101	57.2	57.3	88.4	78.9	55.1	40.4	37.9	79.1	0.0	90.0	137.6	98.8	70.2	56.0	55.2	96.2	120.8	162.3	417.2	117.4
103	120.4	157.7	84.1	174.3	100.3	92.3	90.5	165.4	90.0	0.0	159.8	71.1	154.5	94.5	136.4	101.7	112.1	88.3	316.8	76.5
22	125.7	126.6	128.5	133.5	112.3	144.4	136.7	125.0	137.6	159.8	0.0	192.4	122.6	93.2	122.1	58.1	129.7	189.3	374.9	128.7
29	160.1	179.8	87.0	202.6	152.4	122.6	109.4	203.2	98.8	71.1	192.4	0.0	187.2	132.1	163.1	141.2	114.5	88.1	292.6	104.7
8	30.9	27.4	111.9	21.2	61.6	64.9	68.5	20.5	70.2	154.5	122.6	187.2	0.0	81.4	40.9	120.6	182.6	220.3	471.3	159.6
166	45.3	83.4	93.3	101.2	29.2	39.4	32.6	92.2	56.0	94.5	93.2	132.1	81.4	0.0	62.4	48.0	101.8	152.4	389.9	89.3
10	41.8	41.6	126.6	48.8	43.5	63.4	63.3	33.9	55.2	136.4	122.1	163.1	40.9	62.4	0.0	108.4	140.5	207.9	458.9	144.6
2	86.8	123.7	70.9	140.4	77.2	85.4	81.5	130.7	96.2	101.7	58.1	141.2	120.6	48.0	108.4	0.0	71.6	154.0	350.7	85.0
40	142.7	185.8	174.6	202.4	113.0	126.5	124.9	193.3	120.8	112.1	129.7	114.5	182.6	101.8	140.5	71.6	0.0	136.9	285.5	139.0
38	186.1	223.4	107.4	240.1	158.6	159.9	155.6	231.1	162.3	88.3	189.3	88.1	220.3	152.4	207.9	154.0	136.9	0.0	251.1	79.8
204	437.2	474.5	357.1	491.1	409.7	410.9	407.3	482.2	417.2	316.8	374.9	292.6	471.3	389.9	458.9	350.7	285.5	251.1	0.0	311.7
117	125.5	162.8	72.7	179.4	100.1	102.5	102.0	170.4	117.4	76.5	128.7	104.7	159.6	89.3	144.6	85.0	139.0	79.8	311.7	0.0

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

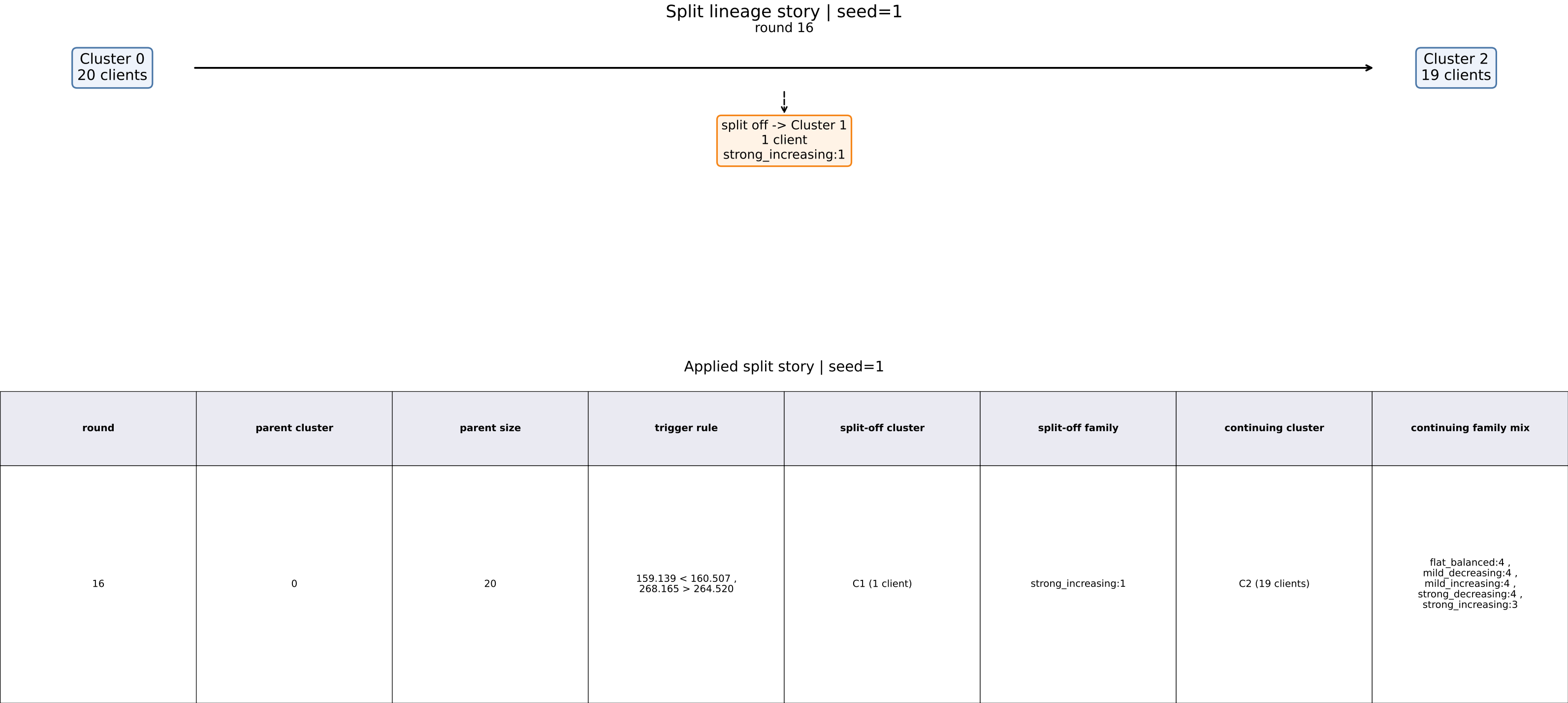


Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	16	16	1, 2	1:1, 2:19

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

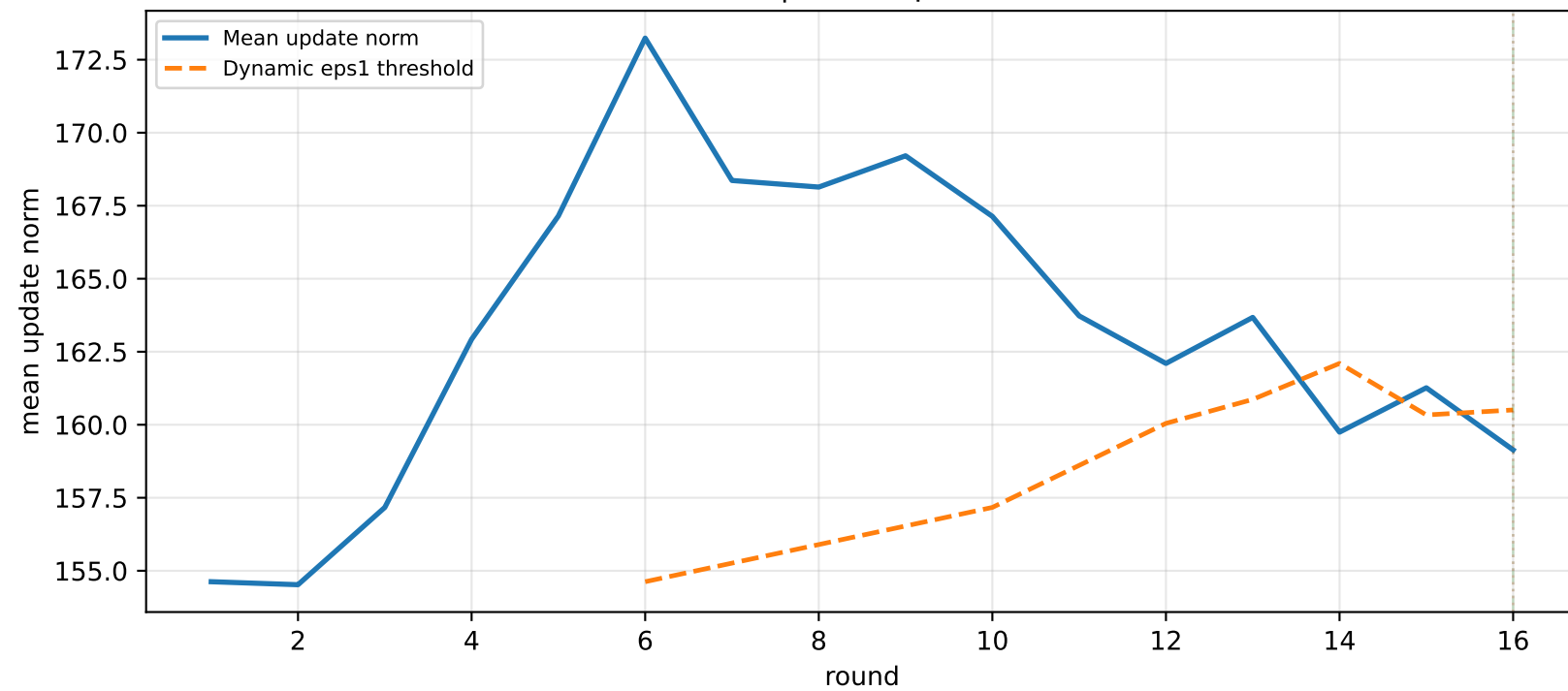
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	204	18	strong_increasing:1
2	19	178, 96, 176, 19, 174, 175, 172, 95, 101, 103, 22, 29, 8, 166, 10, 2, 40, 38, 117	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3

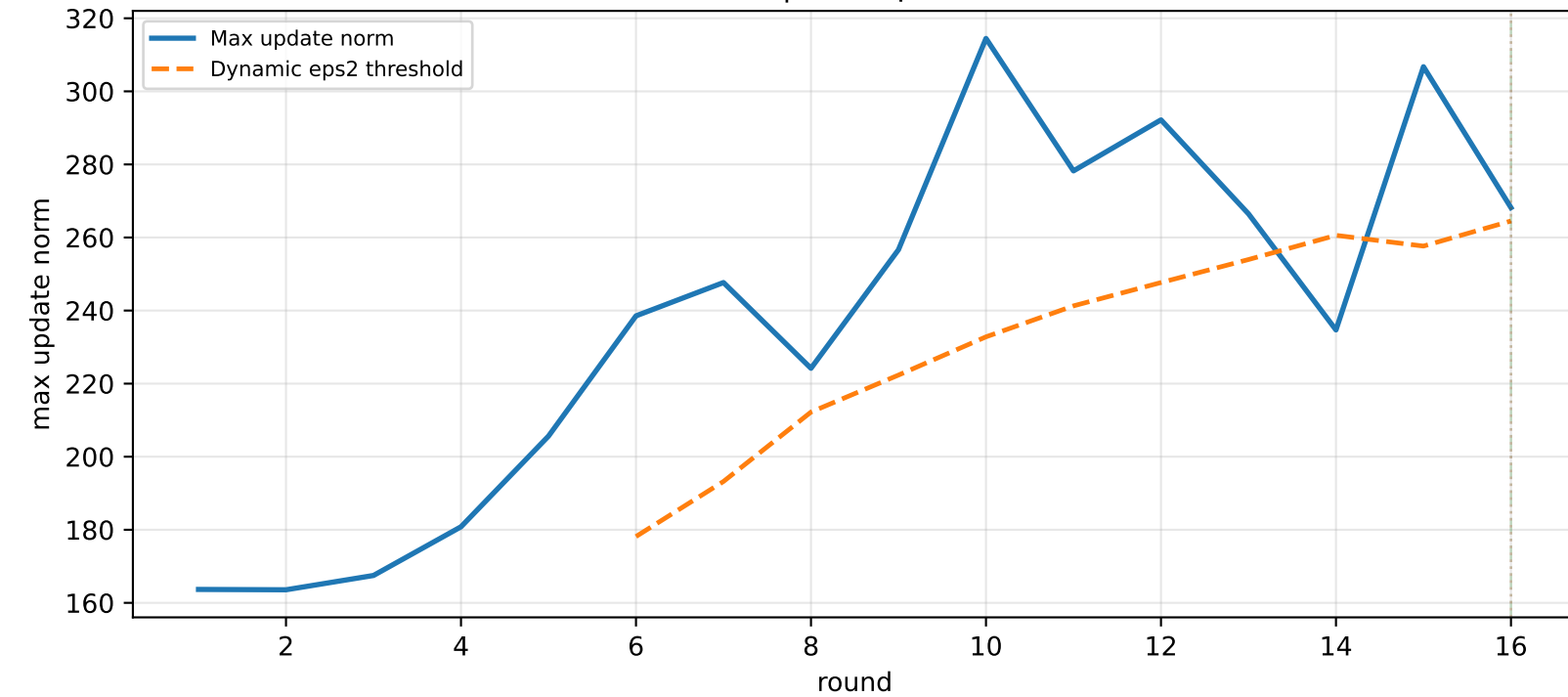
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

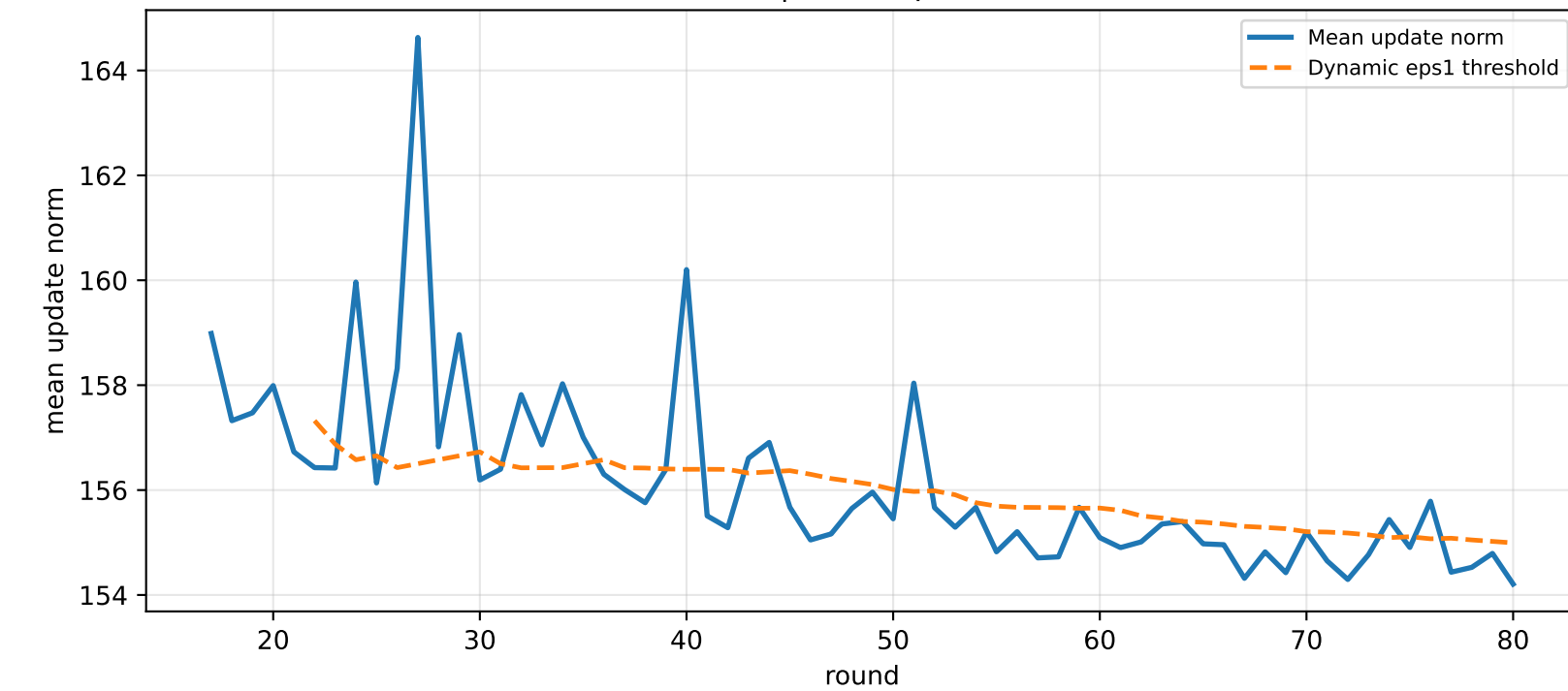
Cluster 0 | Mean update norm



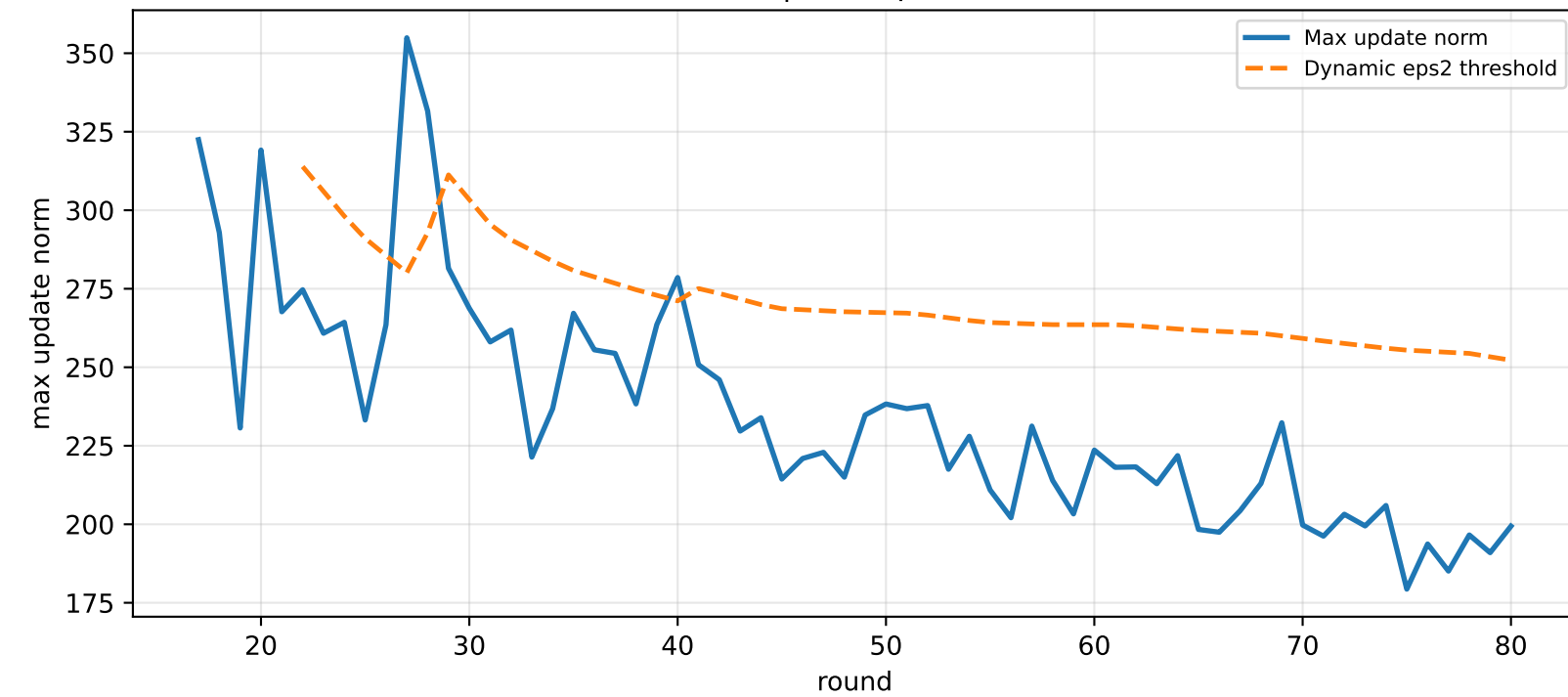
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



Cluster 2 | Max update norm



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

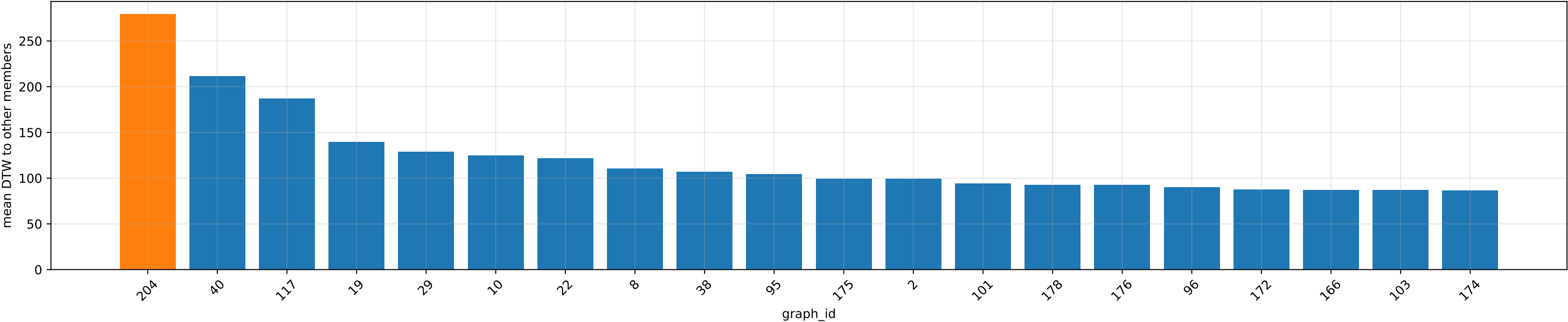
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	13	0	1 vs 19	204	204	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
279.213	nan	103.966

Per-member mean DTW



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

First applied split pairwise DTW metadata | seed=2

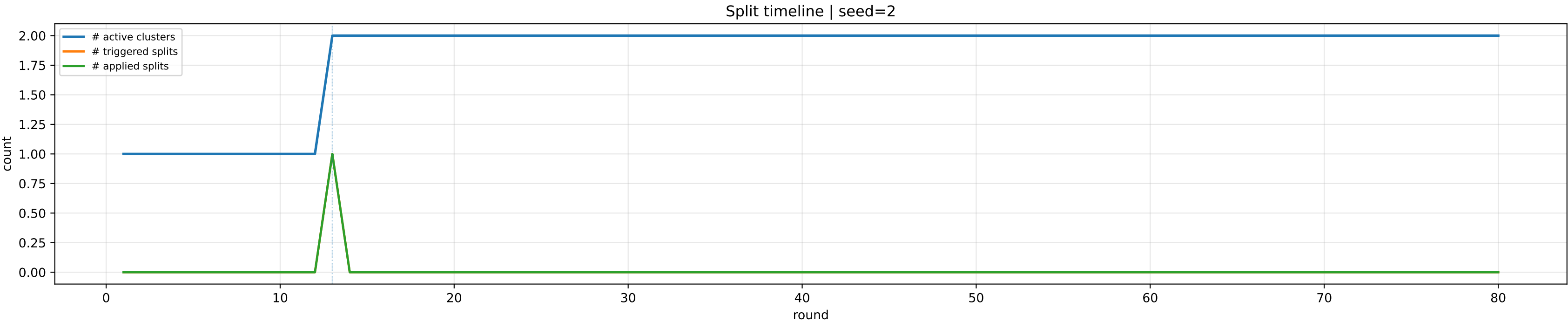
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	13	0	20	204.0	0.074	279.213	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 382.4 at (10, 204)

graph_id	178	96	176	19	174	175	172	95	101	103	22	29	8	166	10	2	40	38	204	117
178	0.0	49.4	46.6	104.5	54.6	73.6	66.6	71.8	57.9	35.0	98.8	92.7	88.4	57.1	102.2	61.8	201.4	42.6	286.9	170.9
96	49.4	0.0	36.9	87.3	40.4	59.2	41.0	57.2	75.2	38.5	81.1	121.0	52.3	43.6	68.7	65.3	221.3	74.9	305.5	195.2
176	46.6	36.9	0.0	70.8	58.8	77.9	57.9	74.3	77.7	46.4	51.1	132.5	87.1	45.6	103.8	96.4	209.9	55.3	256.1	172.0
19	104.5	87.3	70.8	0.0	120.5	140.7	111.3	89.2	146.5	101.3	75.9	200.1	87.4	122.6	86.7	153.8	272.3	151.6	301.6	226.0
174	54.6	40.4	58.8	120.5	0.0	36.4	24.3	52.3	49.4	19.8	92.7	89.6	51.8	19.1	73.8	55.7	229.9	71.3	304.3	199.4
175	73.6	59.2	77.9	140.7	36.4	0.0	59.8	88.8	86.6	53.6	97.2	116.5	87.2	48.4	109.7	83.6	193.3	64.2	244.2	163.5
172	66.6	41.0	57.9	111.3	24.3	59.8	0.0	32.1	44.0	35.7	82.2	110.7	30.4	13.0	51.2	68.0	237.2	83.4	311.7	206.8
95	71.8	57.2	74.3	89.2	52.3	88.8	32.1	0.0	76.7	49.7	97.5	119.7	39.6	45.1	30.9	91.2	269.3	111.8	343.8	238.9
101	57.9	75.2	77.7	146.5	49.4	86.6	44.0	76.7	0.0	60.4	116.4	79.6	79.8	41.3	105.2	34.0	154.7	77.0	281.6	147.8
103	35.0	38.5	46.4	101.3	19.8	53.6	35.7	49.7	60.4	0.0	85.7	93.6	59.7	29.9	73.5	57.9	231.5	72.9	306.0	201.1
22	98.8	81.1	51.1	75.9	92.7	97.2	82.2	97.5	116.4	85.7	0.0	151.7	87.9	83.9	106.1	121.4	280.9	87.6	284.4	225.3
29	92.7	121.0	132.5	200.1	89.6	116.5	110.7	119.7	79.6	93.6	151.7	0.0	134.1	109.2	138.5	47.5	166.6	128.4	246.8	170.6
8	88.4	52.3	87.1	87.4	51.8	87.2	30.4	39.6	79.8	59.7	87.9	134.1	0.0	53.0	22.8	78.6	294.1	133.1	368.6	263.7
166	57.1	43.6	45.6	122.6	19.1	48.4	13.0	45.1	41.3	29.9	83.9	109.2	53.0	0.0	82.3	69.9	224.2	72.1	298.7	193.8
10	102.2	68.7	103.8	86.7	73.8	109.7	51.2	30.9	105.2	73.5	106.1	138.5	22.8	82.3	0.0	92.5	307.9	150.4	382.4	277.5
2	61.8	65.3	96.4	153.8	55.7	83.6	68.0	91.2	34.0	57.9	121.4	47.5	78.6	69.9	92.5	0.0	166.0	86.6	293.2	159.4
40	201.4	221.3	209.9	272.3	229.9	193.3	237.2	269.3	154.7	231.5	280.9	166.6	294.1	224.2	307.9	166.0	0.0	177.3	115.2	62.5
38	42.6	74.9	55.3	151.6	71.3	64.2	83.4	111.8	77.0	72.9	87.6	128.4	133.1	72.1	150.4	86.6	177.3	0.0	242.0	148.1
204	286.9	305.5	256.1	301.6	304.3	244.2	311.7	343.8	281.6	306.0	284.4	246.8	368.6	298.7	382.4	293.2	115.2	242.0	0.0	132.1
117	170.9	195.2	172.0	226.0	199.4	163.5	206.8	238.9	147.8	201.1	225.3	170.6	263.7	193.8	277.5	159.4	62.5	148.1	132.1	0.0

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

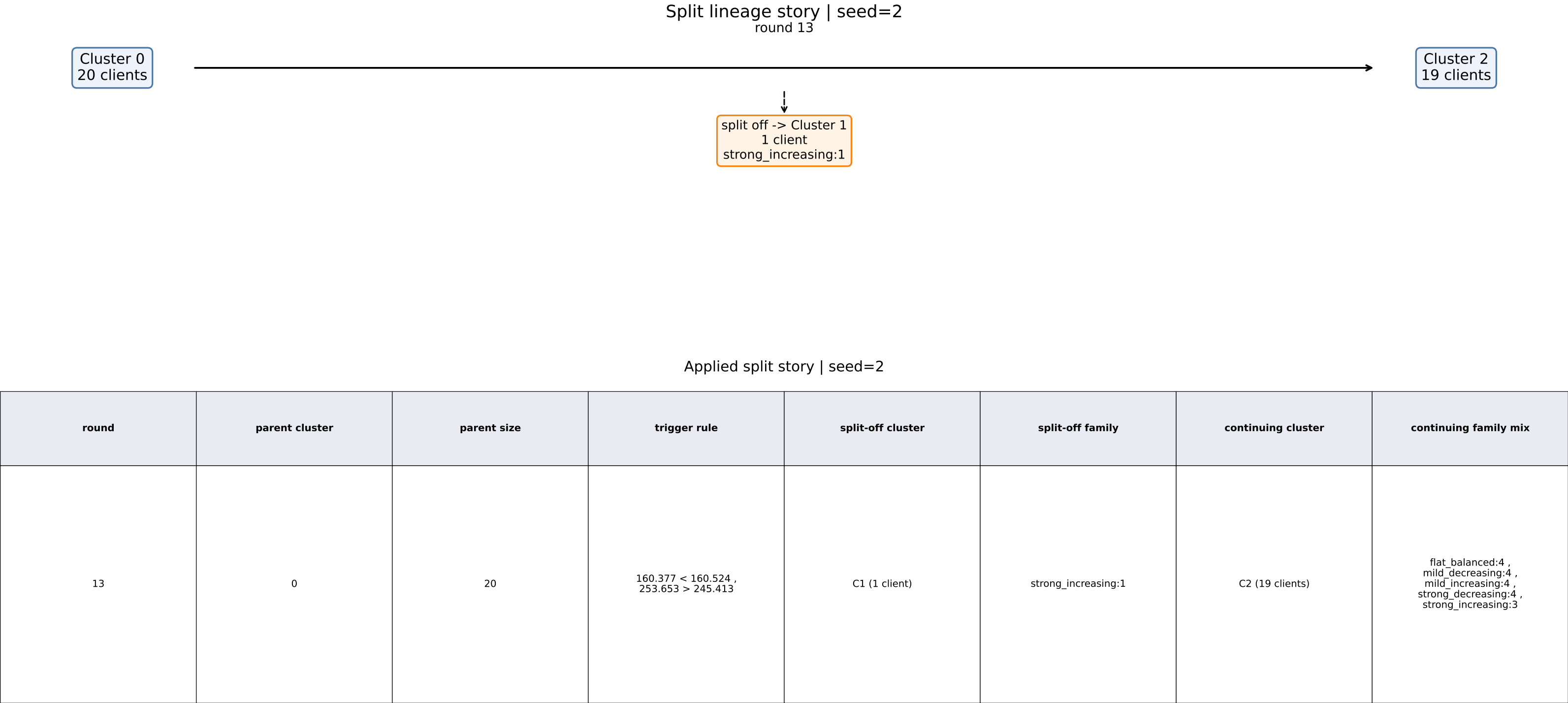


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	13	13	1, 2	1:1, 2:19

Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

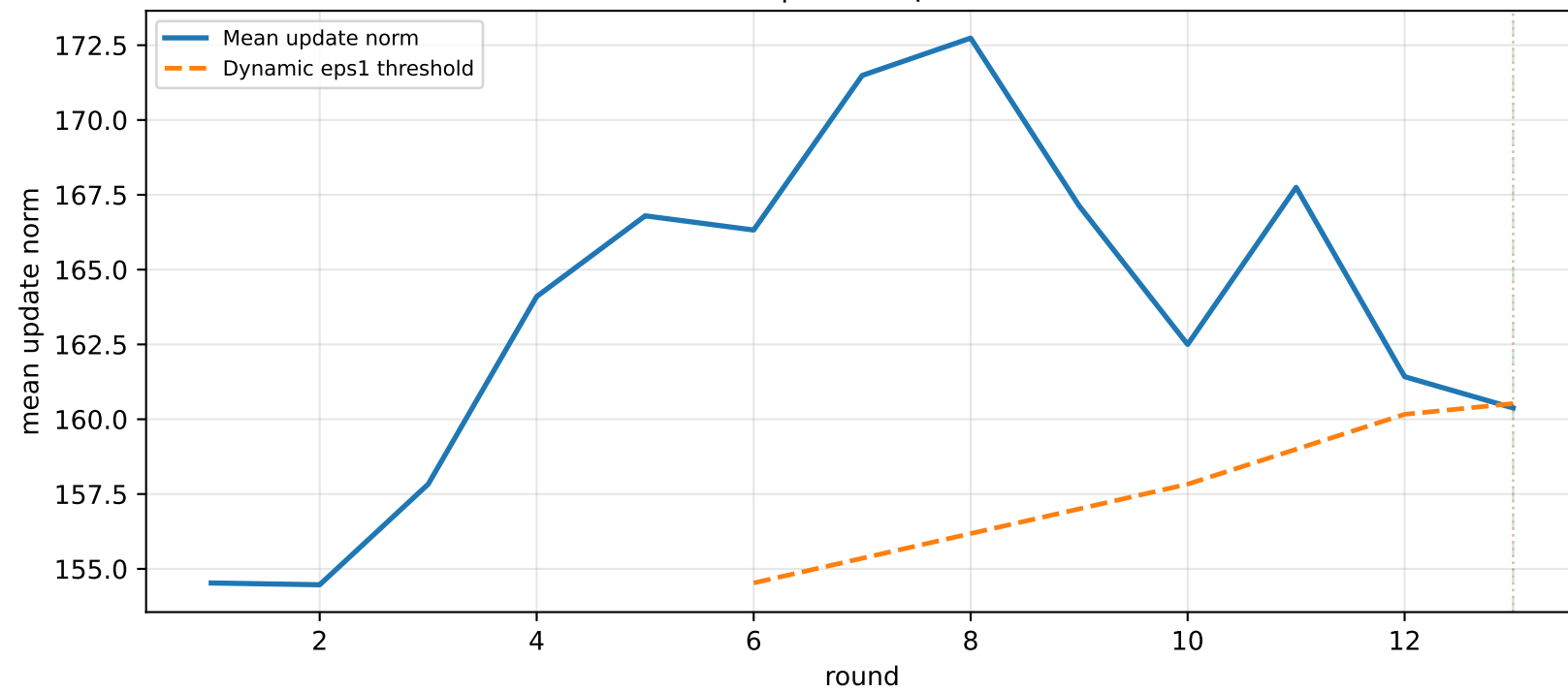
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	204	18	strong_increasing:1
2	19	178, 96, 176, 19, 174, 175, 172, 95, 101, 103, 22, 29, 8, 166, 10, 2, 40, 38, 117	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 19	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3

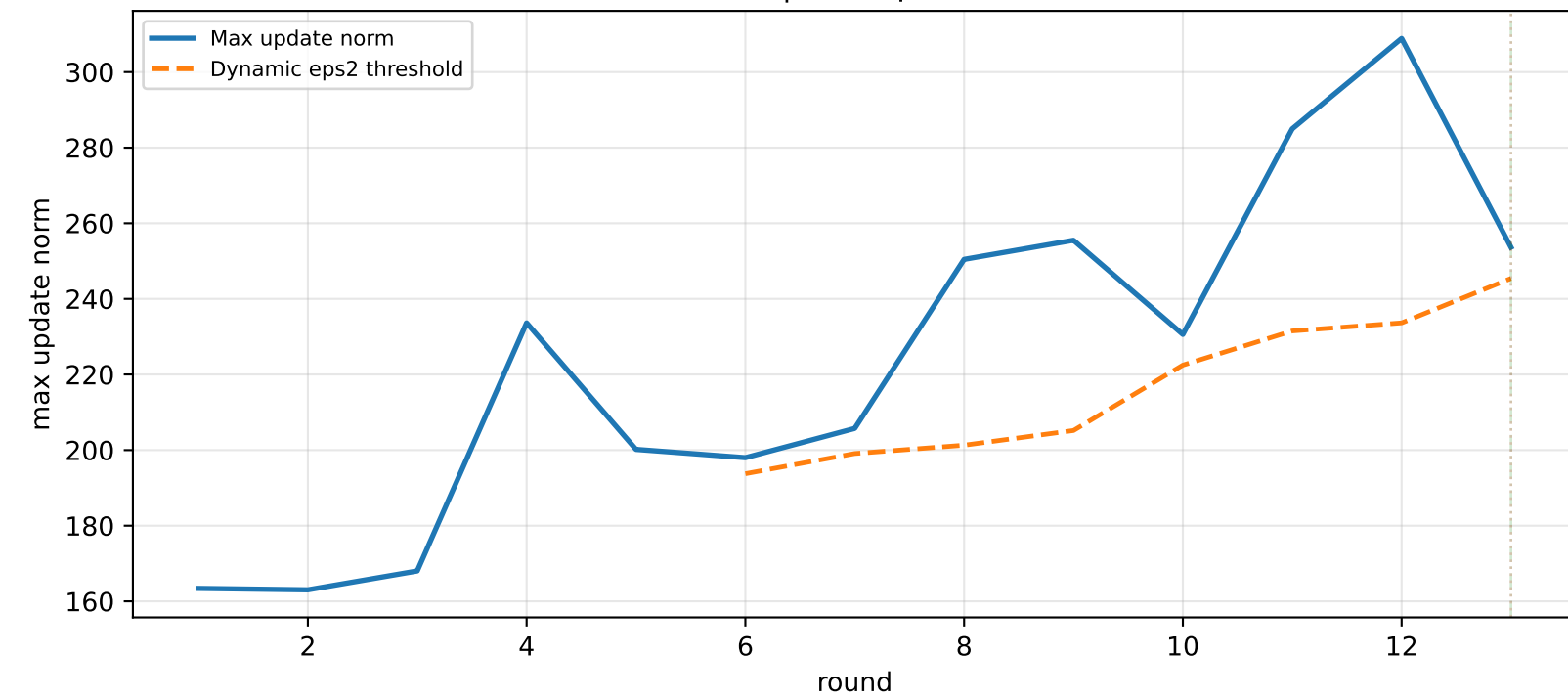
Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

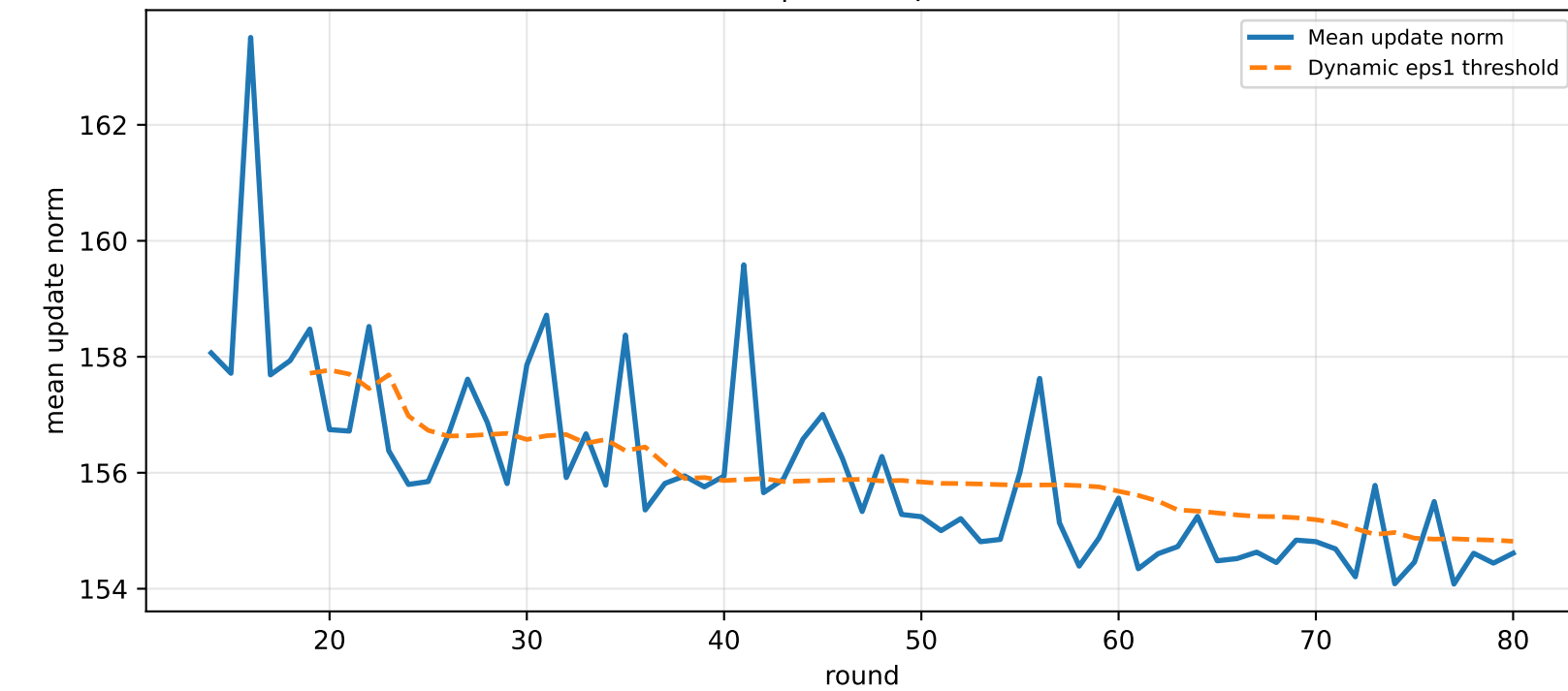
Cluster 0 | Mean update norm



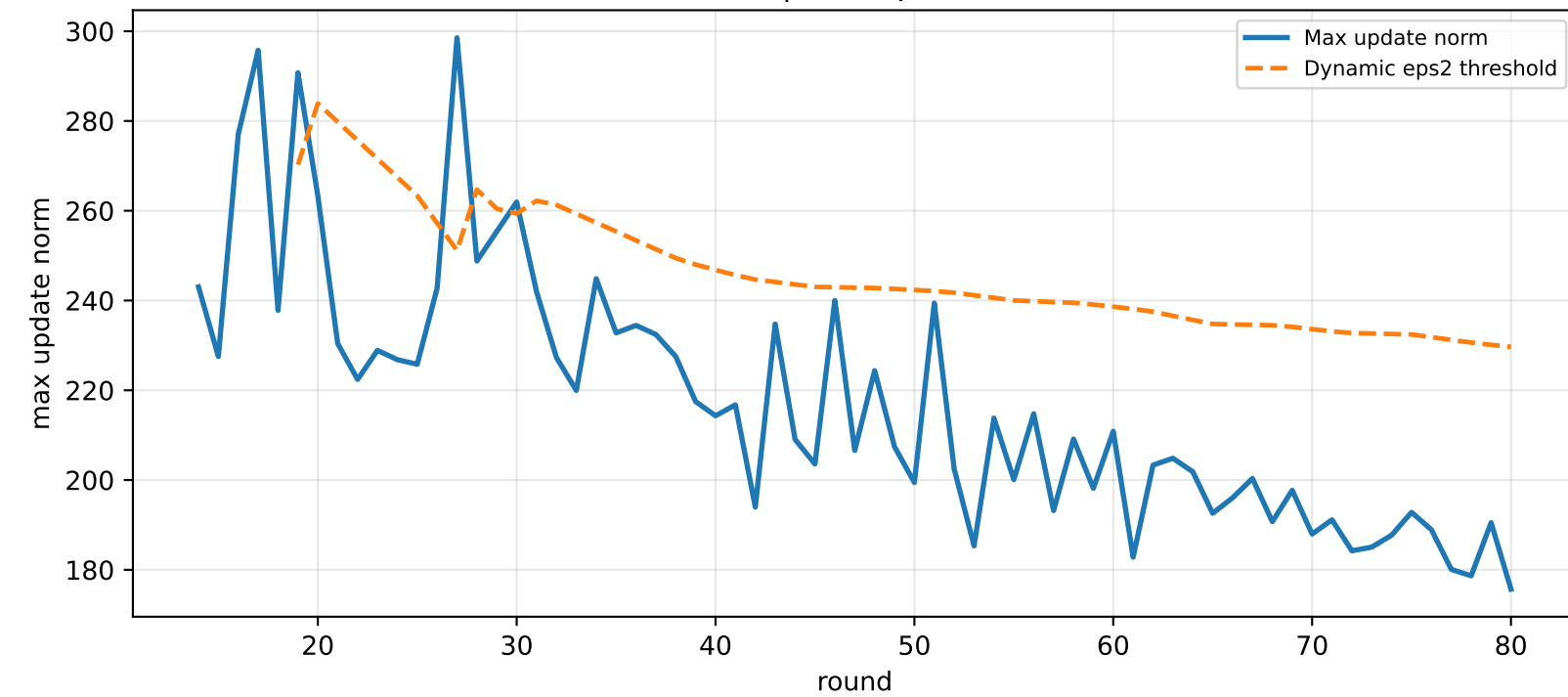
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



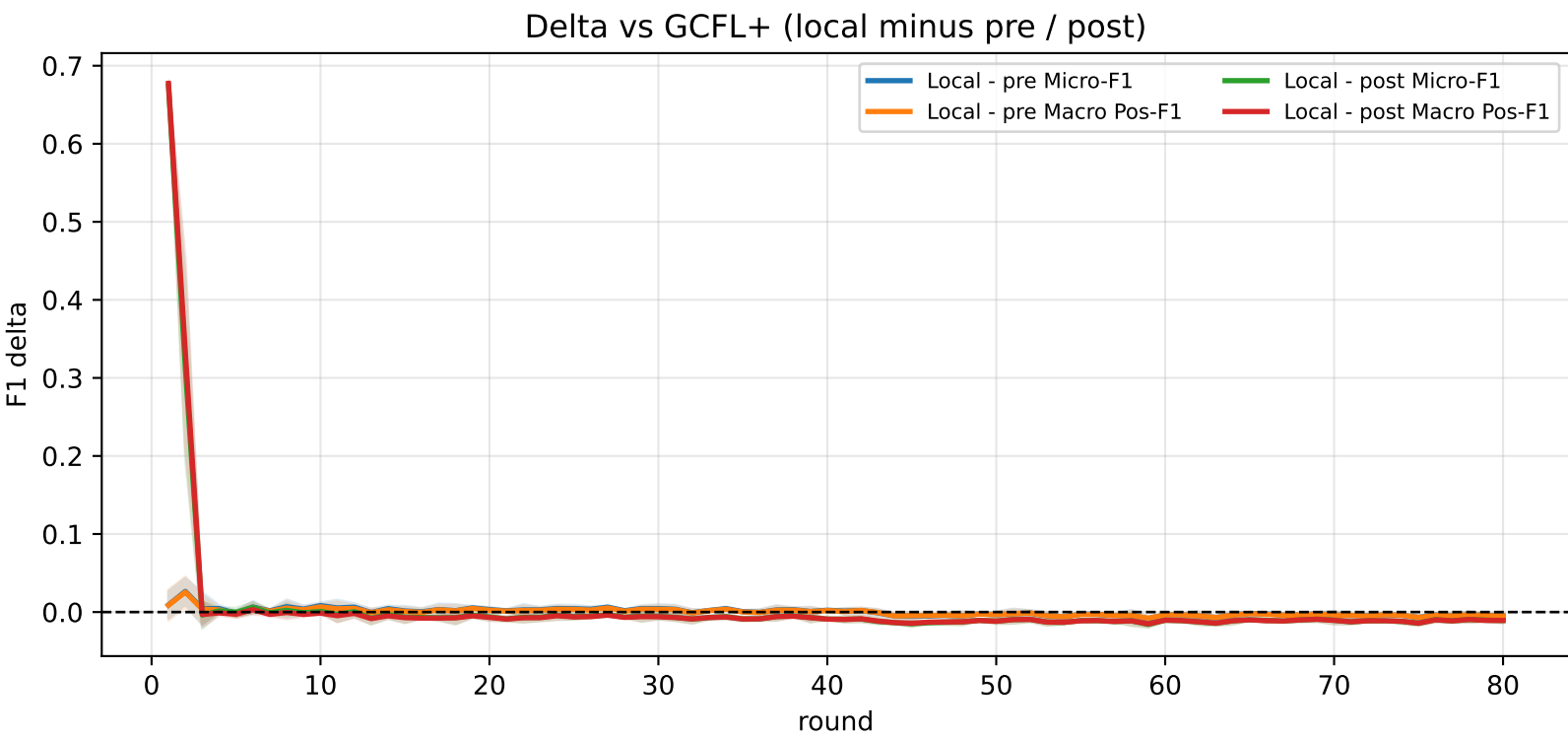
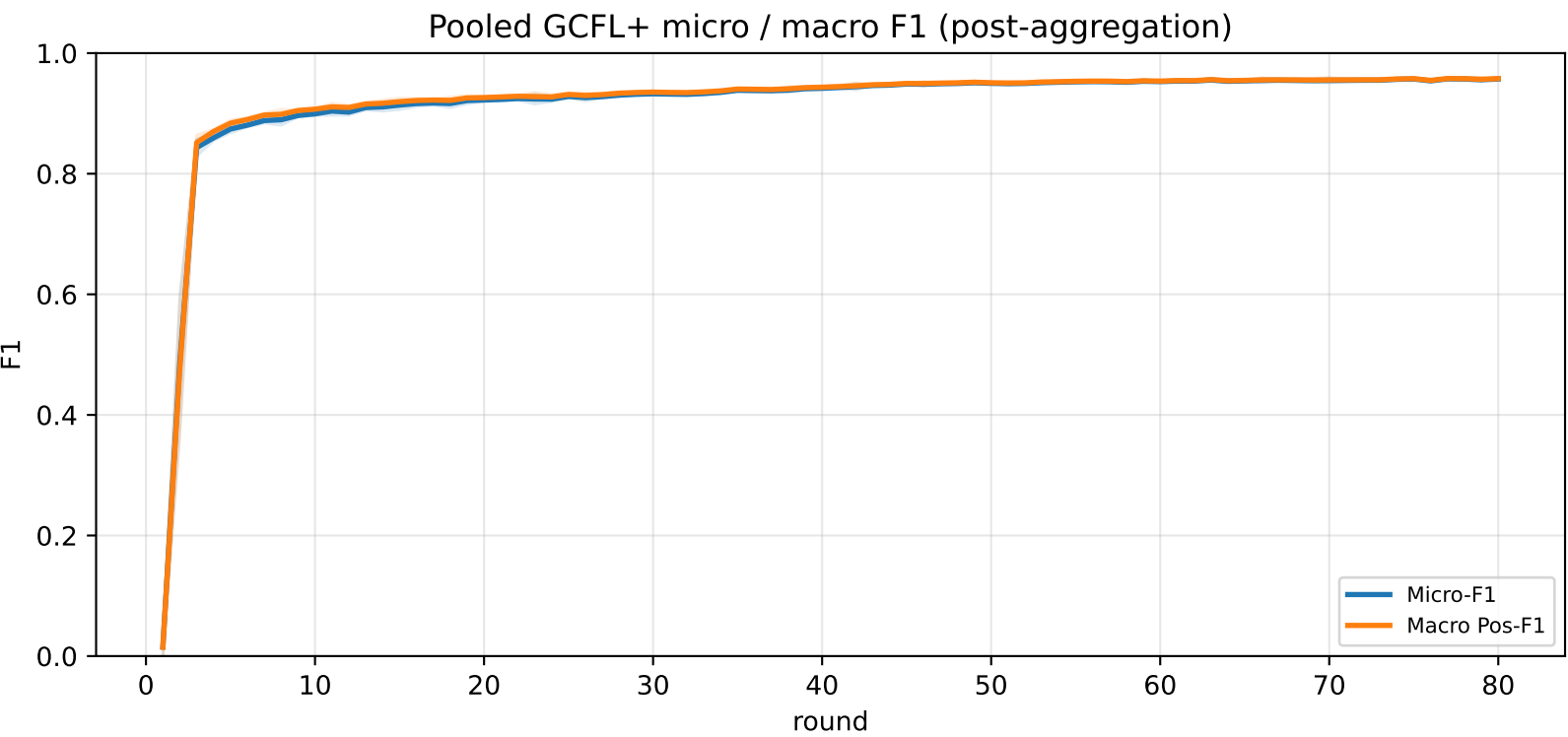
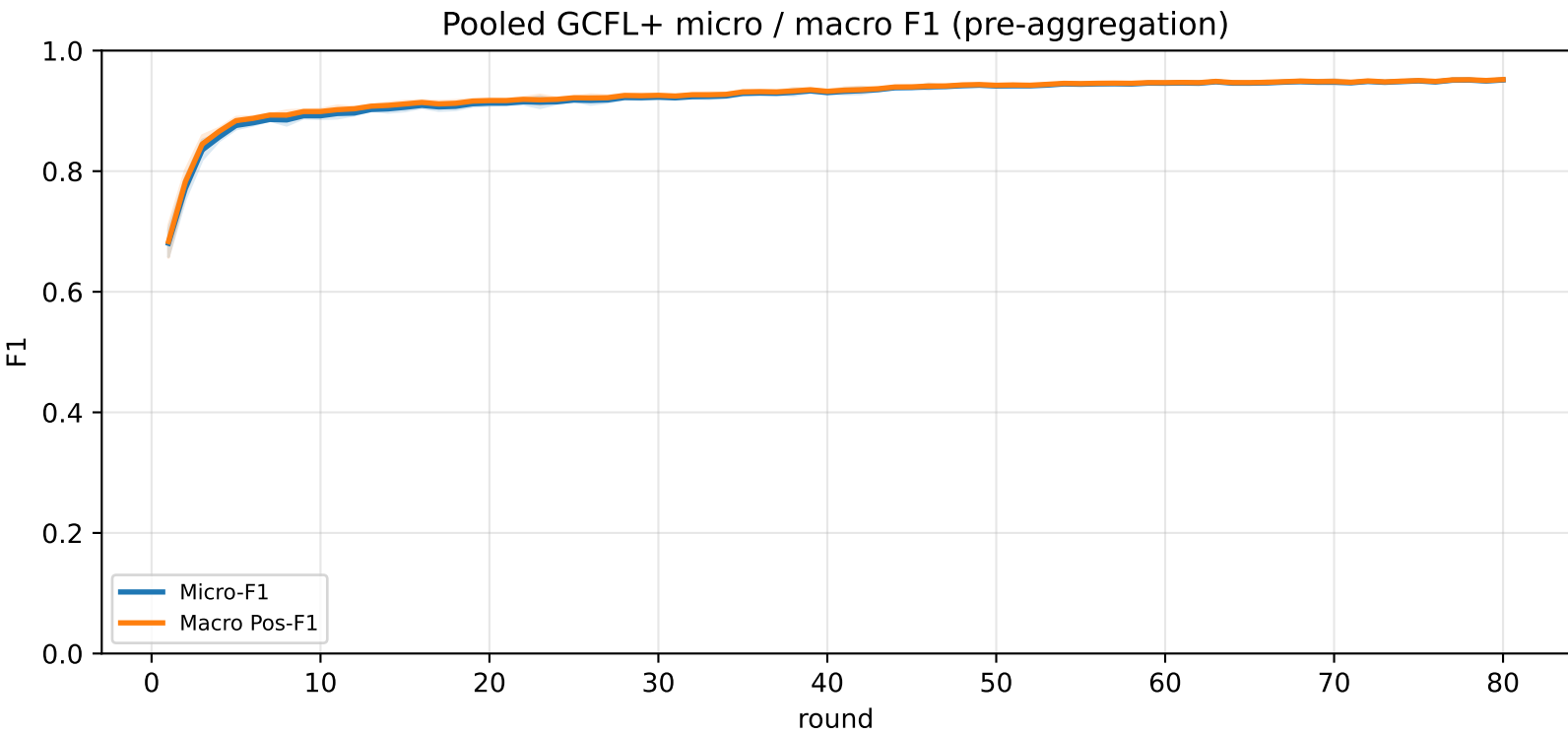
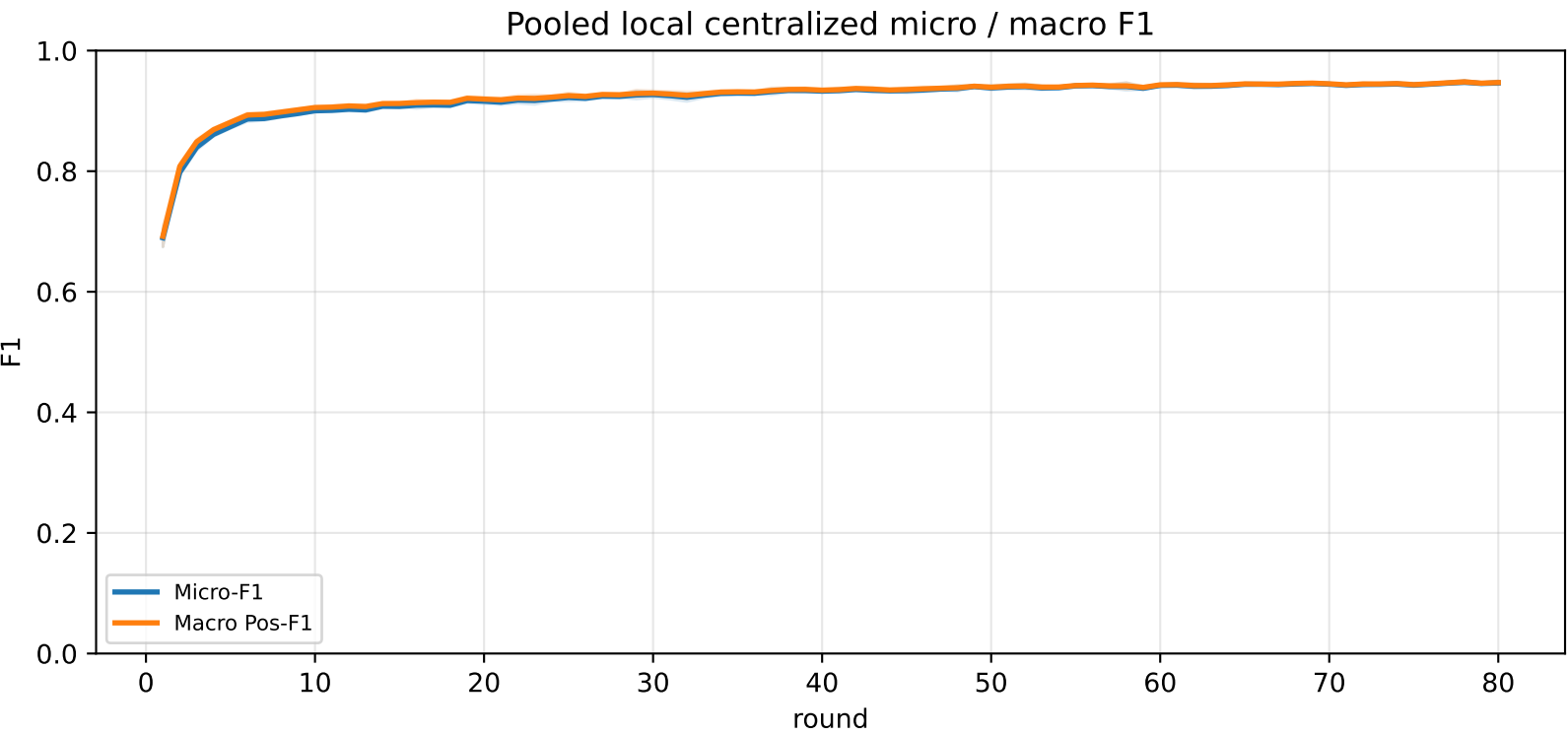
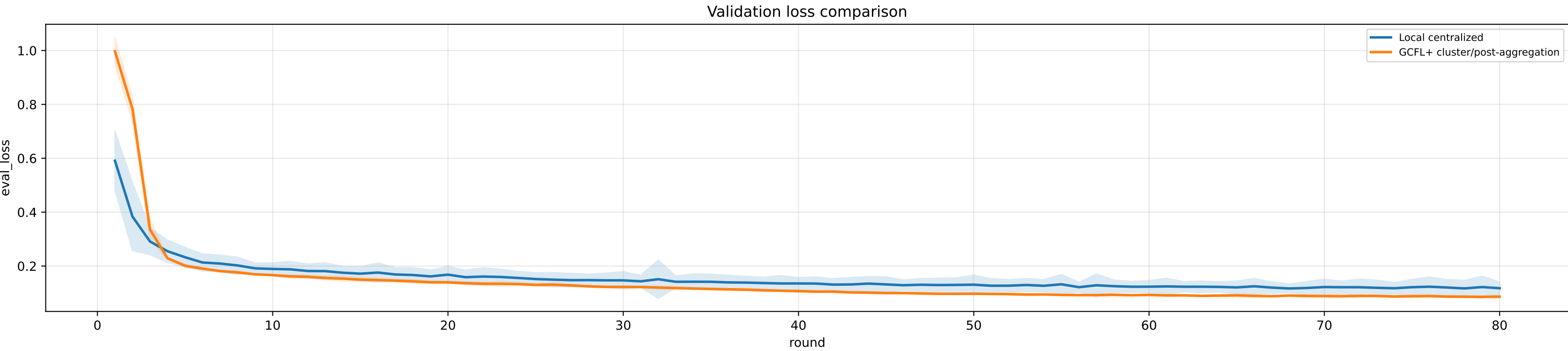
Cluster 2 | Max update norm



Heterogeneity level: Low | GCFL+ diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

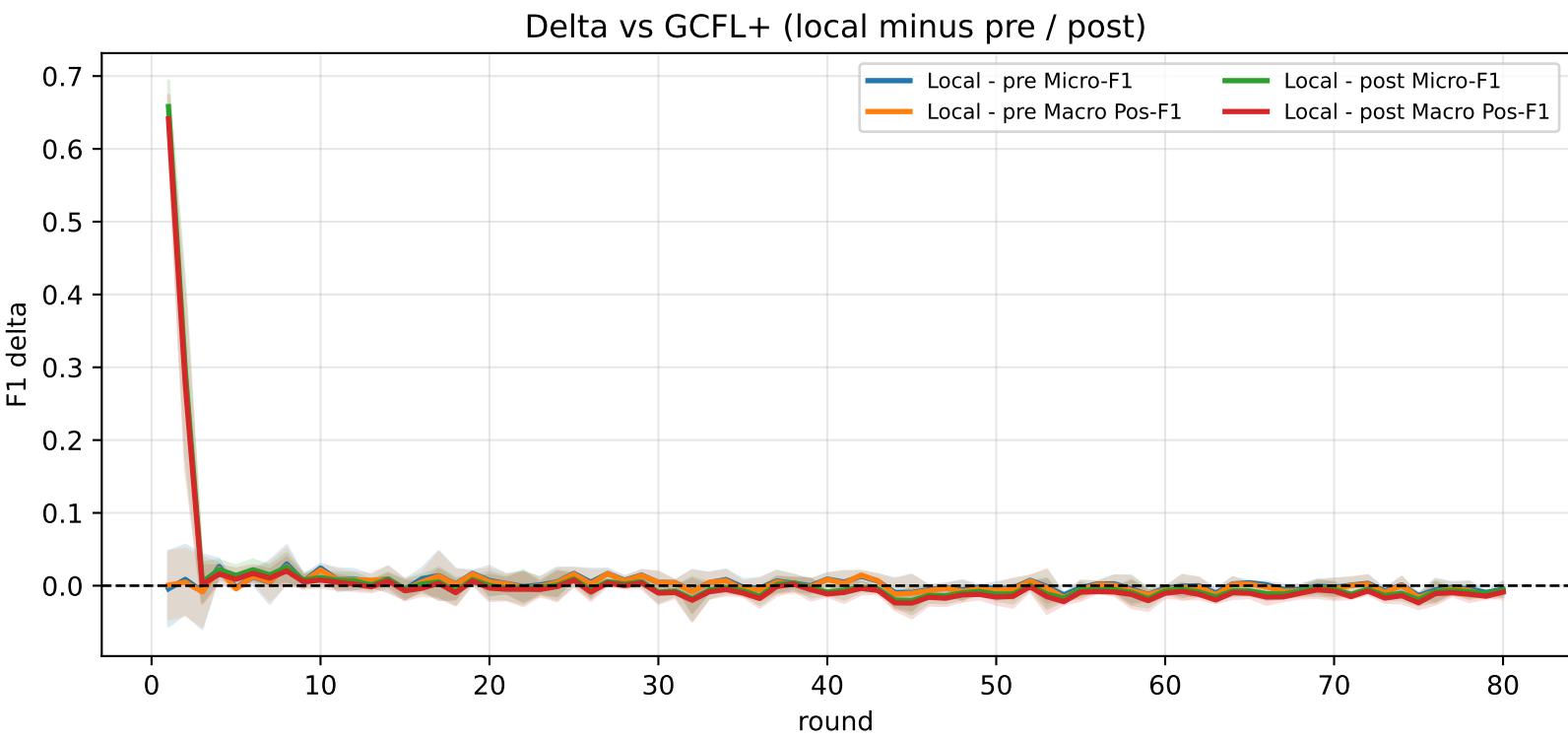
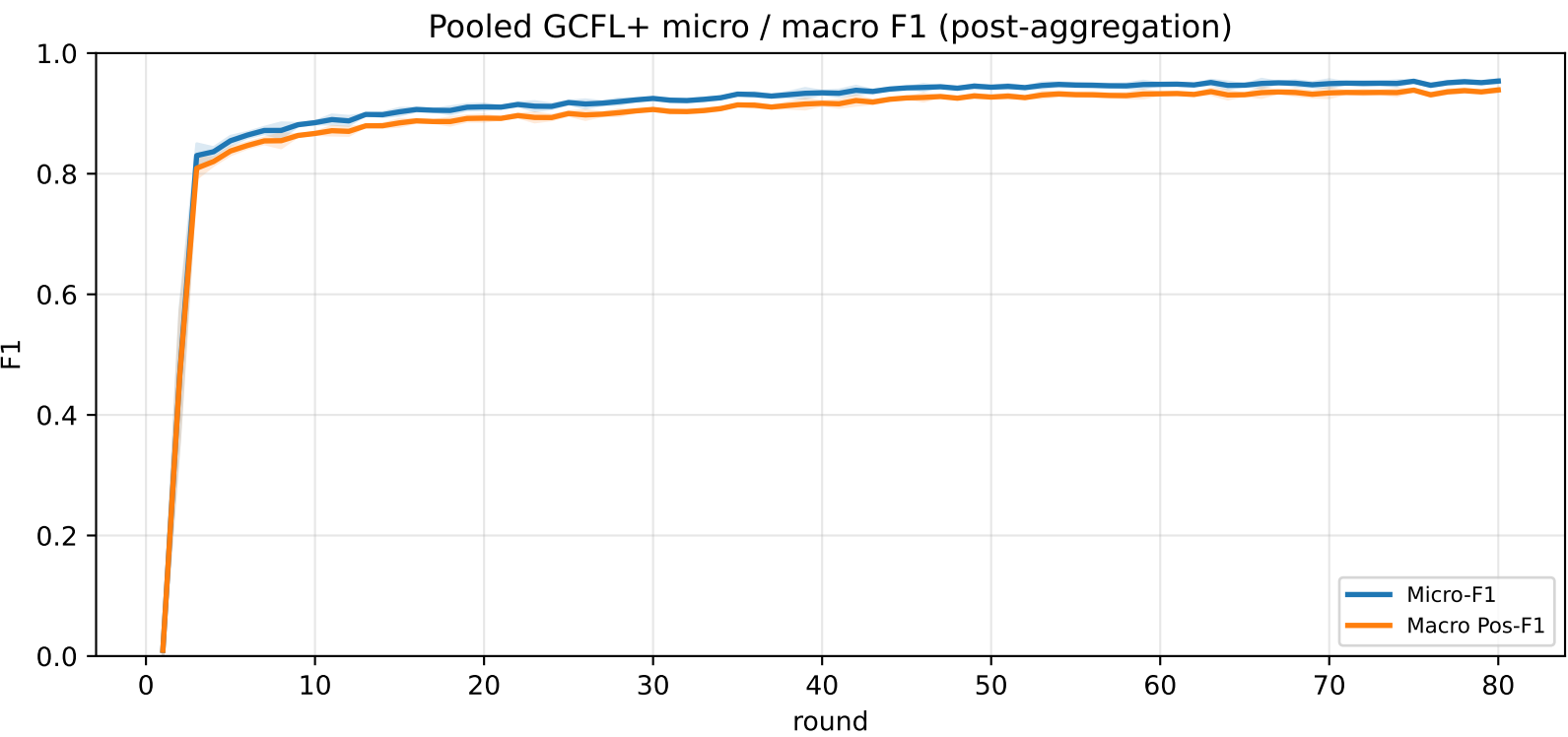
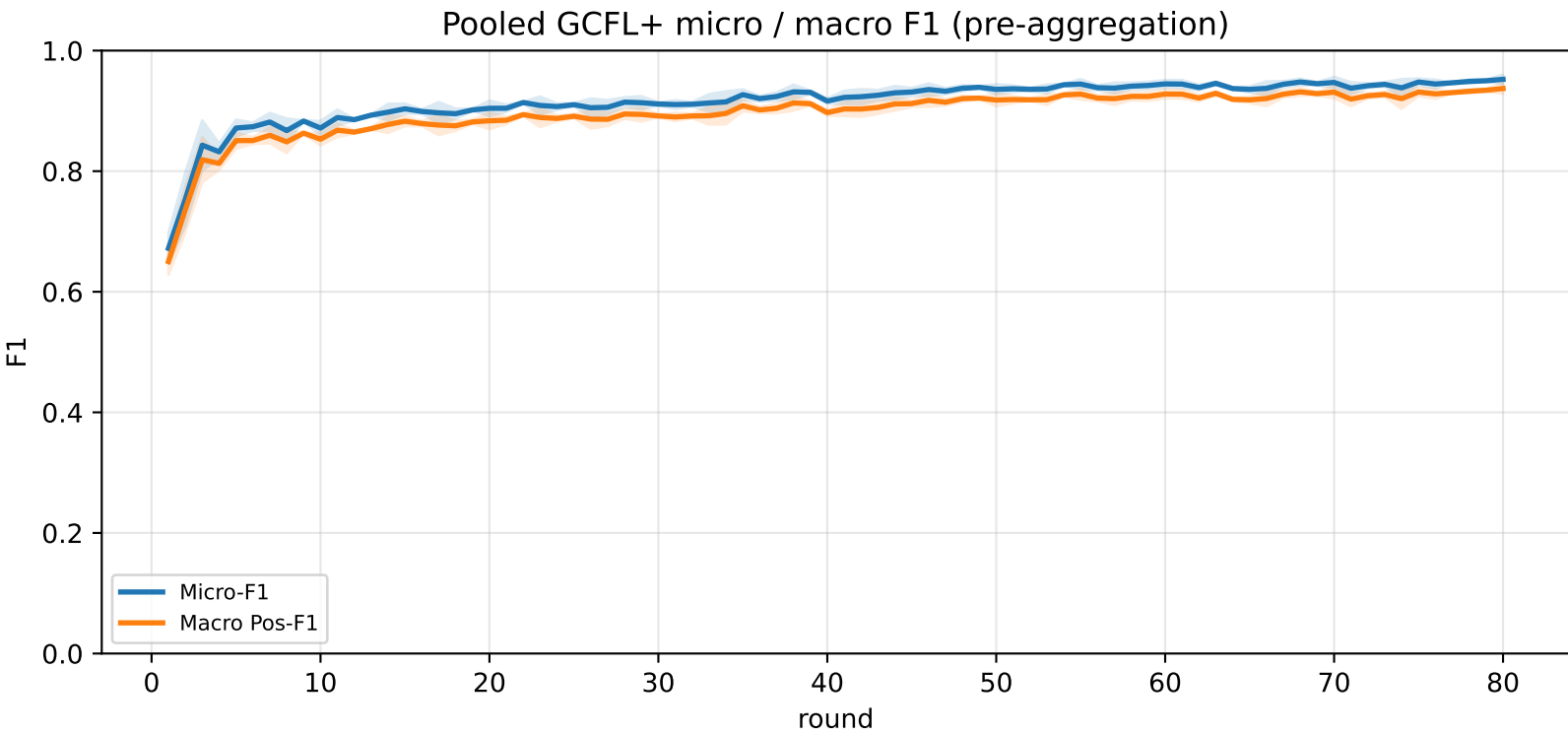
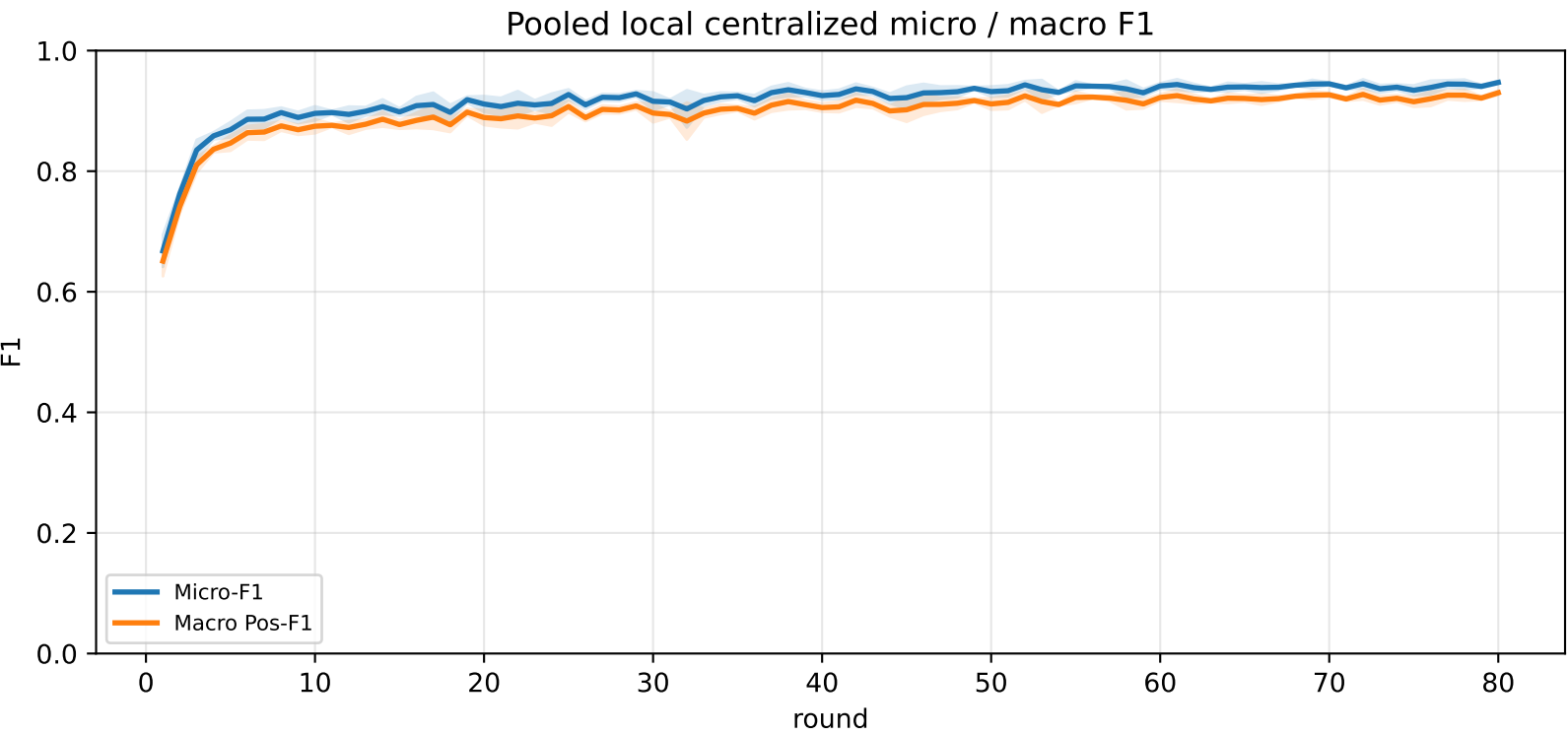
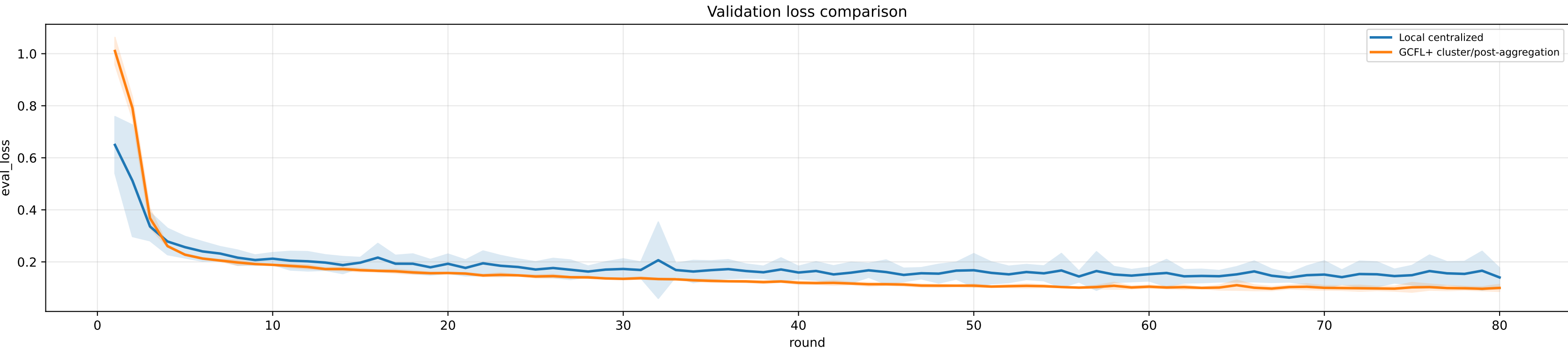
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.7% ± 0.2%	94.4% ± 0.2%	99.2% ± 0.0%	98.6% ± 0.1%	96.8% ± 0.2%	93.1% ± 0.2%	86.2% ± 0.5%
GCFL+	95.6% ± 0.2%	95.3% ± 0.2%	99.3% ± 0.0%	98.7% ± 0.1%	97.2% ± 0.3%	94.8% ± 0.4%	88.4% ± 0.4%



Heterogeneity level: Low | GCFL+ diagnostics | family=strong_decreasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=35.9%±5.1% | cycle3=31.5%±1.3% | cycle4=25.3%±2.1% | cycle5=19.6%±3.4% | cycle6=15.3%±3.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

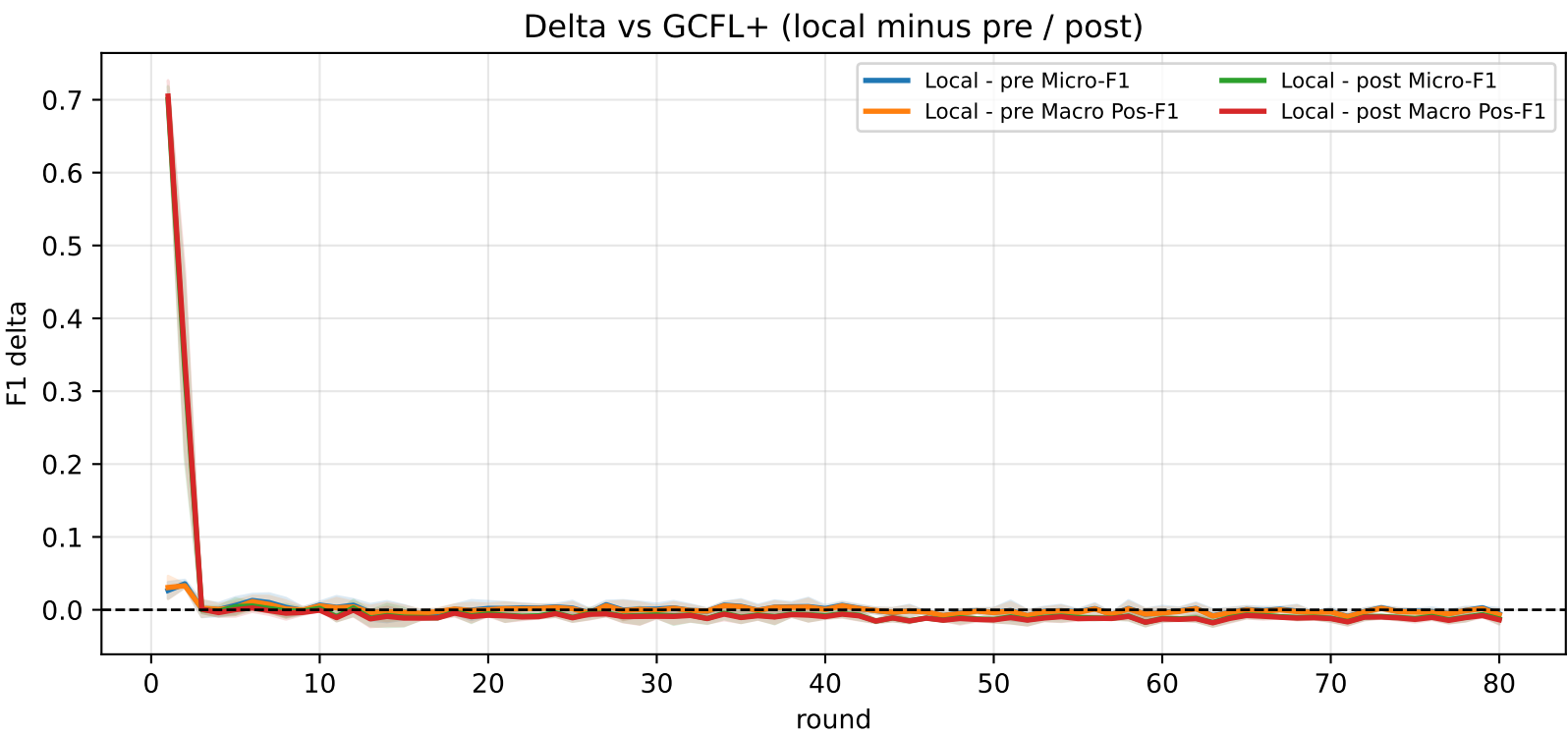
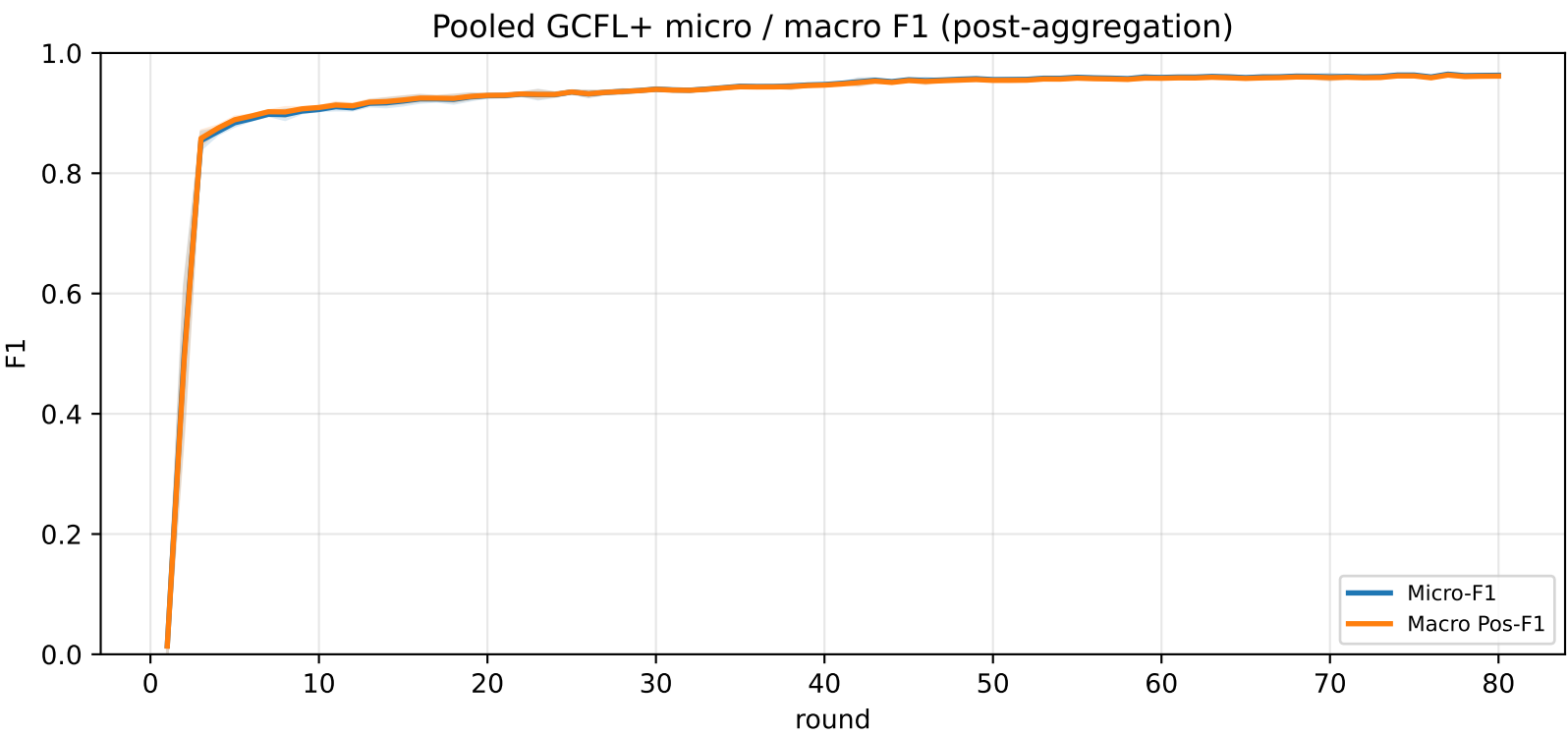
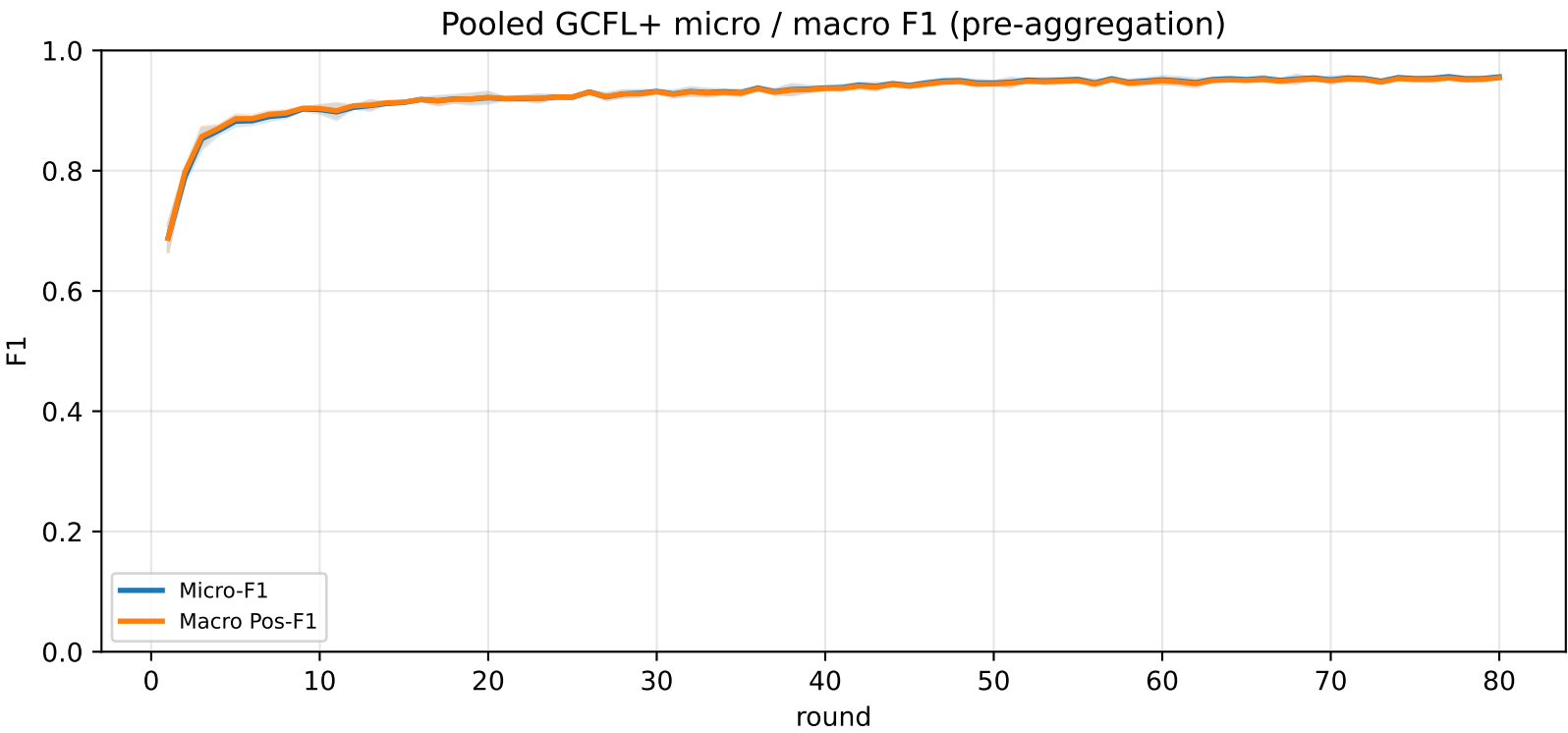
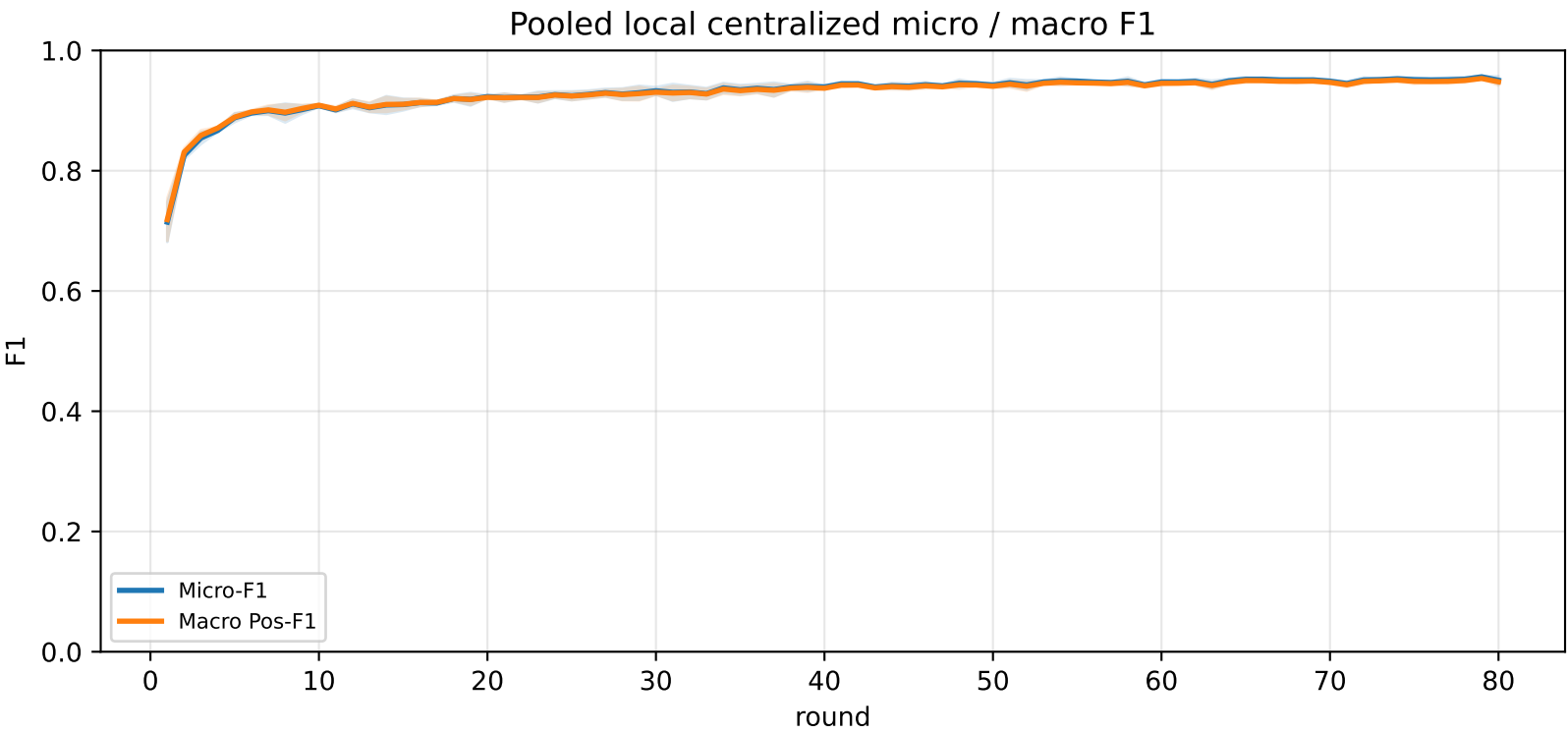
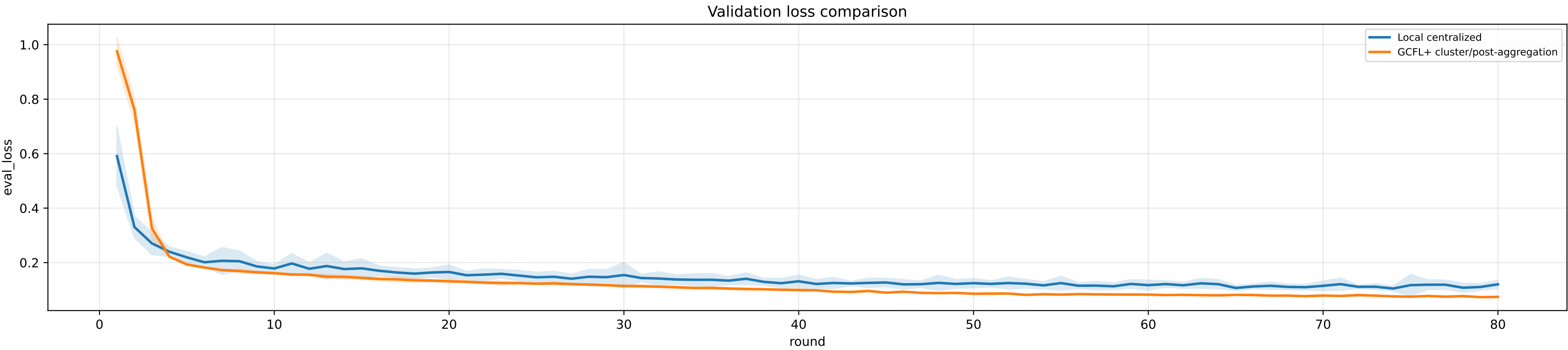
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.2% ± 0.3%	91.8% ± 0.3%	99.6% ± 0.1%	99.0% ± 0.1%	96.5% ± 0.2%	90.1% ± 1.0%	76.7% ± 0.6%
GCFL+	95.0% ± 0.2%	92.8% ± 0.2%	99.6% ± 0.0%	99.3% ± 0.1%	97.2% ± 0.4%	92.1% ± 0.7%	78.8% ± 0.7%



Heterogeneity level: Low | GCFL+ diagnostics | family=mild_decreasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=28.4%±0.4% | cycle3=28.5%±0.9% | cycle4=28.1%±0.7% | cycle5=25.6%±0.3% | cycle6=23.4%±0.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

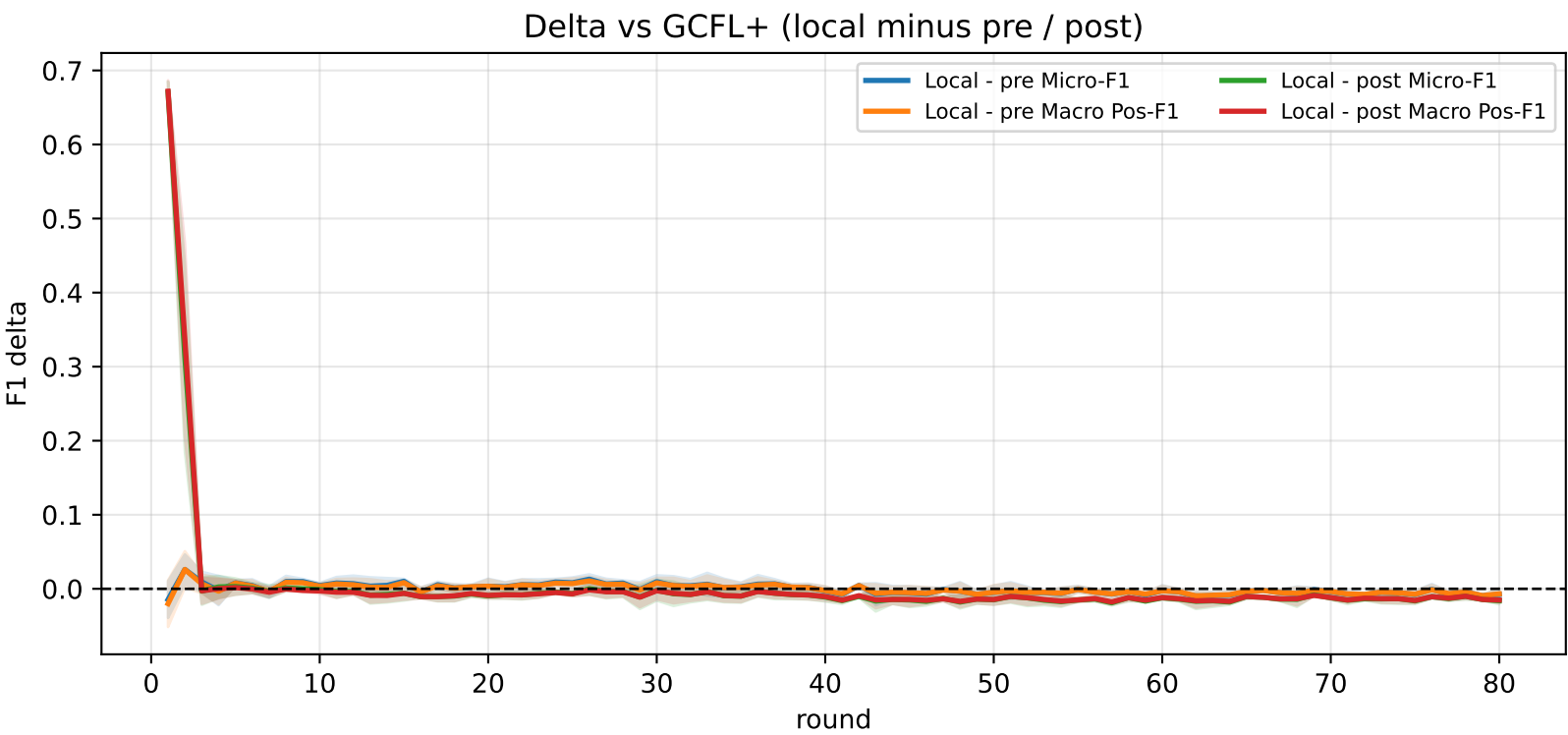
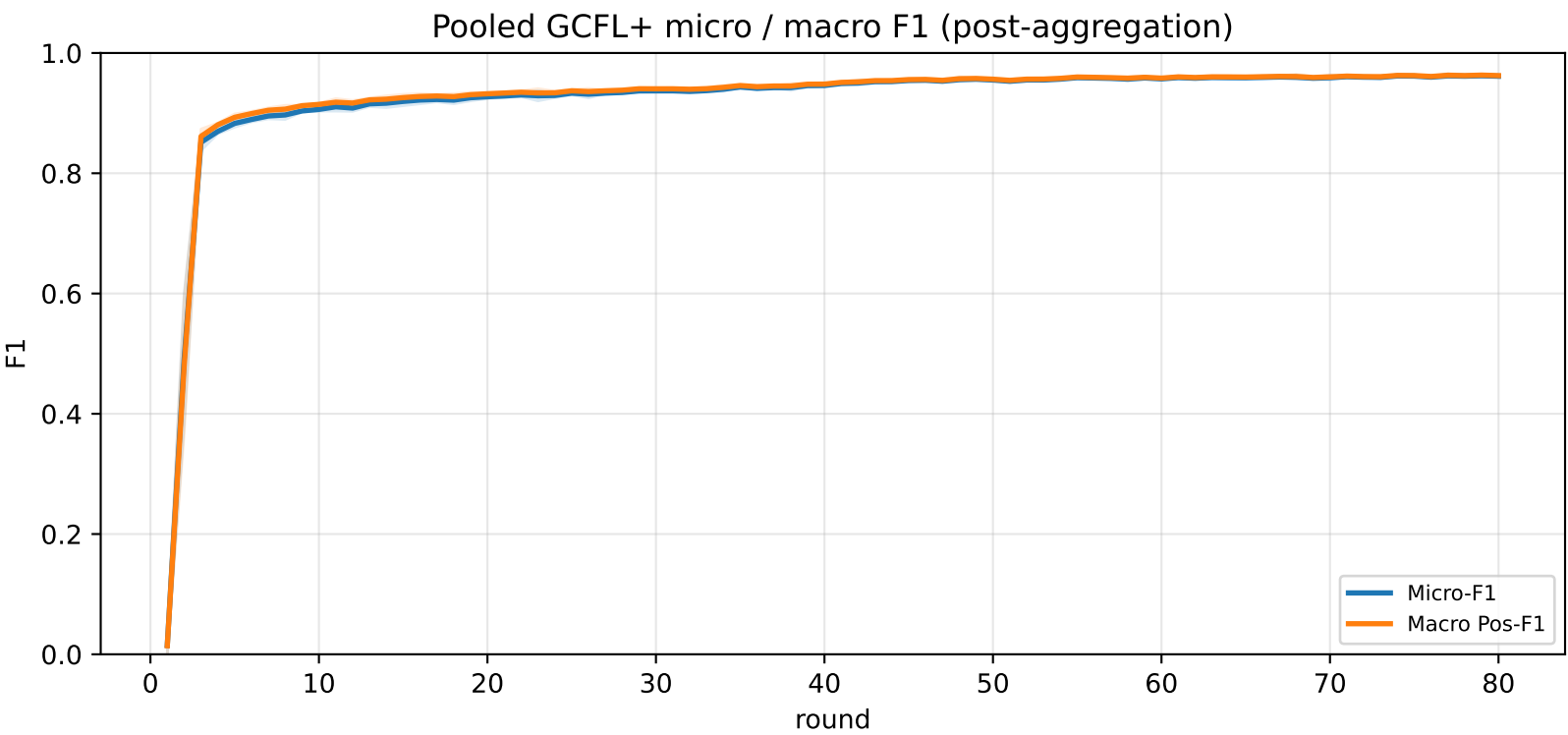
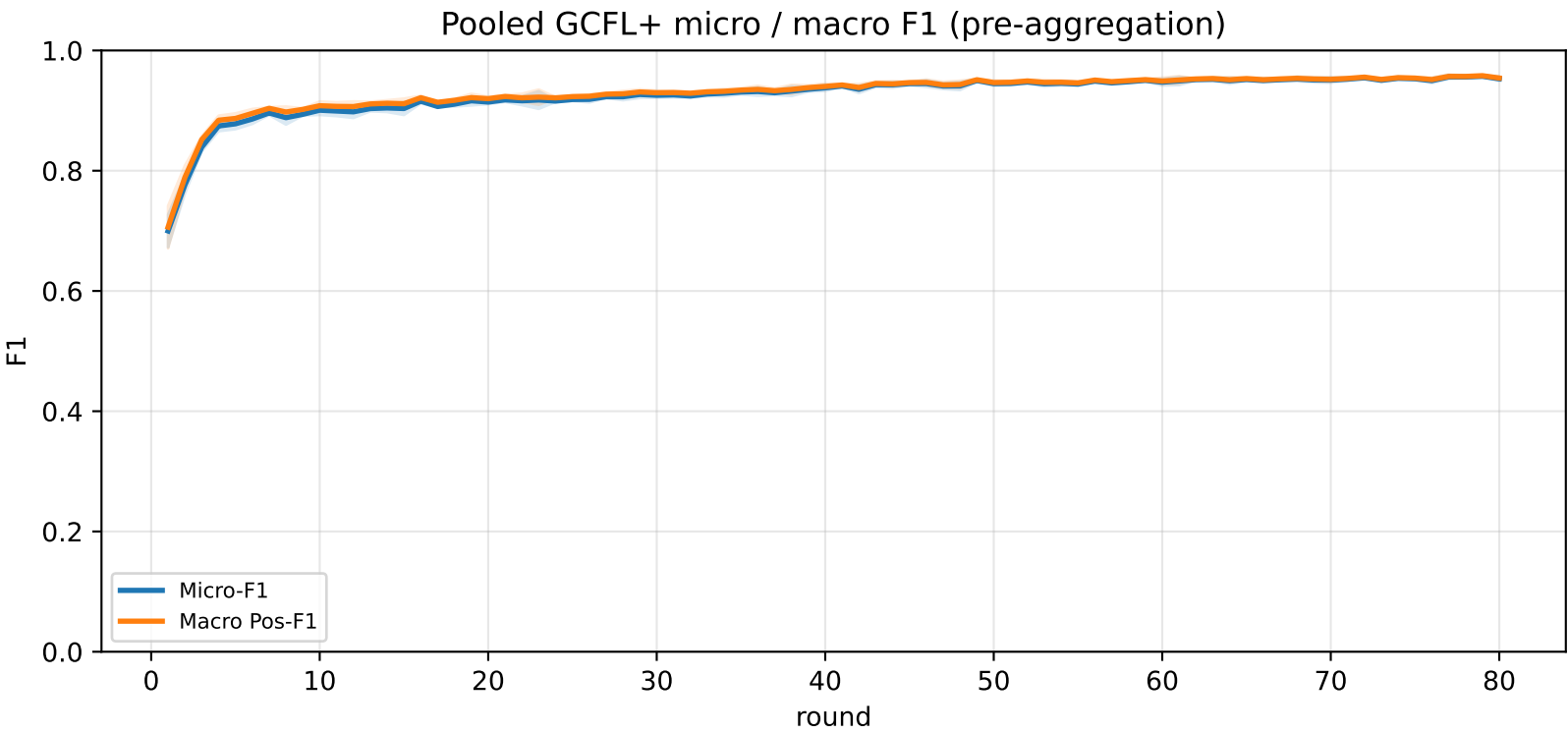
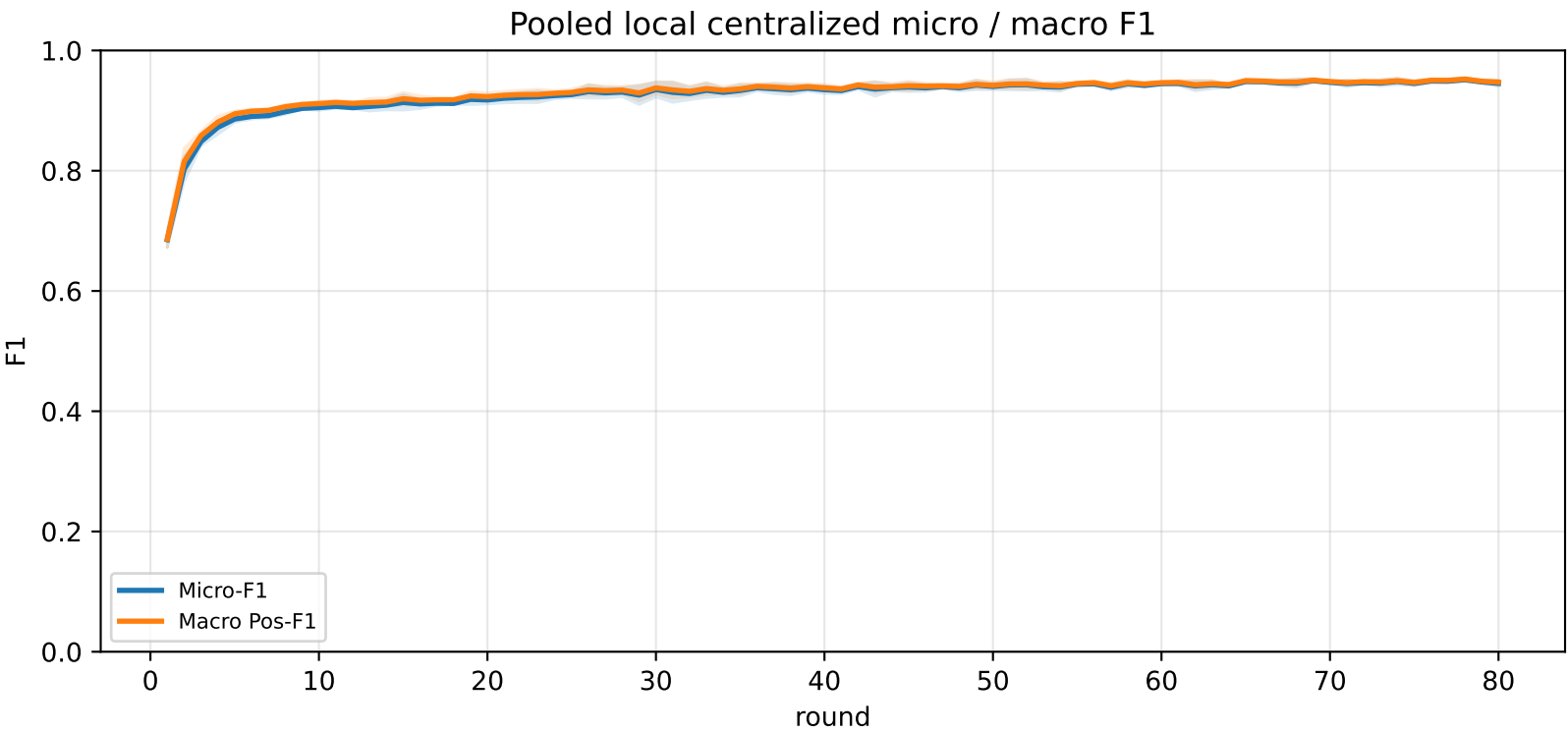
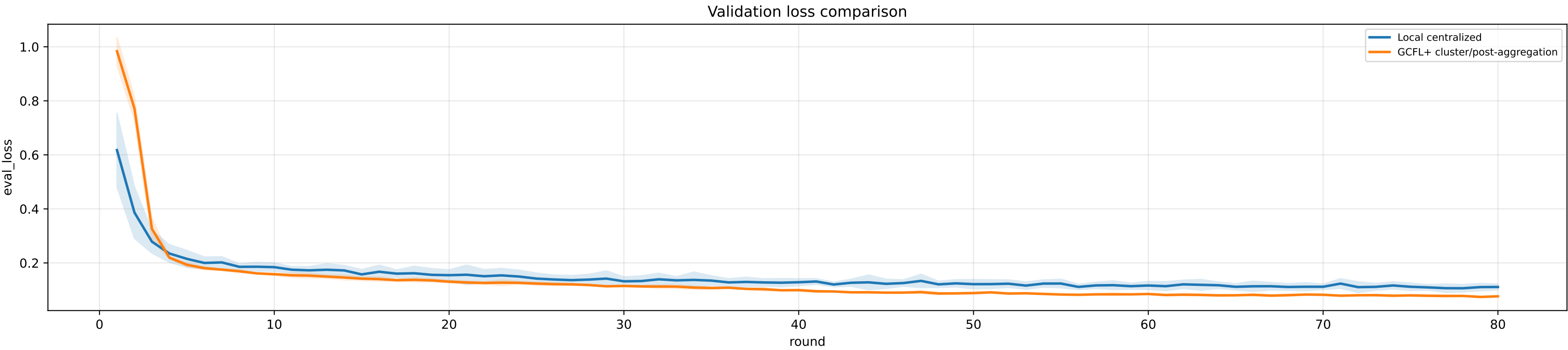
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.3% ± 0.2%	95.2% ± 0.2%	99.3% ± 0.1%	98.9% ± 0.1%	97.4% ± 0.3%	93.3% ± 0.4%	86.9% ± 0.7%
GCFL+	96.2% ± 0.0%	96.1% ± 0.0%	99.4% ± 0.0%	99.1% ± 0.1%	98.0% ± 0.1%	95.5% ± 0.1%	88.4% ± 0.2%



Heterogeneity level: Low | GCFL+ diagnostics | family=flat_balanced |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
GCFL+	96.2% ± 0.1%	96.3% ± 0.1%	99.2% ± 0.0%	98.9% ± 0.1%	97.7% ± 0.1%	95.8% ± 0.2%	89.8% ± 0.4%



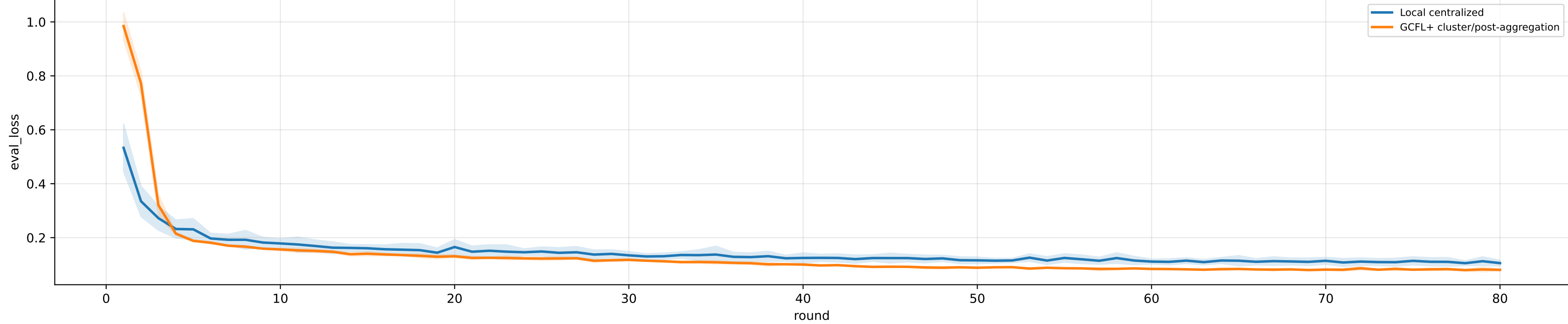
Heterogeneity level: Low | GCFL+ diagnostics | family=mild_increasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=22.3%±1.0% | cycle3=24.2%±0.4% | cycle4=25.3%±1.1% | cycle5=27.0%±0.5% | cycle6=27.3%±1.3%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

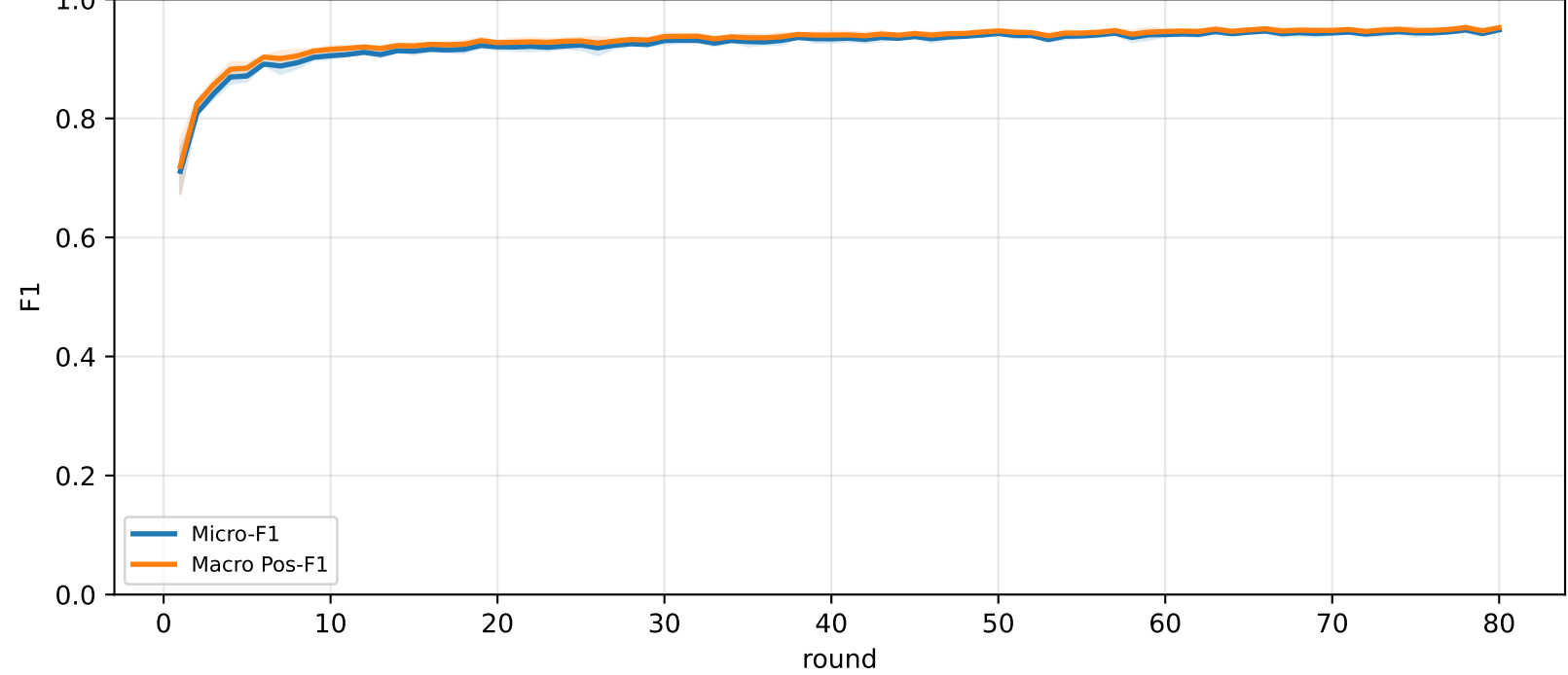
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.8% ± 0.3%	95.2% ± 0.3%	98.9% ± 0.1%	98.4% ± 0.2%	96.7% ± 0.3%	94.2% ± 0.2%	87.9% ± 0.9%
GCFL+	95.7% ± 0.4%	96.0% ± 0.4%	99.0% ± 0.1%	98.4% ± 0.2%	97.2% ± 0.3%	95.4% ± 0.8%	90.1% ± 0.8%

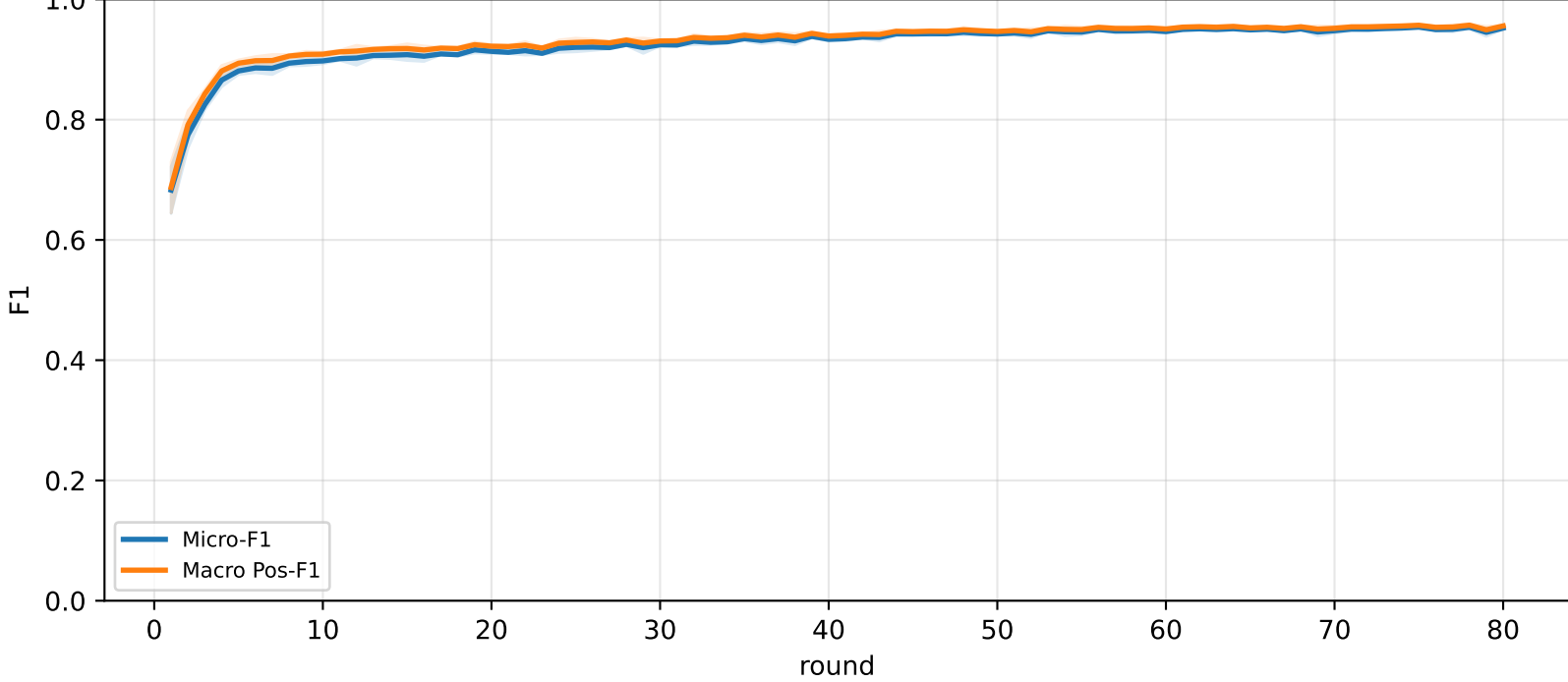
Validation loss comparison



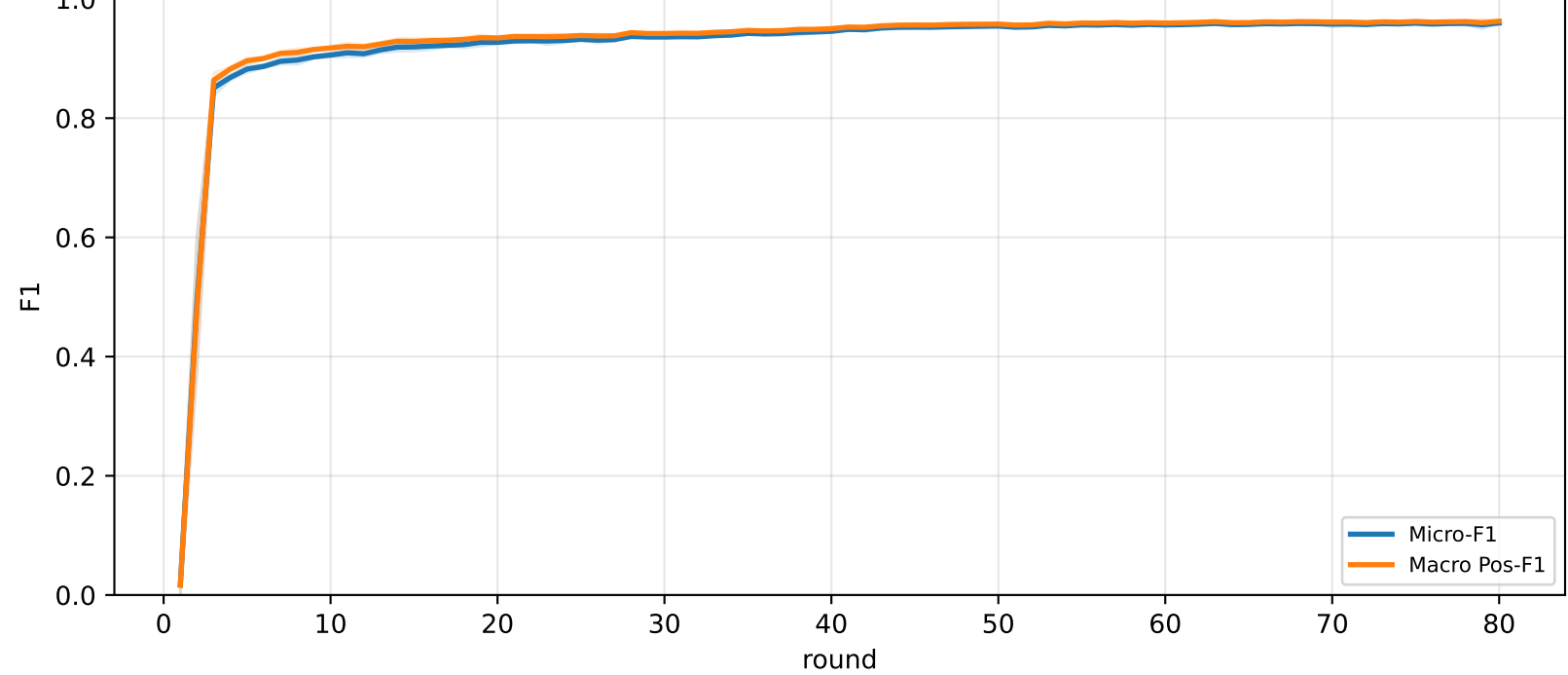
Pooled local centralized micro / macro F1



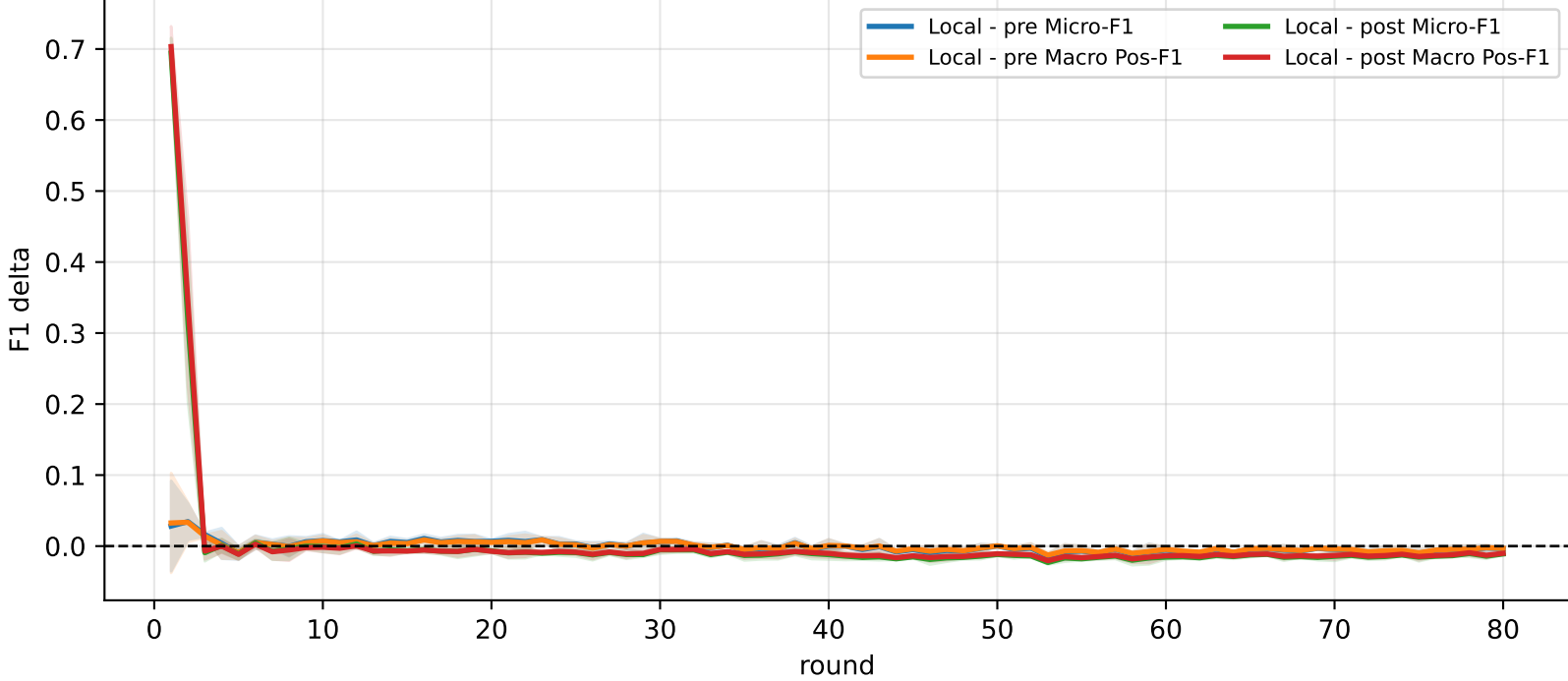
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



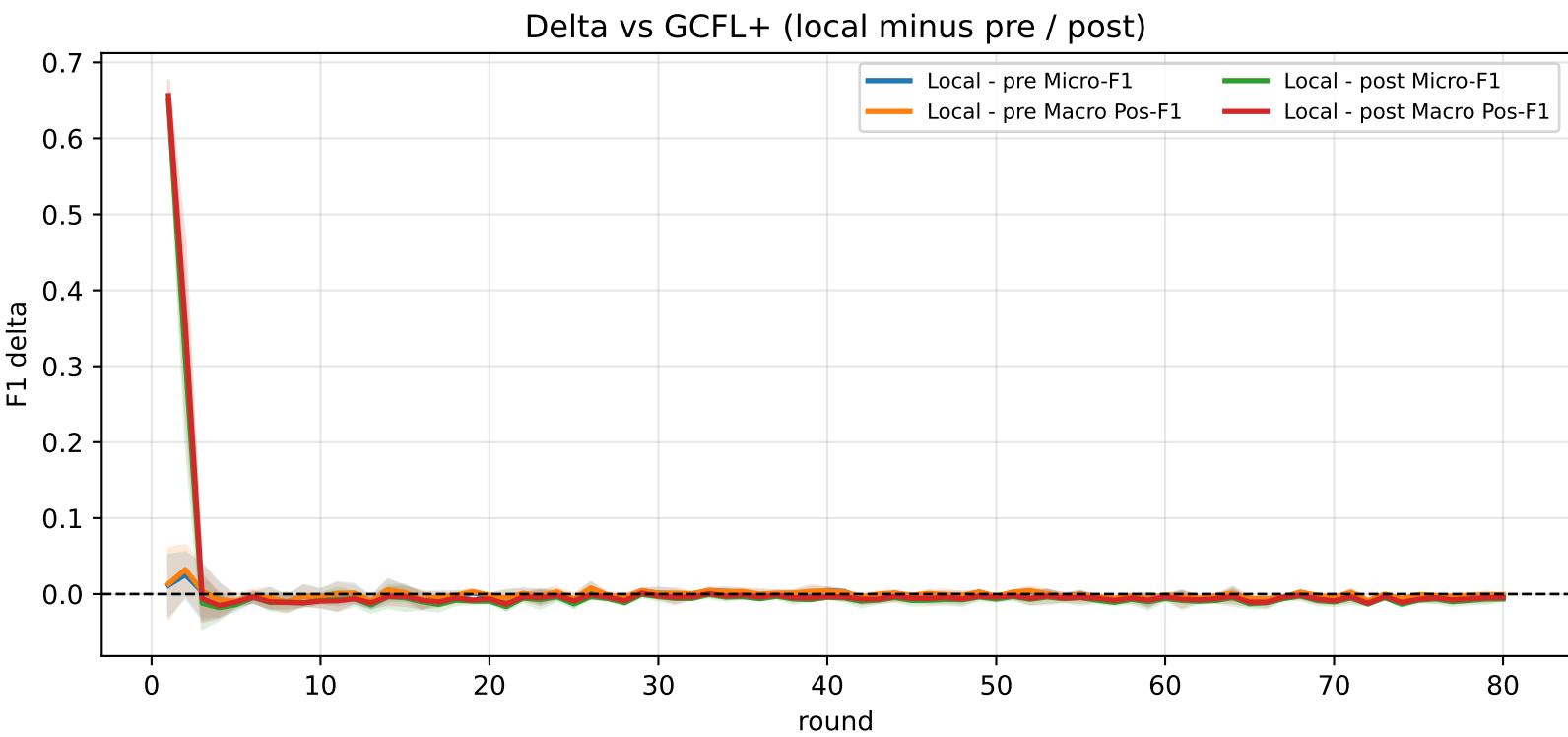
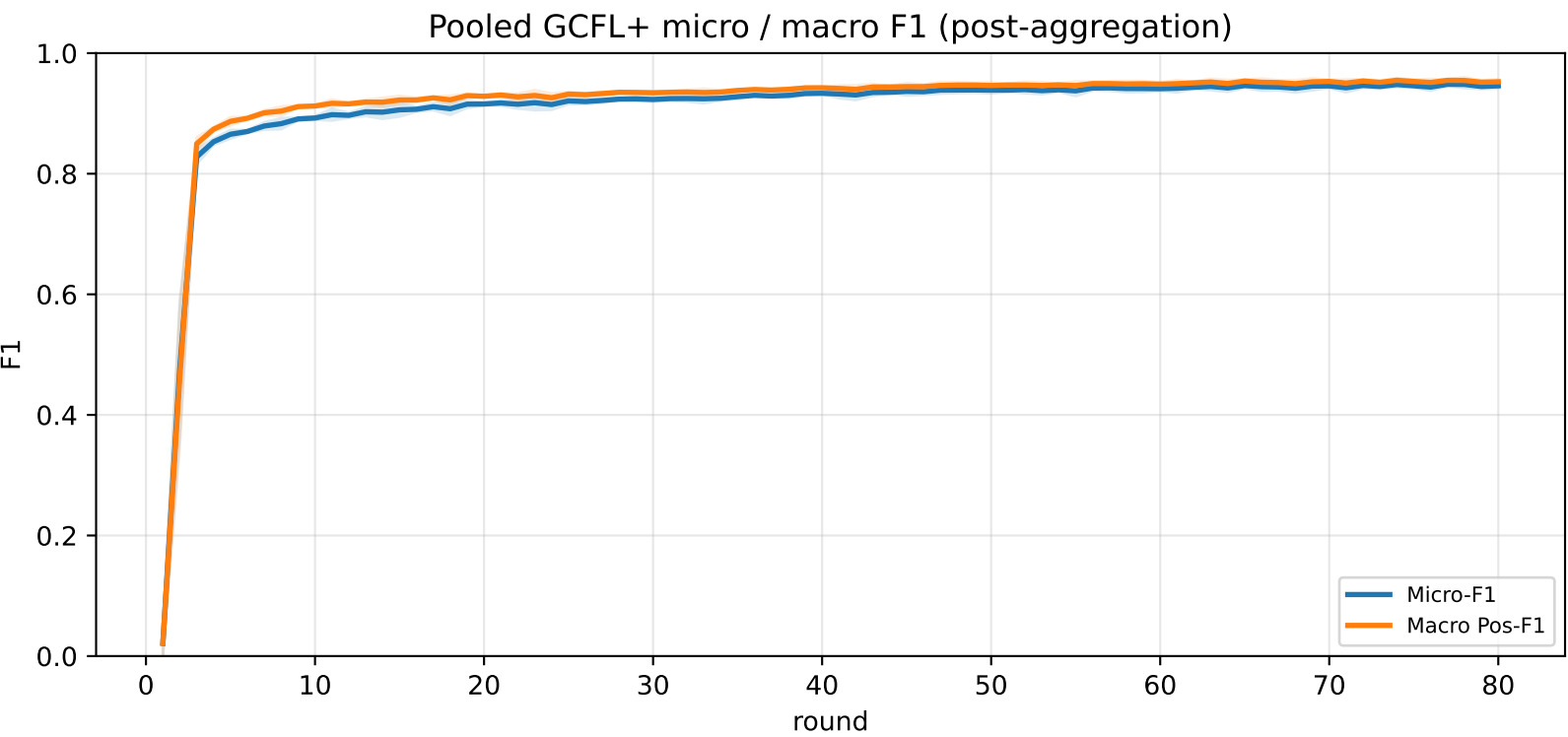
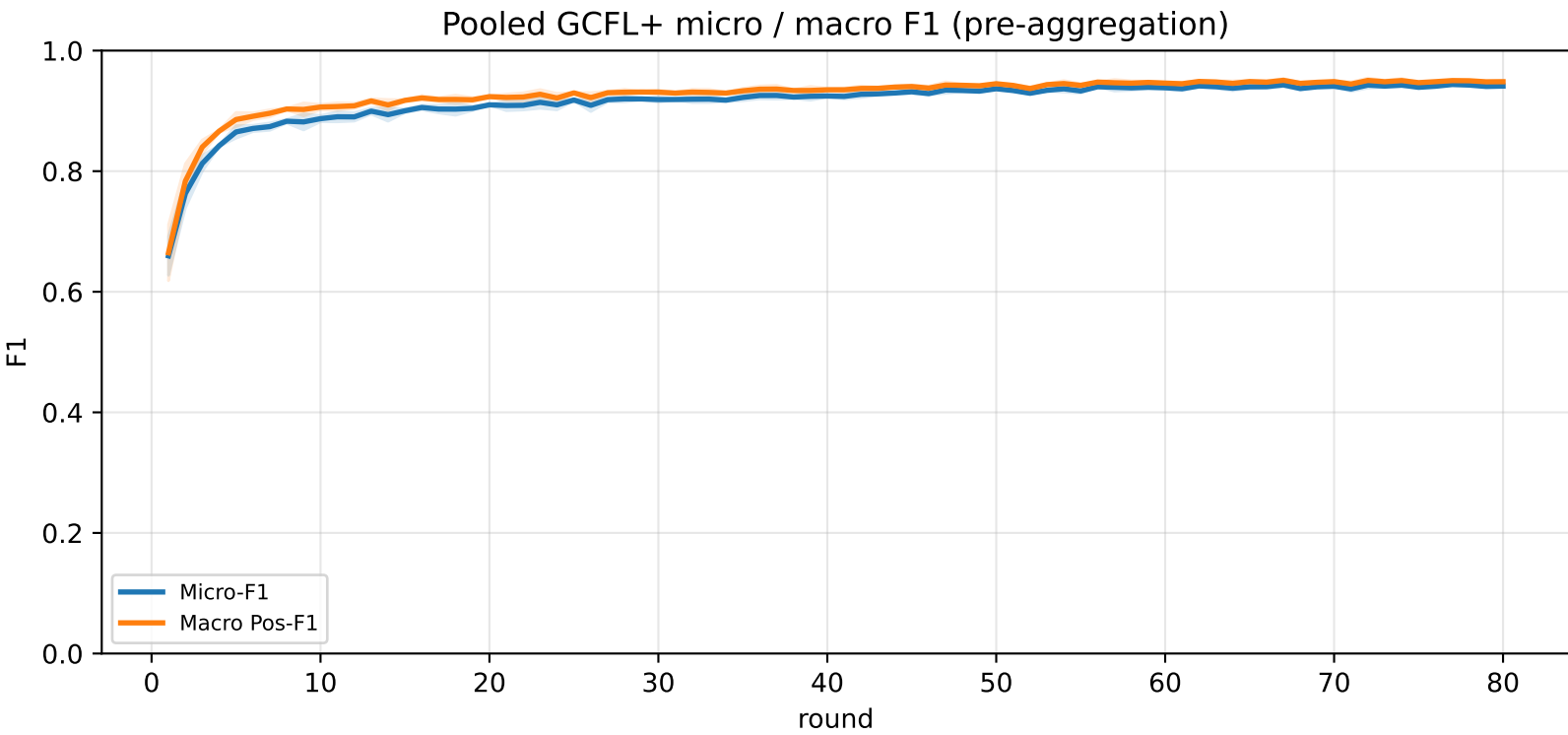
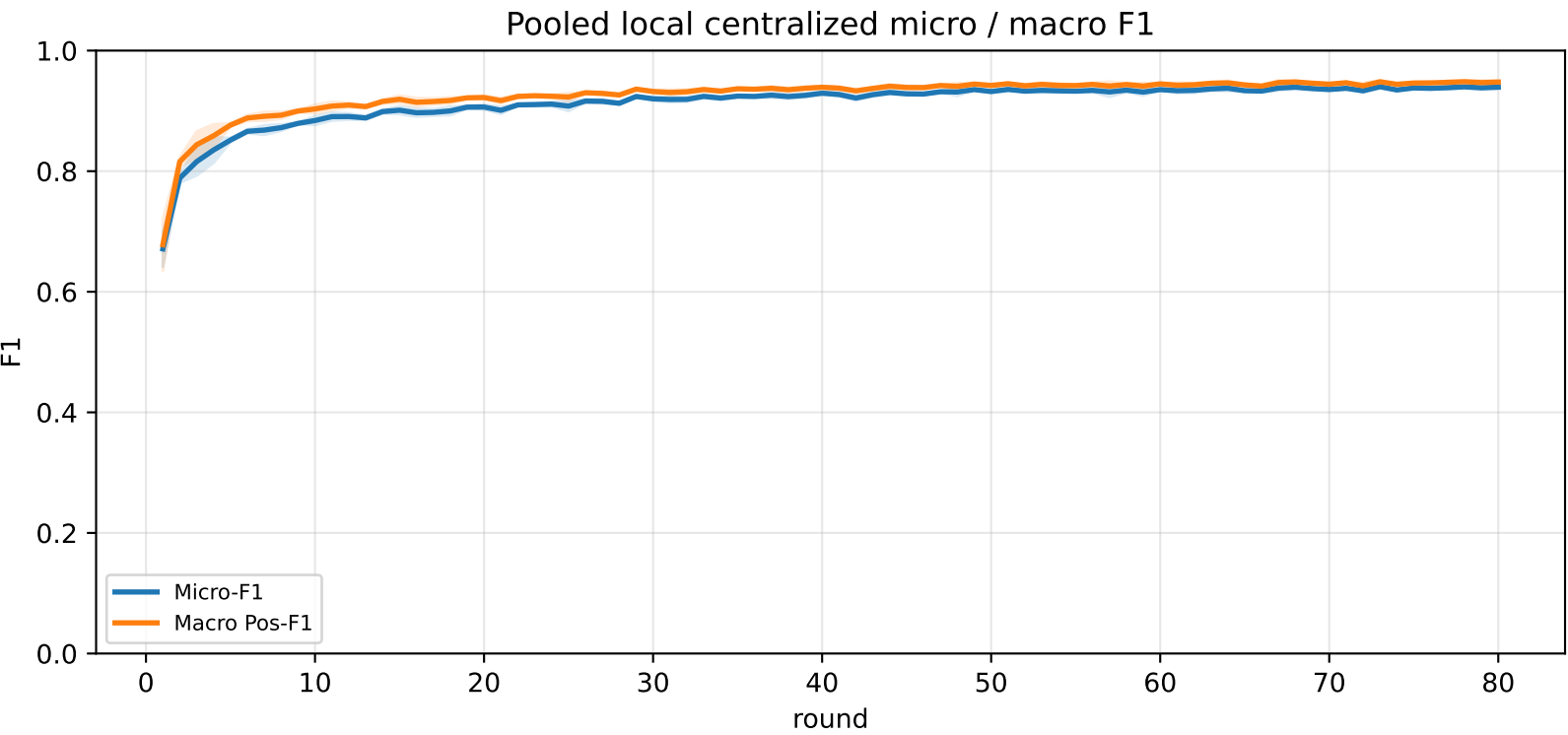
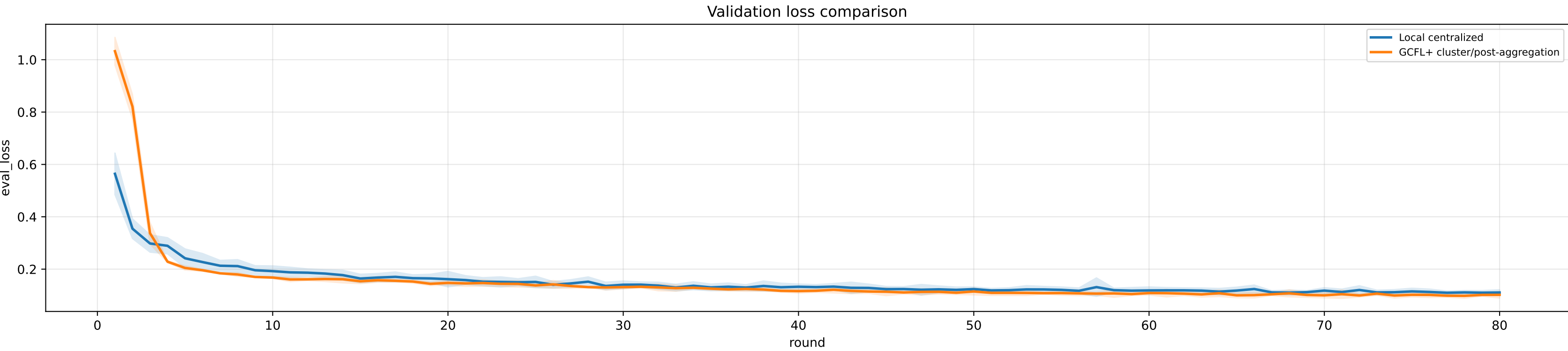
Delta vs GCFL+ (local minus pre / post)



Heterogeneity level: Low | GCFL+ diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=15.0%±1.0% | cycle3=19.4%±0.7% | cycle4=22.9%±1.2% | cycle5=25.6%±1.2% | cycle6=27.6%±0.4%
heterogeneity=low | task_jsd_mean=0.009 | task_jsd_median=0.004 | task_jsd_max=0.075 | family_centroid_jsd_mean=0.009
per-task pos-rate std | cycle2_std=0.073 | cycle3_std=0.042 | cycle4_std=0.021 | cycle5_std=0.032 | cycle6_std=0.050

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.2%	94.8% ± 0.2%	98.3% ± 0.3%	97.7% ± 0.2%	96.0% ± 0.1%	92.9% ± 0.3%	88.9% ± 0.5%
GCFL+	94.7% ± 0.6%	95.4% ± 0.5%	98.6% ± 0.2%	97.5% ± 0.3%	95.9% ± 0.9%	94.3% ± 0.4%	91.0% ± 0.9%



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

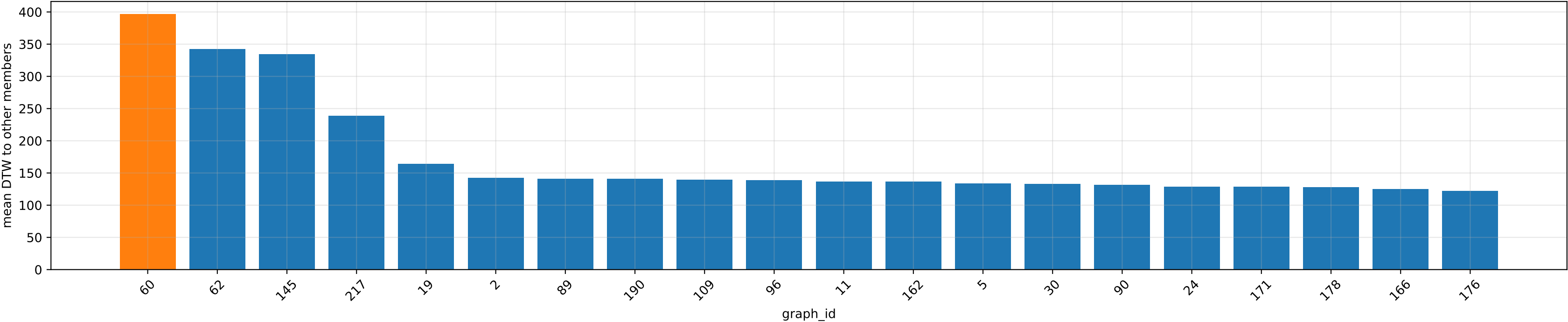
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	14	0	1 vs 19	60	60	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
396.454	nan	149.368

Per-member mean DTW



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

First applied split pairwise DTW metadata | seed=0

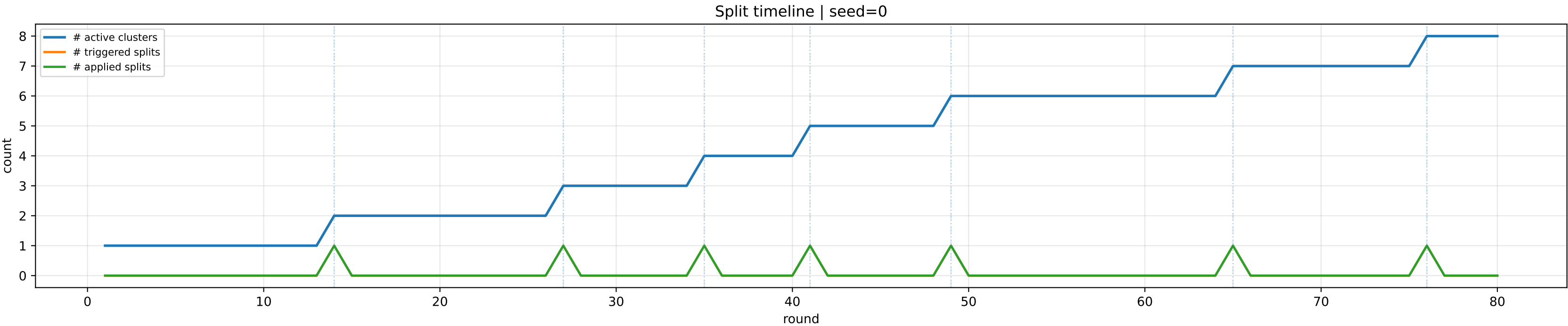
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	14	0	20	60.0	0.052	396.454	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 513.3 at (19, 60)

graph_id	178	96	19	176	171	89	90	11	190	109	24	30	166	5	2	162	62	217	145	60
178	0.0	49.5	81.9	53.8	53.2	53.5	55.0	50.8	102.9	120.6	45.7	87.8	70.3	60.3	71.3	90.8	360.9	245.3	350.2	425.0
96	49.5	0.0	46.4	67.5	56.1	31.6	34.5	45.5	129.7	142.4	31.8	87.3	73.1	31.4	72.7	122.4	416.0	298.7	416.8	478.5
19	81.9	46.4	0.0	79.7	60.4	35.6	71.9	60.5	144.7	160.8	44.6	122.0	122.0	61.7	127.1	152.8	450.8	333.5	451.6	513.3
176	53.8	67.5	79.7	0.0	23.5	50.8	47.7	46.0	85.6	77.4	38.9	43.5	55.7	49.1	97.0	73.2	371.2	253.9	371.9	433.6
171	53.2	56.1	60.4	23.5	0.0	33.6	41.3	41.2	91.0	100.6	18.7	64.1	83.2	57.6	111.1	92.7	390.7	273.4	391.4	453.1
89	53.5	31.6	35.6	50.8	33.6	0.0	48.1	54.7	120.2	134.2	14.3	91.8	89.3	50.0	104.0	126.3	424.3	307.0	425.0	486.7
90	55.0	34.5	71.9	47.7	41.3	48.1	0.0	54.3	99.2	124.8	43.9	76.5	72.7	50.2	90.8	97.4	387.8	270.5	388.5	450.2
11	50.8	45.5	60.5	46.0	41.2	54.7	54.3	0.0	104.8	108.3	49.4	81.0	86.8	41.2	107.2	106.0	404.0	286.7	404.7	466.4
190	102.9	129.7	144.7	85.6	91.0	120.2	99.2	104.8	0.0	41.6	103.8	133.7	117.8	132.5	154.6	86.2	257.8	174.4	223.2	368.6
109	120.6	142.4	160.8	77.4	100.6	134.2	124.8	108.3	41.6	0.0	115.7	94.6	93.8	128.3	152.5	82.7	259.5	158.3	199.6	356.9
24	45.7	31.8	44.6	38.9	18.7	14.3	43.9	49.4	103.8	115.7	0.0	78.6	71.3	39.4	90.3	103.7	401.7	284.4	402.4	464.1
30	87.8	87.3	122.0	43.5	64.1	91.8	76.5	81.0	133.7	94.6	78.6	0.0	51.7	75.1	50.0	59.7	348.2	211.5	371.7	391.2
166	70.3	73.1	122.0	55.7	83.2	89.3	72.7	86.8	117.8	93.8	71.3	51.7	0.0	53.9	48.2	51.1	319.1	203.2	322.0	382.9
5	60.3	31.4	61.7	49.1	57.6	50.0	50.2	41.2	132.5	128.3	39.4	75.1	53.9	0.0	68.7	105.2	397.1	279.8	397.8	459.5
2	71.3	72.7	127.1	97.0	111.1	104.0	90.8	107.2	154.6	152.5	90.3	50.0	48.2	68.7	0.0	69.6	333.7	203.6	371.6	383.4
162	90.8	122.4	152.8	73.2	92.7	126.3	97.4	106.0	86.2	82.7	103.7	59.7	51.1	105.2	69.6	0.0	312.7	180.7	322.4	360.4
62	360.9	416.0	450.8	371.2	390.7	424.3	387.8	404.0	257.8	259.5	401.7	348.2	319.1	397.1	333.7	312.7	0.0	203.8	195.8	274.3
217	245.3	298.7	333.5	253.9	273.4	307.0	270.5	286.7	174.4	158.3	284.4	211.5	203.2	279.8	203.6	180.7	203.8	0.0	162.4	201.0
145	350.2	416.8	451.6	371.9	391.4	425.0	388.5	404.7	223.2	199.6	402.4	371.7	322.0	397.8	371.6	322.4	195.8	162.4	0.0	183.6
60	425.0	478.5	513.3	433.6	453.1	486.7	450.2	466.4	368.6	356.9	464.1	391.2	382.9	459.5	383.4	360.4	274.3	201.0	183.6	0.0

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

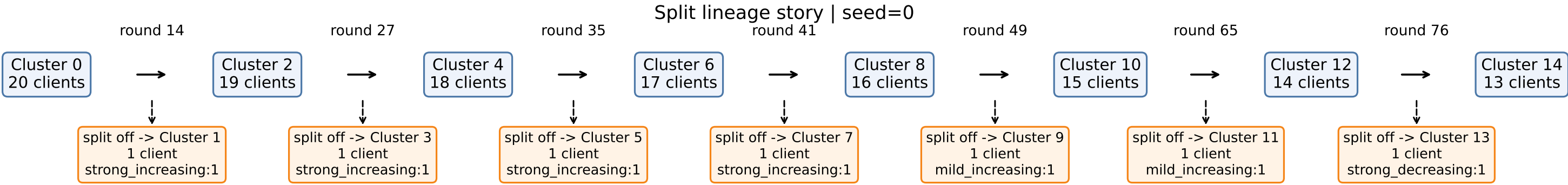


Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	14, 27, 35, 41, 49, 65, 76	14, 27, 35, 41, 49, 65, 76	1, 3, 5, 7, 9, 11, 13, 14	1:1, 3:1, 5:1, 7:1, 9:1, 11:1, 13:1, 14:13

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068



Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
14	0	20	157.922 < 158.968 , 291.622 > 247.884	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
27	2	19	157.073 < 157.529 , 303.467 > 295.007	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
35	4	18	156.973 < 157.120 , 319.517 > 276.306	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1
41	6	17	155.538 < 156.720 , 272.715 > 253.829	C7 (1 client)	strong_increasing:1	C8 (16 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4
49	8	16	155.832 < 156.180 , 230.619 > 225.684	C9 (1 client)	mild_increasing:1	C10 (15 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:3 , strong_decreasing:4
65	10	15	155.582 < 155.739 , 234.950 > 212.143	C11 (1 client)	mild_increasing:1	C12 (14 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:2 , strong_decreasing:4
76	12	14	154.923 < 154.951 , 200.649 > 194.814	C13 (1 client)	strong_decreasing:1	C14 (13 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:2 , strong_decreasing:3

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

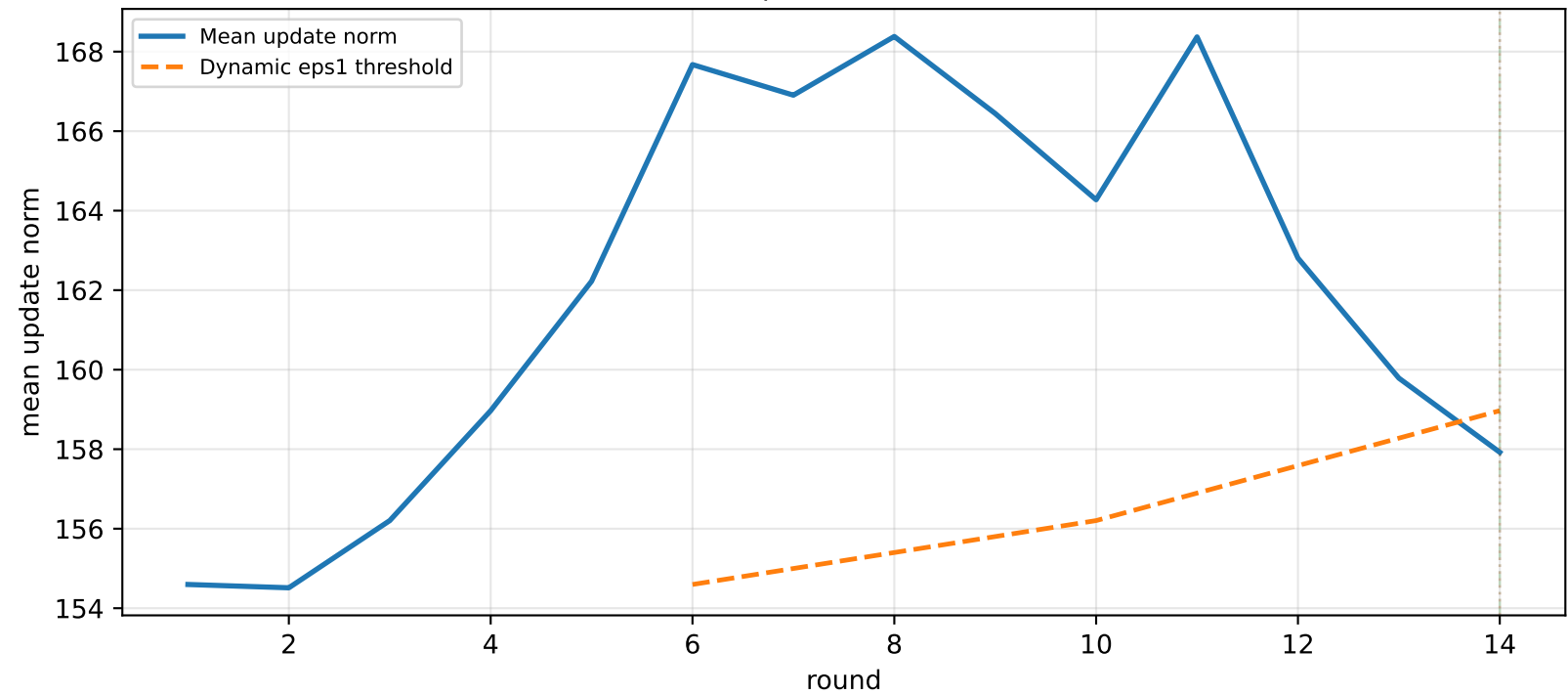
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	60	19	strong_increasing:1
3	1	145	18	strong_increasing:1
5	1	217	17	strong_increasing:1
7	1	62	16	strong_increasing:1
9	1	109	9	mild_increasing:1
11	1	190	8	mild_increasing:1
13	1	162	15	strong_decreasing:1
14	13	178, 96, 19, 176, 171, 89, 90, 11, 24, 30, 166, 5, 2	0, 1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14	flat_balanced:4 , mild_decreasing:4 , mild_increasing:2 , strong_decreasing:3

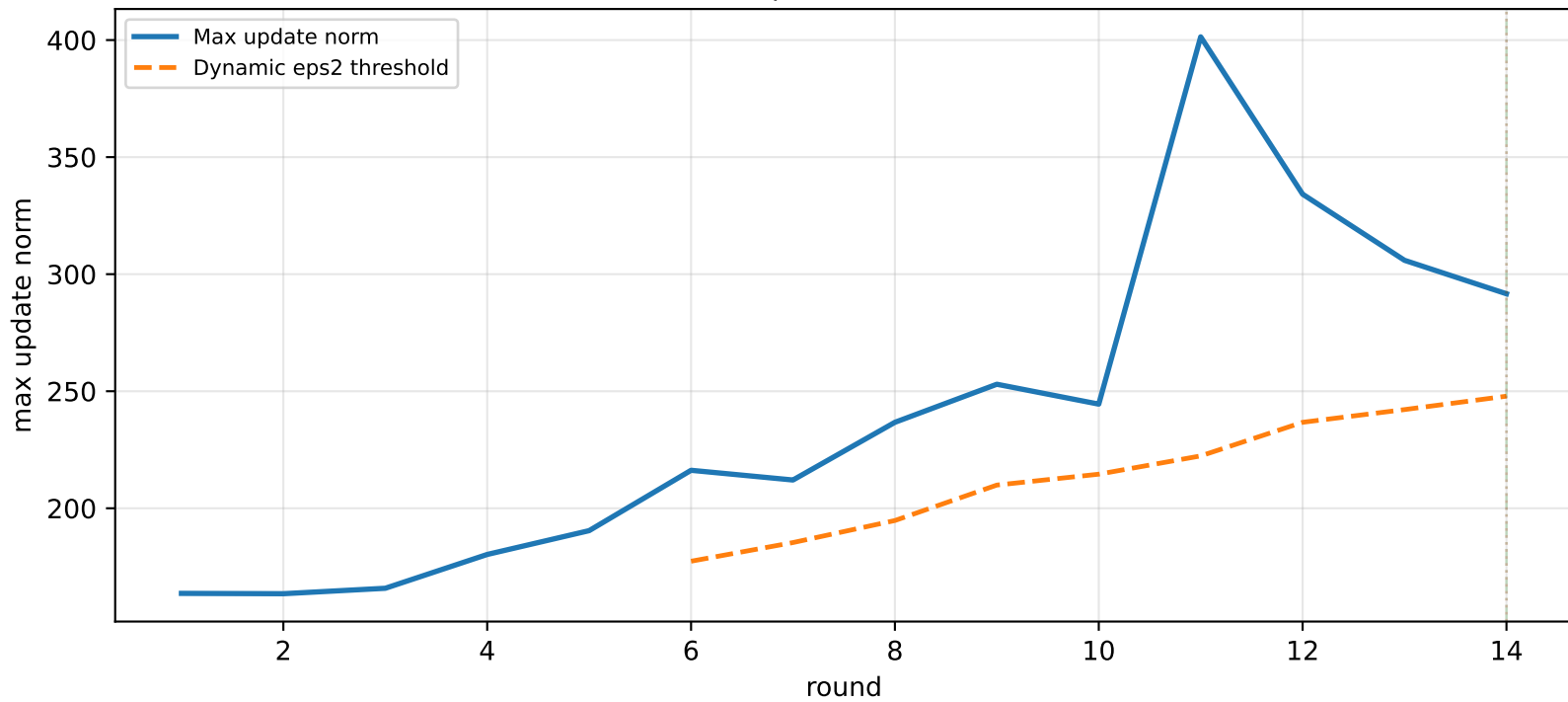
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

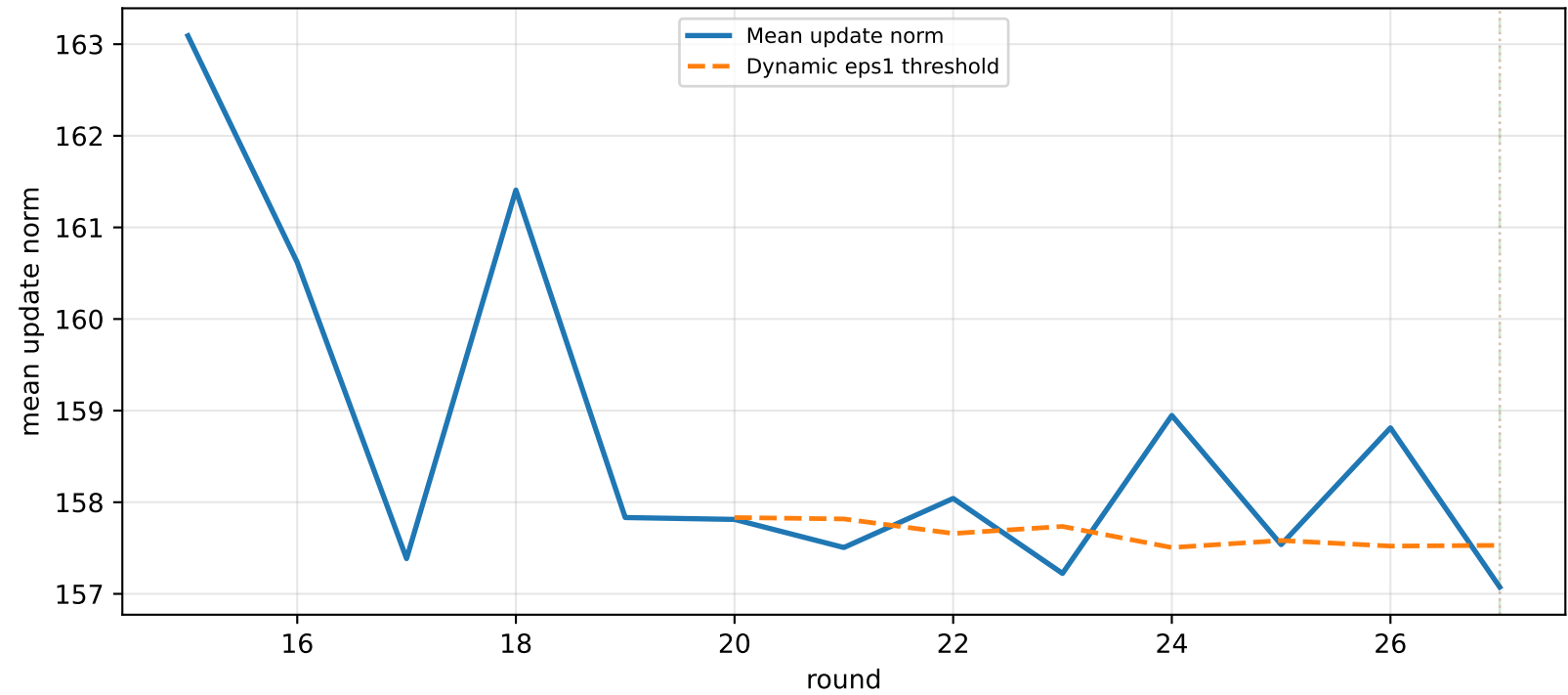
Cluster 0 | Mean update norm



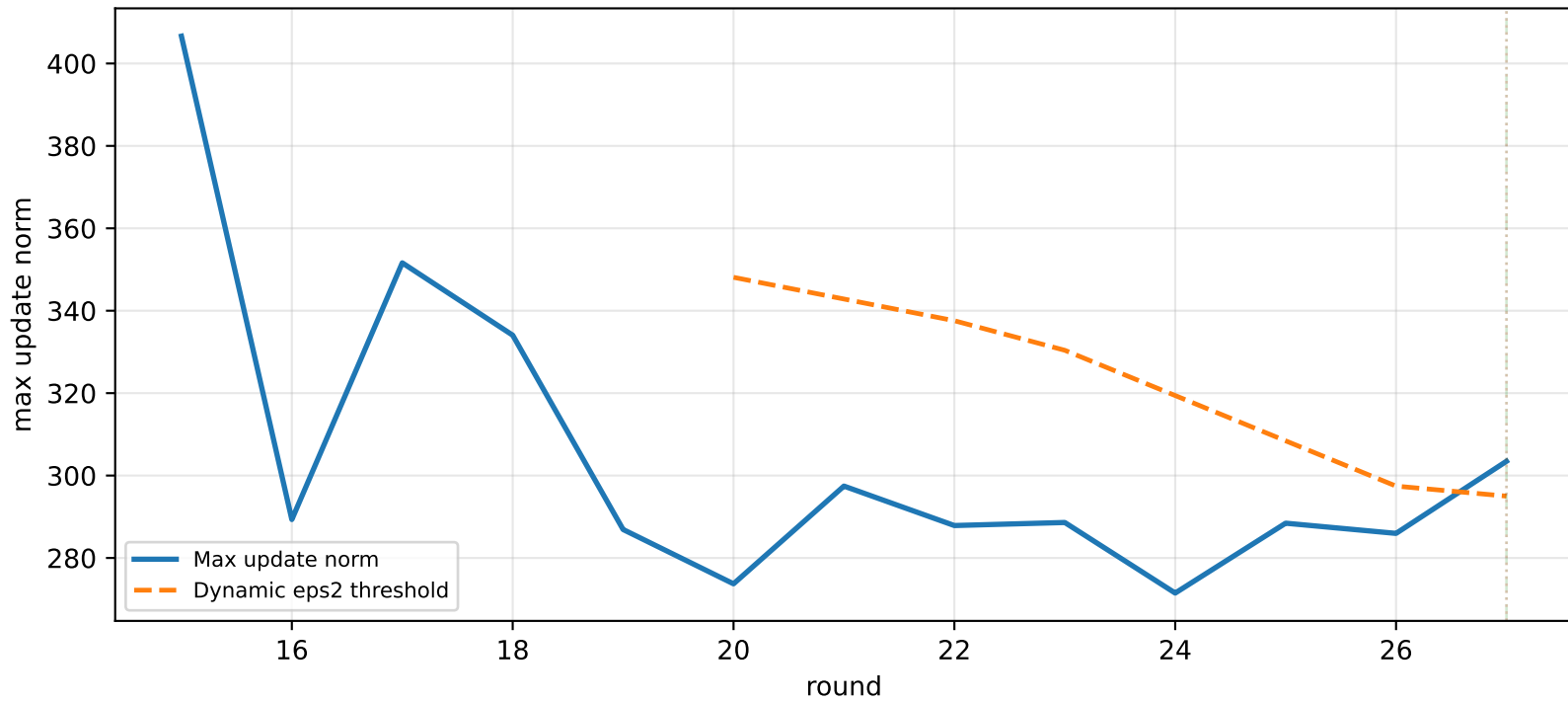
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



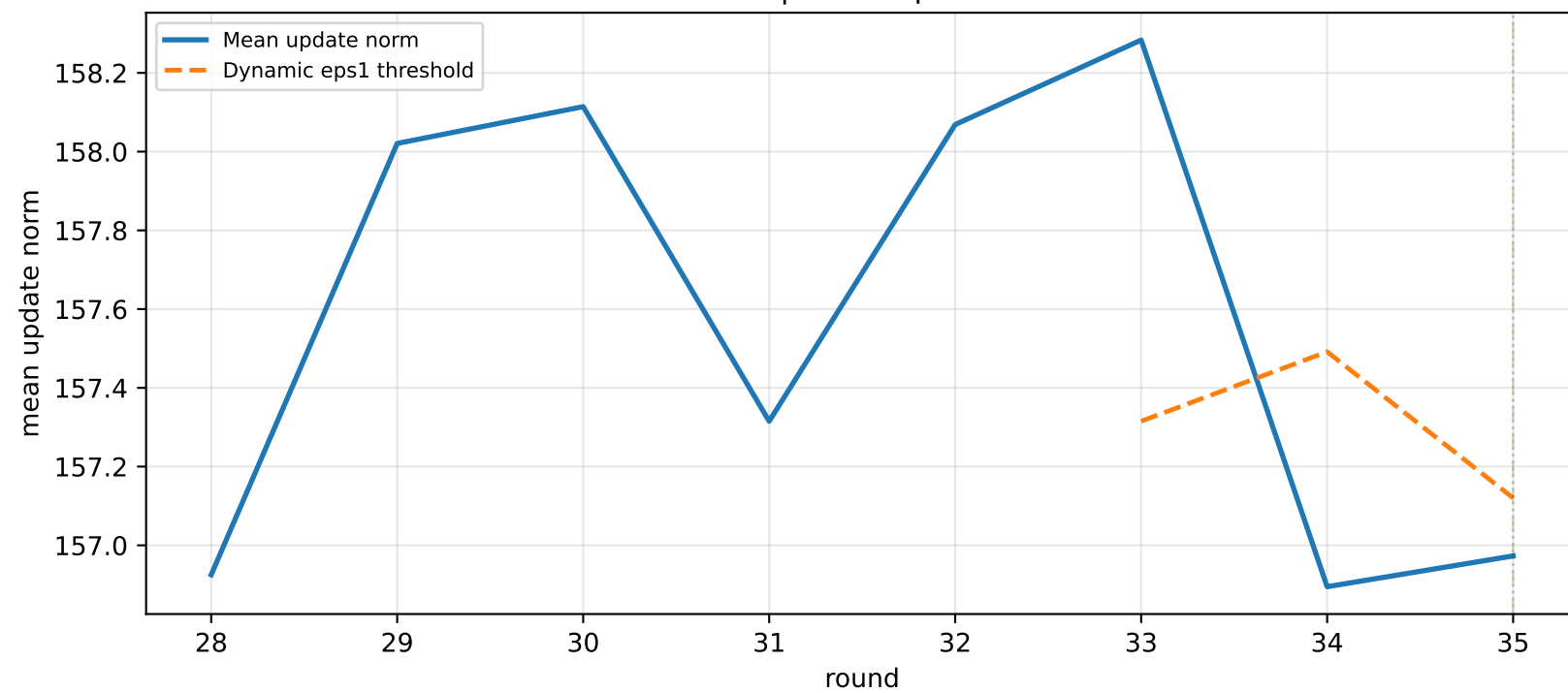
Cluster 2 | Max update norm



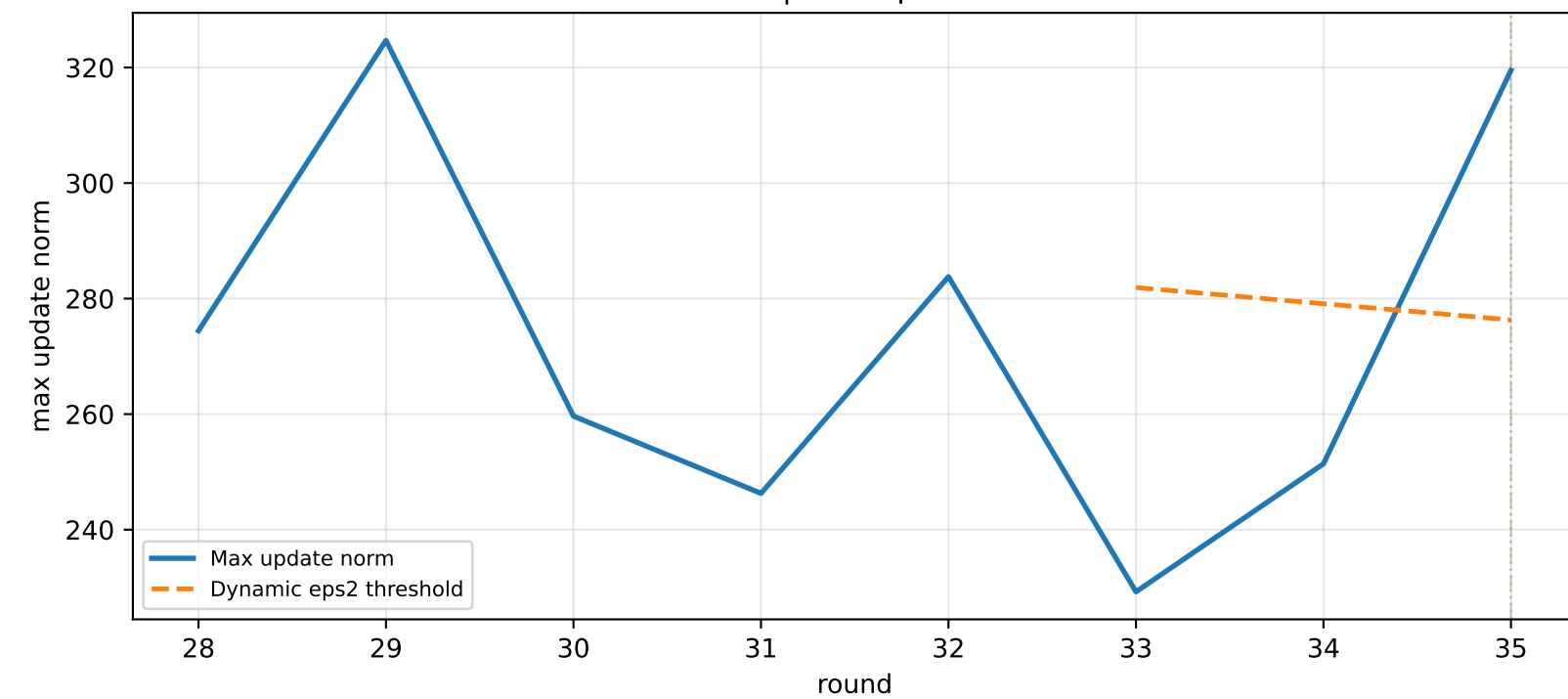
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

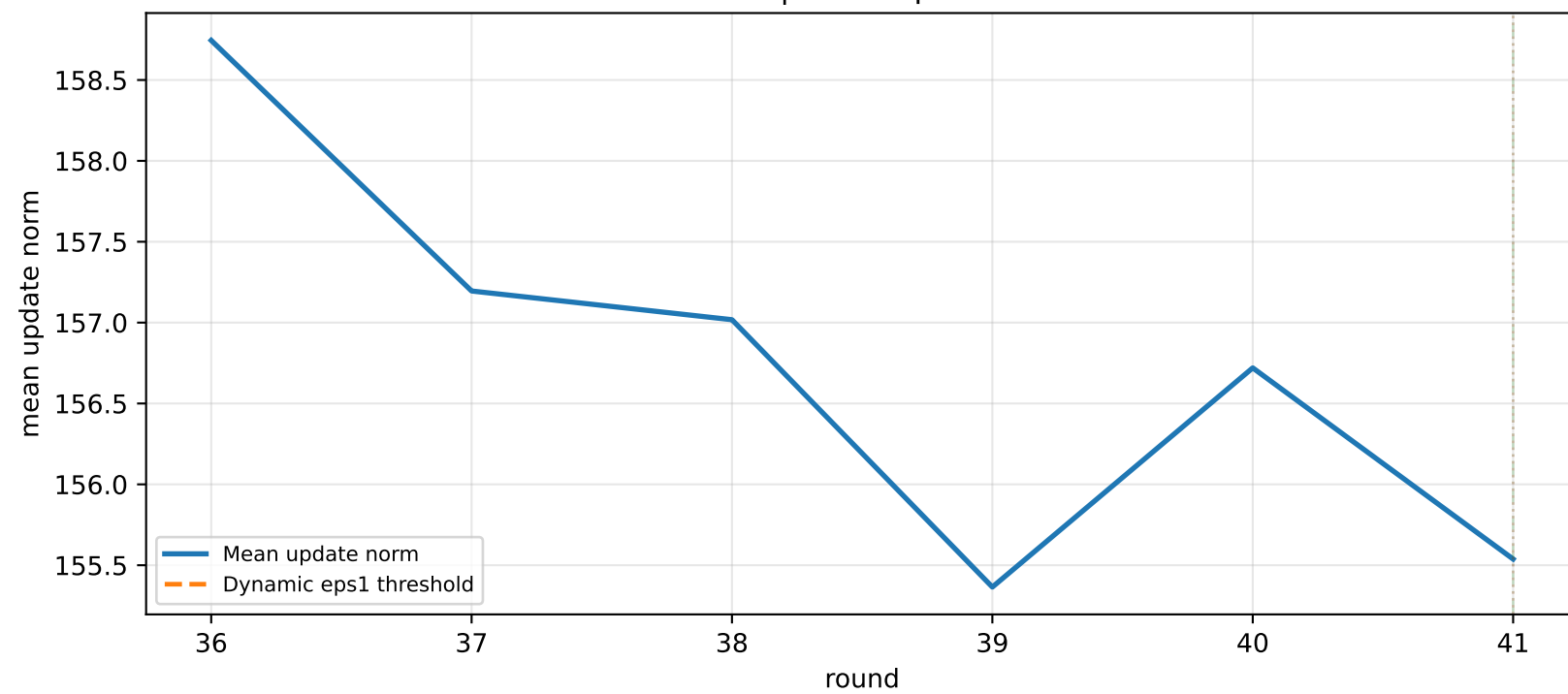
Cluster 4 | Mean update norm



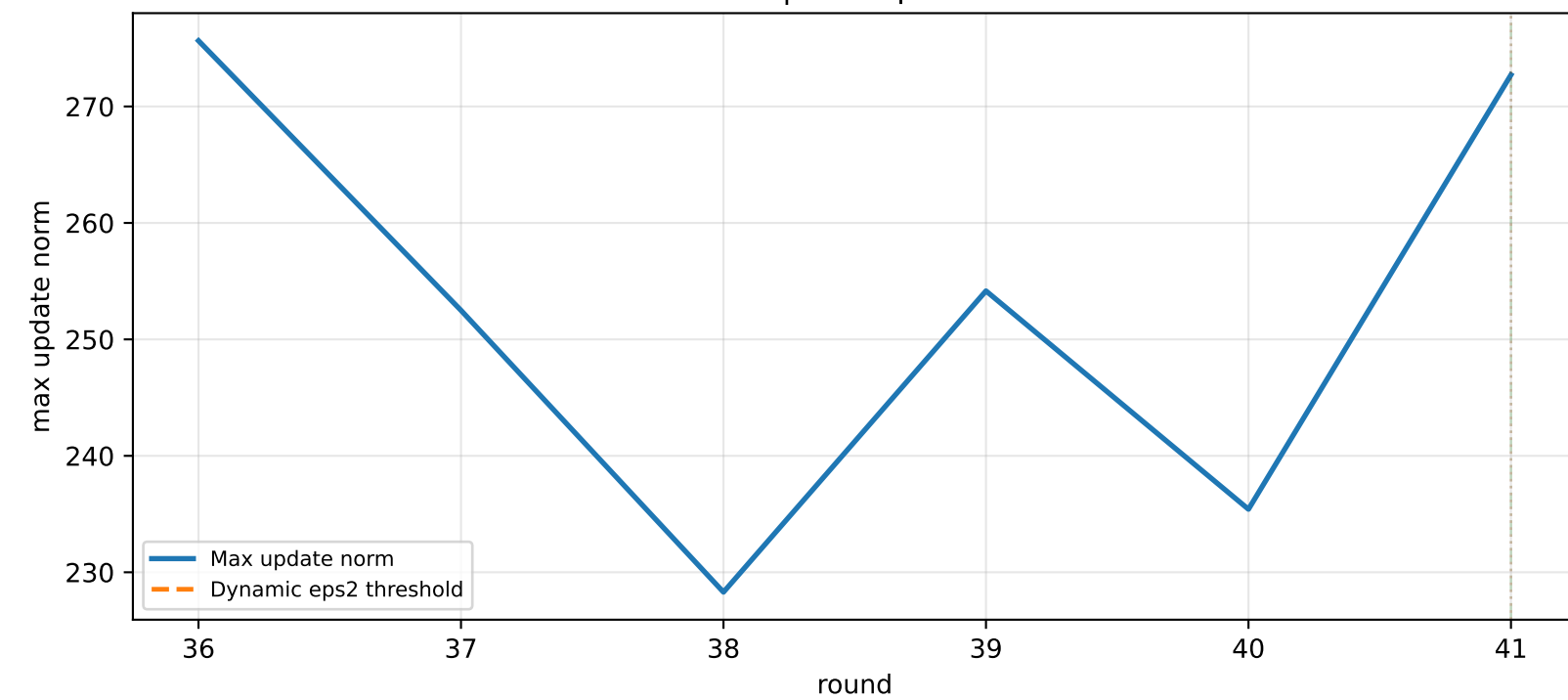
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



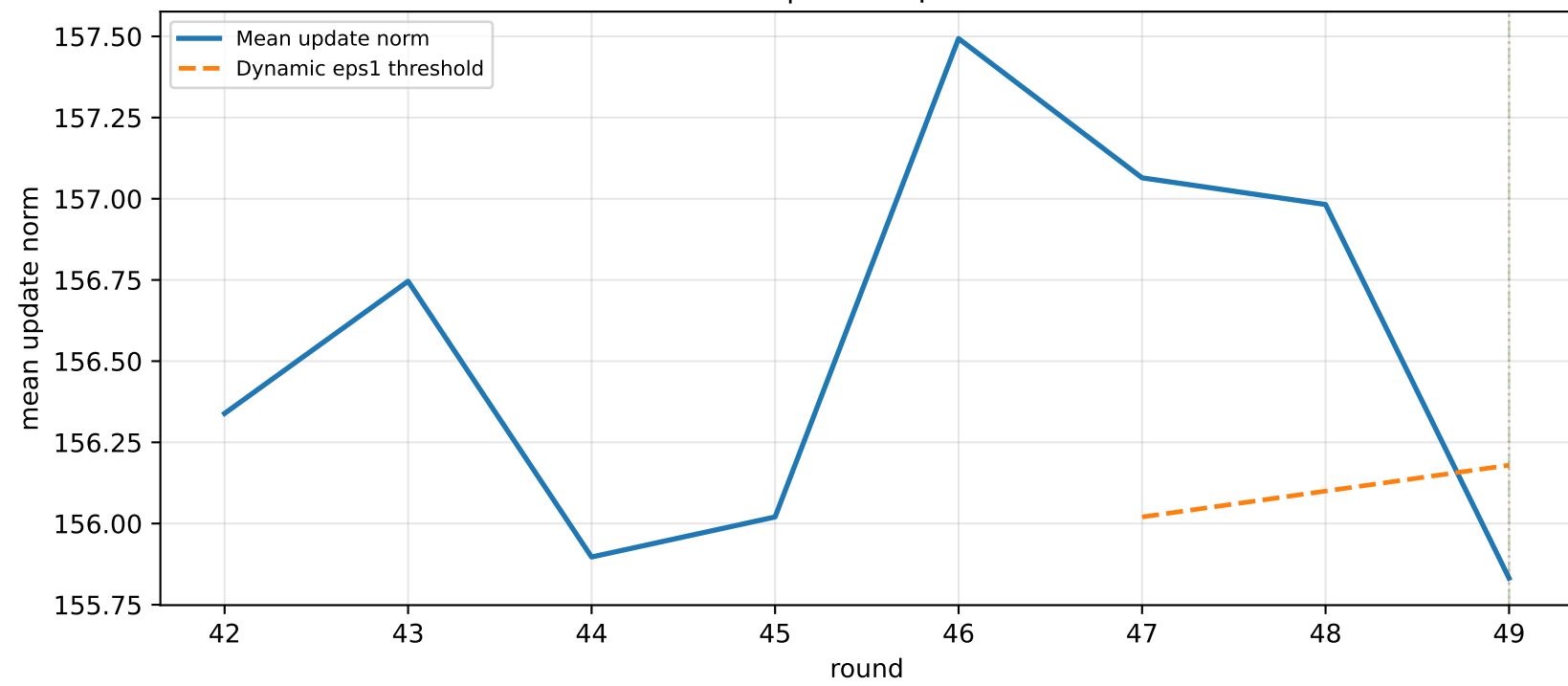
Cluster 6 | Max update norm



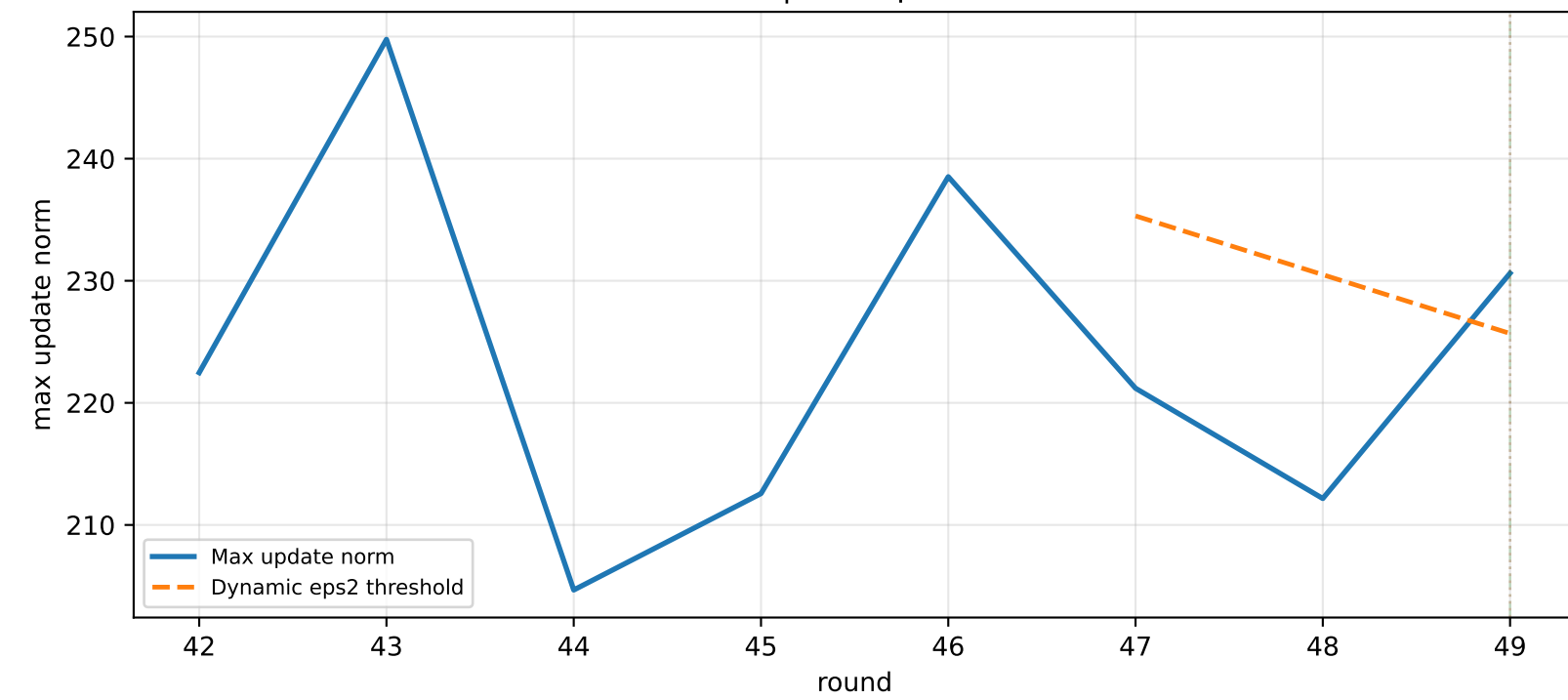
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (8-10)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

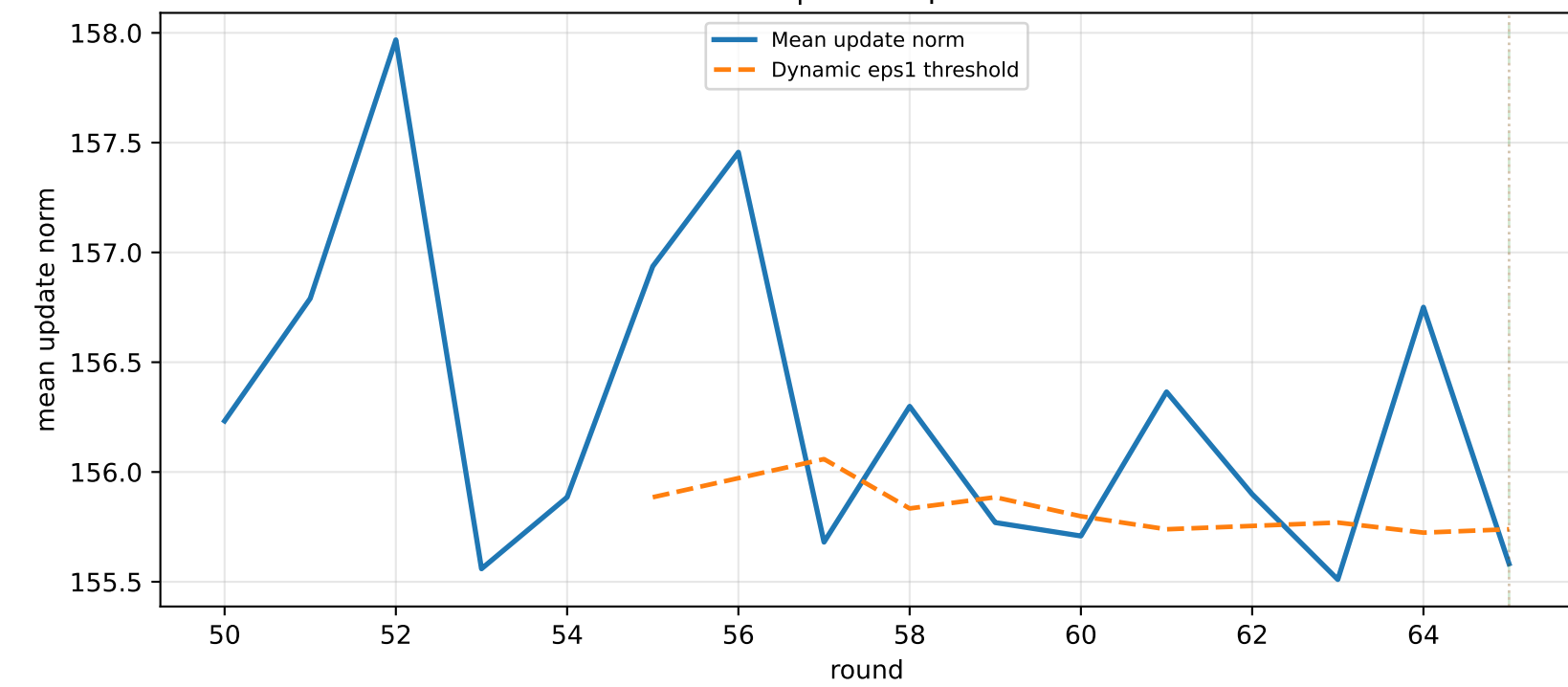
Cluster 8 | Mean update norm



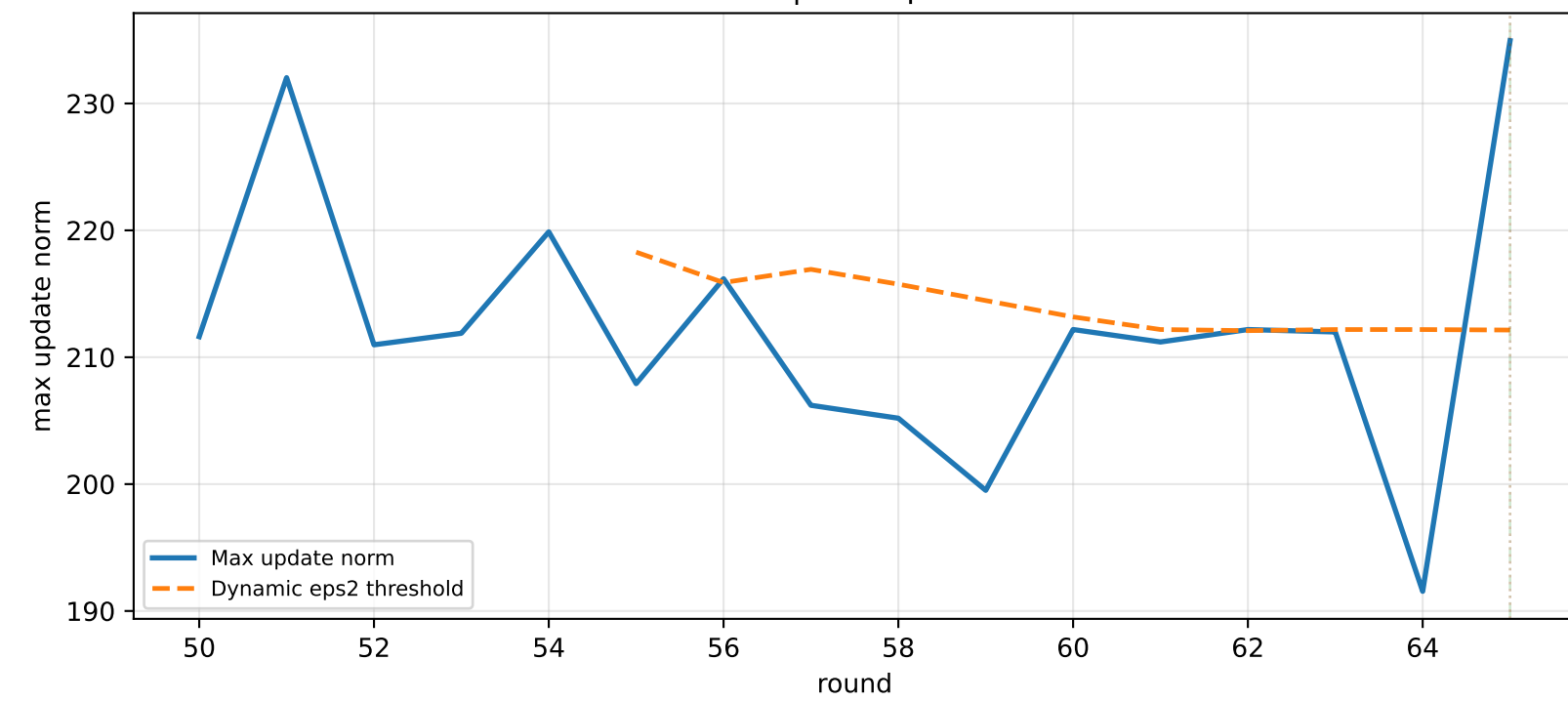
Cluster 8 | Max update norm



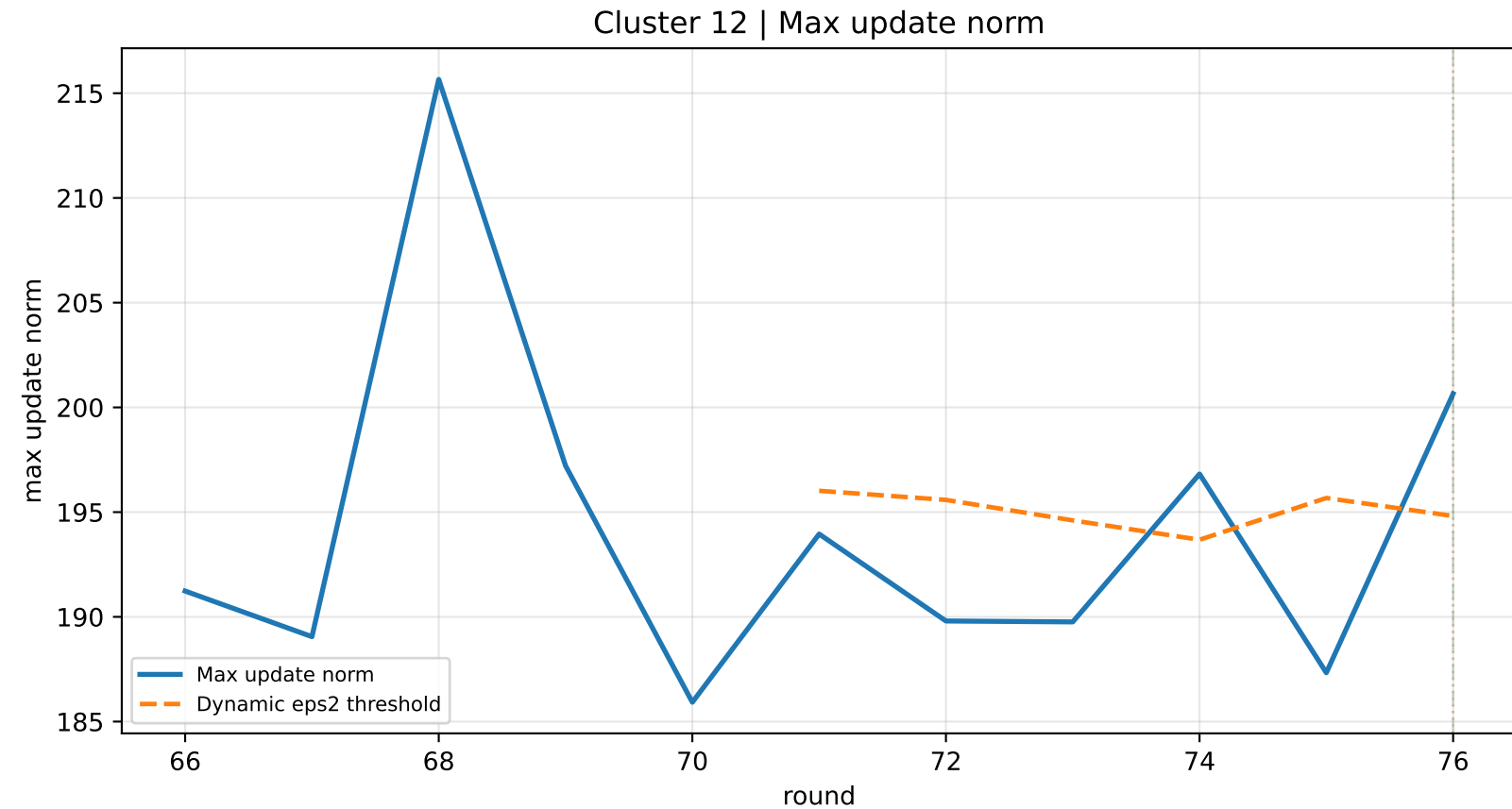
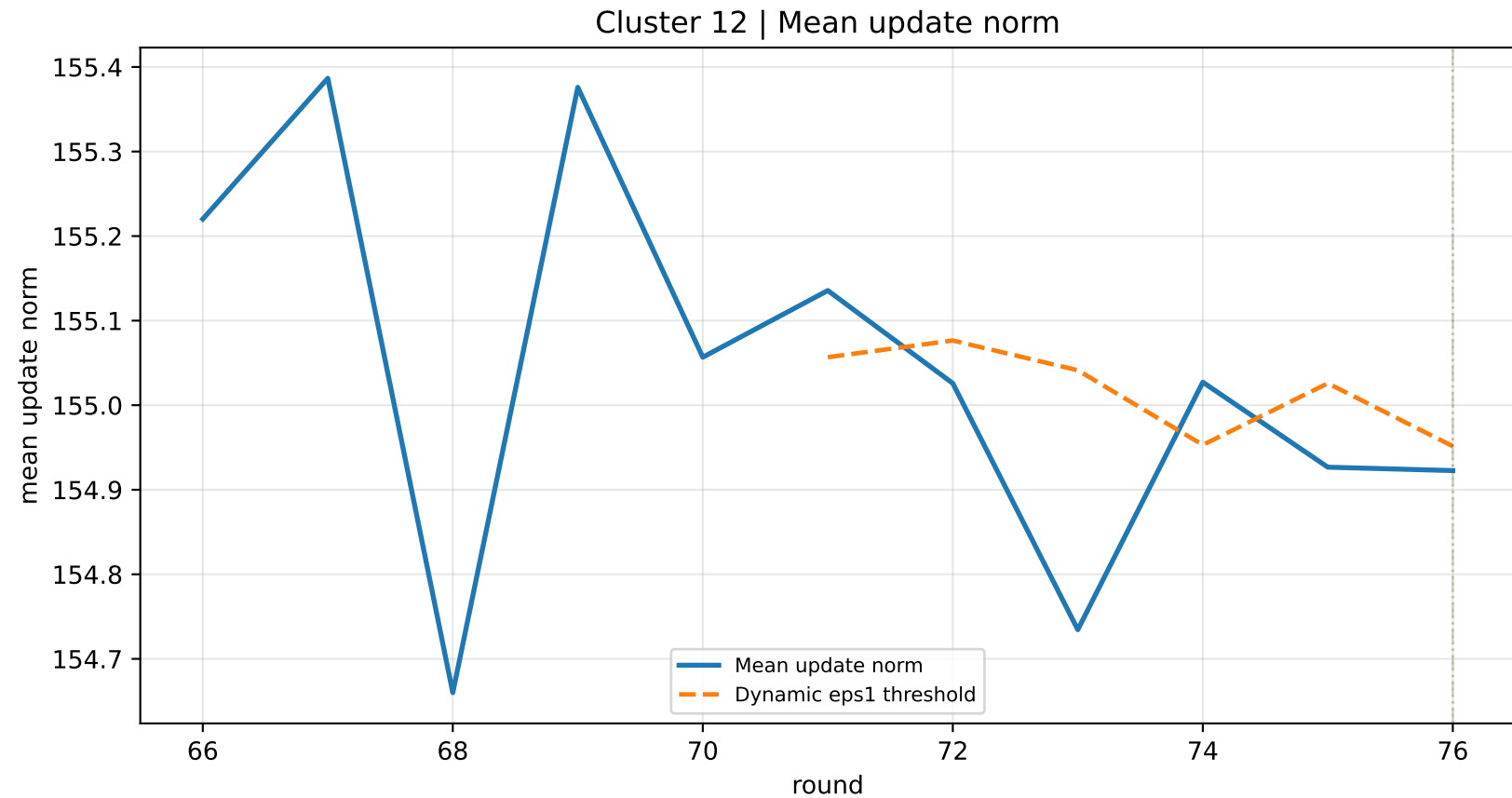
Cluster 10 | Mean update norm



Cluster 10 | Max update norm



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (12-12)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

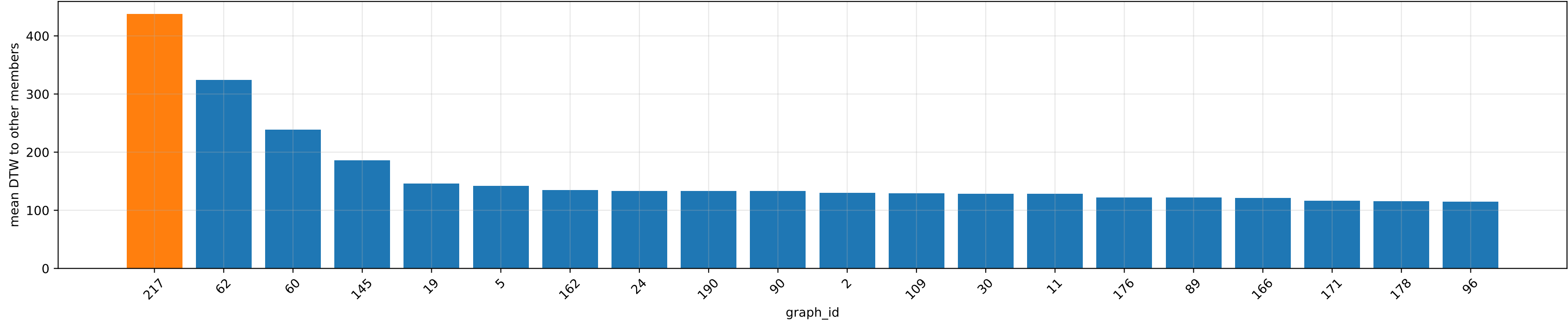
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	14	0	1 vs 19	217	217	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
437.355	nan	130.979

Per-member mean DTW



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

First applied split pairwise DTW metadata | seed=1

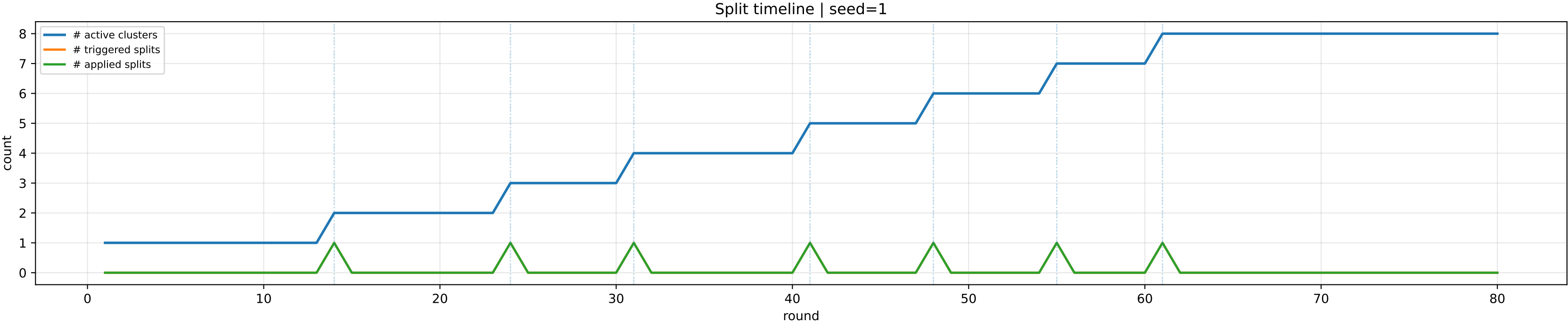
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	14	0	20	217.0	0.045	437.355	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 531.4 at (5, 217)

graph_id	178	96	19	176	171	89	90	11	190	109	24	30	166	5	2	162	62	217	145	60
178	0.0	58.4	97.5	21.6	66.3	85.4	70.9	58.8	91.8	50.8	68.2	67.9	80.8	71.8	53.6	88.5	323.0	441.1	174.9	217.7
96	58.4	0.0	73.4	64.7	32.5	54.0	66.0	38.8	87.2	86.3	68.2	59.9	65.8	58.9	54.9	80.0	351.0	459.8	172.8	250.2
19	97.5	73.4	0.0	98.0	61.6	91.5	47.9	48.4	142.0	69.6	38.9	101.3	113.3	81.8	81.5	141.9	413.0	521.8	234.8	312.2
176	21.6	64.7	98.0	0.0	78.7	100.0	75.0	79.8	117.9	44.2	76.7	70.8	81.1	89.8	67.5	103.9	309.4	436.4	179.1	220.3
171	66.3	32.5	61.6	78.7	0.0	62.7	54.5	45.3	86.1	64.1	60.4	60.2	52.1	65.8	53.4	96.7	359.3	468.1	181.1	258.5
89	85.4	54.0	91.5	100.0	62.7	0.0	92.1	70.5	51.4	72.7	82.4	95.9	92.0	87.1	82.1	77.0	321.6	430.3	143.3	220.7
90	70.9	66.0	47.9	75.0	54.5	92.1	0.0	42.0	136.8	58.7	20.0	97.0	93.3	42.9	27.3	136.1	407.8	516.6	229.6	307.0
11	58.8	38.8	48.4	79.8	45.3	70.5	42.0	0.0	134.4	61.6	50.6	69.6	83.3	34.7	33.9	130.0	405.4	514.2	227.2	304.6
190	91.8	87.2	142.0	117.9	86.1	51.4	136.8	134.4	0.0	100.9	132.8	95.7	74.0	151.6	135.7	56.2	278.7	377.3	93.1	179.3
109	50.8	86.3	69.6	44.2	64.1	72.7	58.7	61.6	100.9	0.0	59.6	42.7	106.2	85.0	65.1	149.2	384.2	461.8	189.9	292.4
24	68.2	68.2	38.9	76.7	60.4	82.4	20.0	50.6	132.8	59.6	0.0	97.2	93.9	52.4	38.8	139.5	403.8	512.6	225.6	303.0
30	67.9	59.9	101.3	70.8	60.2	95.9	97.0	69.6	95.7	42.7	97.2	0.0	95.6	91.3	85.8	119.2	341.1	442.4	162.9	244.9
166	80.8	65.8	113.3	81.1	52.1	92.0	93.3	83.3	74.0	106.2	93.9	95.6	0.0	97.3	85.0	56.2	266.1	447.3	133.6	177.2
5	71.8	58.9	81.8	89.8	65.8	87.1	42.9	34.7	151.6	85.0	52.4	91.3	97.3	0.0	21.0	144.0	422.6	531.4	244.4	321.8
2	53.6	54.9	81.5	67.5	53.4	82.1	27.3	33.9	135.7	65.1	38.8	85.8	85.0	21.0	0.0	132.5	403.6	512.4	225.4	302.8
162	88.5	80.0	141.9	103.9	96.7	77.0	136.1	130.0	56.2	149.2	139.5	119.2	56.2	144.0	132.5	0.0	240.7	410.7	113.6	145.5
62	323.0	351.0	413.0	309.4	359.3	321.6	407.8	405.4	278.7	384.2	403.8	341.1	266.1	422.6	403.6	240.7	0.0	261.5	198.0	62.1
217	441.1	459.8	521.8	436.4	468.1	430.3	516.6	514.2	377.3	461.8	512.6	442.4	447.3	531.4	512.4	410.7	261.5	0.0	276.3	287.7
145	174.9	172.8	234.8	179.1	181.1	143.3	229.6	227.2	93.1	189.9	225.6	162.9	133.6	244.4	225.4	113.6	198.0	276.3	0.0	128.9
60	217.7	250.2	312.2	220.3	258.5	220.7	307.0	304.6	179.3	292.4	303.0	244.9	177.2	321.8	302.8	145.5	62.1	287.7	128.9	0.0

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

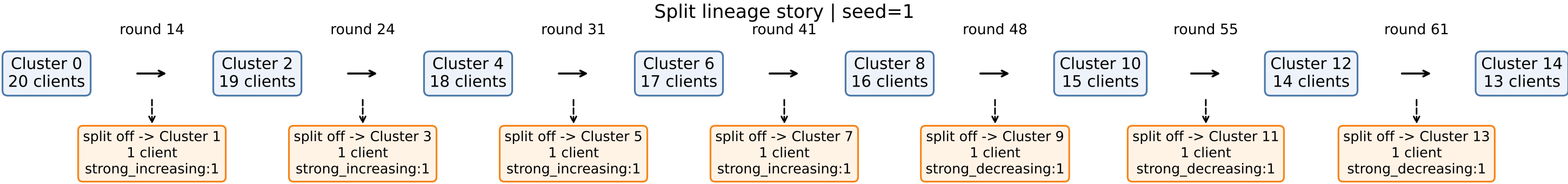


Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	14, 24, 31, 41, 48, 55, 61	14, 24, 31, 41, 48, 55, 61	1, 3, 5, 7, 9, 11, 13, 14	1:1, 3:1, 5:1, 7:1, 9:1, 11:1, 13:1, 14:13

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068



Applied split story | seed=1

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
14	0	20	157.261 < 159.798 , 292.764 > 259.205	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
24	2	19	156.686 < 156.736 , 368.557 > 335.454	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
31	4	18	154.986 < 156.039 , 307.055 > 303.487	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1
41	6	17	155.919 < 156.291 , 246.161 > 245.892	C7 (1 client)	strong_increasing:1	C8 (16 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4
48	8	16	156.141 < 156.164 , 209.324 > 207.950	C9 (1 client)	strong_decreasing:1	C10 (15 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:3
55	10	15	154.962 < 155.099 , 201.381 > 195.333	C11 (1 client)	strong_decreasing:1	C12 (14 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:2
61	12	14	155.756 < 155.832 , 200.840 > 195.392	C13 (1 client)	strong_decreasing:1	C14 (13 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:1

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

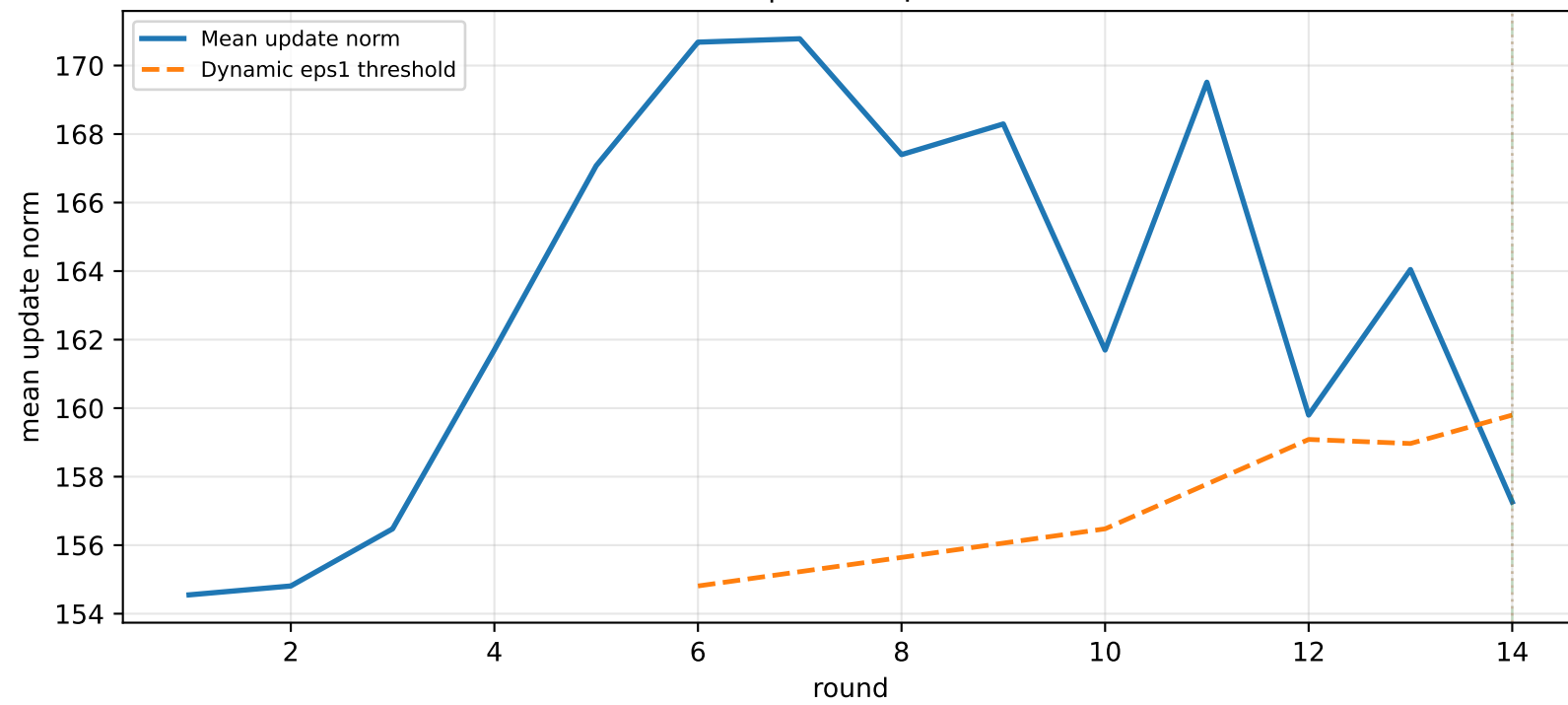
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	217	17	strong_increasing:1
3	1	145	18	strong_increasing:1
5	1	60	19	strong_increasing:1
7	1	62	16	strong_increasing:1
9	1	162	15	strong_decreasing:1
11	1	2	14	strong_decreasing:1
13	1	166	12	strong_decreasing:1
14	13	178, 96, 19, 176, 171, 89, 90, 11, 190, 109, 24, 30, 5	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:1

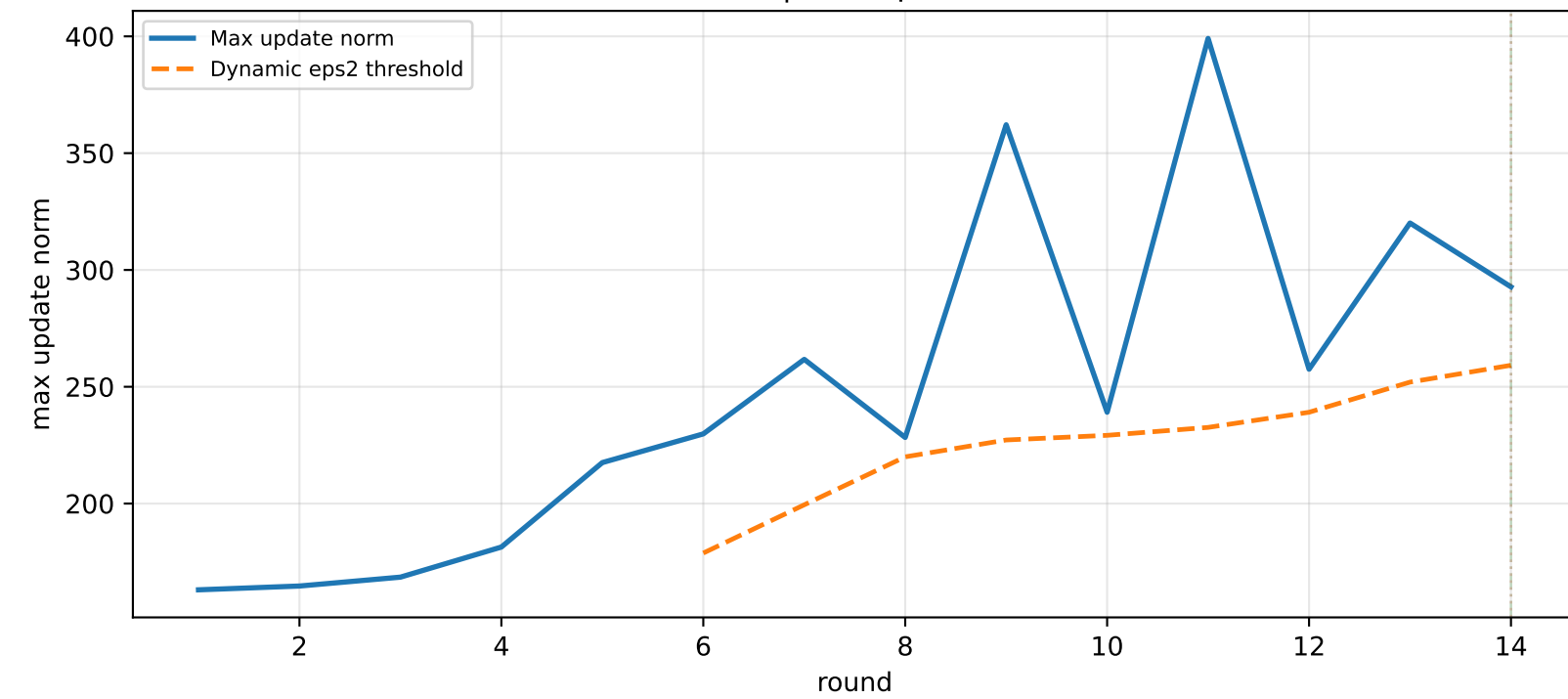
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

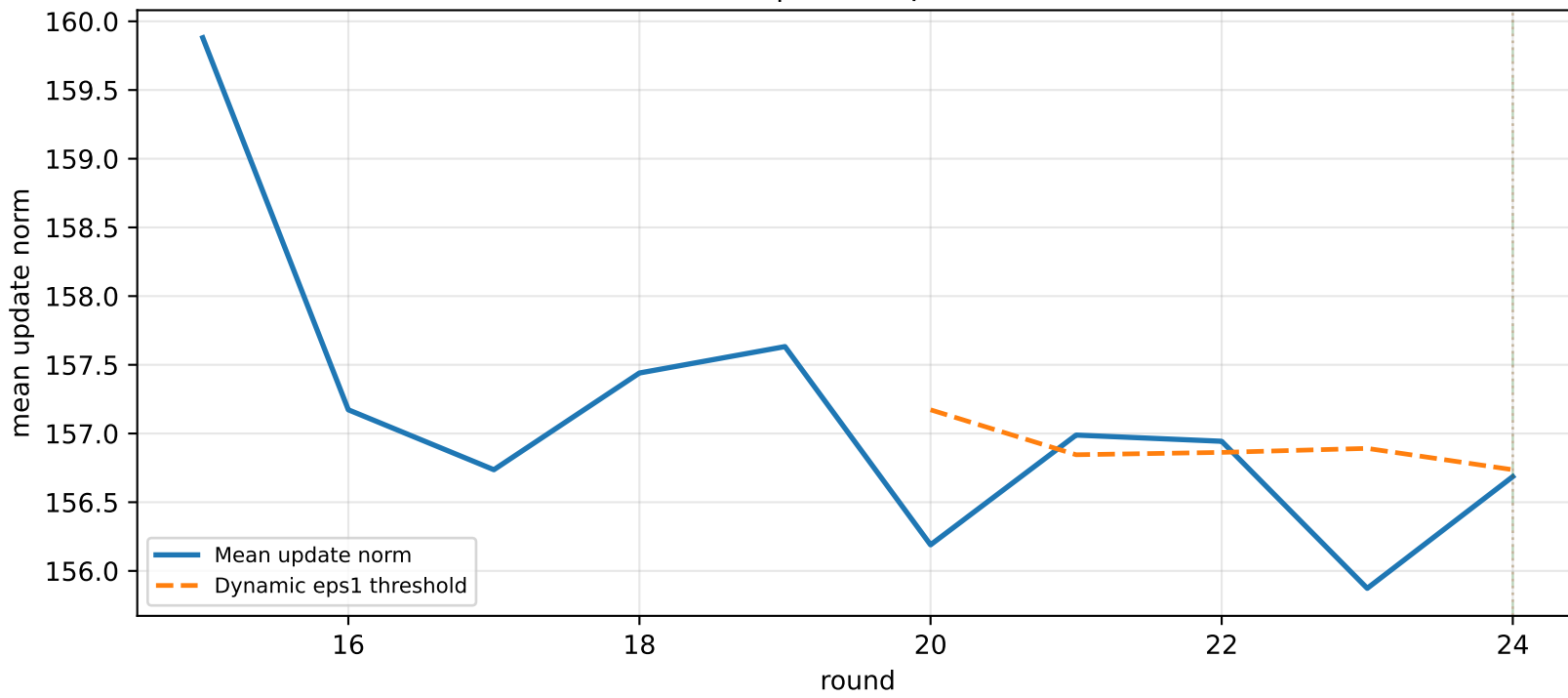
Cluster 0 | Mean update norm



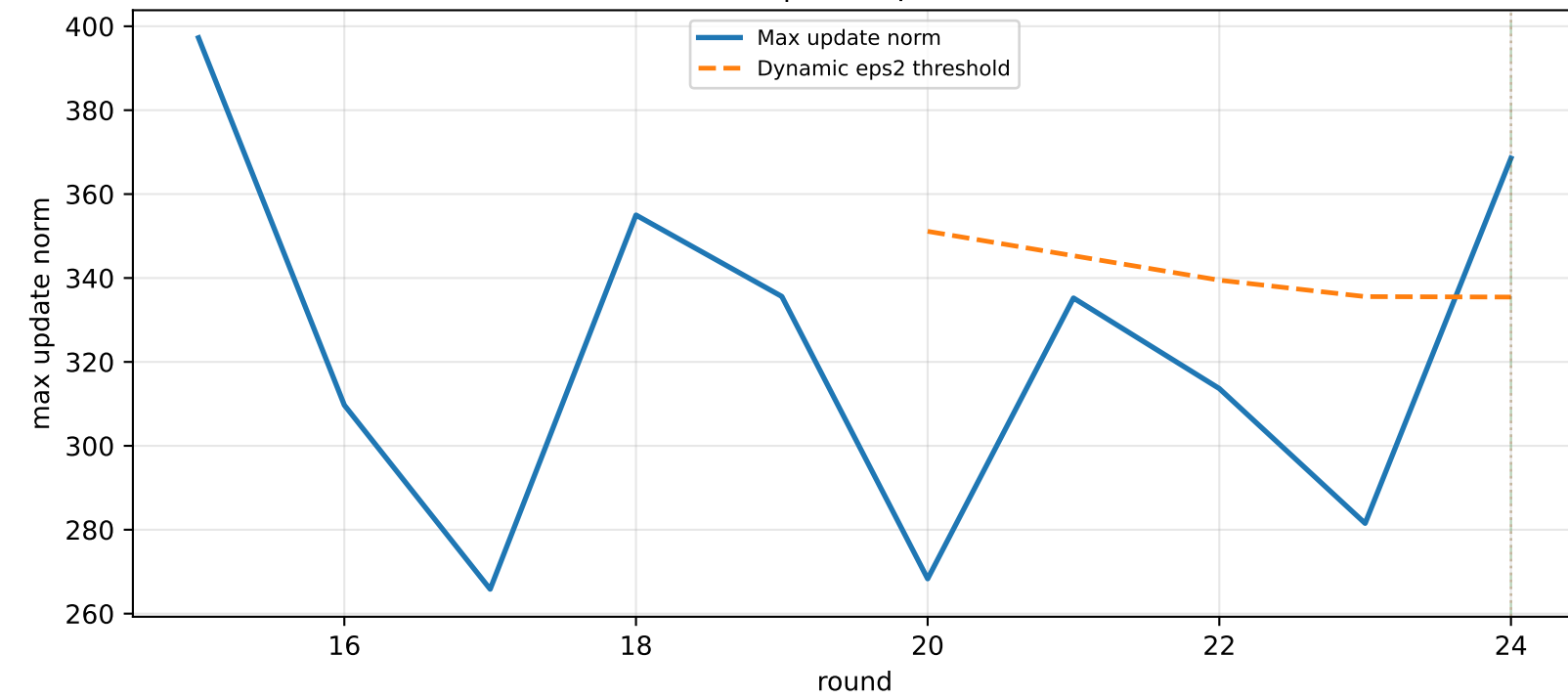
Cluster 0 | Max update norm



Cluster 2 | Mean update norm

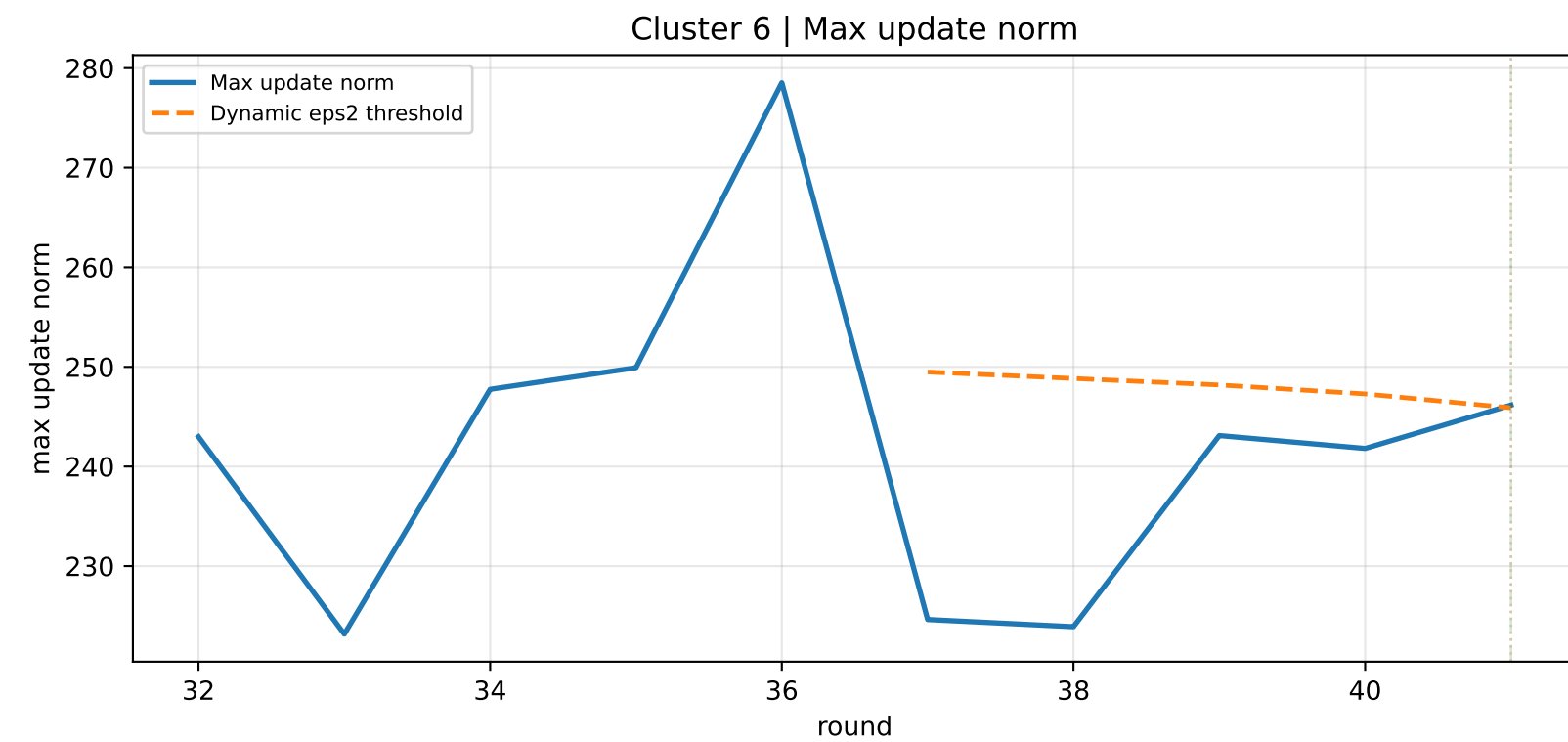
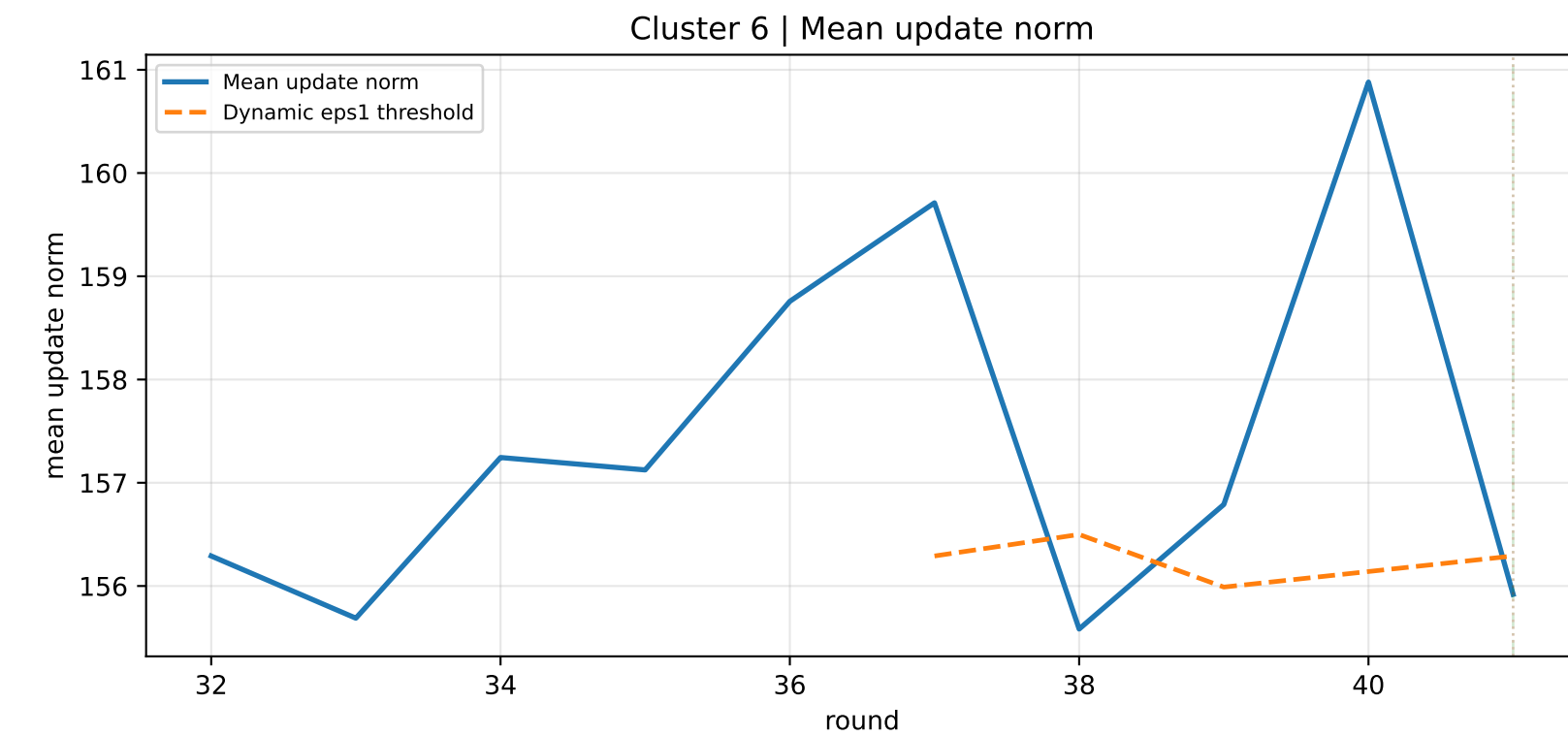
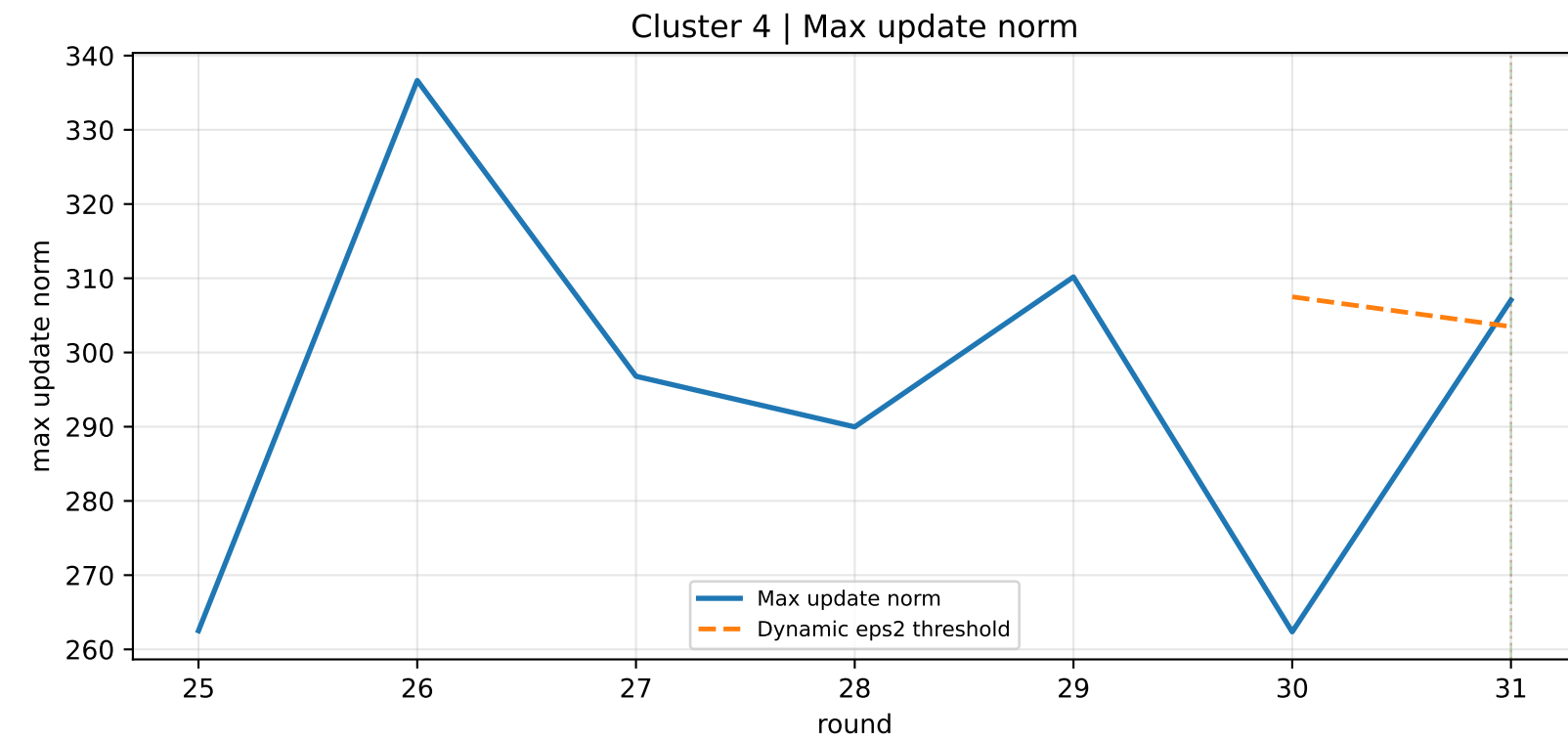
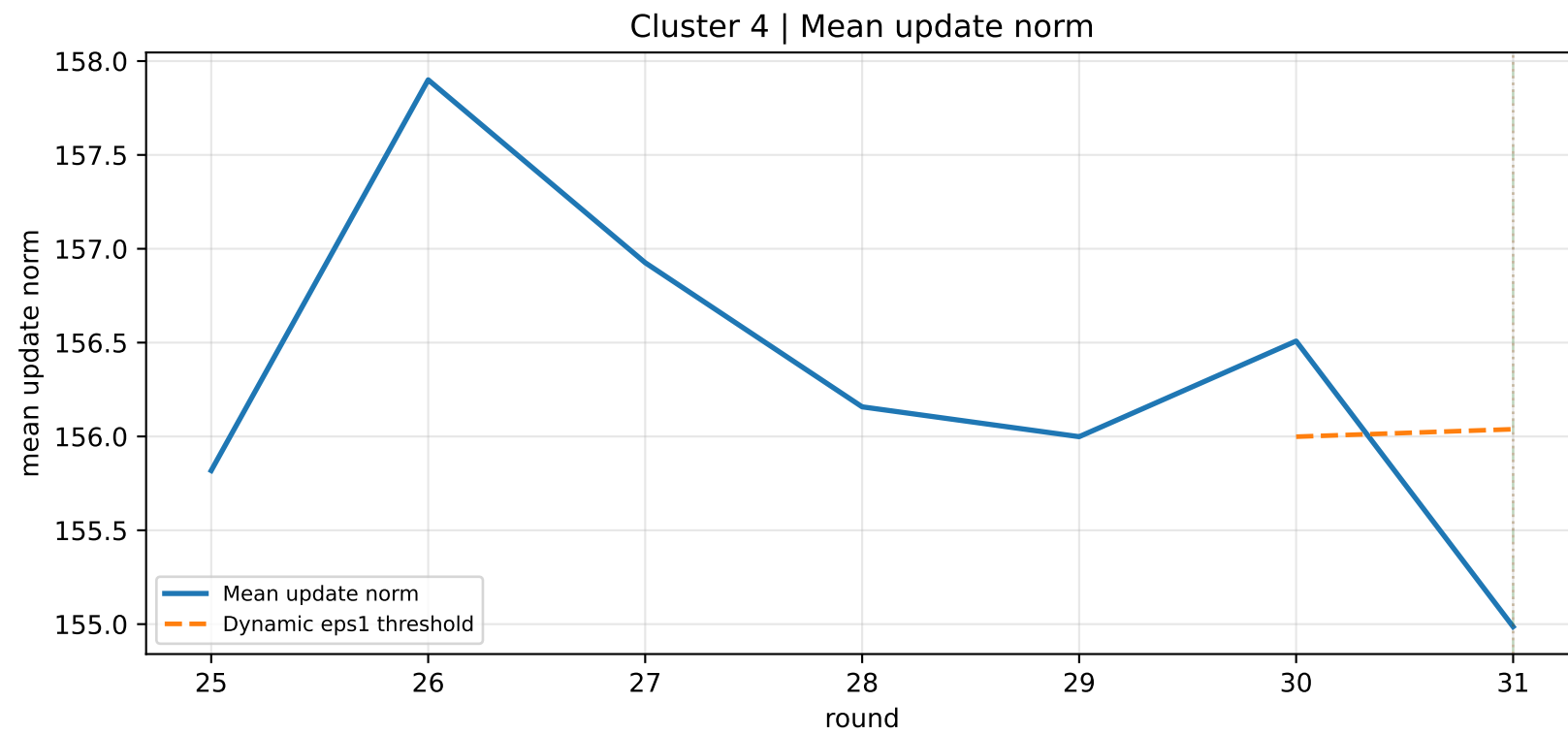


Cluster 2 | Max update norm



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (4-6)

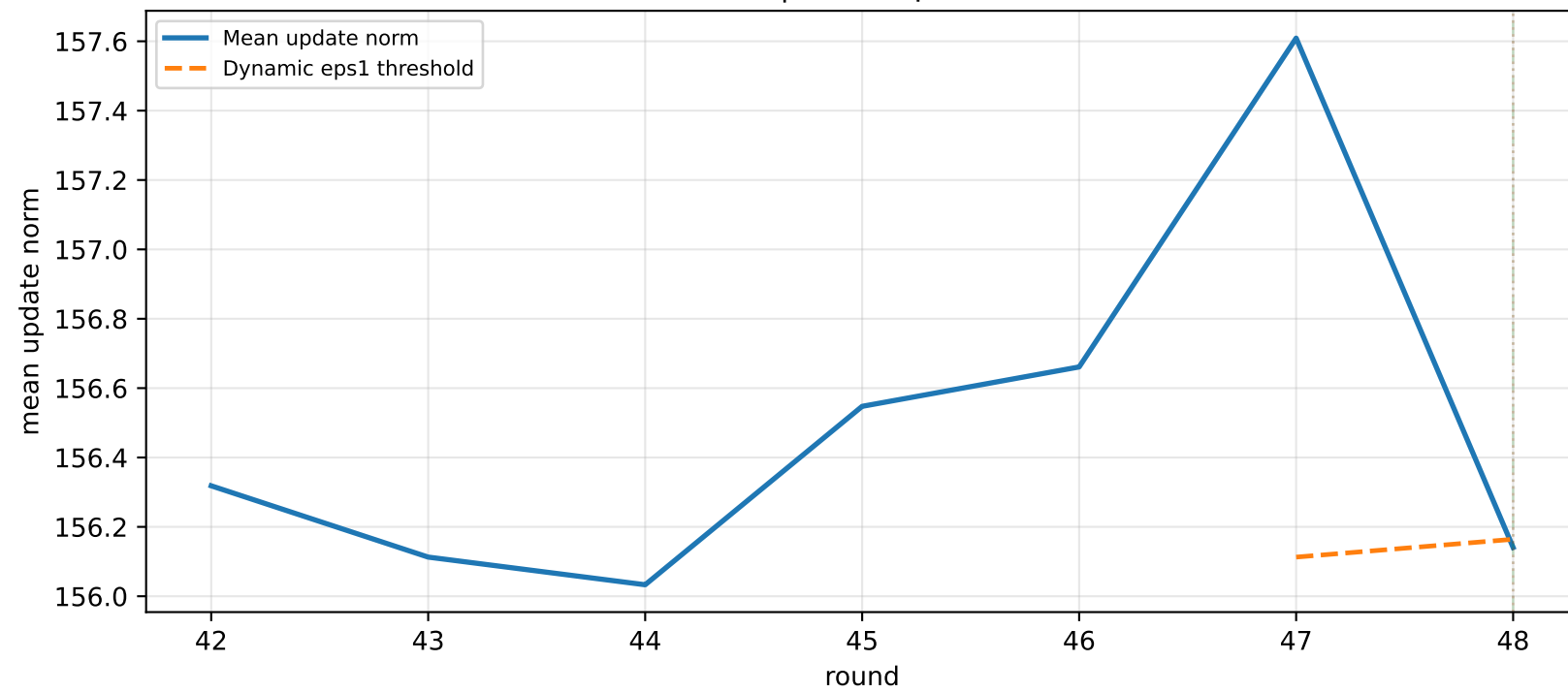
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068



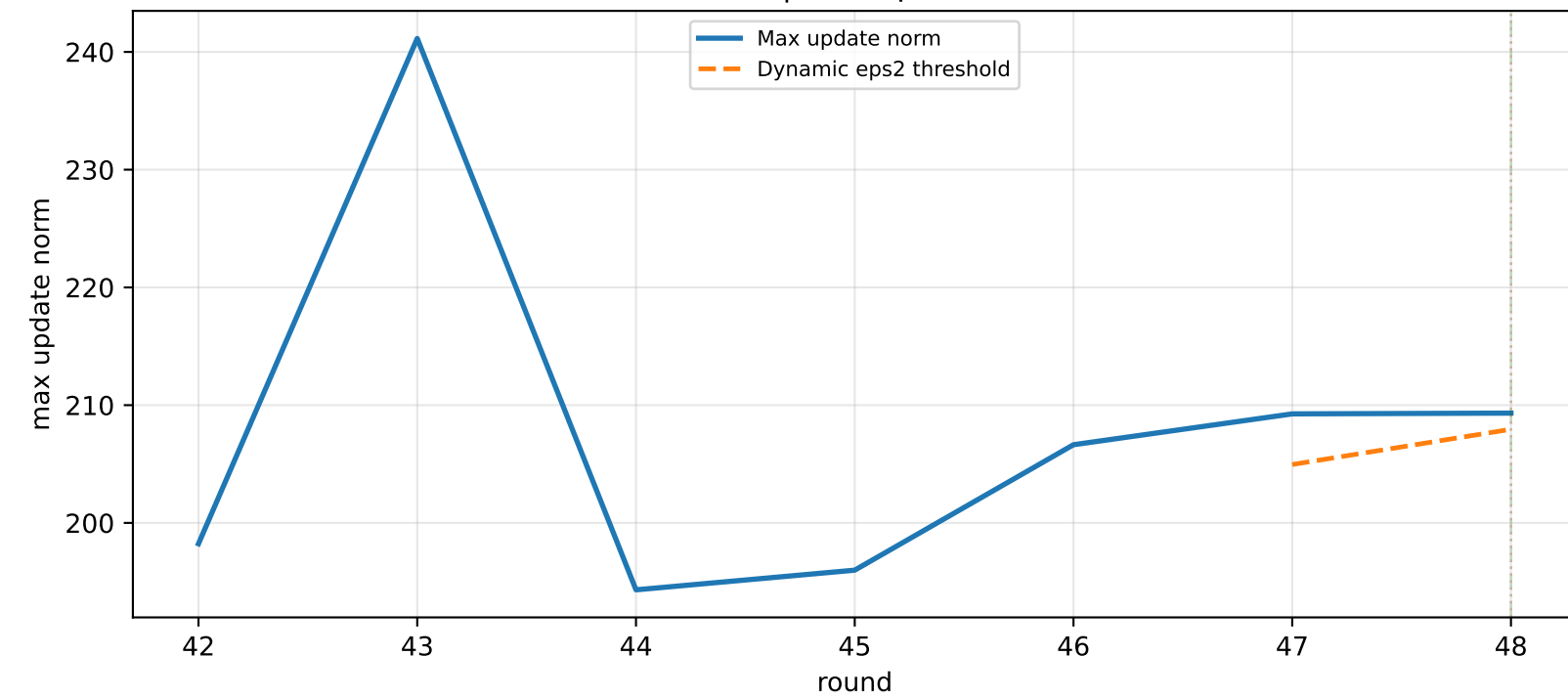
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (8-10)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

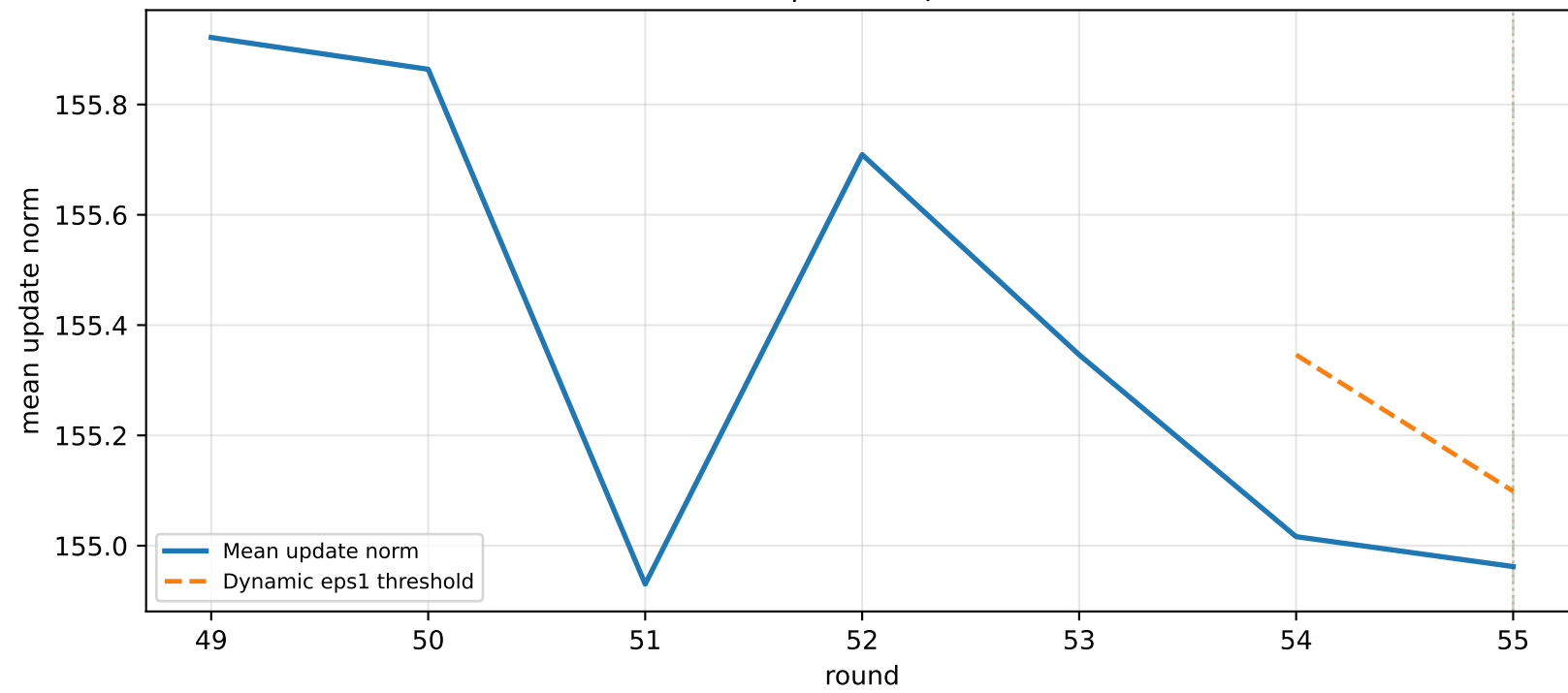
Cluster 8 | Mean update norm



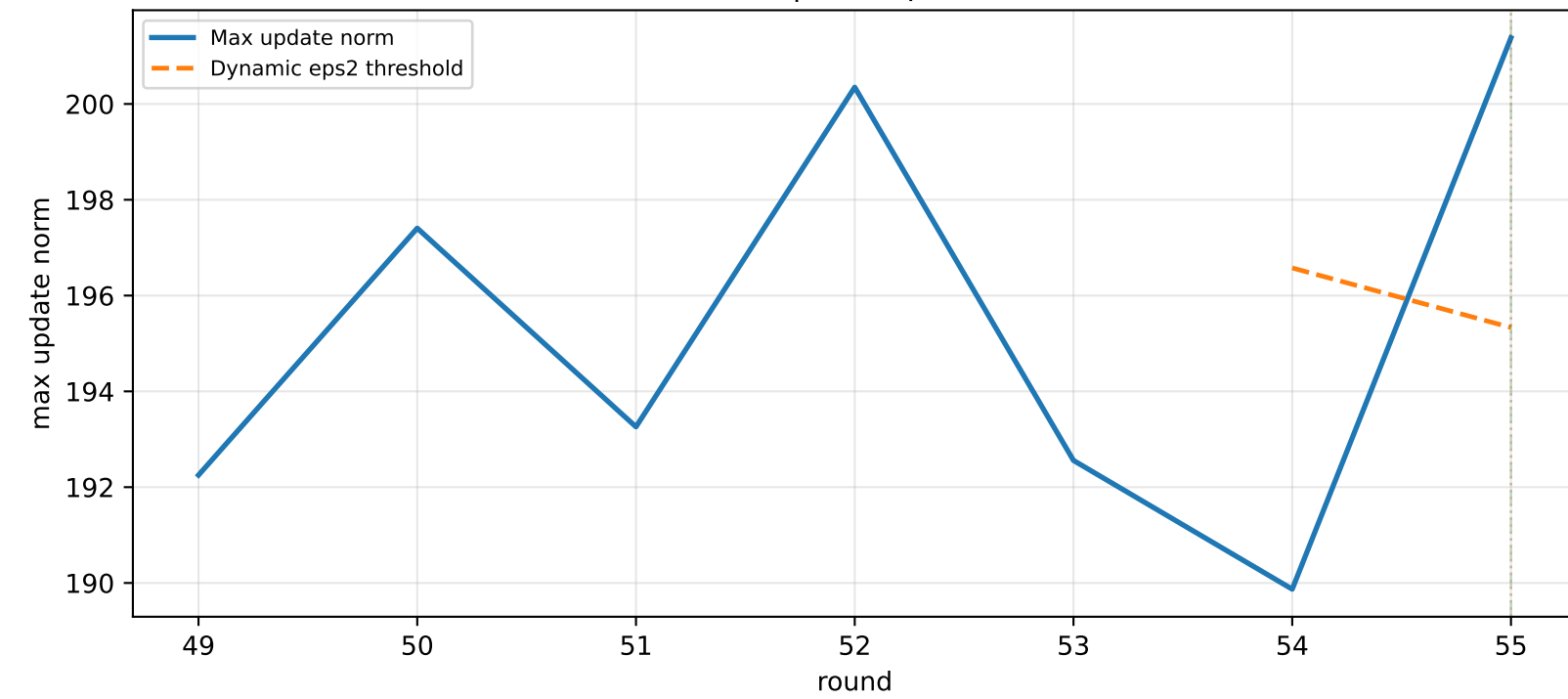
Cluster 8 | Max update norm



Cluster 10 | Mean update norm



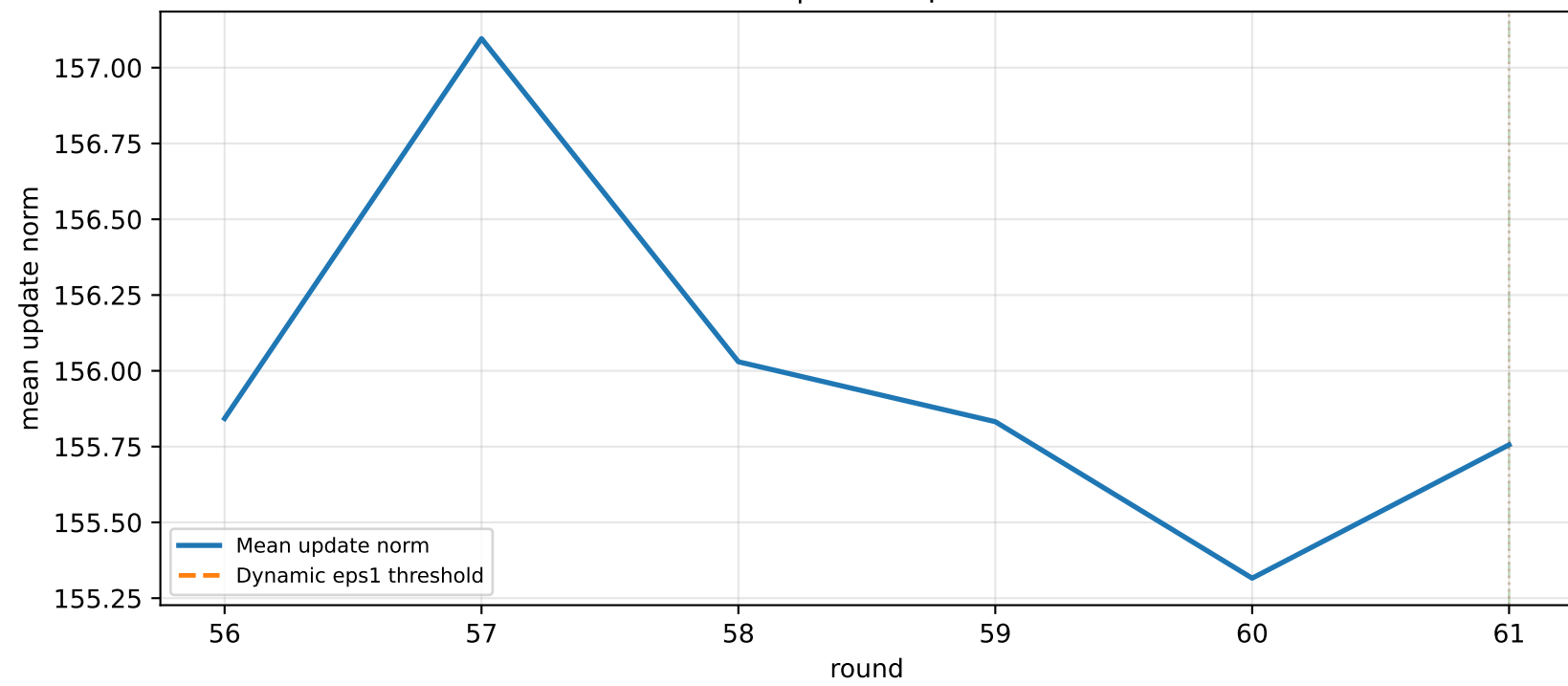
Cluster 10 | Max update norm



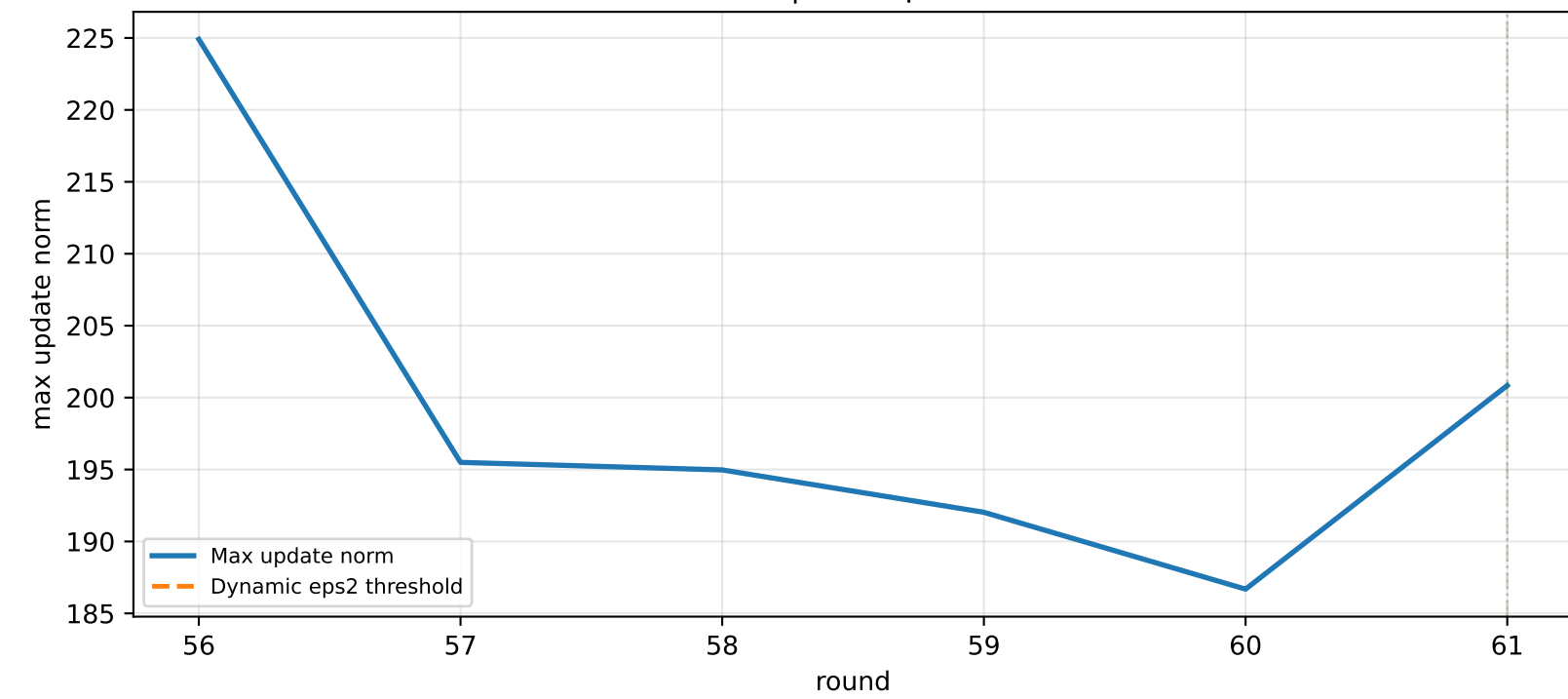
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (12-14)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

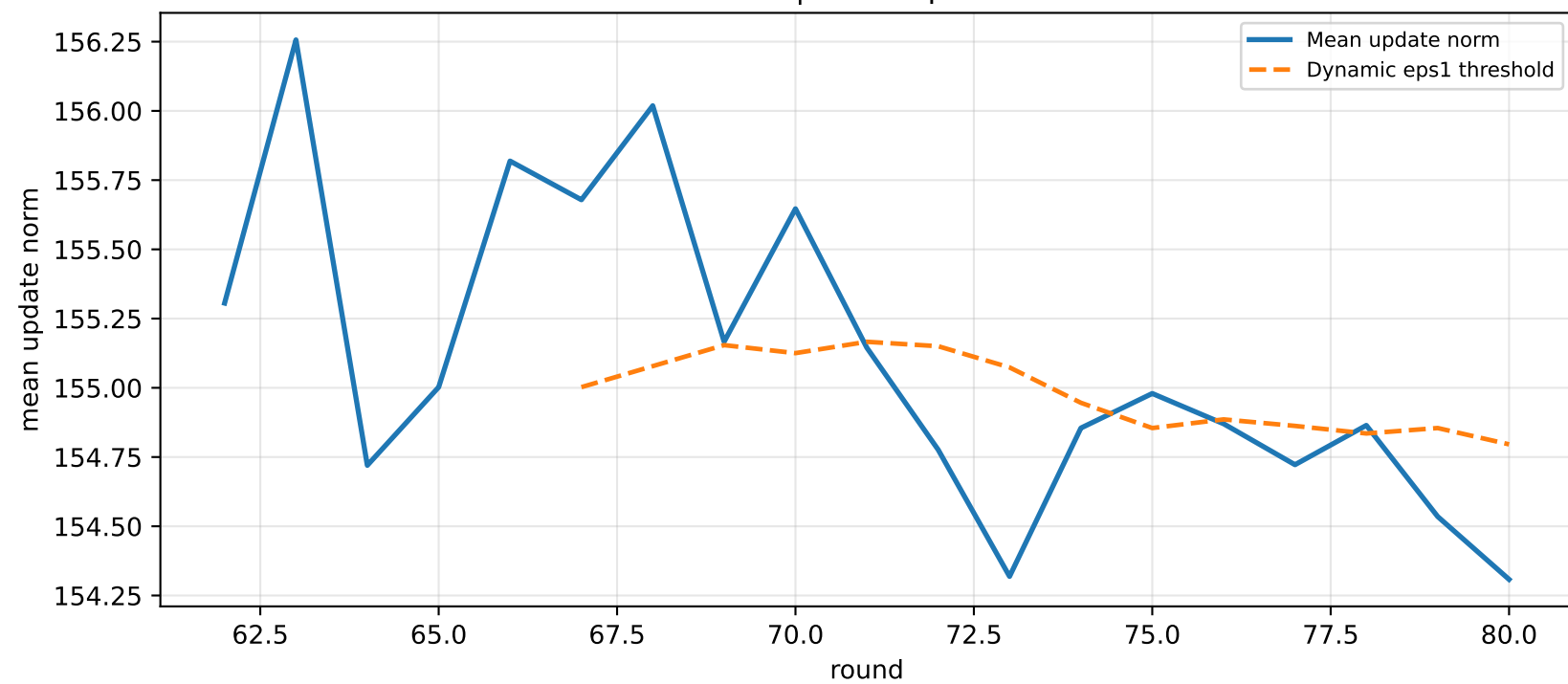
Cluster 12 | Mean update norm



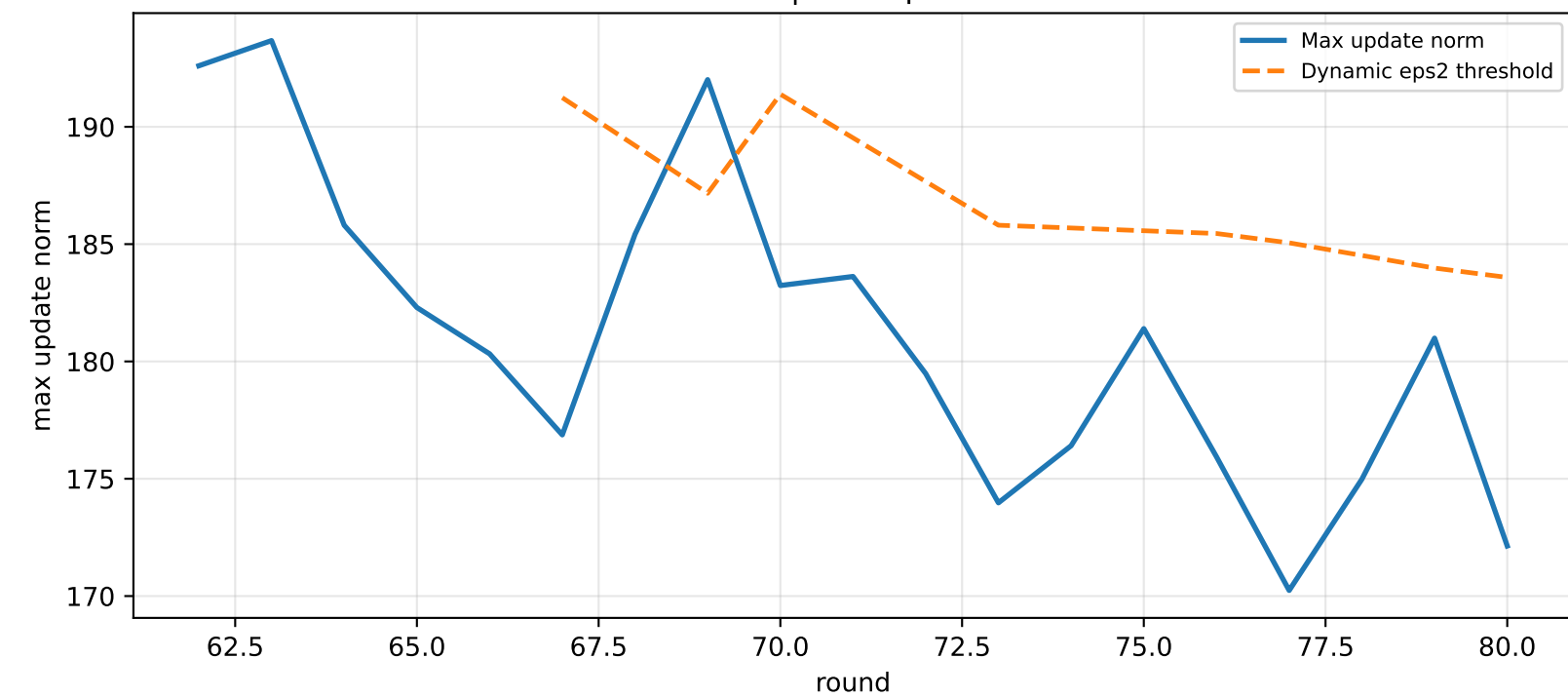
Cluster 12 | Max update norm



Cluster 14 | Mean update norm



Cluster 14 | Max update norm



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

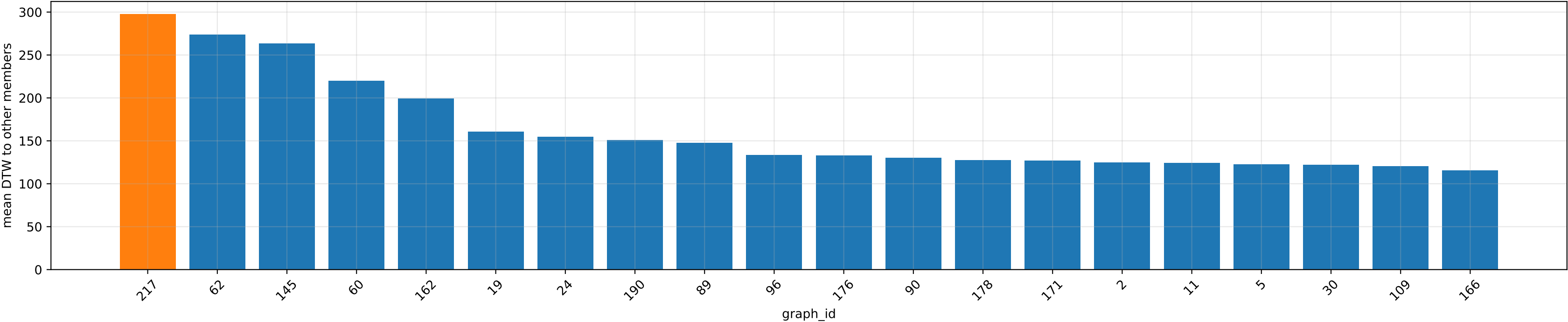
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	25	0	1 vs 19	217	217	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
297.466	nan	147.380

Per-member mean DTW



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

First applied split pairwise DTW metadata | seed=2

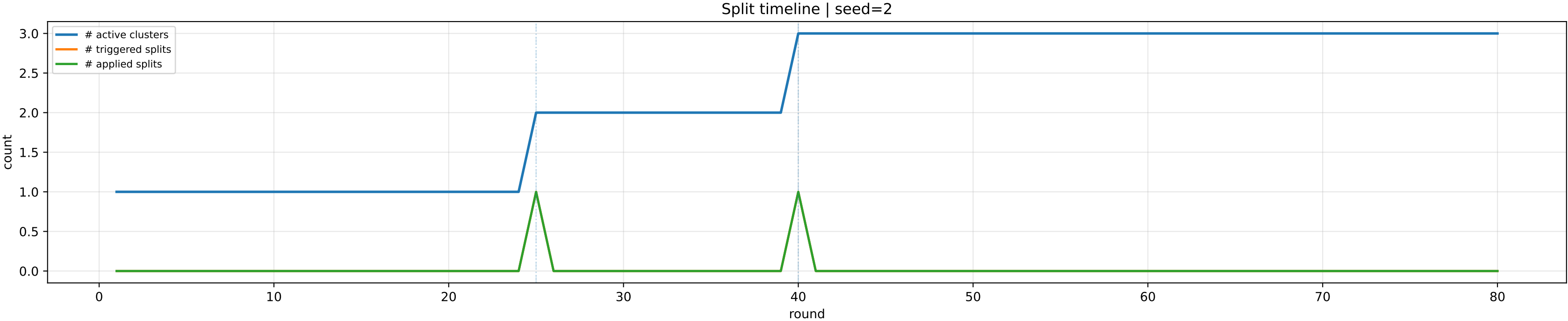
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	25	0	20	217.0	0.071	297.466	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 411.5 at (19, 217)

graph_id	178	96	19	176	171	89	90	11	190	109	24	30	166	5	2	162	62	217	145	60
178	0.0	48.8	57.6	94.8	35.8	34.4	22.4	40.2	163.8	73.2	43.5	68.6	64.6	29.6	81.3	250.1	337.0	363.4	332.1	277.1
96	48.8	0.0	55.6	95.8	44.2	45.9	30.9	45.6	169.1	79.4	47.4	62.8	58.3	54.6	113.0	255.1	341.9	368.4	337.0	282.1
19	57.6	55.6	0.0	133.9	57.0	34.2	58.5	81.3	212.2	110.6	32.3	108.0	110.6	68.8	129.9	298.2	385.1	411.5	380.2	325.2
176	94.8	95.8	133.9	0.0	105.0	121.5	108.8	89.7	122.5	89.9	130.5	81.9	63.6	63.2	95.9	167.6	210.9	307.9	246.2	194.3
171	35.8	44.2	57.0	105.0	0.0	37.4	20.5	58.9	159.1	61.9	52.0	85.8	65.0	53.8	84.9	236.6	323.5	349.9	318.6	263.6
89	34.4	45.9	34.2	121.5	37.4	0.0	36.8	65.6	199.7	94.4	10.4	103.9	107.5	58.2	121.6	285.7	372.6	399.0	367.7	312.7
90	22.4	30.9	58.5	108.8	20.5	36.8	0.0	34.2	182.5	67.4	41.8	85.8	64.3	50.0	90.0	254.1	340.9	367.4	336.0	281.1
11	40.2	45.6	81.3	89.7	58.9	65.6	34.2	0.0	147.5	65.7	73.5	58.0	39.4	57.8	67.8	229.3	303.5	334.9	311.3	256.3
190	163.8	169.1	212.2	122.5	159.1	199.7	182.5	147.5	0.0	110.1	208.7	115.1	112.9	137.0	59.7	106.4	200.9	199.3	156.7	102.5
109	73.2	79.4	110.6	89.9	61.9	94.4	67.4	65.7	110.1	0.0	107.1	50.3	40.5	65.6	69.5	183.9	252.4	283.8	269.6	211.1
24	43.5	47.4	32.3	130.5	52.0	10.4	41.8	73.5	208.7	107.1	0.0	108.7	108.4	62.4	130.7	294.7	381.6	408.1	376.7	321.7
30	68.6	62.8	108.0	81.9	85.8	103.9	85.8	58.0	115.1	50.3	108.7	0.0	24.5	37.4	72.8	192.4	282.6	278.8	277.7	219.8
166	64.6	58.3	110.6	63.6	65.0	107.5	64.3	39.4	112.9	40.5	108.4	24.5	0.0	37.3	66.6	189.9	262.3	293.8	271.8	216.9
5	29.6	54.6	68.8	63.2	53.8	58.2	50.0	57.8	137.0	65.6	62.4	37.4	37.3	0.0	79.6	232.4	319.3	345.7	314.4	259.4
2	81.3	113.0	129.9	95.9	84.9	121.6	90.0	67.8	59.7	69.5	130.7	72.8	66.6	79.6	0.0	160.9	249.4	260.8	246.1	188.1
162	250.1	255.1	298.2	167.6	236.6	285.7	254.1	229.3	106.4	183.9	294.7	192.4	189.9	232.4	160.9	0.0	166.4	146.8	104.7	33.6
62	337.0	341.9	385.1	210.9	323.5	372.6	340.9	303.5	200.9	252.4	381.6	282.6	262.3	319.3	249.4	166.4	0.0	214.6	80.1	172.3
217	363.4	368.4	411.5	307.9	349.9	399.0	367.4	334.9	199.3	283.8	408.1	278.8	293.8	345.7	260.8	146.8	214.6	0.0	166.2	151.5
145	332.1	337.0	380.2	246.2	318.6	367.7	336.0	311.3	156.7	269.6	376.7	277.7	271.8	314.4	246.1	104.7	80.1	166.2	0.0	110.6
60	277.1	282.1	325.2	194.3	263.6	312.7	281.1	256.3	102.5	211.1	321.7	219.8	216.9	259.4	188.1	33.6	172.3	151.5	110.6	0.0

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

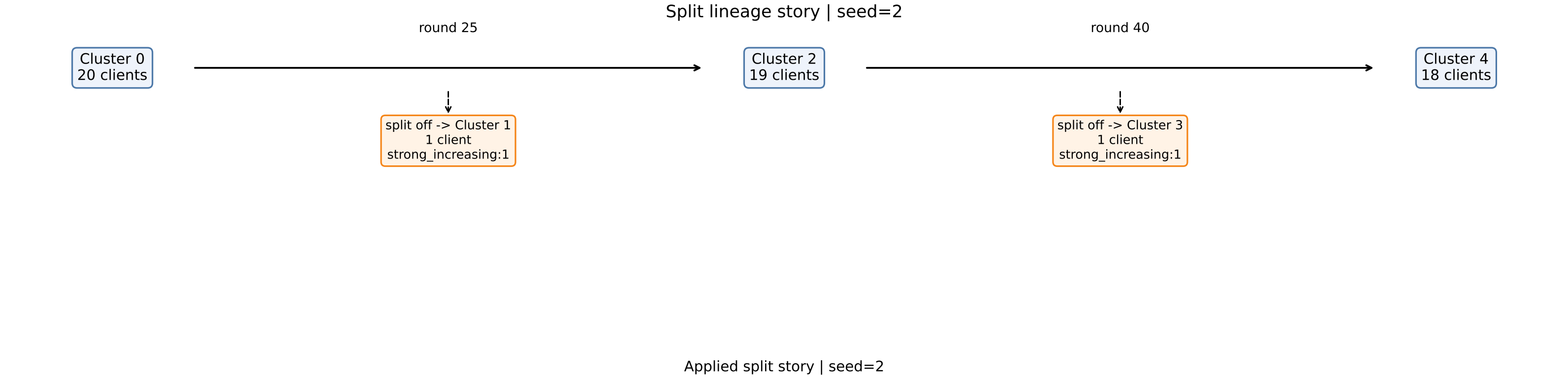


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	25, 40	25, 40	1, 3, 4	1:1, 3:1, 4:18

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068



Applied split story seed=2							
round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
25	0	20	156.879 < 157.856 , 265.069 > 254.237	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
40	2	19	155.518 < 155.968 , 322.993 > 295.290	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2

Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

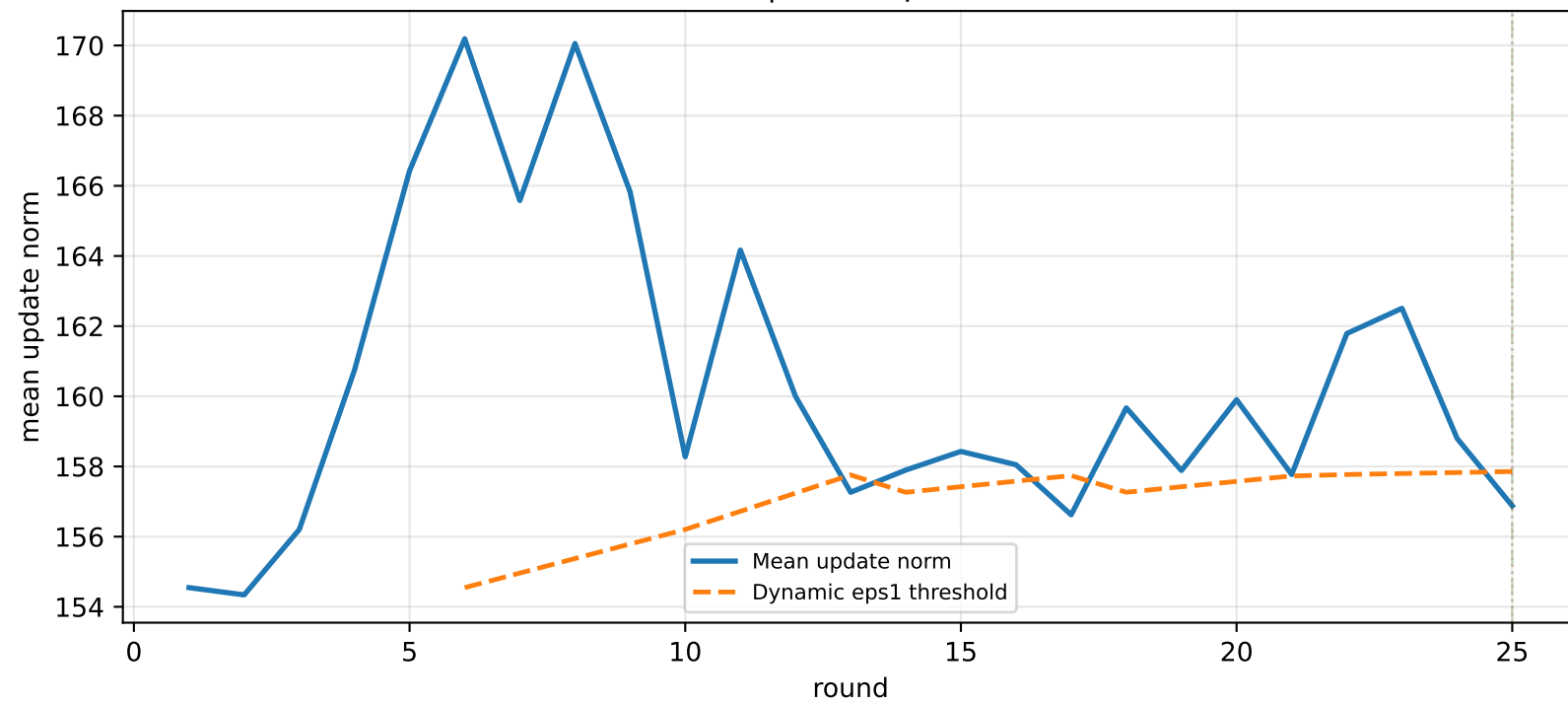
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	217	17	strong_increasing:1
3	1	145	18	strong_increasing:1
4	18	178, 96, 19, 176, 171, 89, 90, 11, 190, 109, 24, 30, 166, 5, 2, 162, 62, 60	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 19	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2

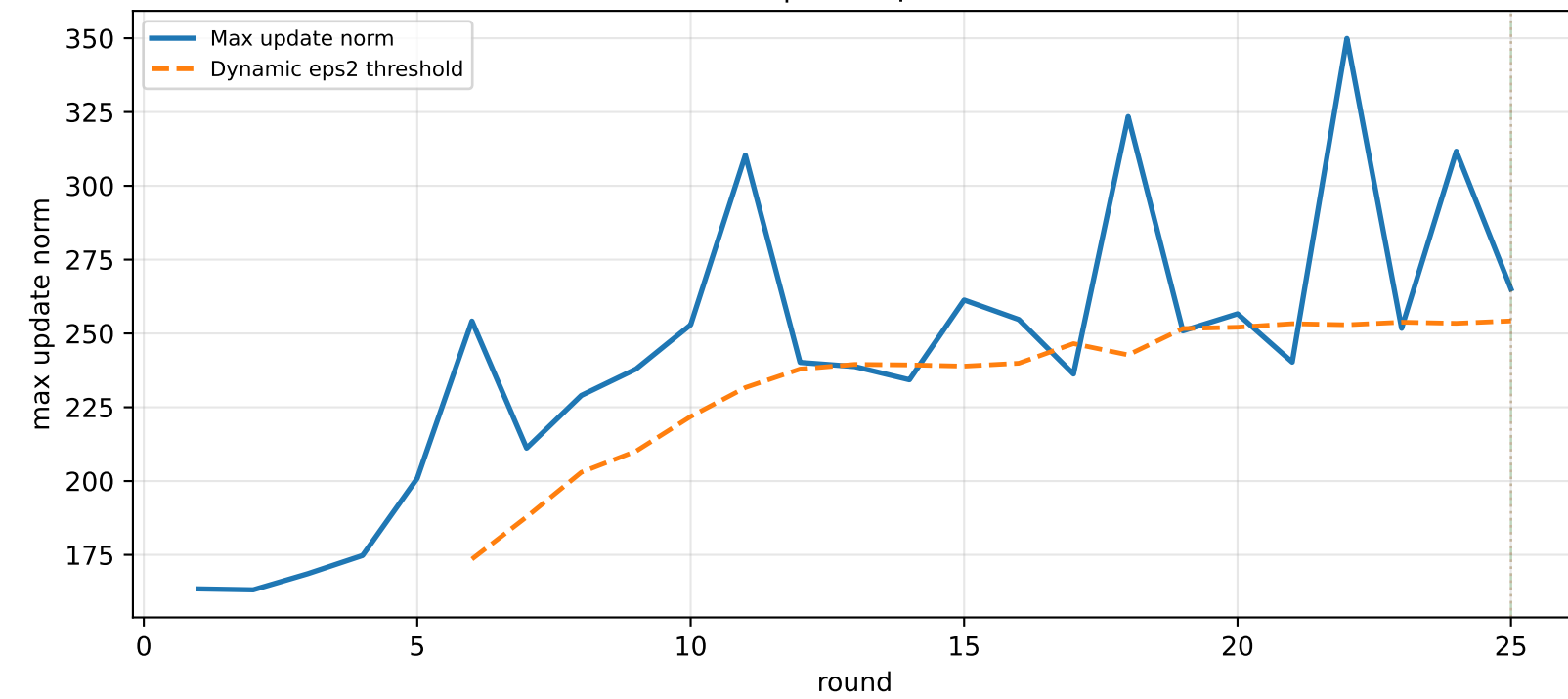
Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

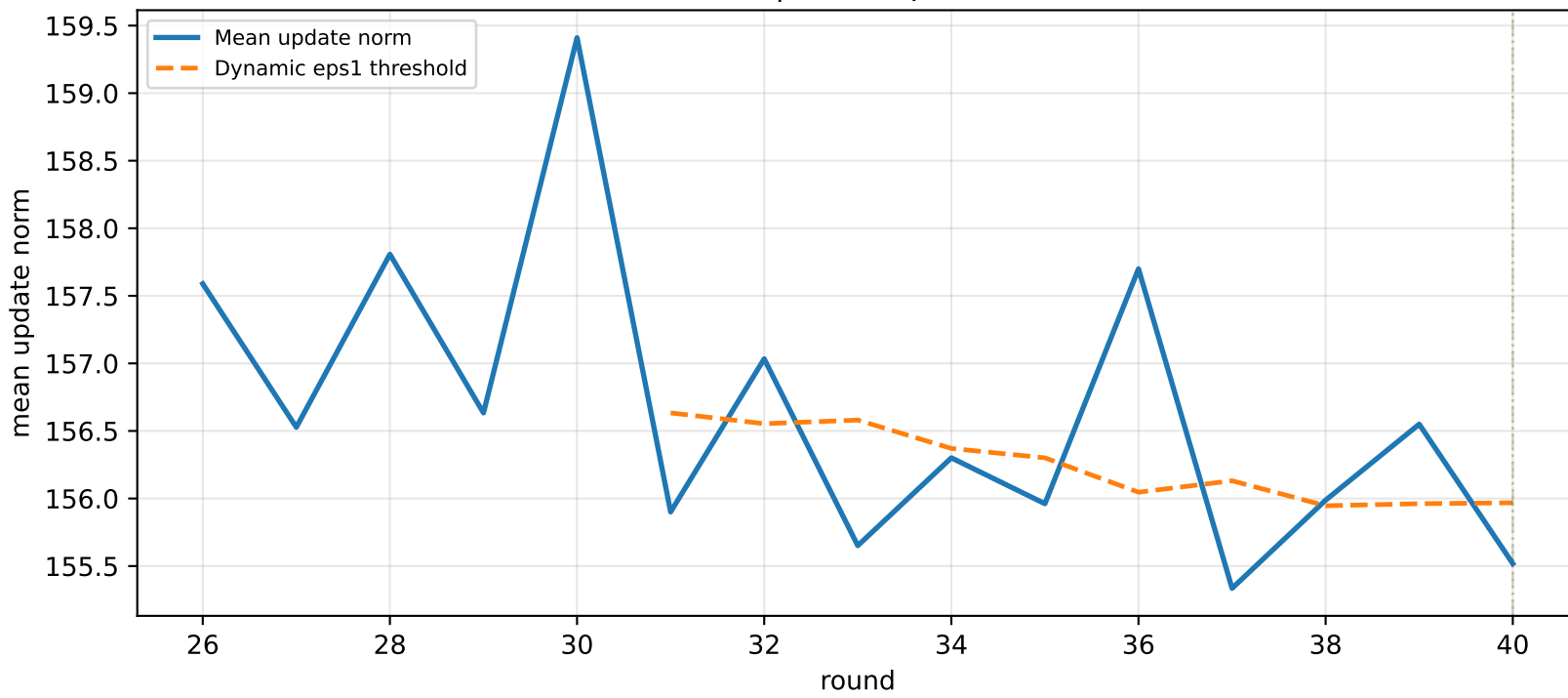
Cluster 0 | Mean update norm



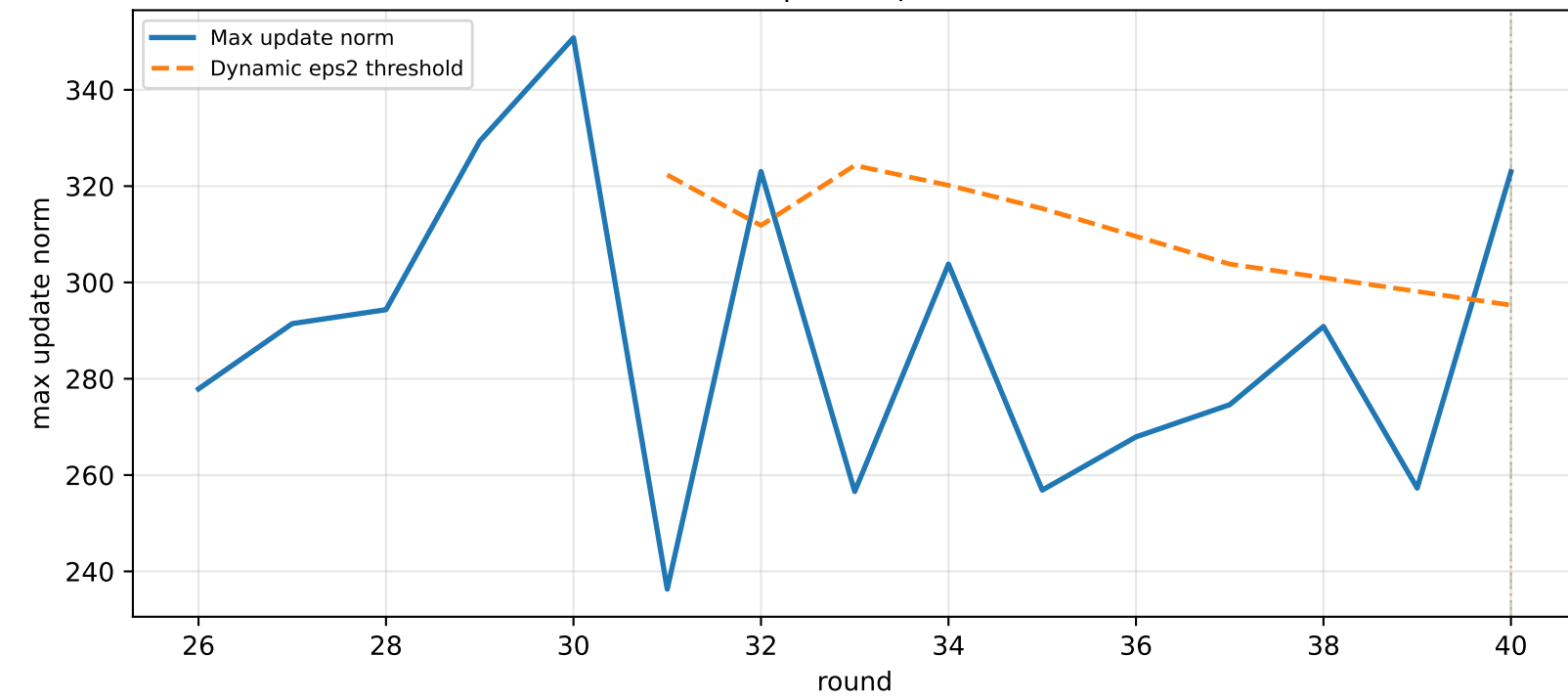
Cluster 0 | Max update norm



Cluster 2 | Mean update norm

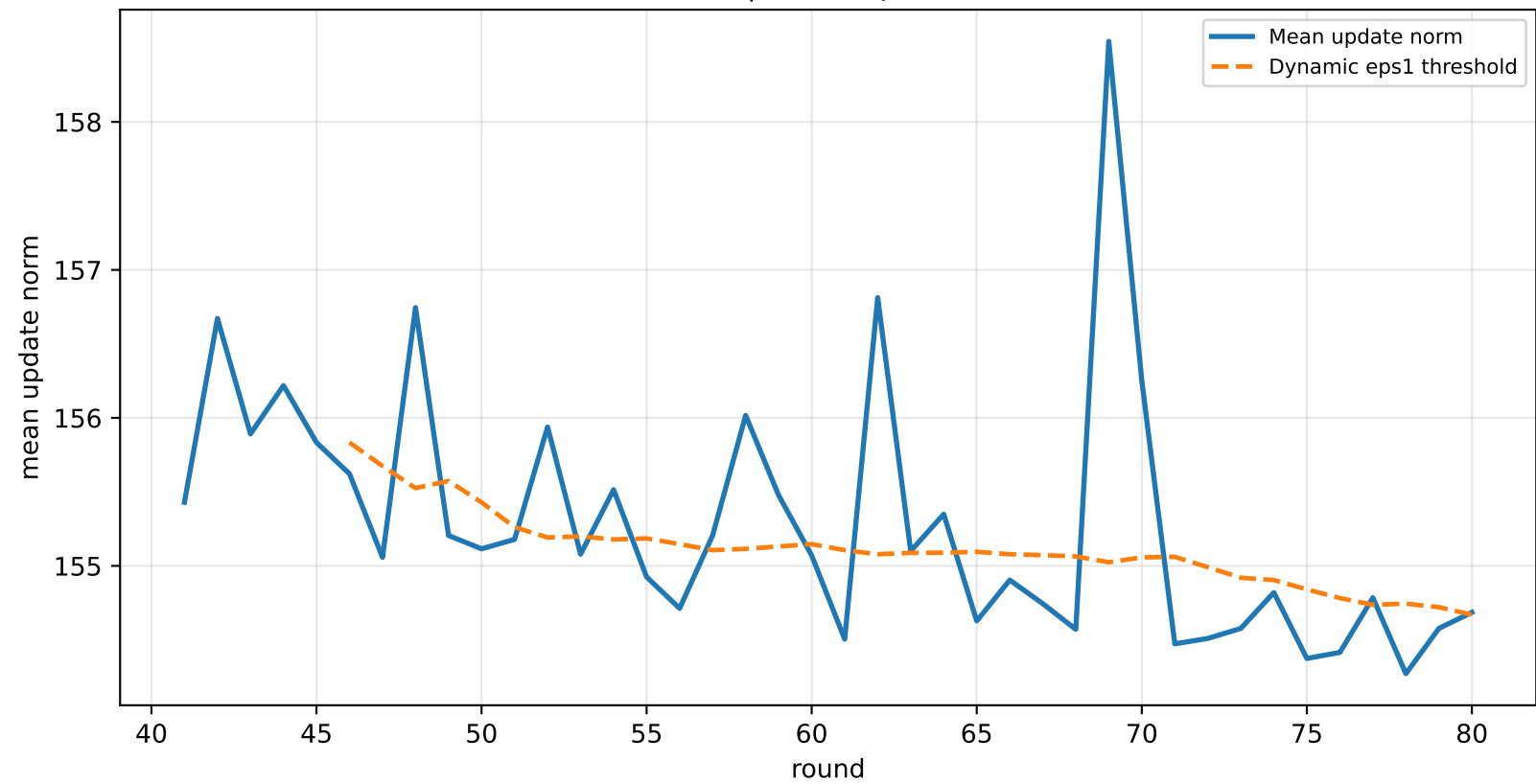


Cluster 2 | Max update norm

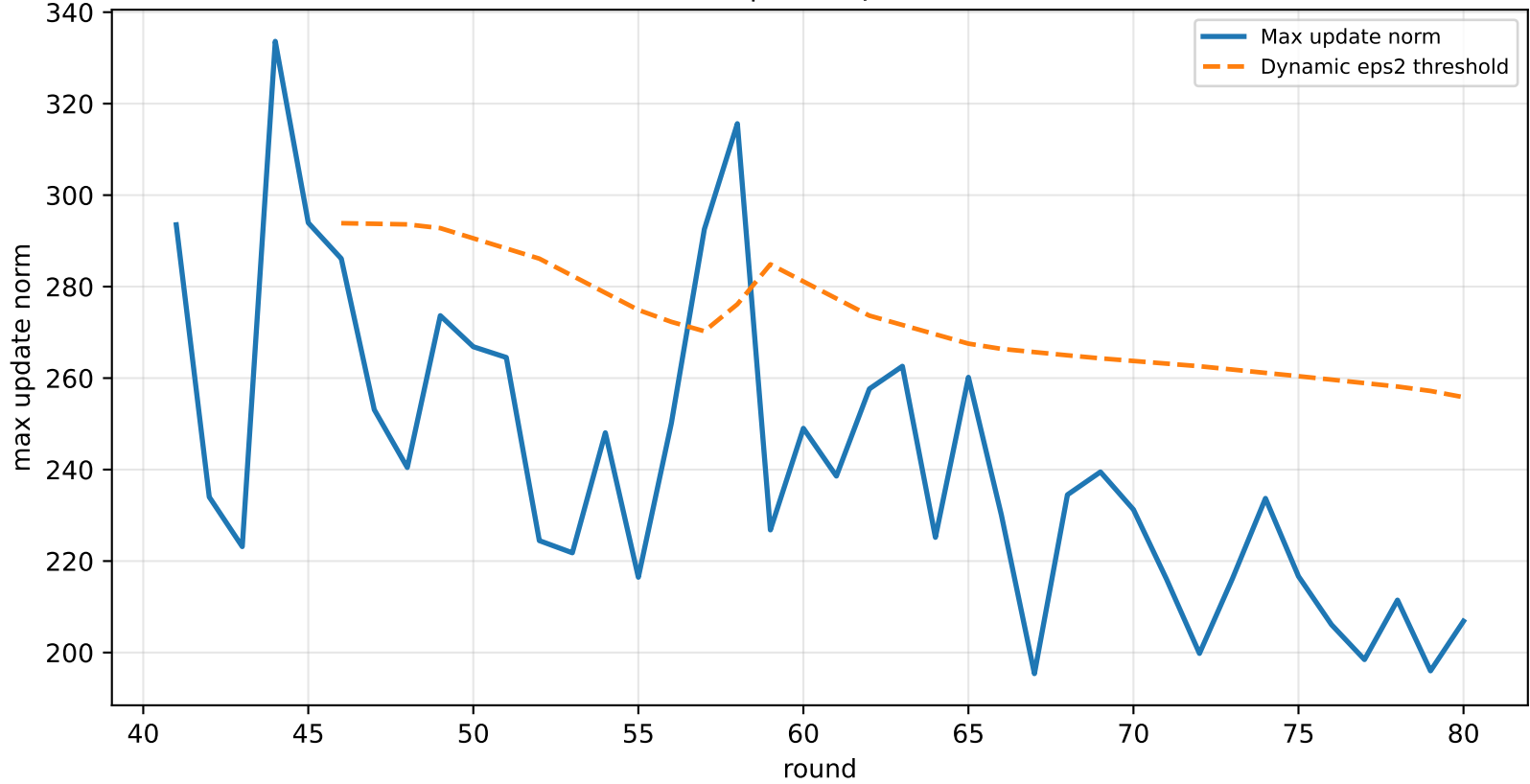


Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (4-4)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Cluster 4 | Mean update norm



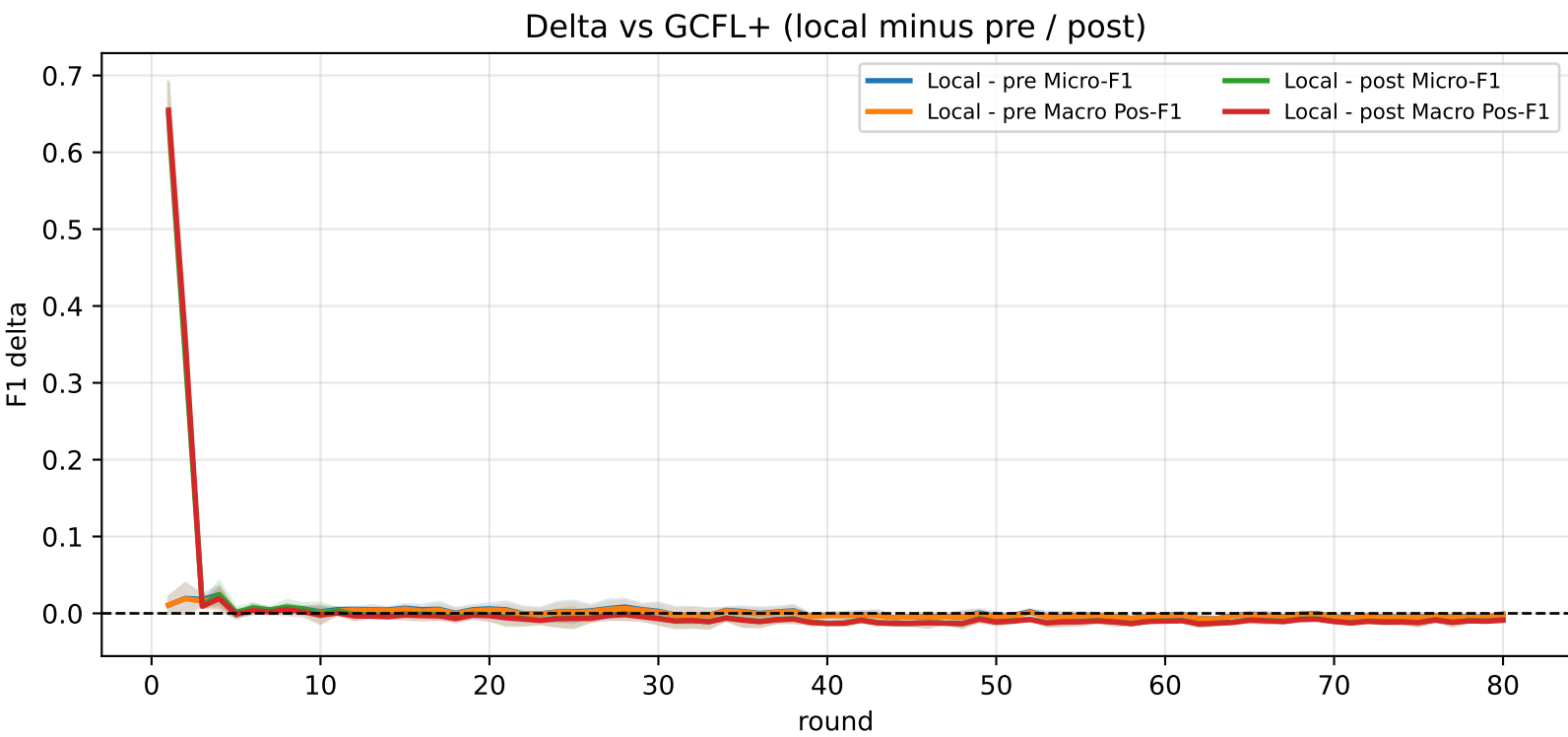
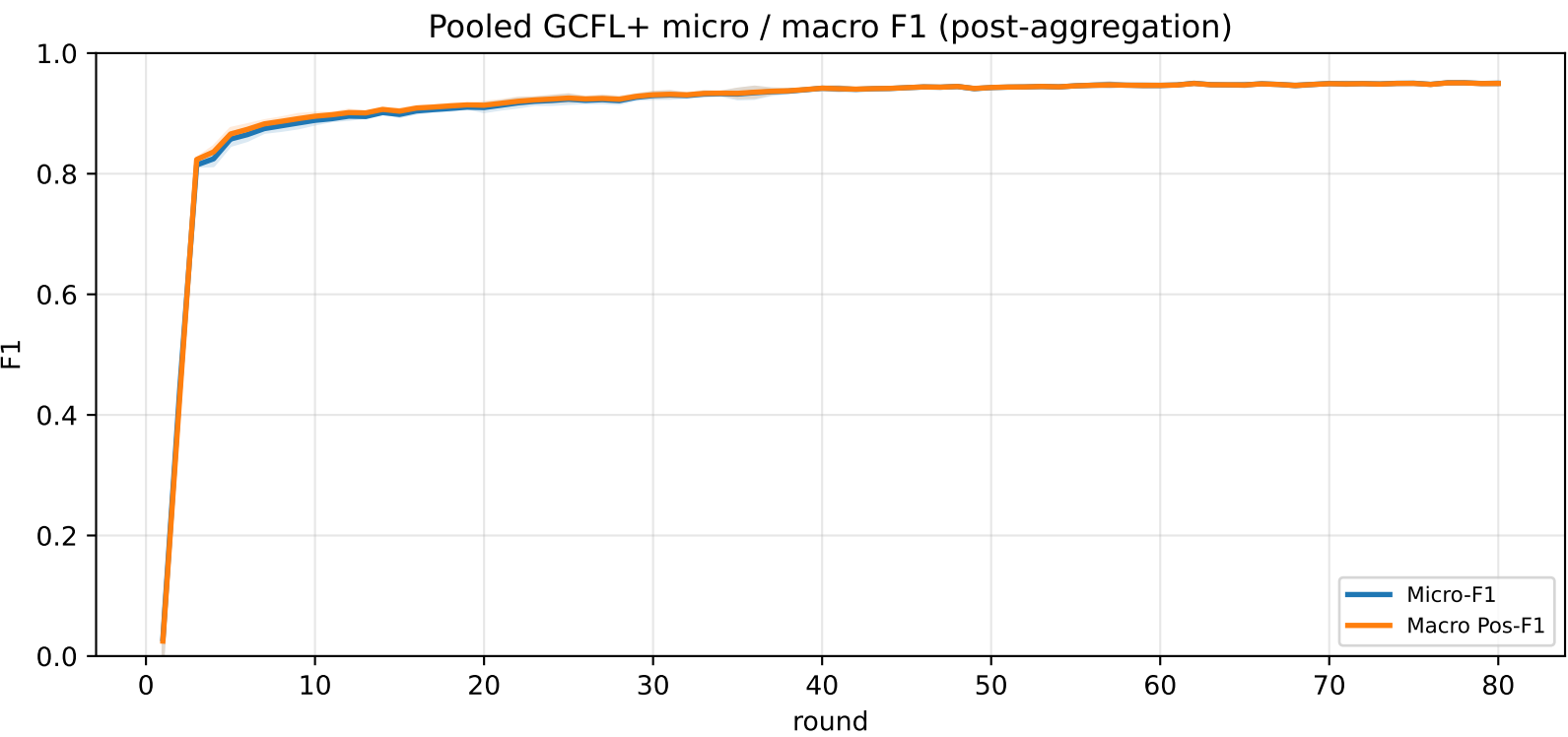
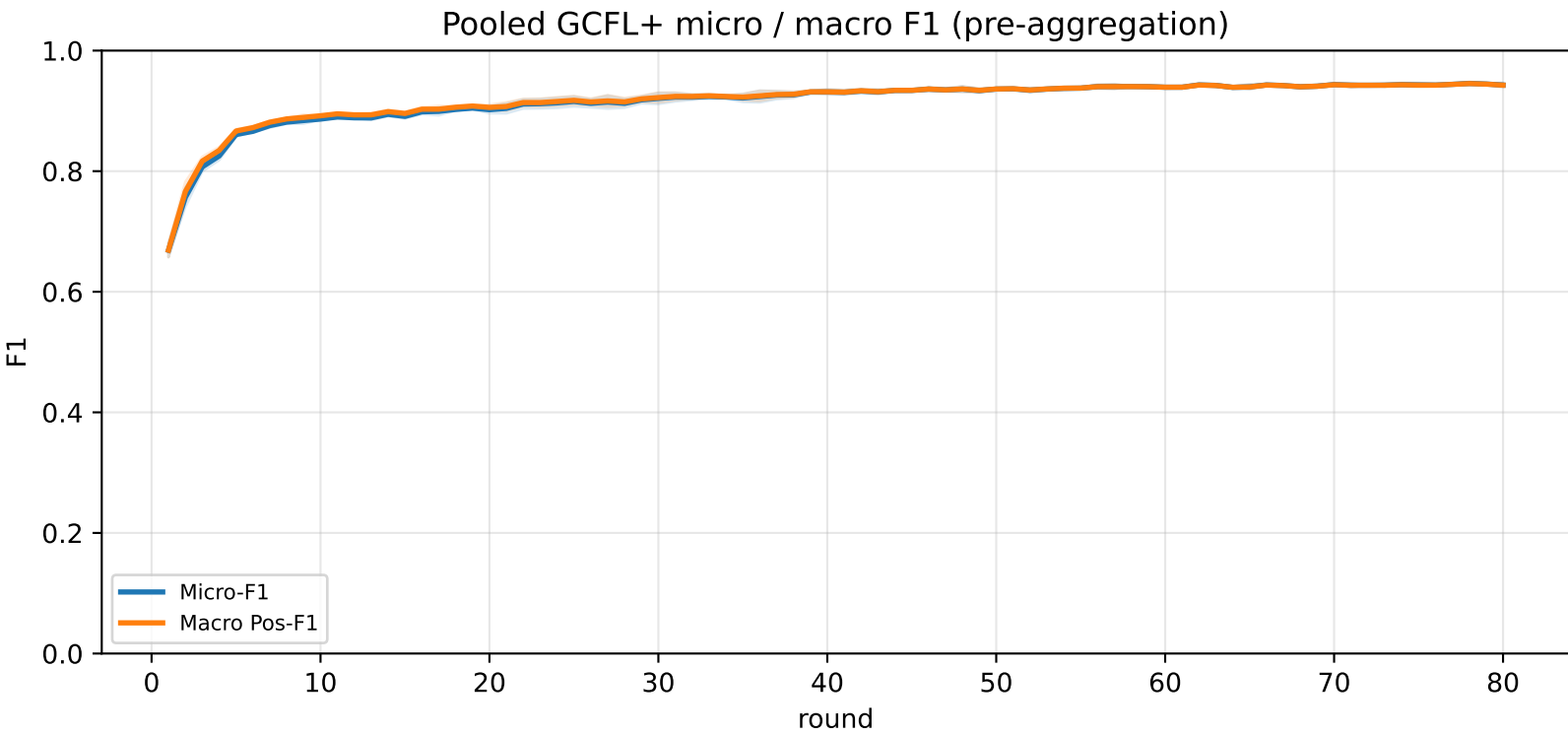
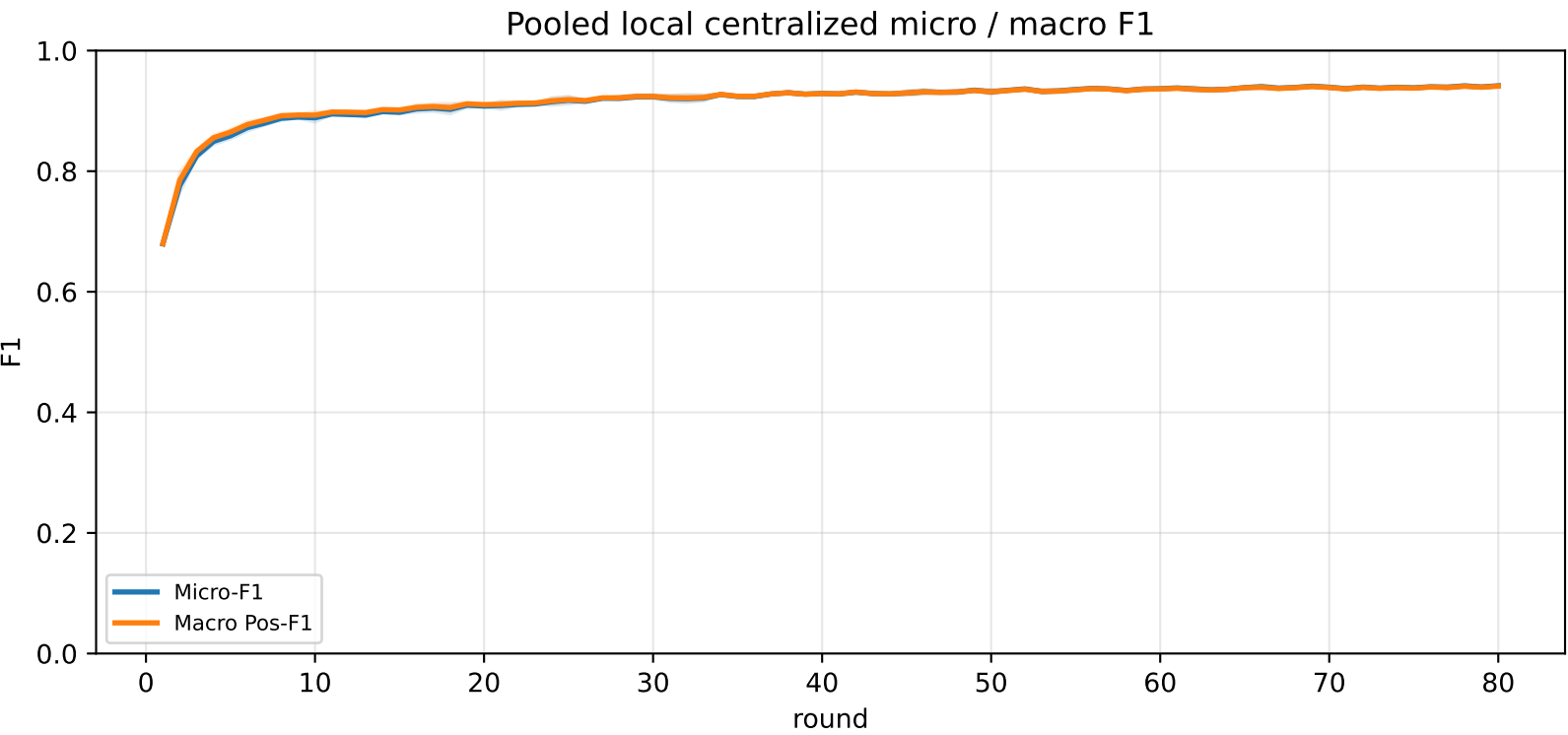
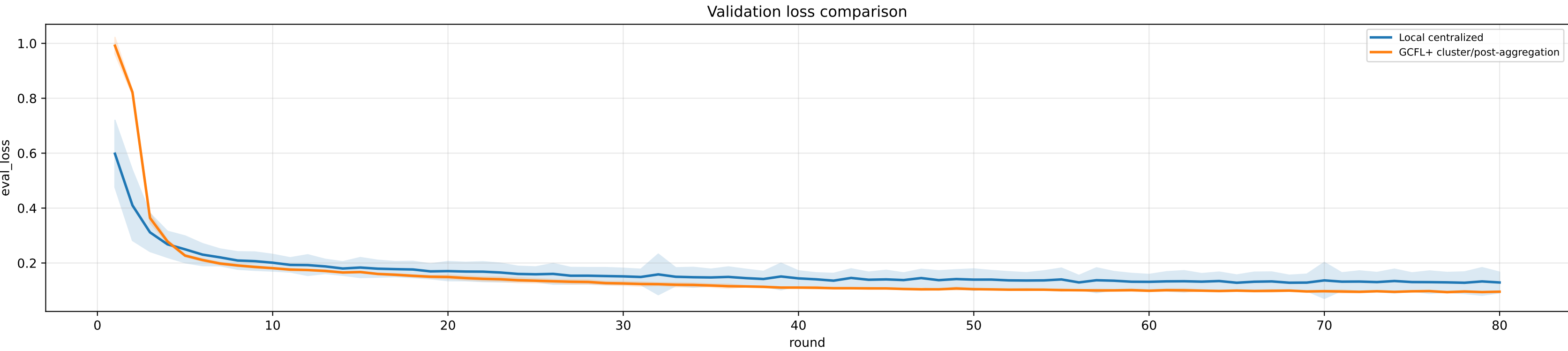
Cluster 4 | Max update norm



Heterogeneity level: Mid | GCFL+ diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

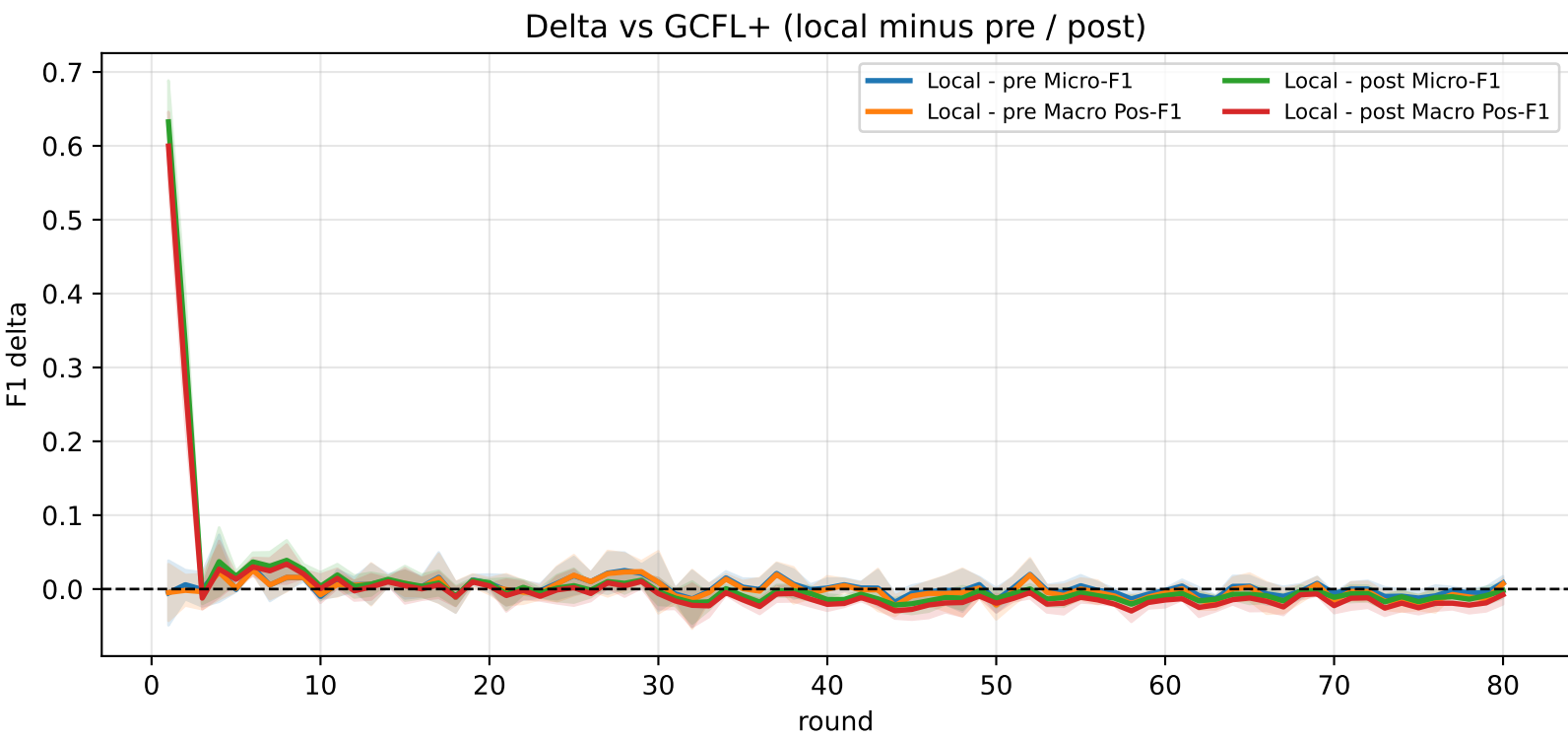
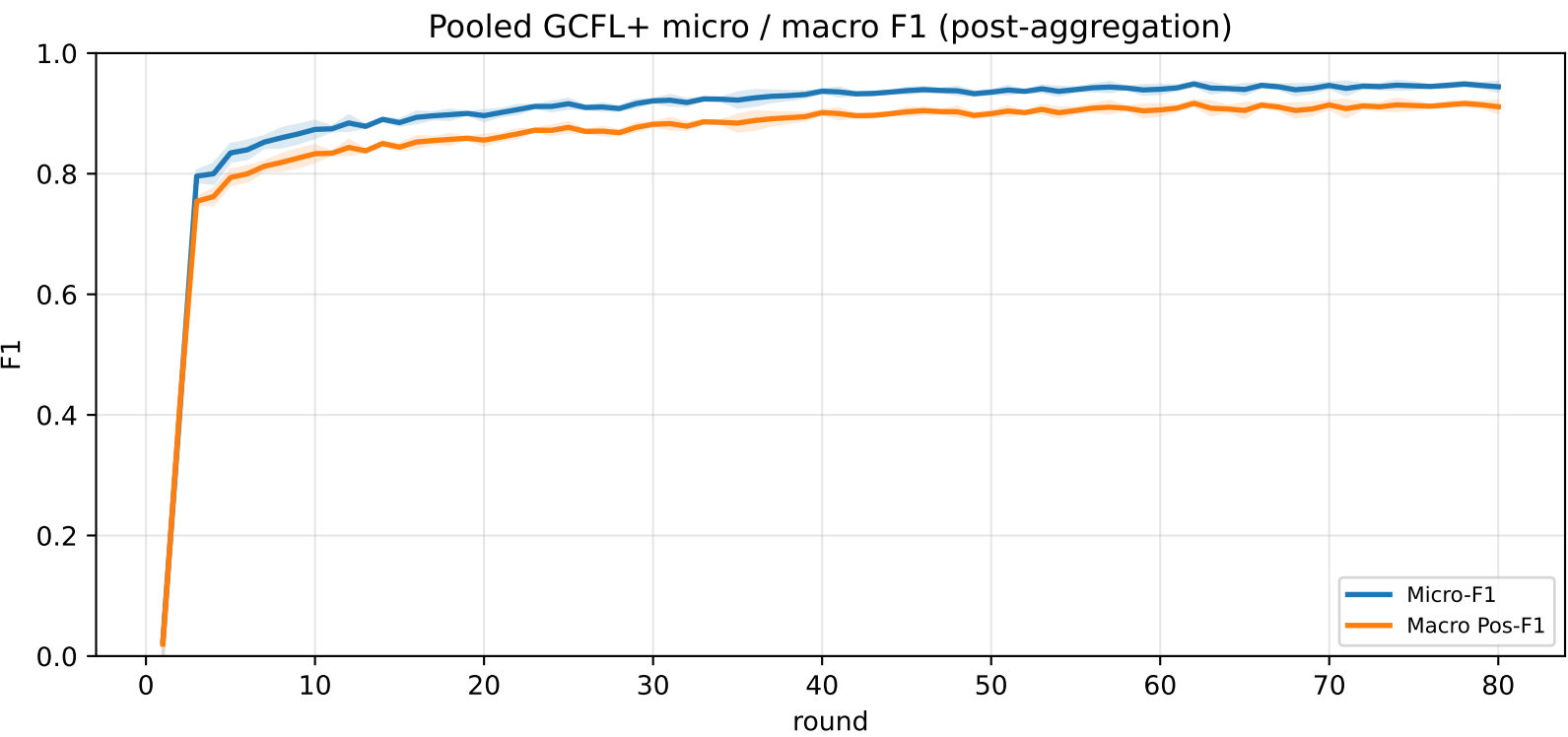
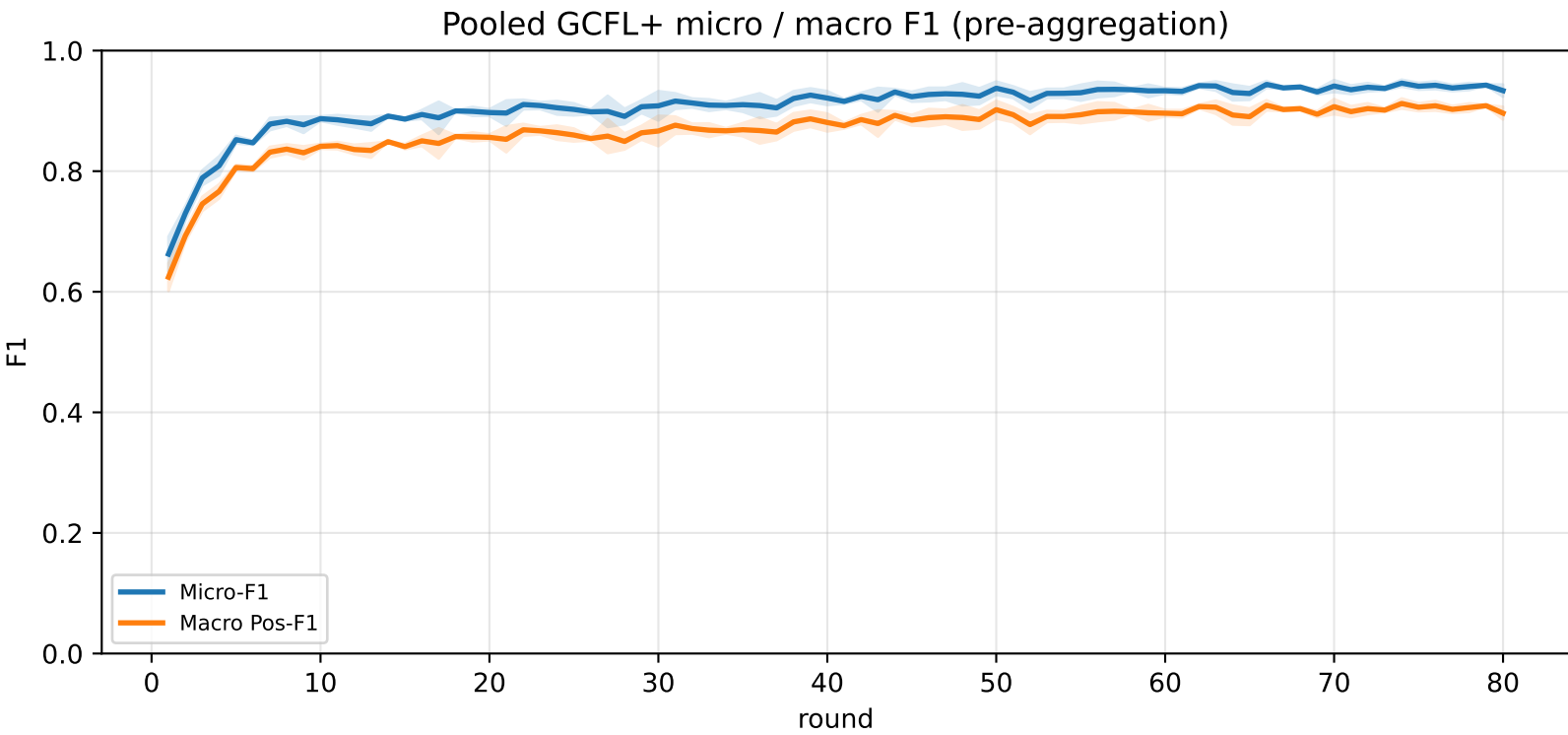
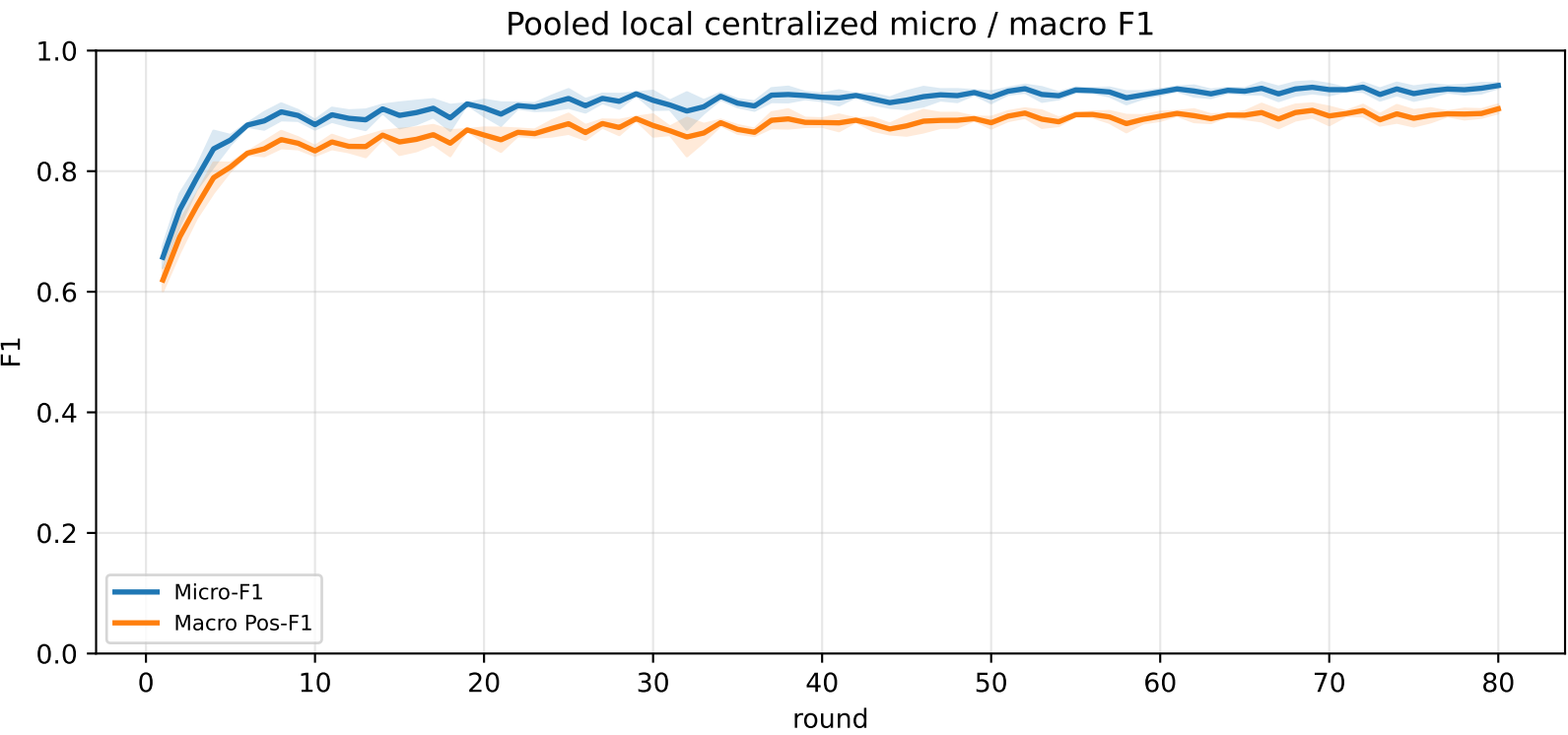
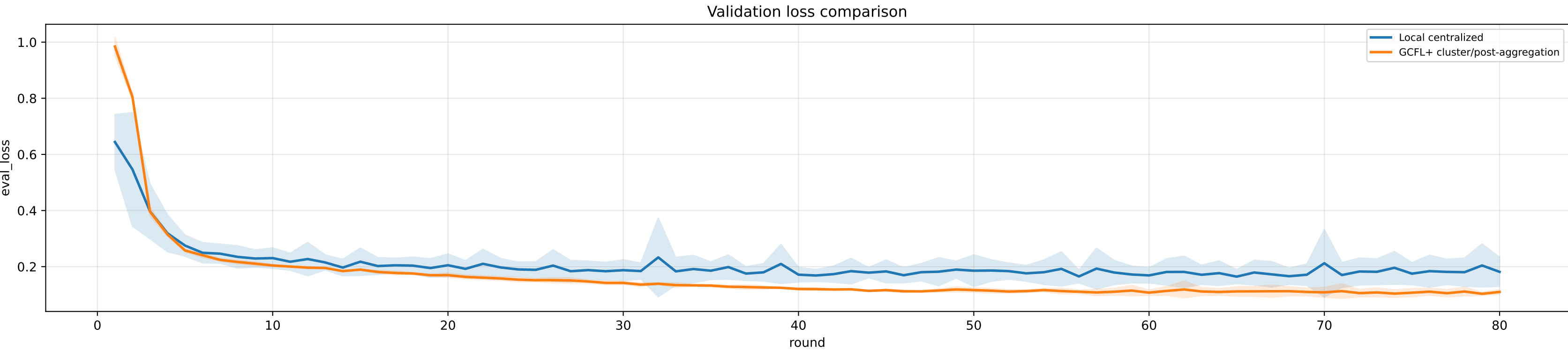
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.0% ± 0.1%	93.5% ± 0.1%	99.2% ± 0.1%	98.5% ± 0.0%	96.4% ± 0.0%	92.1% ± 0.4%	84.6% ± 0.3%
GCFL+	95.0% ± 0.2%	94.6% ± 0.2%	99.3% ± 0.0%	98.7% ± 0.0%	96.8% ± 0.2%	93.8% ± 0.3%	87.2% ± 0.3%



Heterogeneity level: Mid | GCFL+ diagnostics | family=strong_decreasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=40.2%±4.0% | cycle3=32.9%±0.8% | cycle4=24.4%±1.8% | cycle5=16.8%±2.6% | cycle6=11.2%±3.3%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

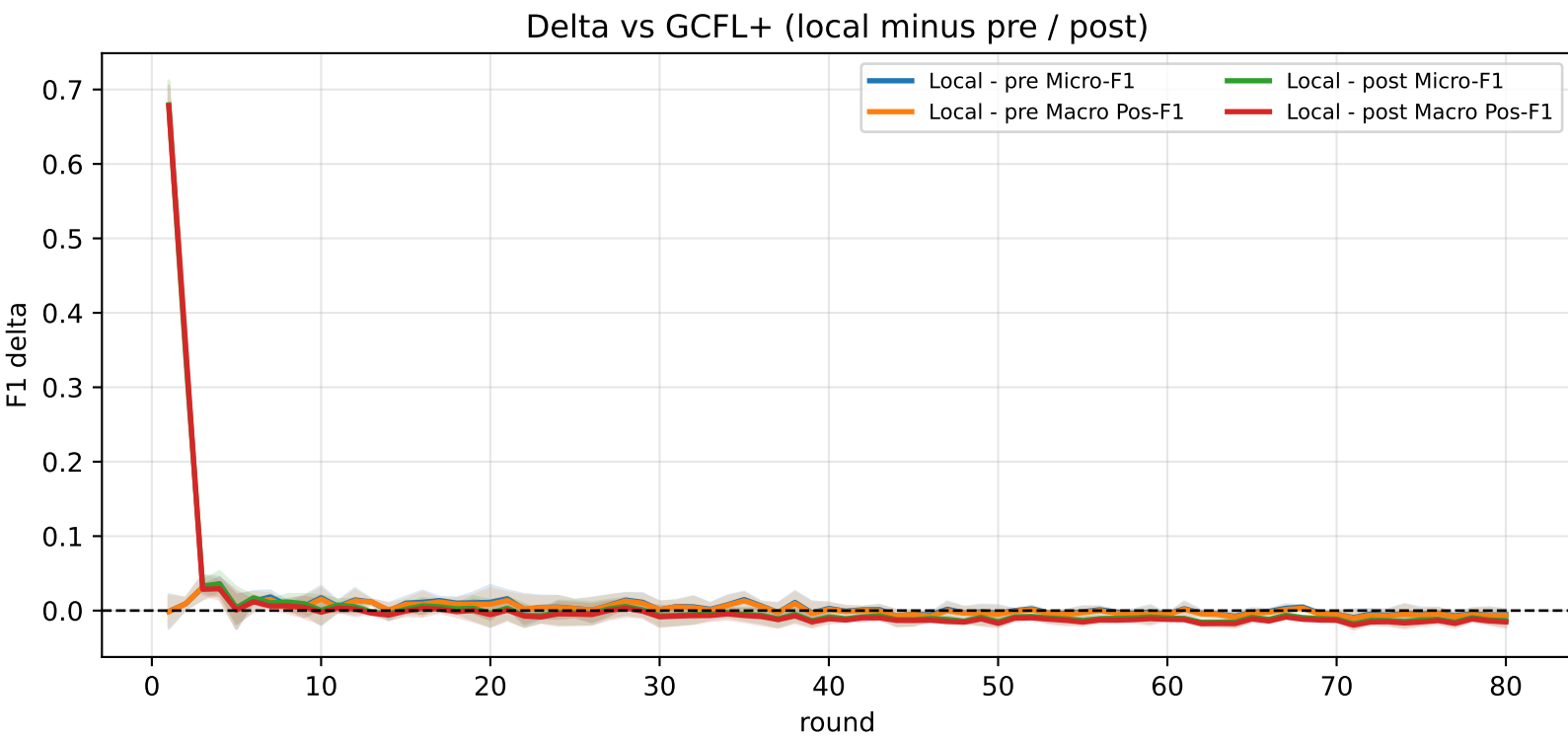
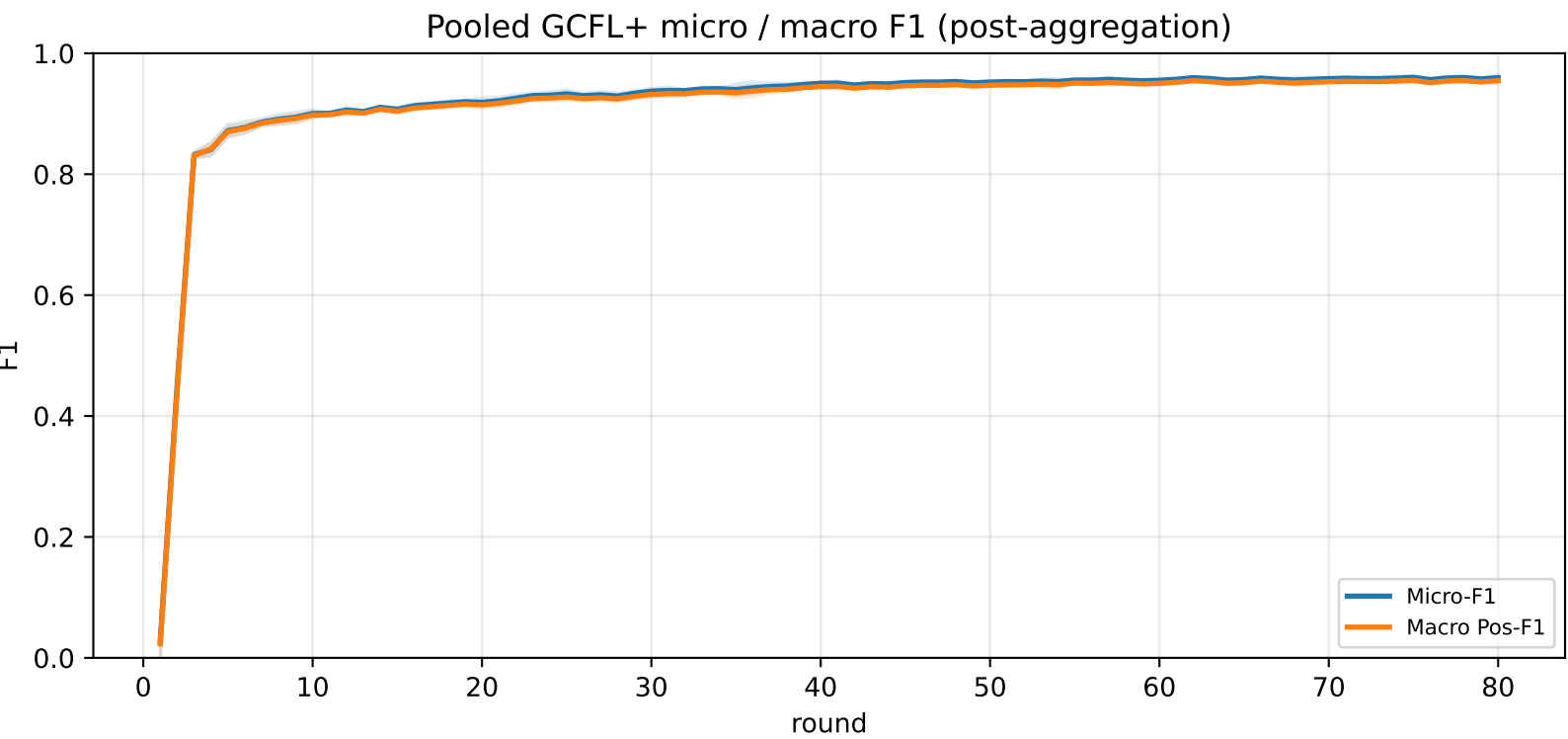
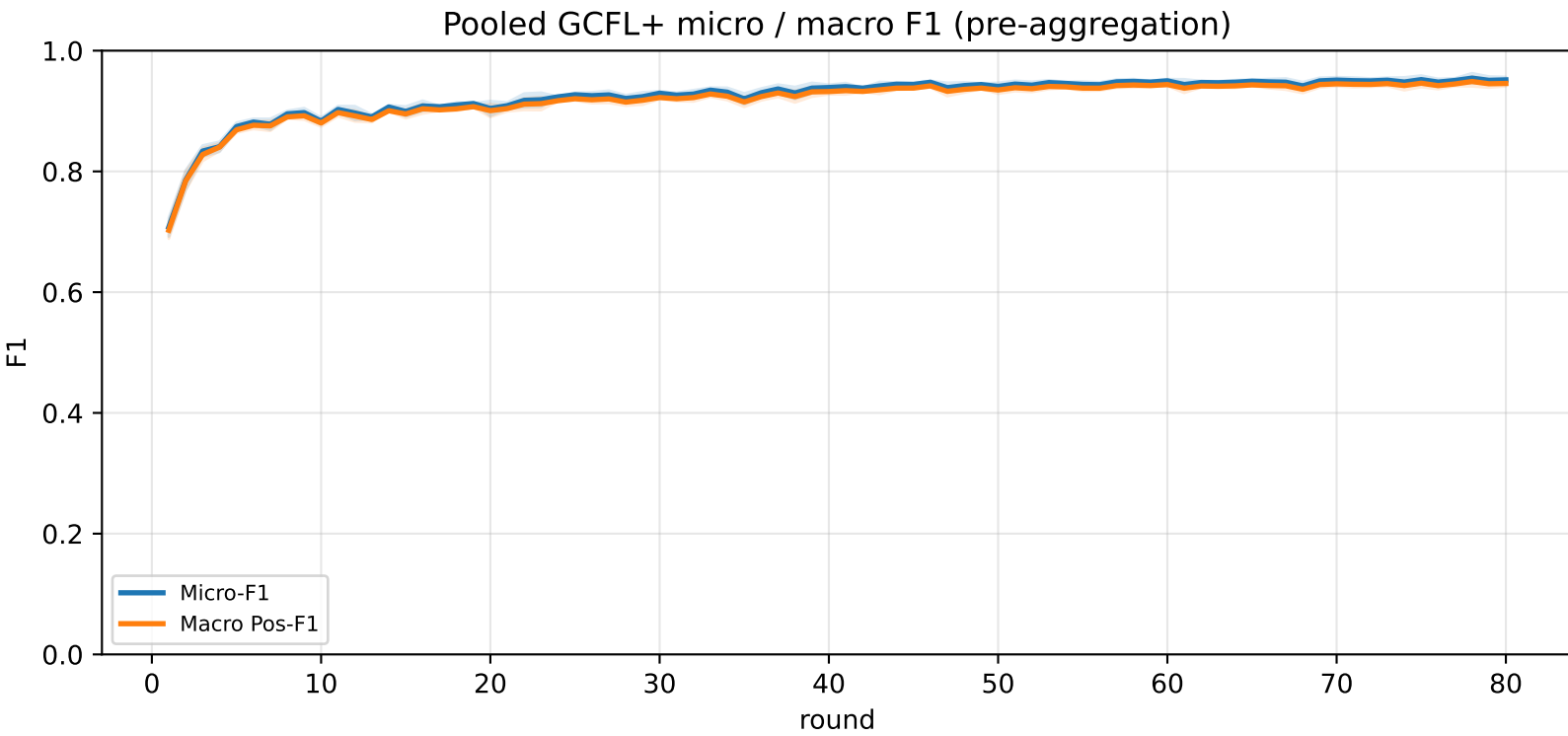
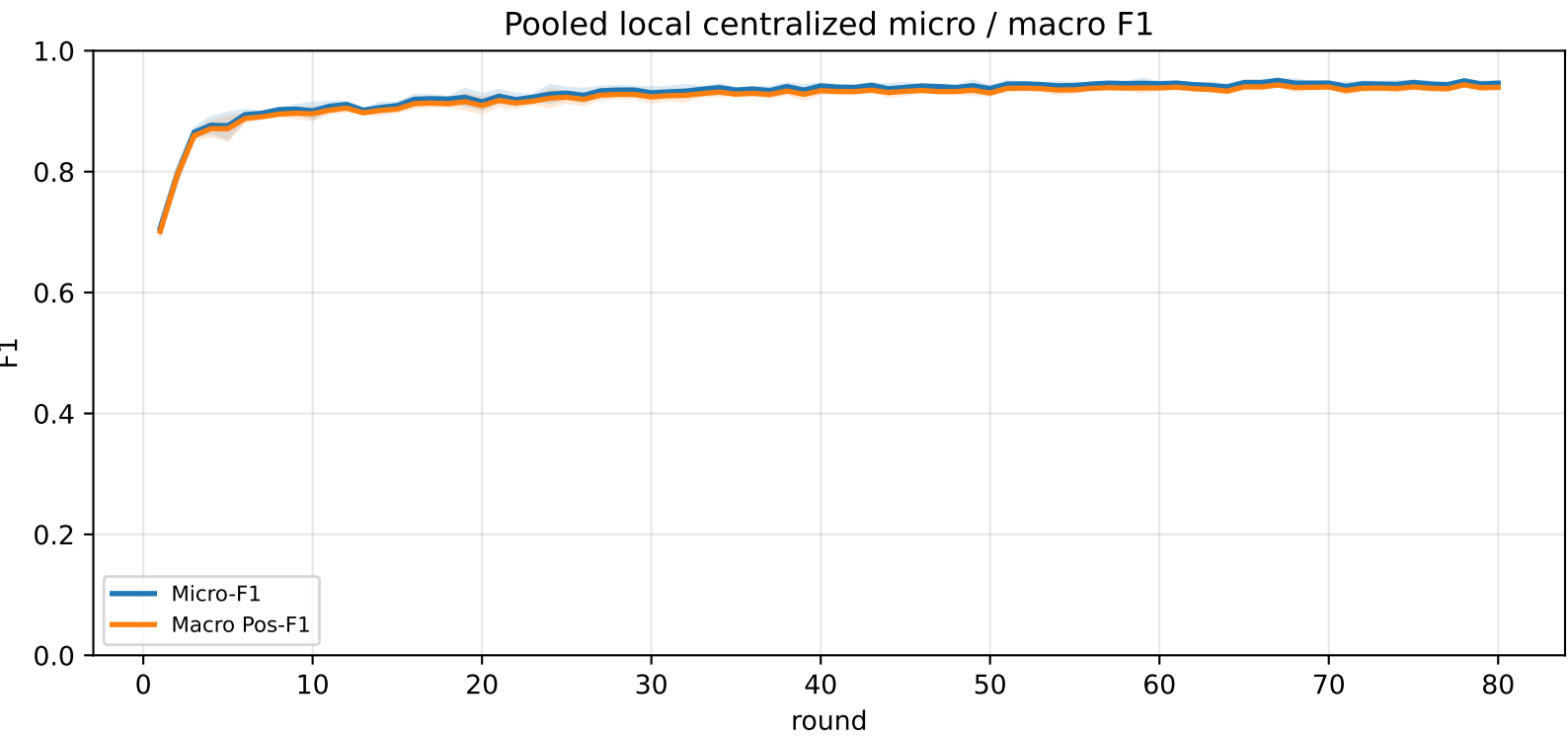
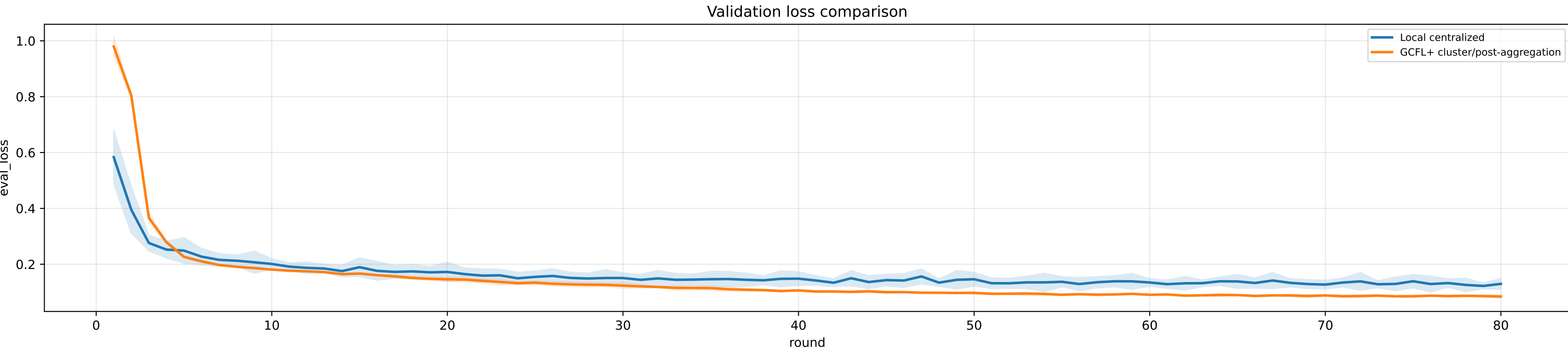
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.1% ± 0.5%	89.3% ± 0.6%	99.6% ± 0.1%	99.0% ± 0.0%	95.8% ± 0.5%	86.3% ± 0.9%	68.1% ± 2.2%
GCFL+	94.9% ± 0.3%	91.7% ± 0.4%	99.8% ± 0.1%	99.4% ± 0.0%	96.5% ± 0.6%	91.0% ± 0.7%	74.3% ± 0.6%



Heterogeneity level: Mid | GCFL+ diagnostics | family=mild_decreasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=31.7%±0.3% | cycle3=30.3%±0.5% | cycle4=28.0%±0.8% | cycle5=25.5%±0.5% | cycle6=20.2%±1.0%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

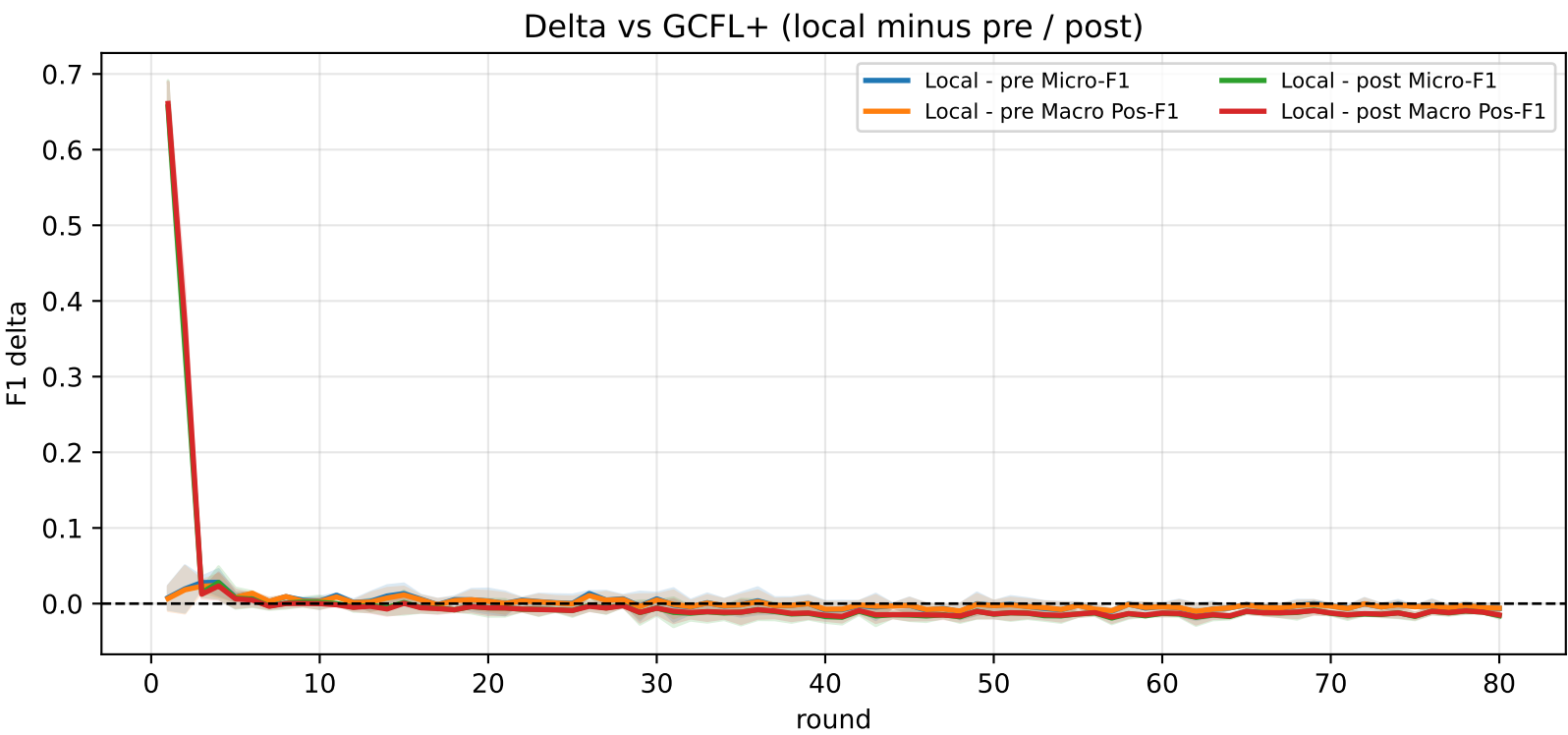
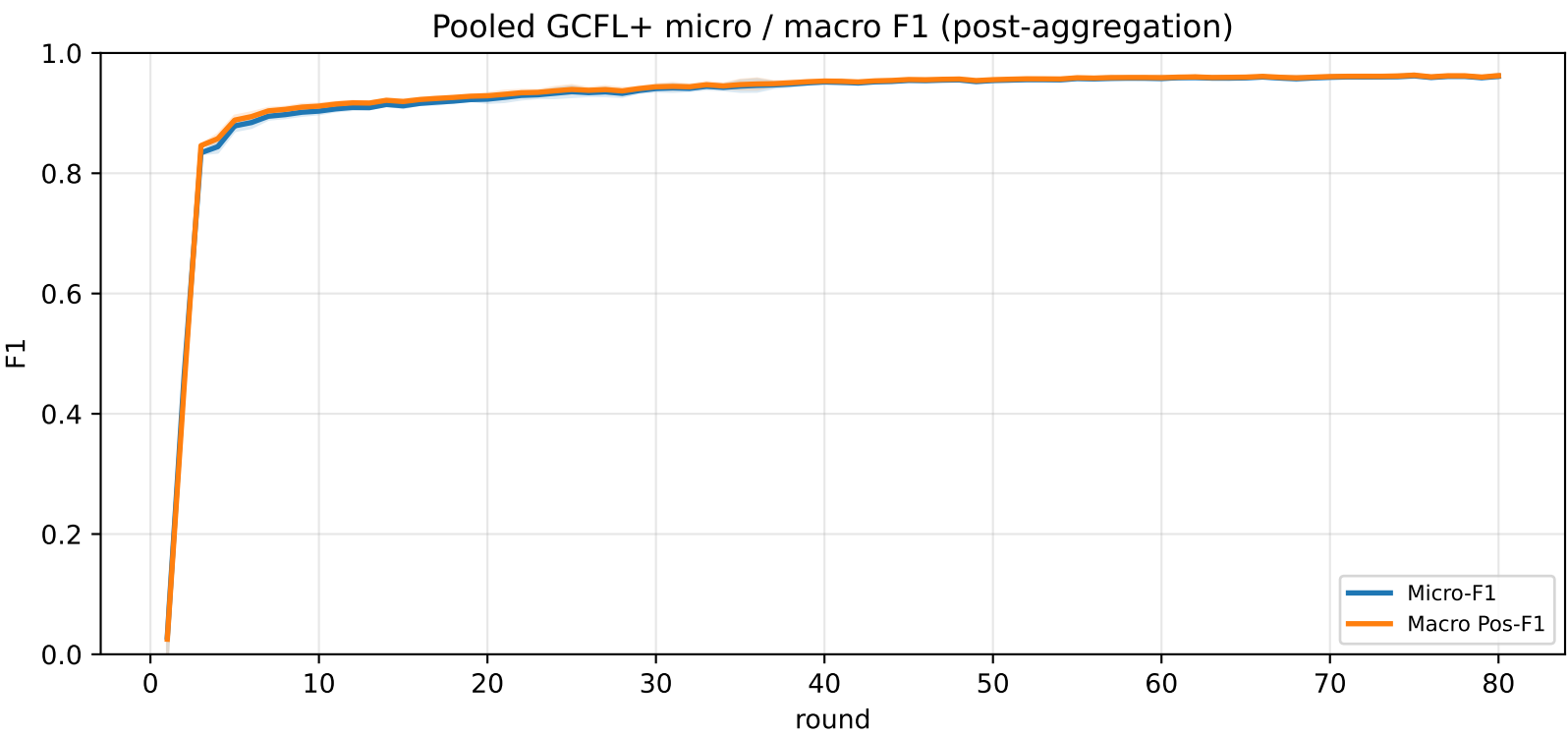
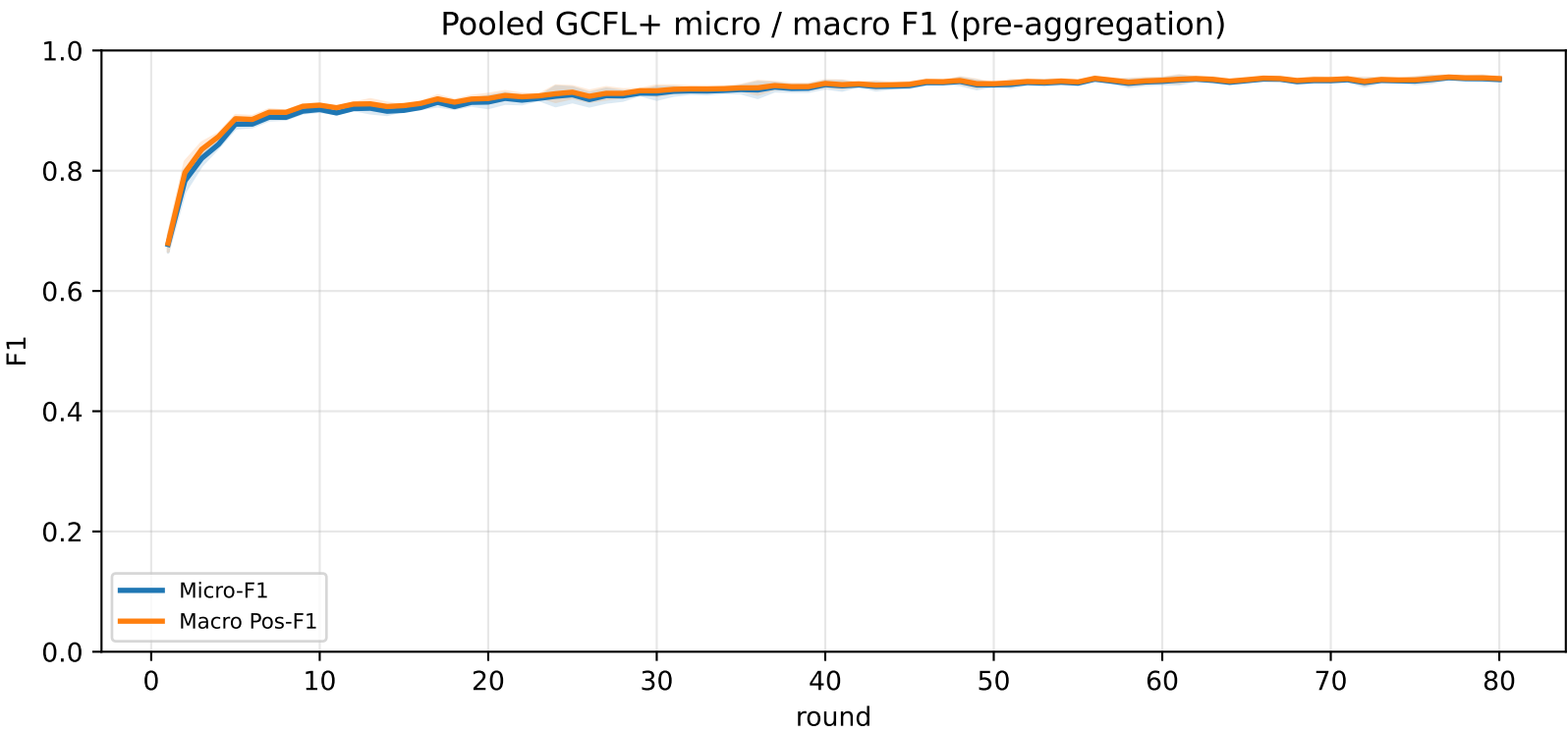
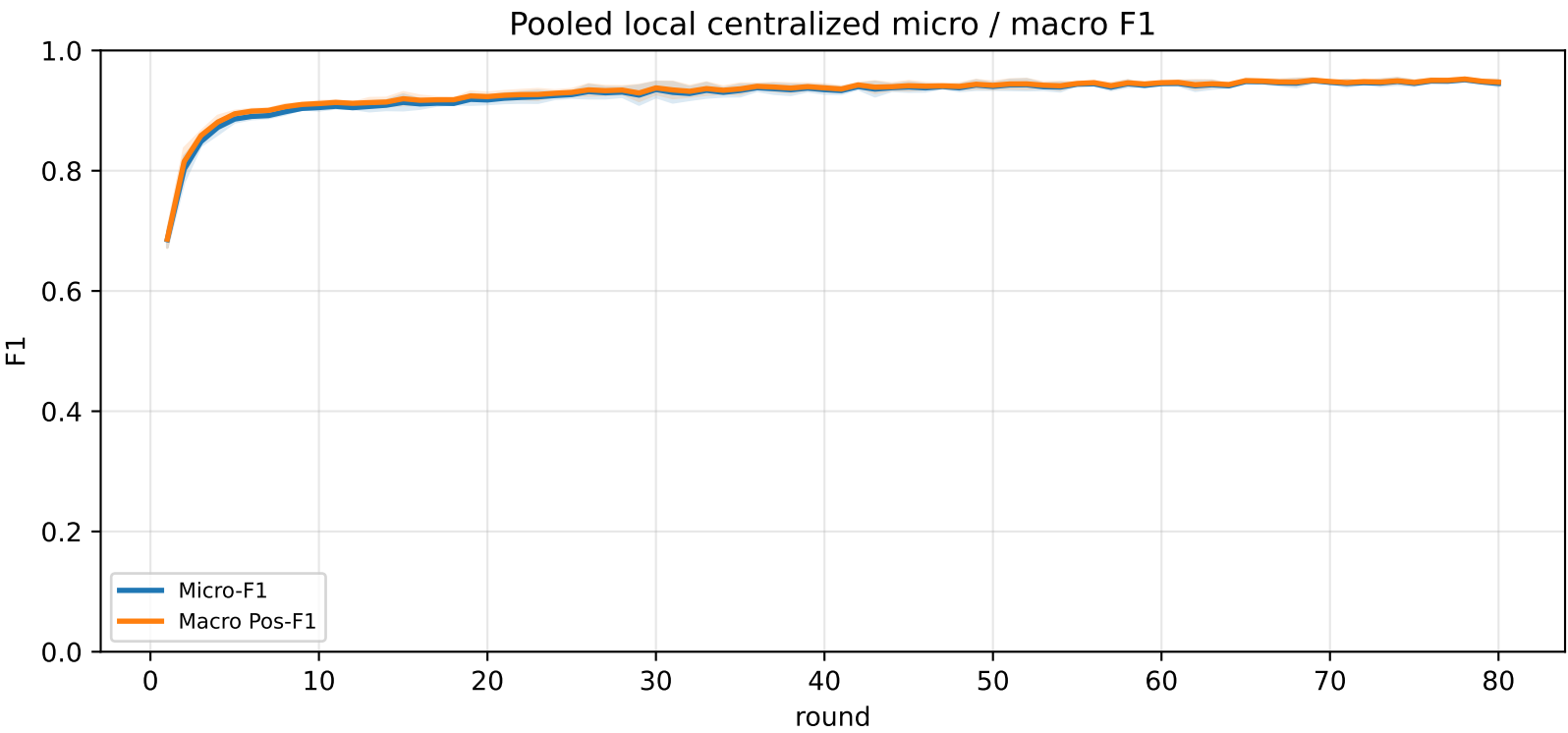
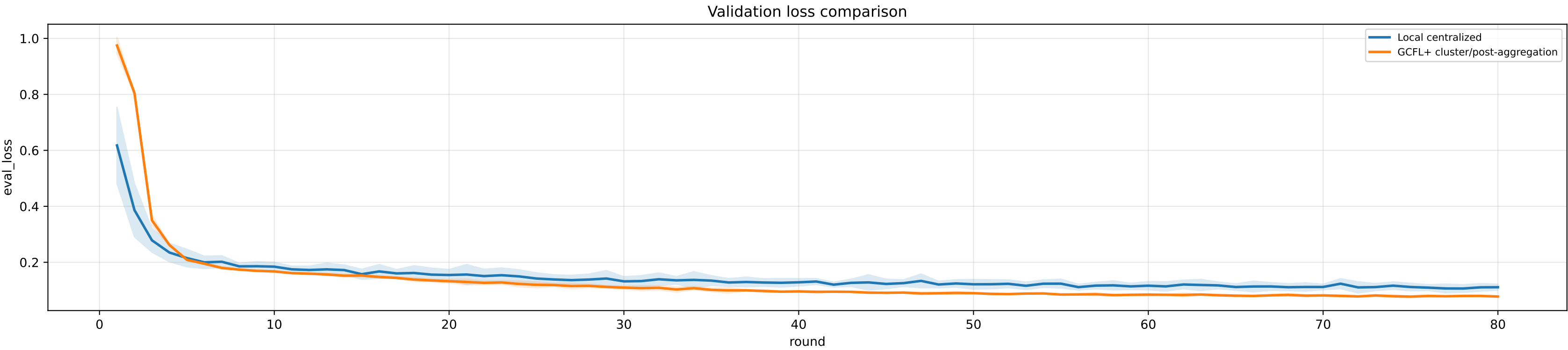
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	94.3% ± 0.2%	99.4% ± 0.0%	99.1% ± 0.0%	97.2% ± 0.1%	92.7% ± 0.7%	83.0% ± 0.3%
GCFL+	96.0% ± 0.4%	95.4% ± 0.4%	99.5% ± 0.1%	99.2% ± 0.1%	98.0% ± 0.2%	94.7% ± 0.5%	85.8% ± 1.0%



Heterogeneity level: Mid | GCFL+ diagnostics | family=flat_balanced |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.3%±0.6% | cycle3=26.4%±0.5% | cycle4=26.9%±0.4% | cycle5=26.6%±0.7% | cycle6=26.2%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.2% ± 0.1%	98.8% ± 0.2%	97.4% ± 0.2%	94.1% ± 0.2%	86.6% ± 0.4%
GCFL+	96.3% ± 0.2%	96.4% ± 0.2%	99.3% ± 0.1%	98.8% ± 0.1%	97.7% ± 0.2%	95.9% ± 0.3%	90.4% ± 0.4%



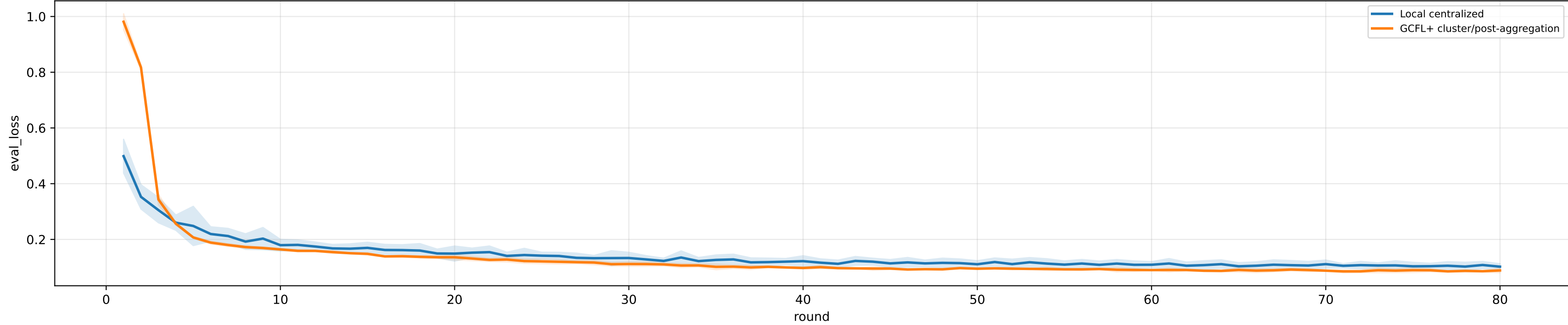
Heterogeneity level: Mid | GCFL+ diagnostics | family=mild_increasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=18.6%±0.8% | cycle3=22.7%±1.2% | cycle4=25.7%±0.5% | cycle5=27.9%±0.9% | cycle6=28.0%±0.4%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

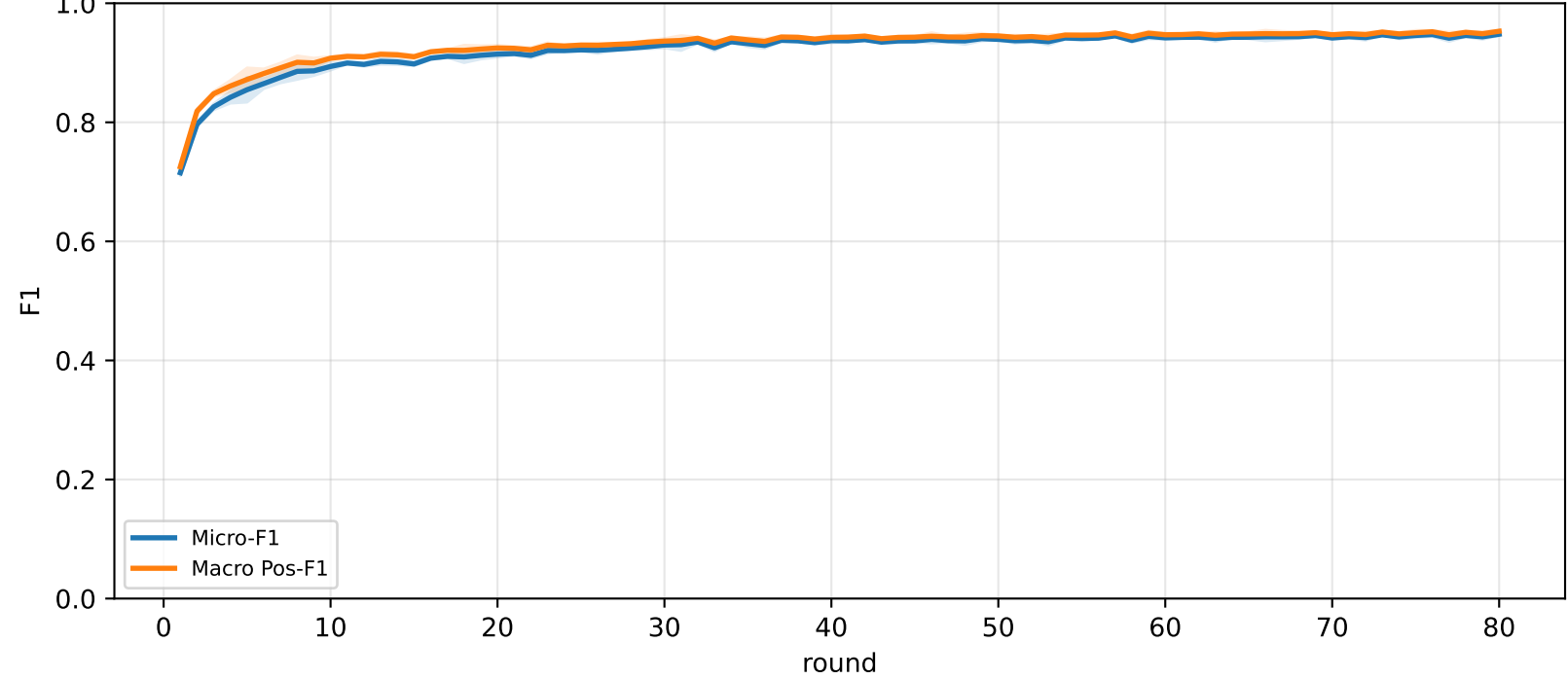
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	95.4% ± 0.2%	98.9% ± 0.1%	98.2% ± 0.2%	96.8% ± 0.3%	94.1% ± 0.5%	89.2% ± 0.3%
GCFL+	95.5% ± 0.3%	95.9% ± 0.3%	98.8% ± 0.2%	98.1% ± 0.1%	97.0% ± 0.2%	94.6% ± 0.6%	91.1% ± 0.5%

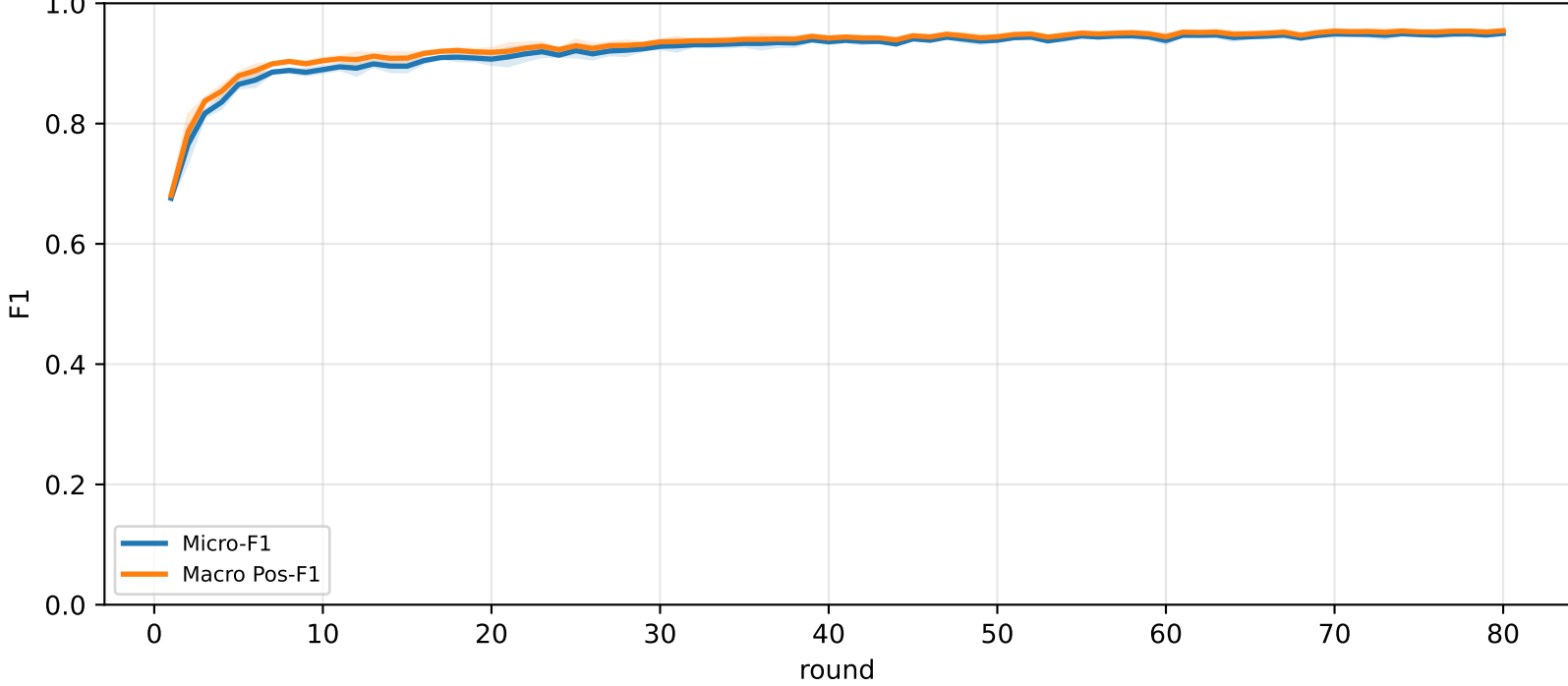
Validation loss comparison



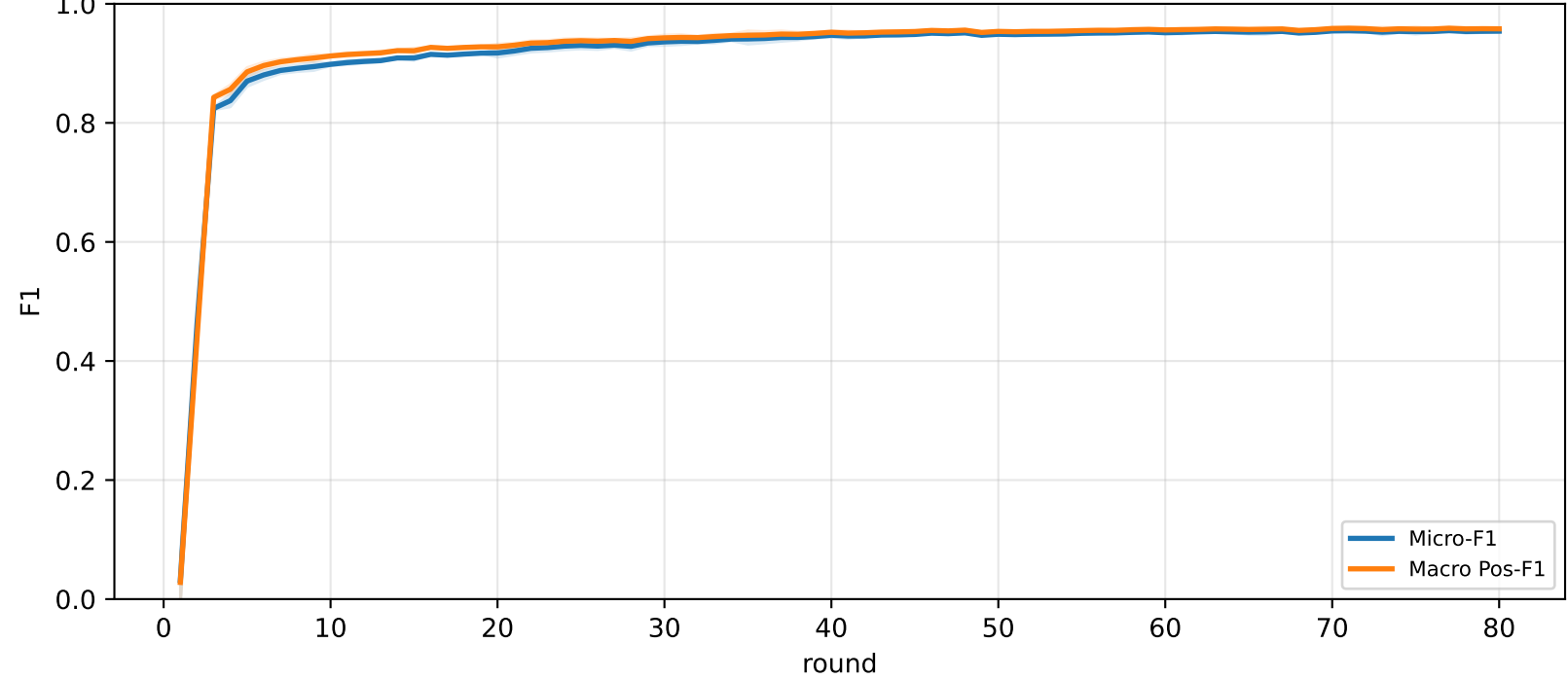
Pooled local centralized micro / macro F1



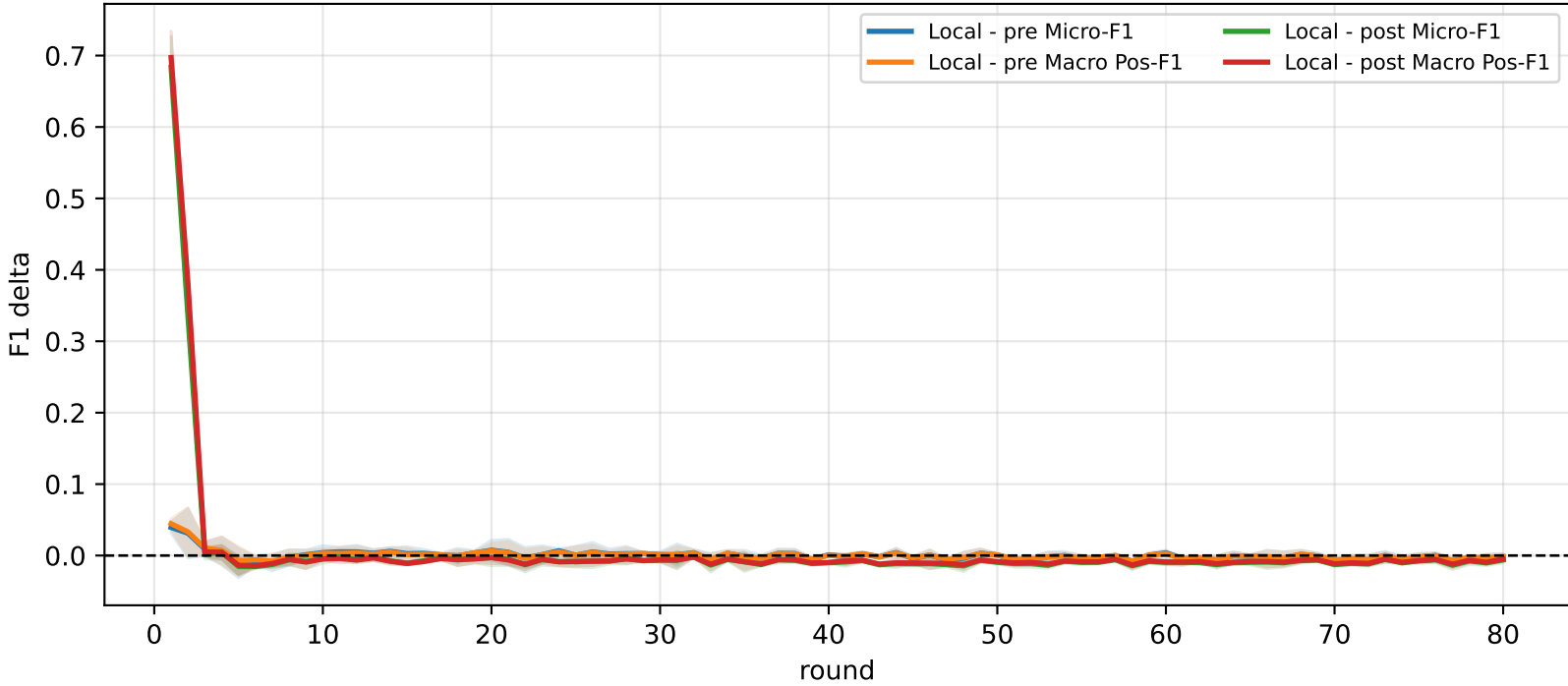
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



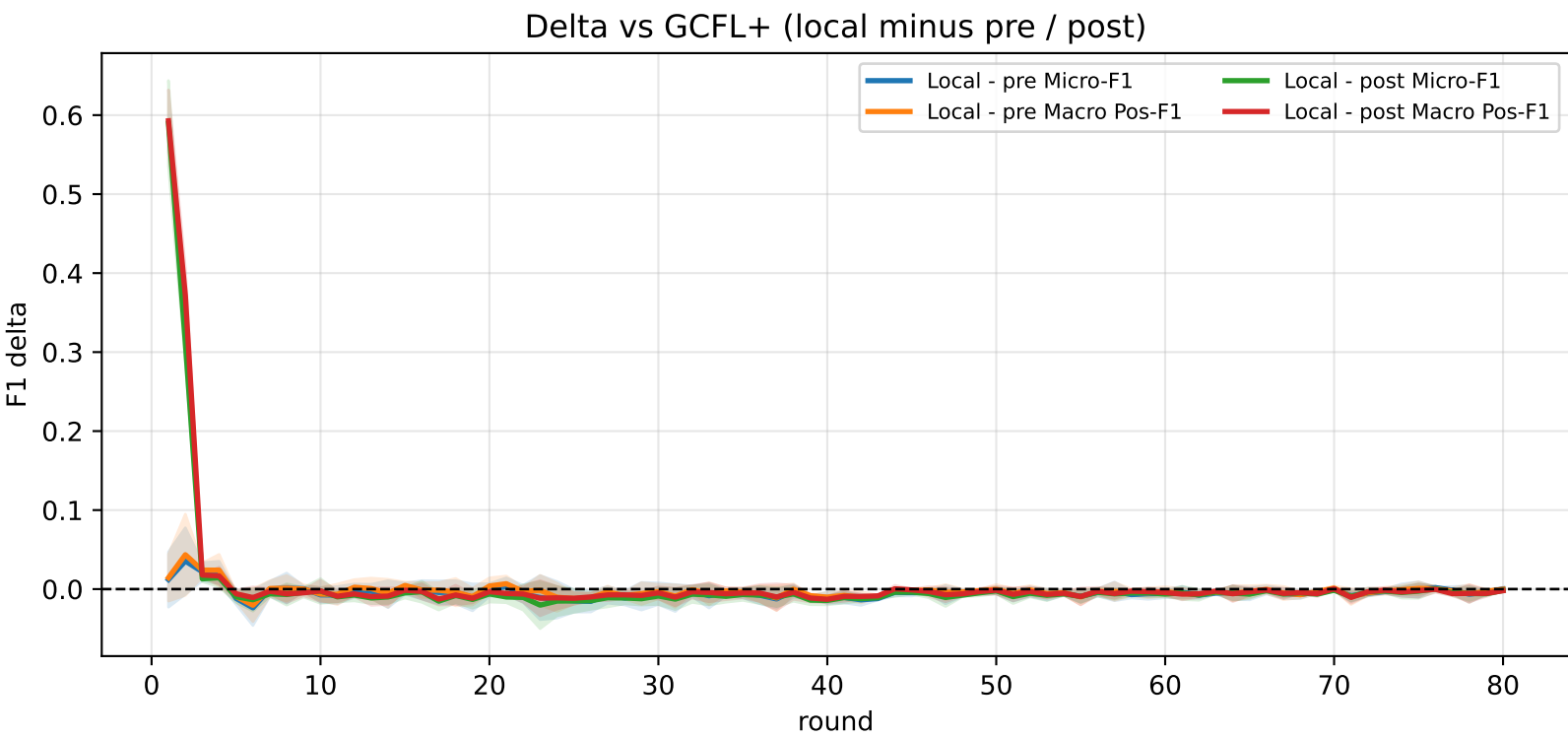
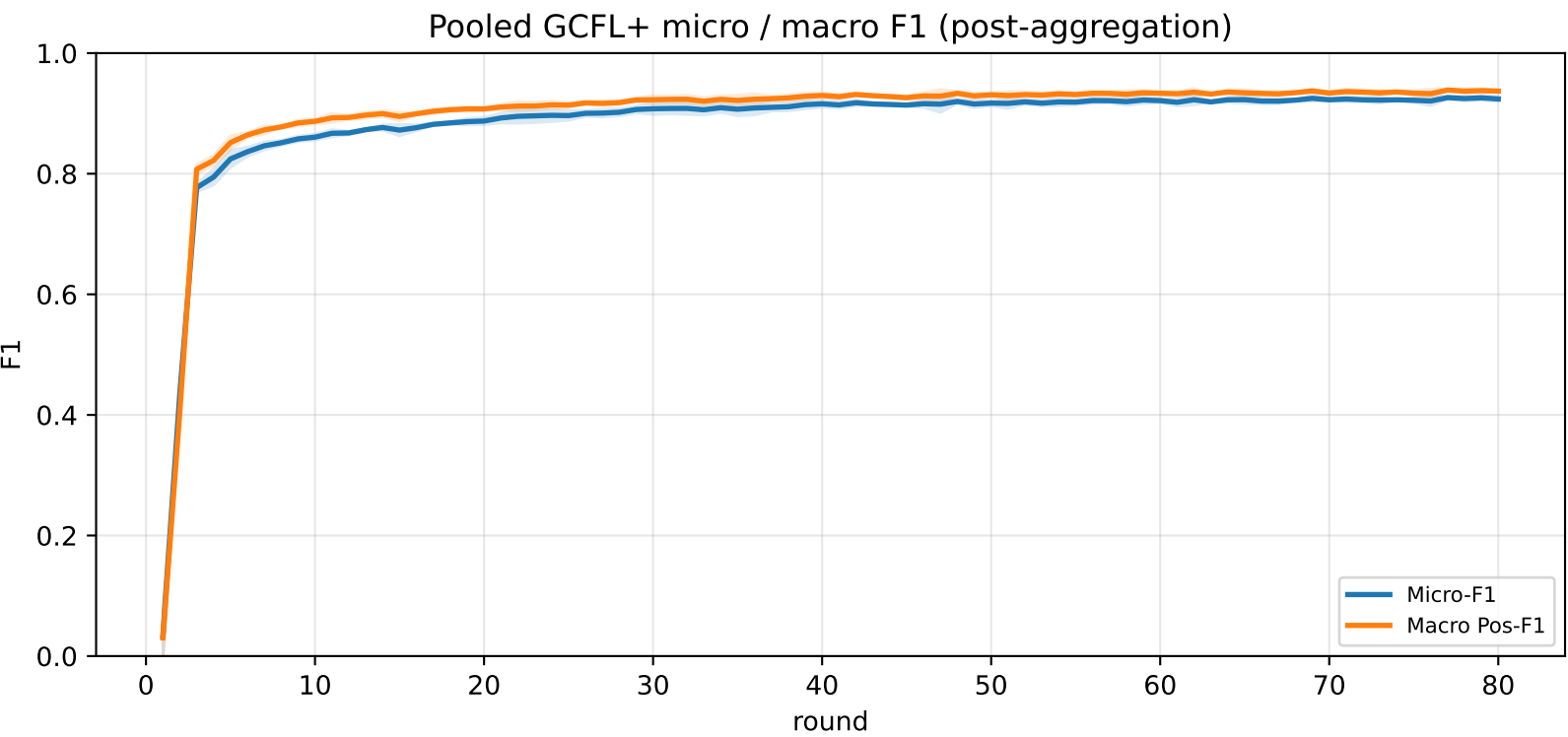
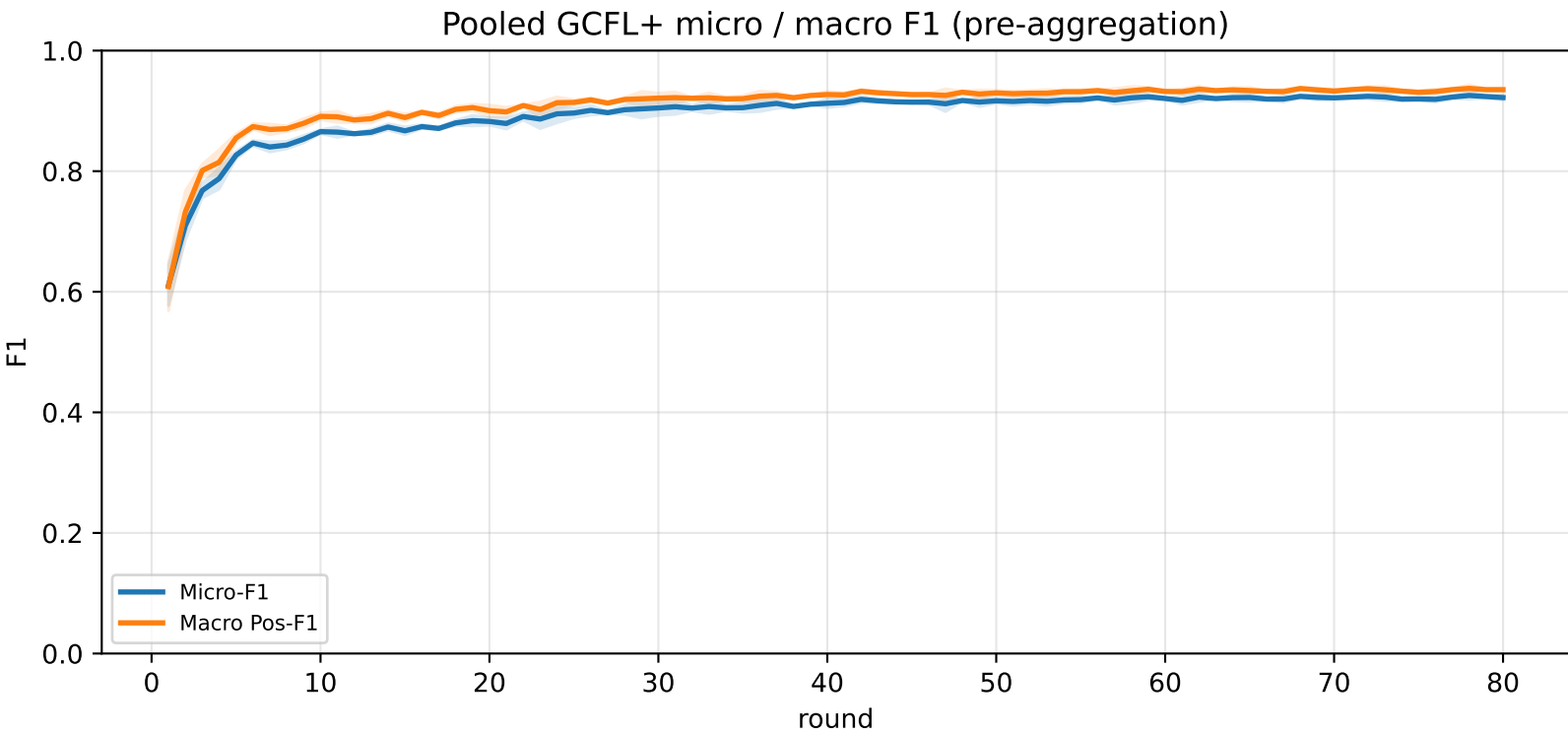
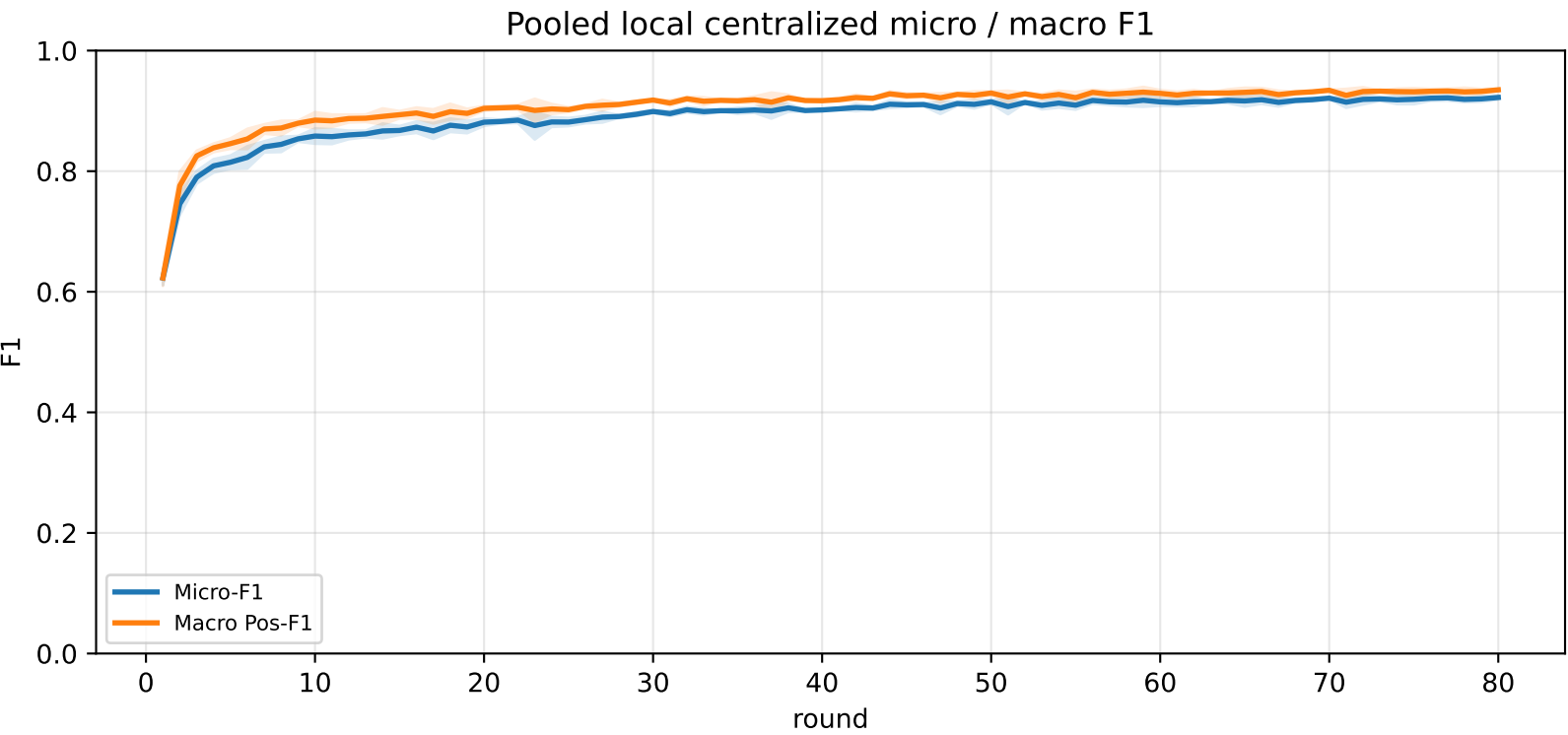
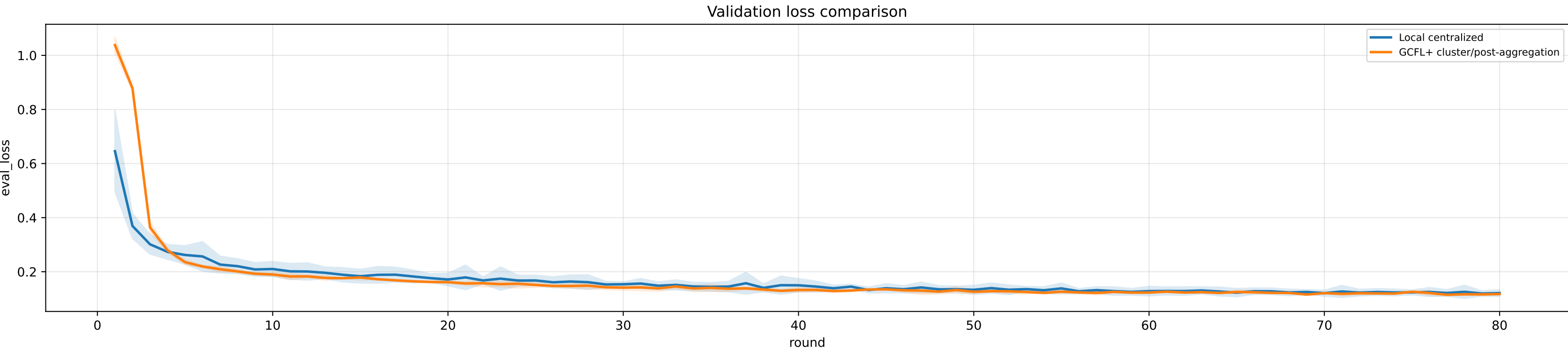
Delta vs GCFL+ (local minus pre / post)



Heterogeneity level: Mid | GCFL+ diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=10.4%±0.6% | cycle3=15.5%±0.4% | cycle4=21.0%±0.3% | cycle5=24.5%±0.8% | cycle6=28.9%±0.9%
heterogeneity=mid | task_jsd_mean=0.019 | task_jsd_median=0.010 | task_jsd_max=0.100 | family_centroid_jsd_mean=0.021
per-task pos-rate std | cycle2_std=0.105 | cycle3_std=0.062 | cycle4_std=0.026 | cycle5_std=0.041 | cycle6_std=0.068

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.3%	93.4% ± 0.2%	97.9% ± 0.3%	96.5% ± 0.2%	93.9% ± 0.2%	91.1% ± 0.7%	87.5% ± 0.7%
GCFL+	92.4% ± 0.3%	93.6% ± 0.2%	98.1% ± 0.3%	96.6% ± 0.1%	94.0% ± 0.3%	91.6% ± 0.5%	87.8% ± 0.8%



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

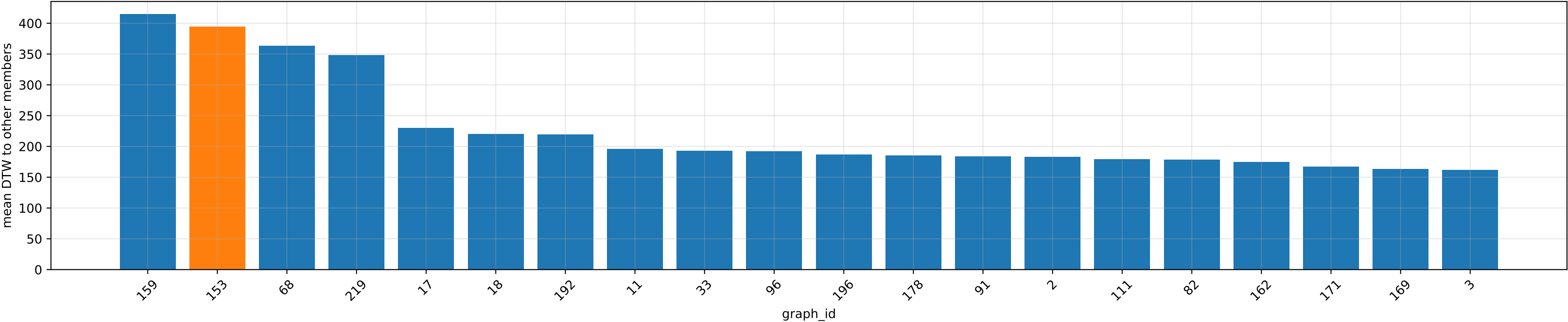
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	15	0	1 vs 19	153	159	no

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
394.433	nan	208.037

Per-member mean DTW



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

First applied split pairwise DTW metadata | seed=0

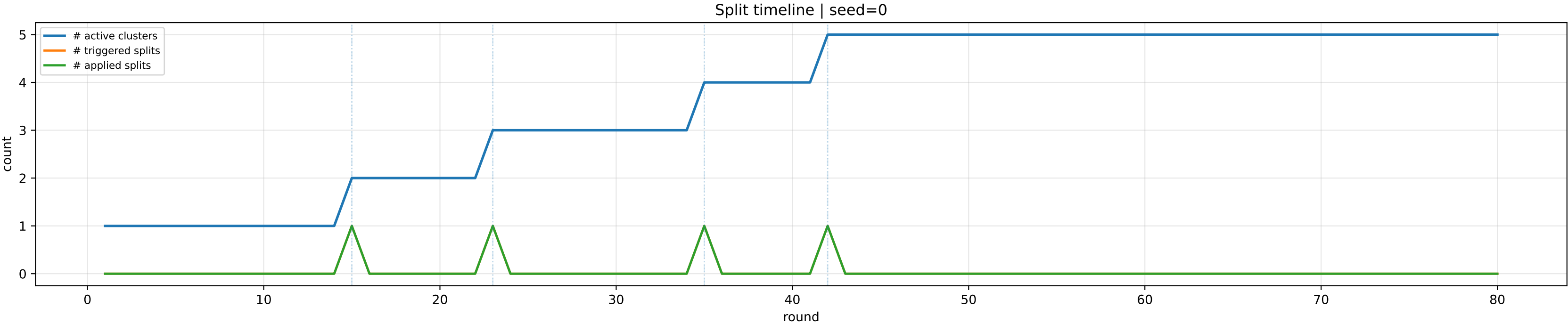
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	15	0	20	153.0	0.054	394.433	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 589.6 at (17, 159)

graph_id	178	96	17	18	171	169	91	11	192	33	111	196	2	162	82	3	153	68	219	159
178	0.0	53.3	67.1	119.0	63.1	54.8	38.3	53.8	242.4	95.7	46.9	192.7	151.2	161.8	173.7	72.8	506.3	447.8	456.4	522.5
96	53.3	0.0	54.1	89.3	64.7	65.4	40.9	39.8	257.8	102.6	42.5	208.0	171.6	177.1	189.1	98.6	521.7	463.1	471.8	537.9
17	67.1	54.1	0.0	77.2	106.4	117.8	73.1	49.1	309.5	154.4	77.1	259.8	204.7	228.9	240.8	152.0	573.4	514.9	523.5	589.6
18	119.0	89.3	77.2	0.0	129.6	145.3	111.1	69.1	269.2	185.8	77.4	188.2	164.5	172.1	200.5	174.5	496.3	474.6	483.2	549.3
171	63.1	64.7	106.4	129.6	0.0	26.3	41.0	79.8	240.8	124.3	73.0	130.1	100.4	75.6	108.1	52.7	432.3	423.8	417.1	483.2
169	54.8	65.4	117.8	145.3	26.3	0.0	51.3	72.9	230.1	122.1	69.8	123.0	103.3	56.7	101.7	44.2	433.7	403.9	405.8	471.8
91	38.3	40.9	73.1	111.1	41.0	51.3	0.0	56.7	229.8	101.3	50.6	190.9	163.2	160.1	172.0	88.9	504.6	446.1	454.7	520.8
11	53.8	39.8	49.1	69.1	79.8	72.9	56.7	0.0	266.1	116.7	33.7	216.4	154.0	185.5	197.4	103.1	530.0	471.5	480.1	546.2
192	242.4	257.8	309.5	269.2	240.8	230.1	229.8	266.1	0.0	94.6	232.4	119.4	200.8	140.1	167.2	219.2	313.0	188.6	172.0	280.1
33	95.7	102.6	154.4	185.8	124.3	122.1	101.3	116.7	94.6	0.0	117.9	166.7	230.0	167.8	173.7	133.9	488.6	287.7	369.1	435.2
111	46.9	42.5	77.1	77.4	73.0	69.8	50.6	33.7	232.4	117.9	0.0	182.6	136.5	131.8	163.7	82.3	496.3	437.8	446.4	512.5
196	192.7	208.0	259.8	188.2	130.1	123.0	190.9	216.4	119.4	166.7	182.6	0.0	161.3	50.8	93.8	98.2	313.6	255.4	263.7	329.8
2	151.2	171.6	204.7	164.5	100.4	103.3	163.2	154.0	200.8	230.0	136.5	161.3	0.0	142.4	80.8	66.8	270.3	314.3	269.1	384.7
162	161.8	177.1	228.9	172.1	75.6	56.7	160.1	185.5	140.1	167.8	131.8	50.8	142.4	0.0	76.5	77.4	342.3	309.7	294.6	360.7
82	173.7	189.1	240.8	200.5	108.1	101.7	172.0	197.4	167.2	173.7	163.7	93.8	80.8	76.5	0.0	54.0	254.9	308.8	278.0	348.8
3	72.8	98.6	152.0	174.5	52.7	44.2	88.9	103.1	219.2	133.9	82.3	98.2	66.8	77.4	54.0	0.0	343.3	403.7	371.5	437.6
153	506.3	521.7	573.4	496.3	432.3	433.7	504.6	530.0	313.0	488.6	496.3	313.6	270.3	342.3	254.9	343.3	0.0	306.8	148.4	218.5
68	447.8	463.1	514.9	474.6	423.8	403.9	446.1	471.5	188.6	287.7	437.8	255.4	314.3	309.7	308.8	403.7	306.8	0.0	203.8	245.3
219	456.4	471.8	523.5	483.2	417.1	405.8	454.7	480.1	172.0	369.1	446.4	263.7	269.1	294.6	278.0	371.5	148.4	203.8	0.0	103.0
159	522.5	537.9	589.6	549.3	483.2	471.8	520.8	546.2	280.1	435.2	512.5	329.8	384.7	360.7	348.8	437.6	218.5	245.3	103.0	0.0

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

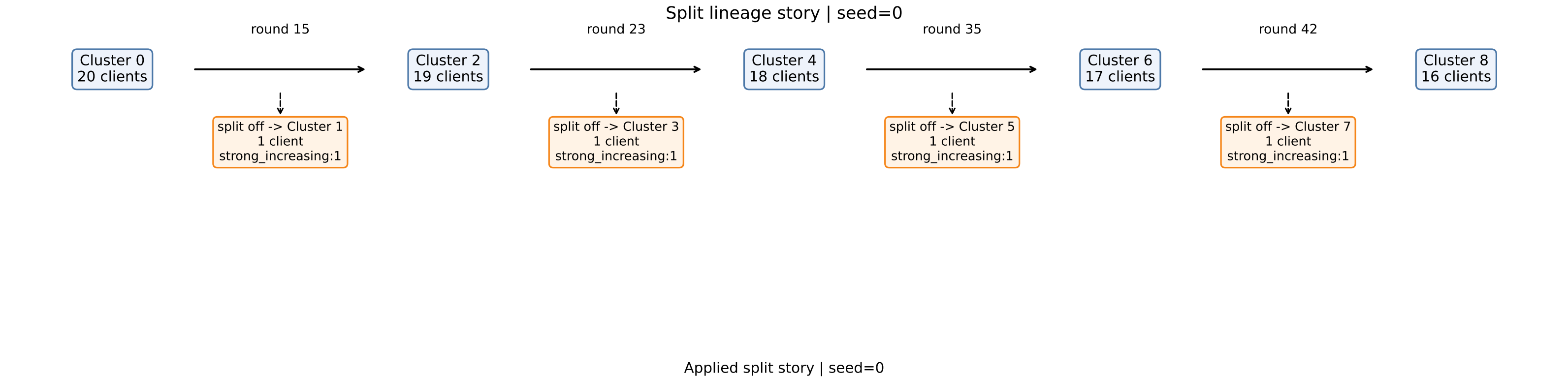


Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	15, 23, 35, 42	15, 23, 35, 42	1, 3, 5, 7, 8	1:1, 3:1, 5:1, 7:1, 8:16

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087



round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
15	0	20	158.461 < 159.842 , 335.525 > 302.597	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
23	2	19	156.969 < 158.416 , 388.349 > 368.440	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
35	4	18	157.495 < 157.540 , 355.067 > 332.376	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1
42	6	17	156.787 < 156.864 , 365.625 > 315.074	C7 (1 client)	strong_increasing:1	C8 (16 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

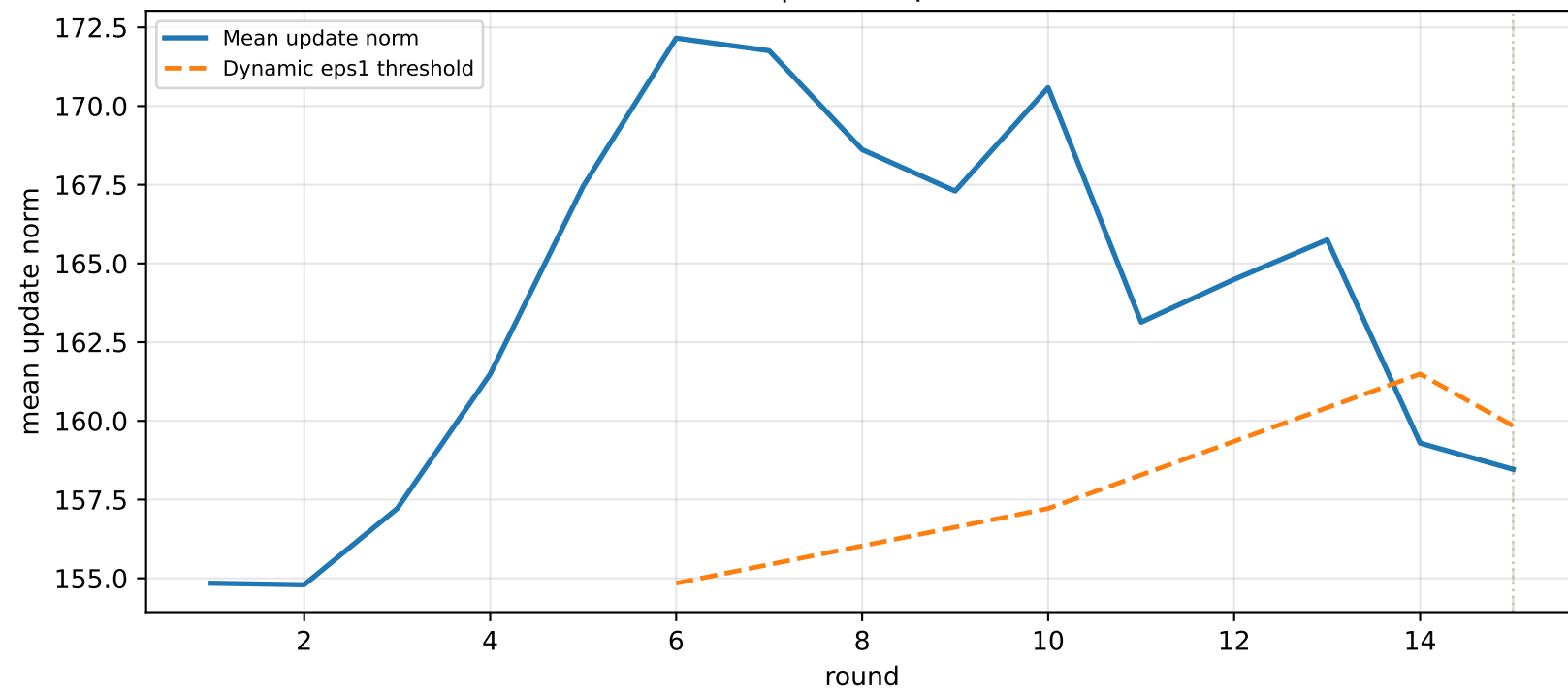
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	153	16	strong_increasing:1
3	1	159	19	strong_increasing:1
5	1	219	18	strong_increasing:1
7	1	68	17	strong_increasing:1
8	16	178, 96, 17, 18, 171, 169, 91, 11, 192, 33, 111, 196, 2, 162, 82, 3	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4

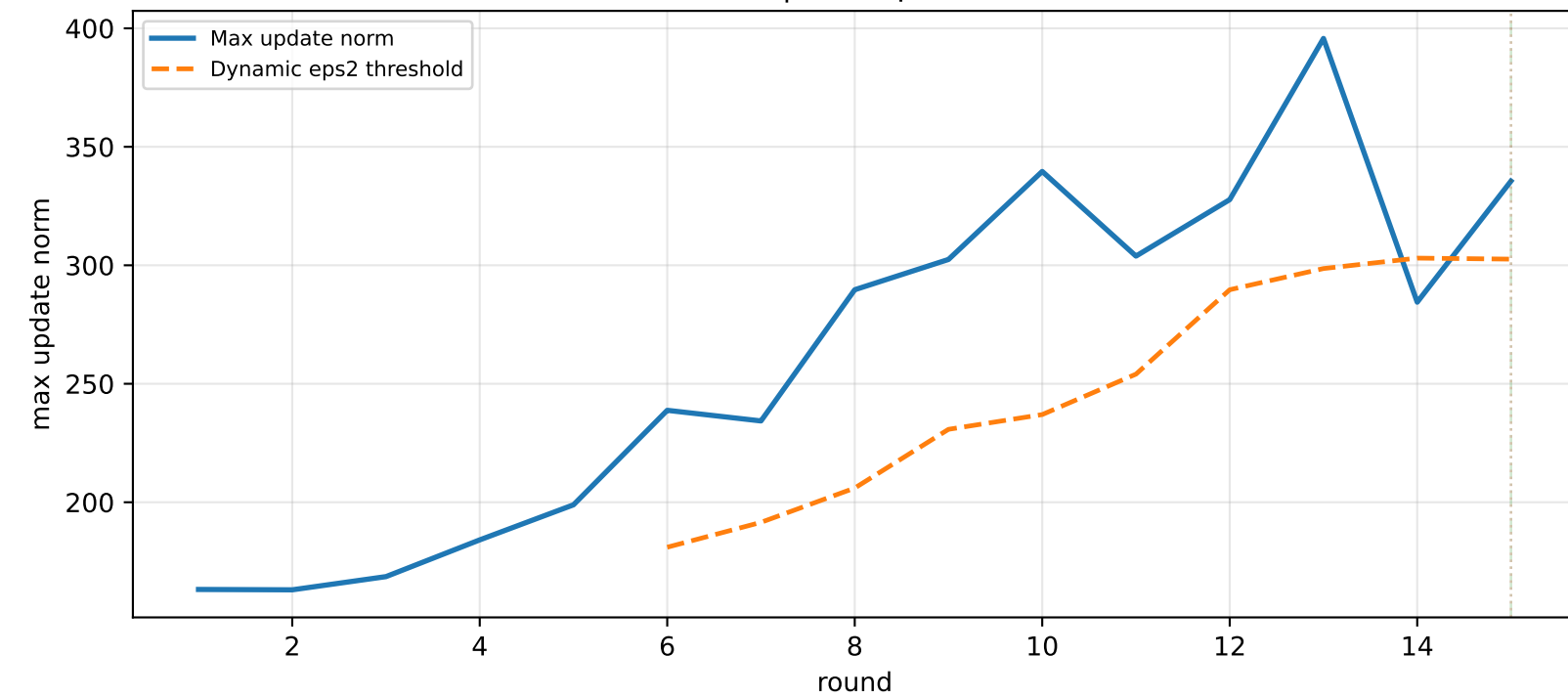
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

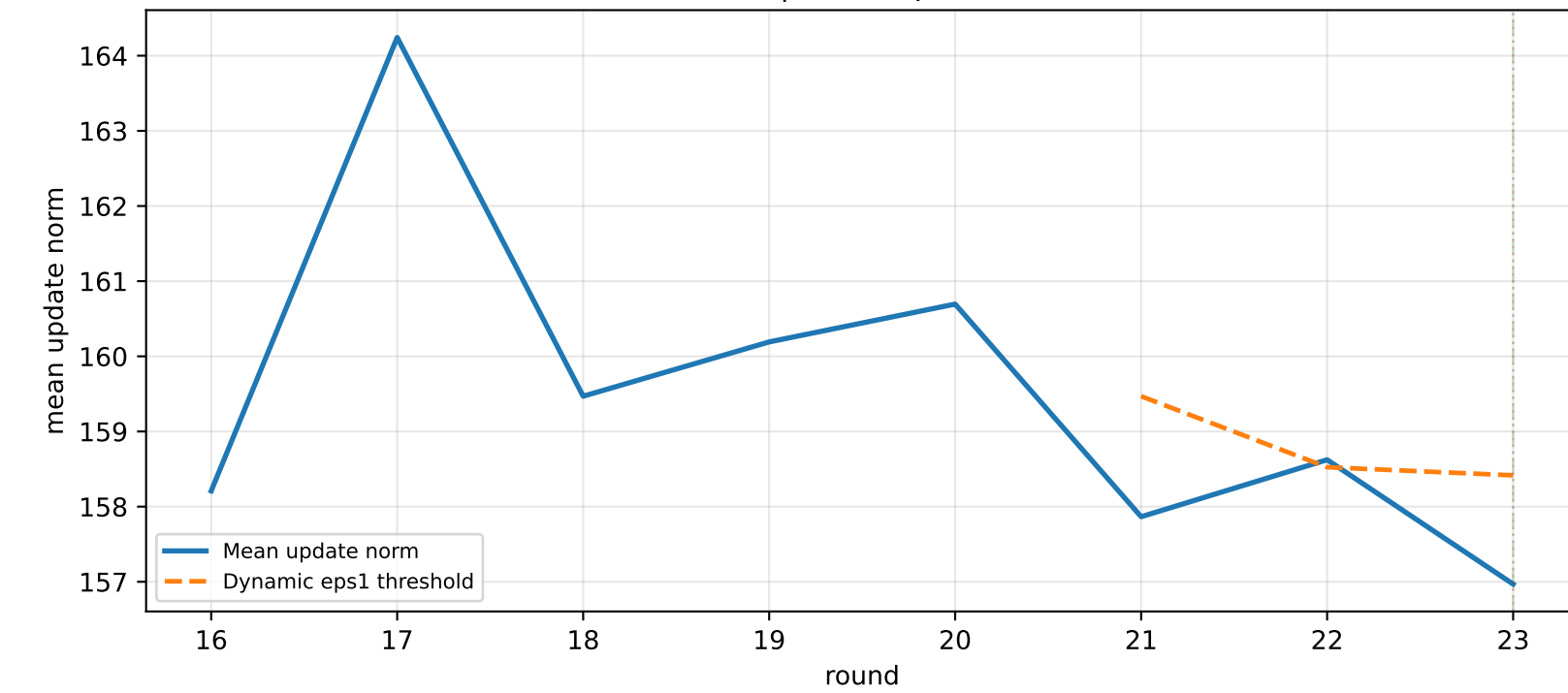
Cluster 0 | Mean update norm



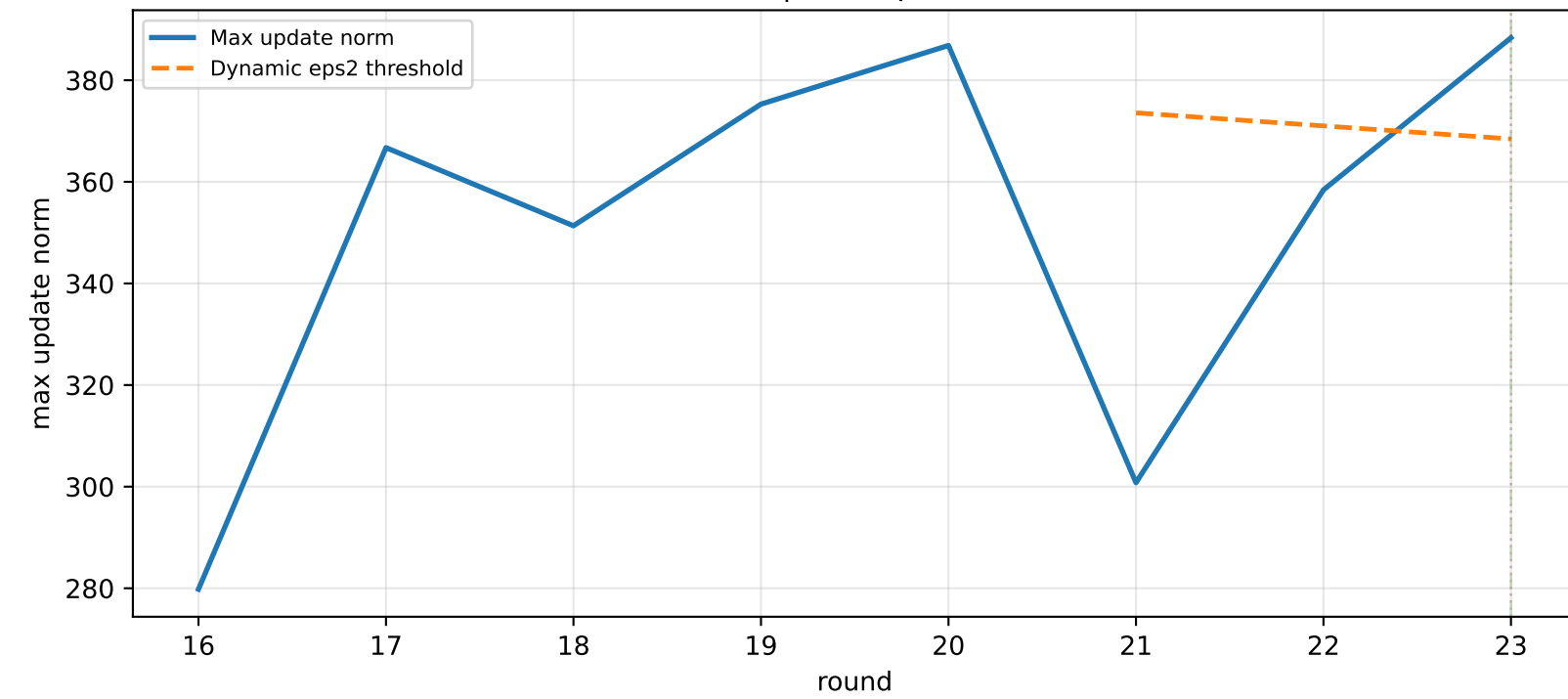
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



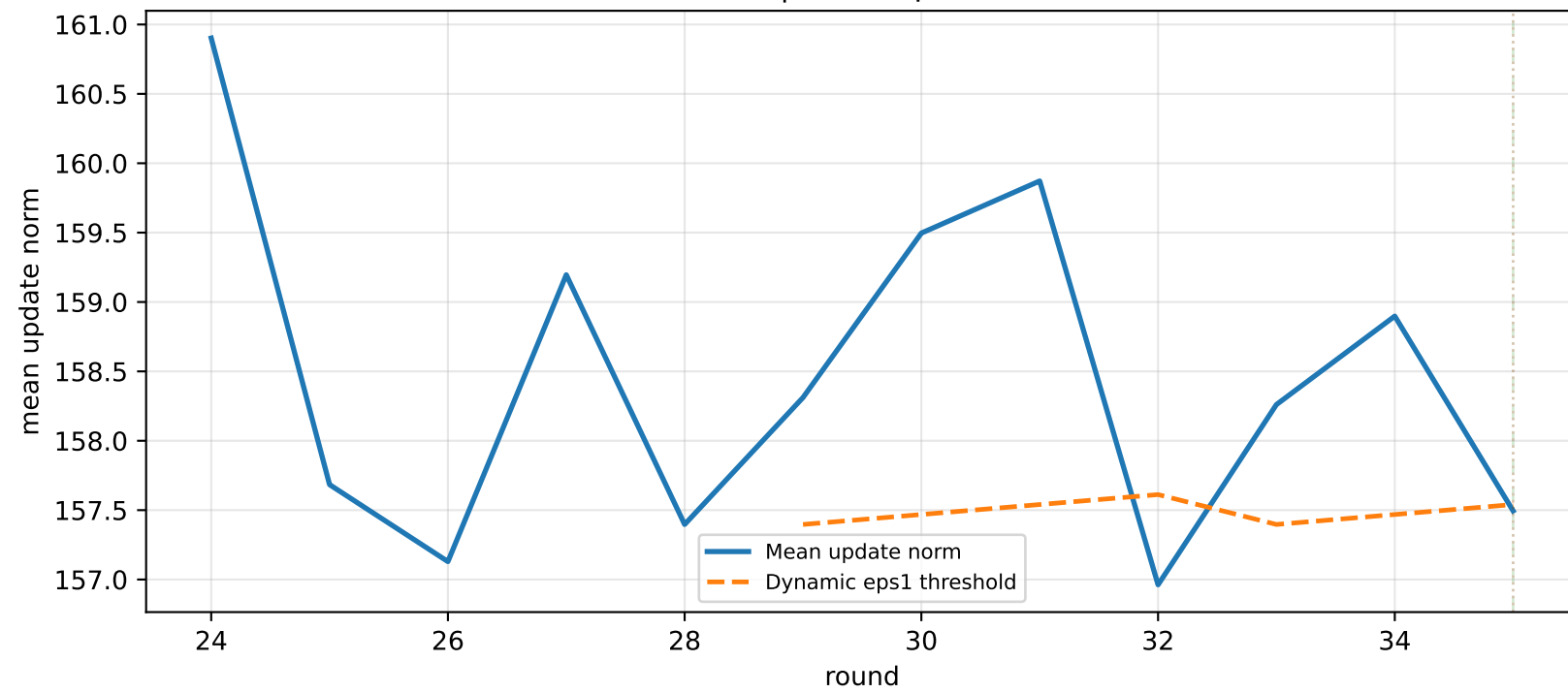
Cluster 2 | Max update norm



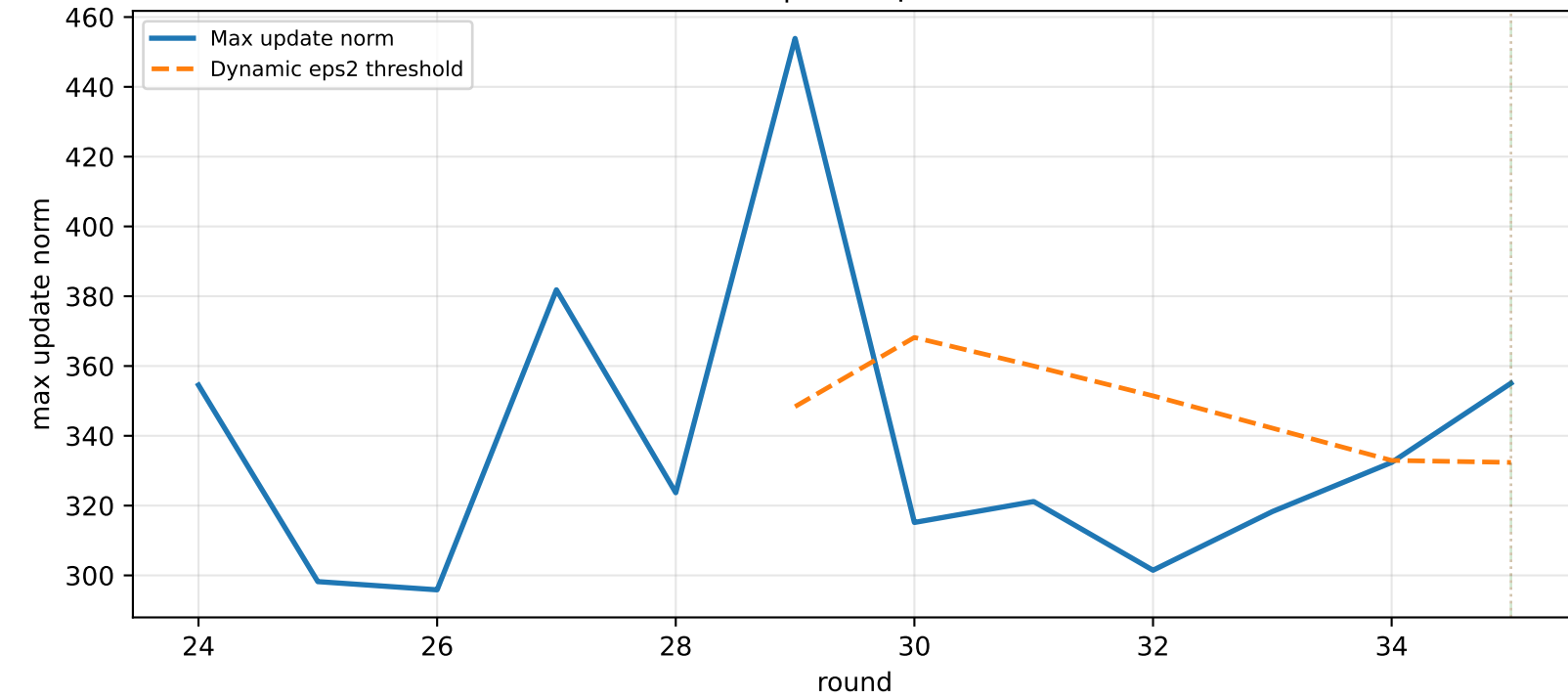
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

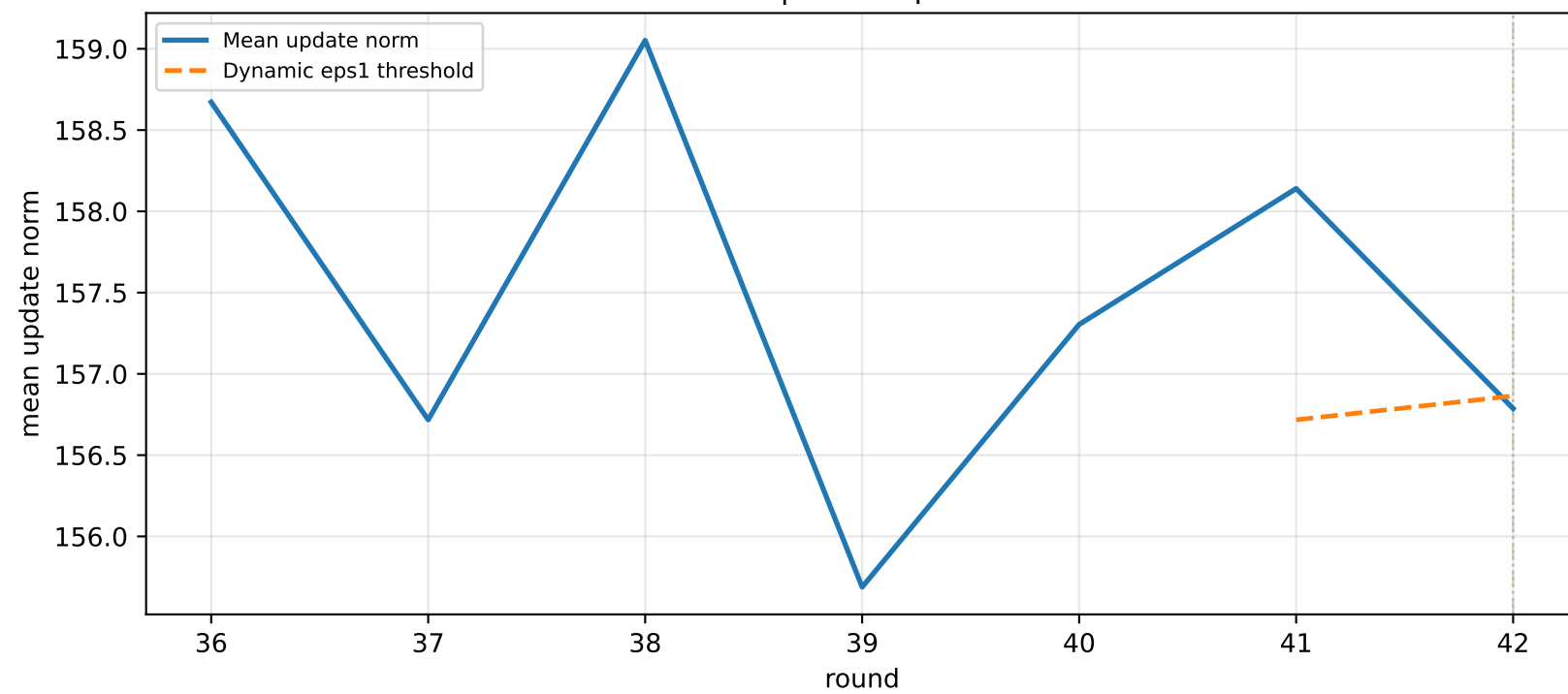
Cluster 4 | Mean update norm



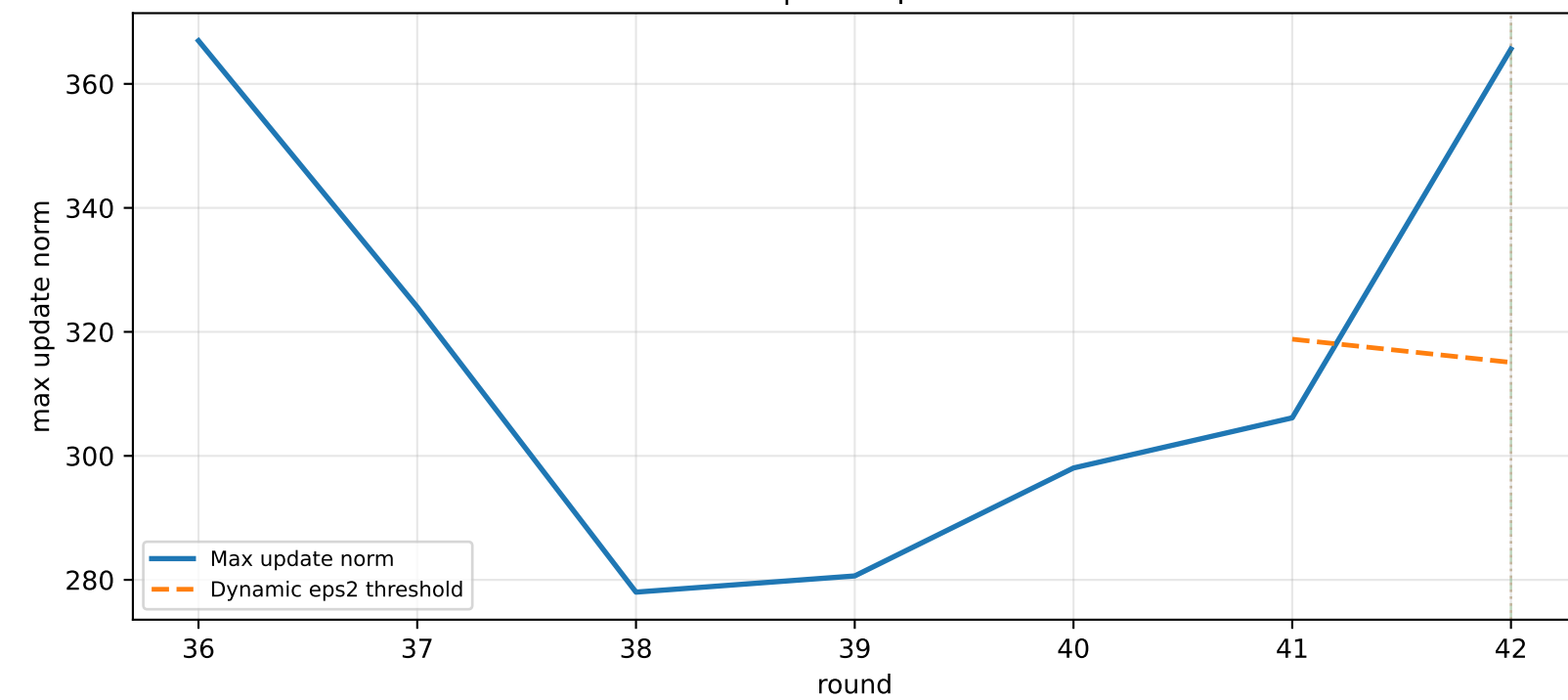
Cluster 4 | Max update norm



Cluster 6 | Mean update norm

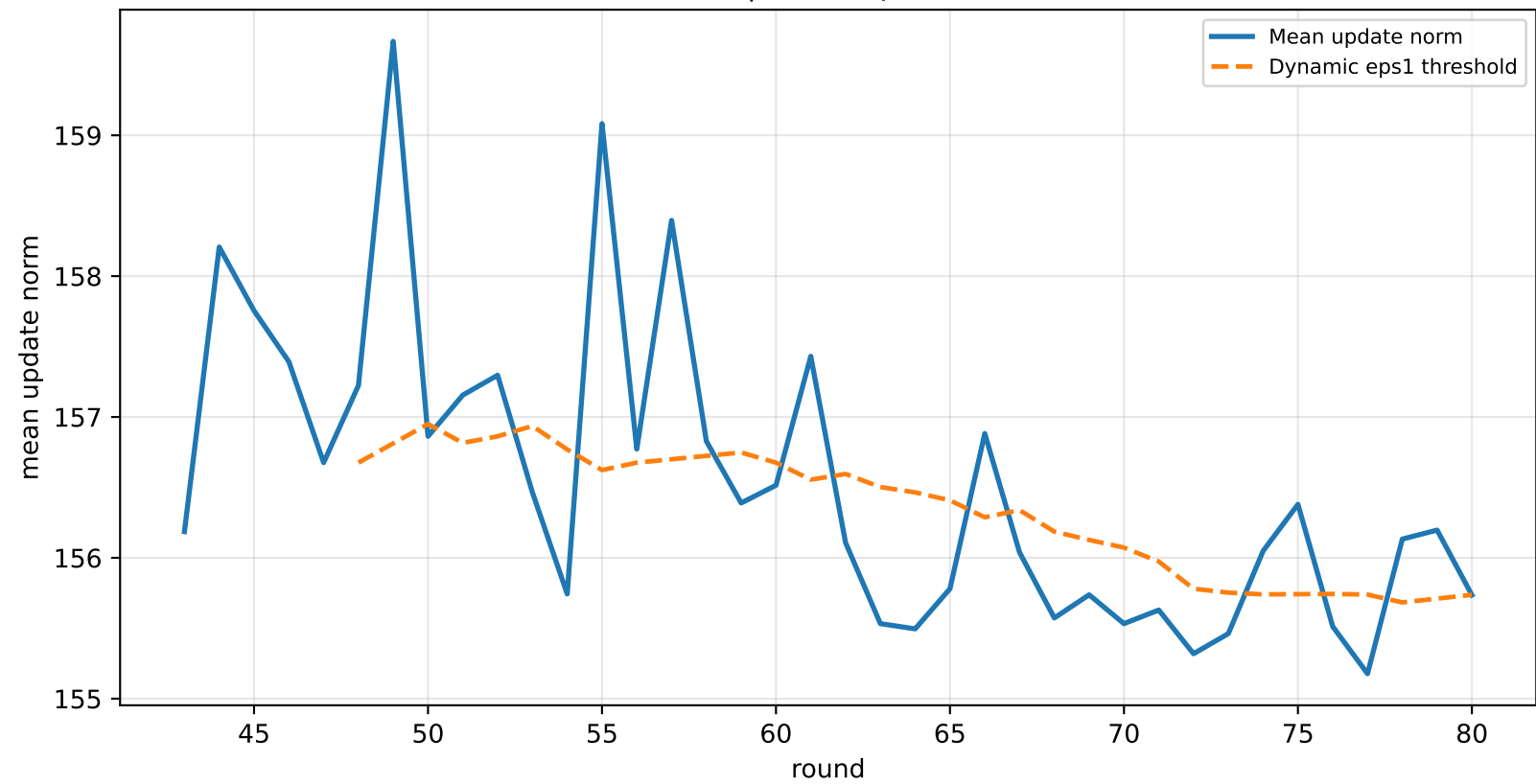


Cluster 6 | Max update norm

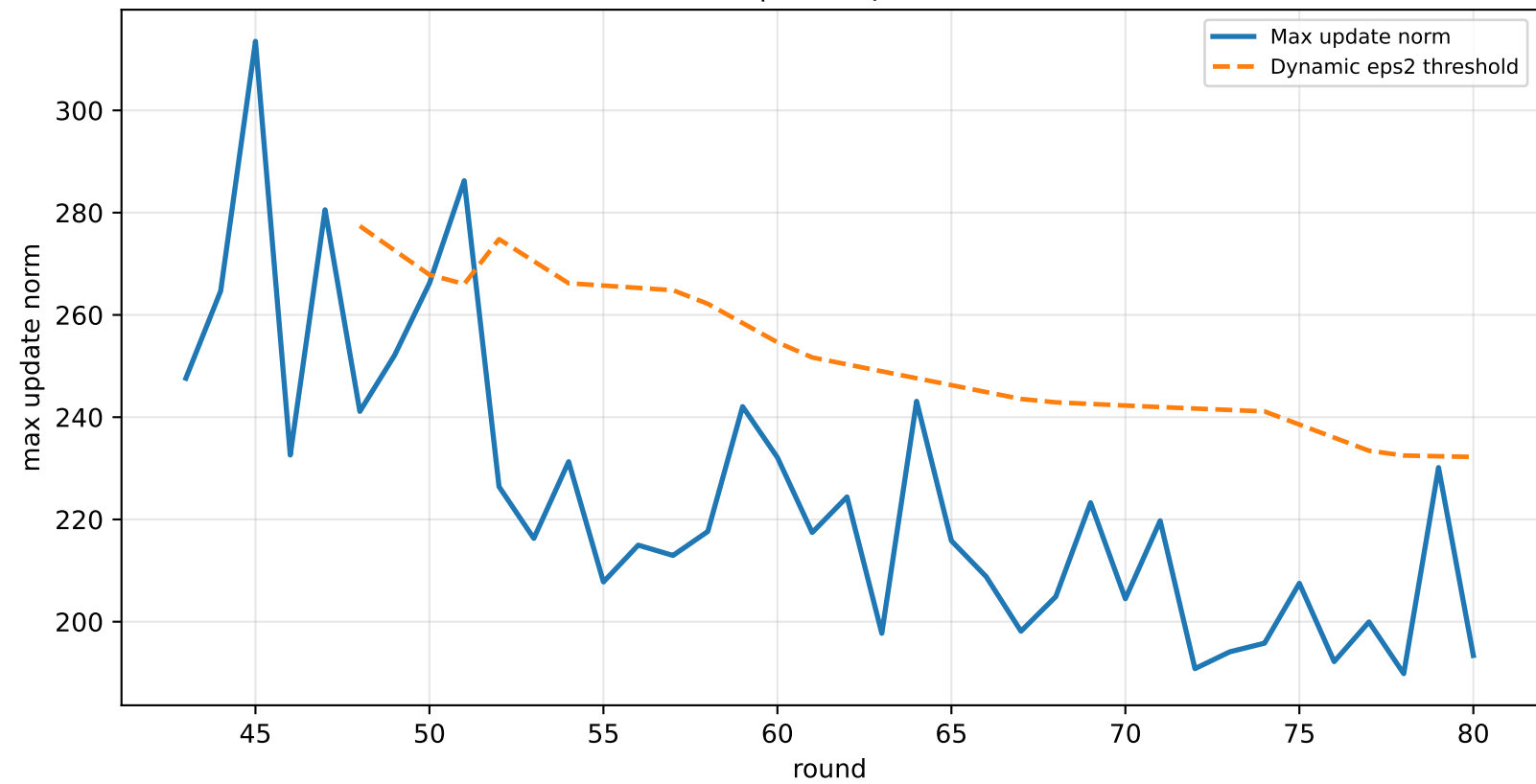


Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (8-8)
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Cluster 8 | Mean update norm



Cluster 8 | Max update norm



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

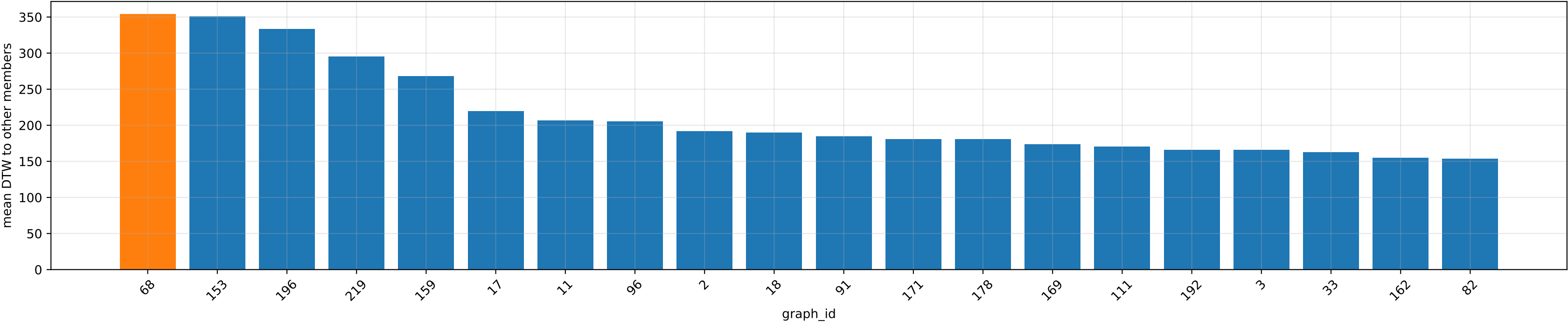
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	13	0	1 vs 19	68	68	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
353.866	nan	200.020

Per-member mean DTW



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

First applied split pairwise DTW metadata | seed=1

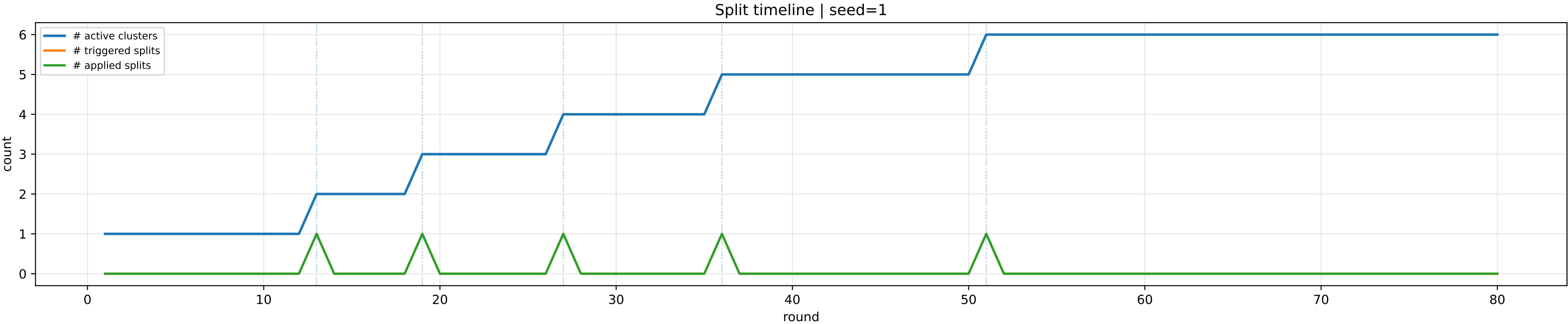
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	13	0	20	68.0	0.056	353.866	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 502.6 at (17, 153)

graph_id	178	96	17	18	171	169	91	11	192	33	111	196	2	162	82	3	153	68	219	159
178	0.0	187.9	43.8	106.0	15.7	20.3	45.7	43.9	145.7	89.3	190.7	423.7	179.8	72.7	112.7	130.2	451.0	421.1	383.2	369.0
96	187.9	0.0	179.6	239.8	197.0	187.4	197.4	233.0	126.5	145.4	120.7	276.3	262.7	159.1	141.8	134.5	274.8	413.0	226.7	192.7
17	43.8	179.6	0.0	120.2	64.3	72.8	59.7	43.9	197.3	142.9	242.3	475.3	214.8	133.9	173.5	182.7	502.6	472.7	434.8	420.6
18	106.0	239.8	120.2	0.0	109.1	106.6	88.5	119.1	157.5	130.4	149.5	392.9	81.7	112.2	104.0	130.9	423.3	357.2	381.1	301.7
171	15.7	197.0	64.3	109.1	0.0	14.6	55.0	36.7	146.5	90.6	191.5	424.5	175.5	62.4	103.1	118.5	451.8	421.9	384.0	369.8
169	20.3	187.4	72.8	106.6	14.6	0.0	59.0	50.7	130.0	71.1	176.9	409.9	170.1	53.5	93.9	109.9	437.2	407.3	369.4	355.2
91	45.7	197.4	59.7	88.5	55.0	59.0	0.0	53.9	162.5	108.5	192.7	431.6	129.8	66.5	104.9	102.3	458.9	429.0	391.0	376.8
11	43.9	233.0	43.9	119.1	36.7	50.7	53.9	0.0	184.2	129.7	229.2	462.2	173.3	106.9	148.0	130.7	489.5	459.6	421.7	407.4
192	145.7	126.5	197.3	157.5	146.5	130.0	162.5	184.2	0.0	62.2	93.3	278.0	152.2	81.1	86.7	119.2	308.3	275.4	237.5	212.8
33	89.3	145.4	142.9	130.4	90.6	71.1	108.5	129.7	62.2	0.0	117.4	332.5	142.0	62.2	92.7	112.7	359.8	329.9	291.9	277.7
111	190.7	120.7	242.3	149.5	191.5	176.9	192.7	229.2	93.3	117.4	0.0	250.0	110.8	105.4	73.5	87.3	298.3	306.5	180.9	125.7
196	423.7	276.3	475.3	392.9	424.5	409.9	431.6	462.2	278.0	332.5	250.0	0.0	343.7	354.0	289.8	309.4	170.1	307.5	177.2	221.2
2	179.8	262.7	214.8	81.7	175.5	170.1	129.8	173.3	152.2	142.0	110.8	343.7	0.0	110.7	123.9	121.8	364.4	289.0	325.1	171.6
162	72.7	159.1	133.9	112.2	62.4	53.5	66.5	106.9	81.1	62.2	105.4	354.0	110.7	0.0	49.2	72.2	381.3	348.9	313.4	299.2
82	112.7	141.8	173.5	104.0	103.1	93.9	104.9	148.0	86.7	92.7	73.5	289.8	123.9	49.2	0.0	50.5	370.5	313.6	273.1	214.0
3	130.2	134.5	182.7	130.9	118.5	109.9	102.3	130.7	119.2	112.7	87.3	309.4	121.8	72.2	50.5	0.0	405.6	368.6	272.0	191.5
153	451.0	274.8	502.6	423.3	451.8	437.2	458.9	489.5	308.3	359.8	298.3	170.1	364.4	381.3	370.5	405.6	0.0	226.4	130.9	156.5
68	421.1	413.0	472.7	357.2	421.9	407.3	429.0	459.6	275.4	329.9	306.5	307.5	289.0	348.9	313.6	368.6	226.4	0.0	282.1	293.6
219	383.2	226.7	434.8	381.1	384.0	369.4	391.0	421.7	237.5	291.9	180.9	177.2	325.1	313.4	273.1	272.0	130.9	282.1	0.0	137.9
159	369.0	192.7	420.6	301.7	369.8	355.2	376.8	407.4	212.8	277.7	125.7	221.2	171.6	299.2	214.0	191.5	156.5	293.6	137.9	0.0

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

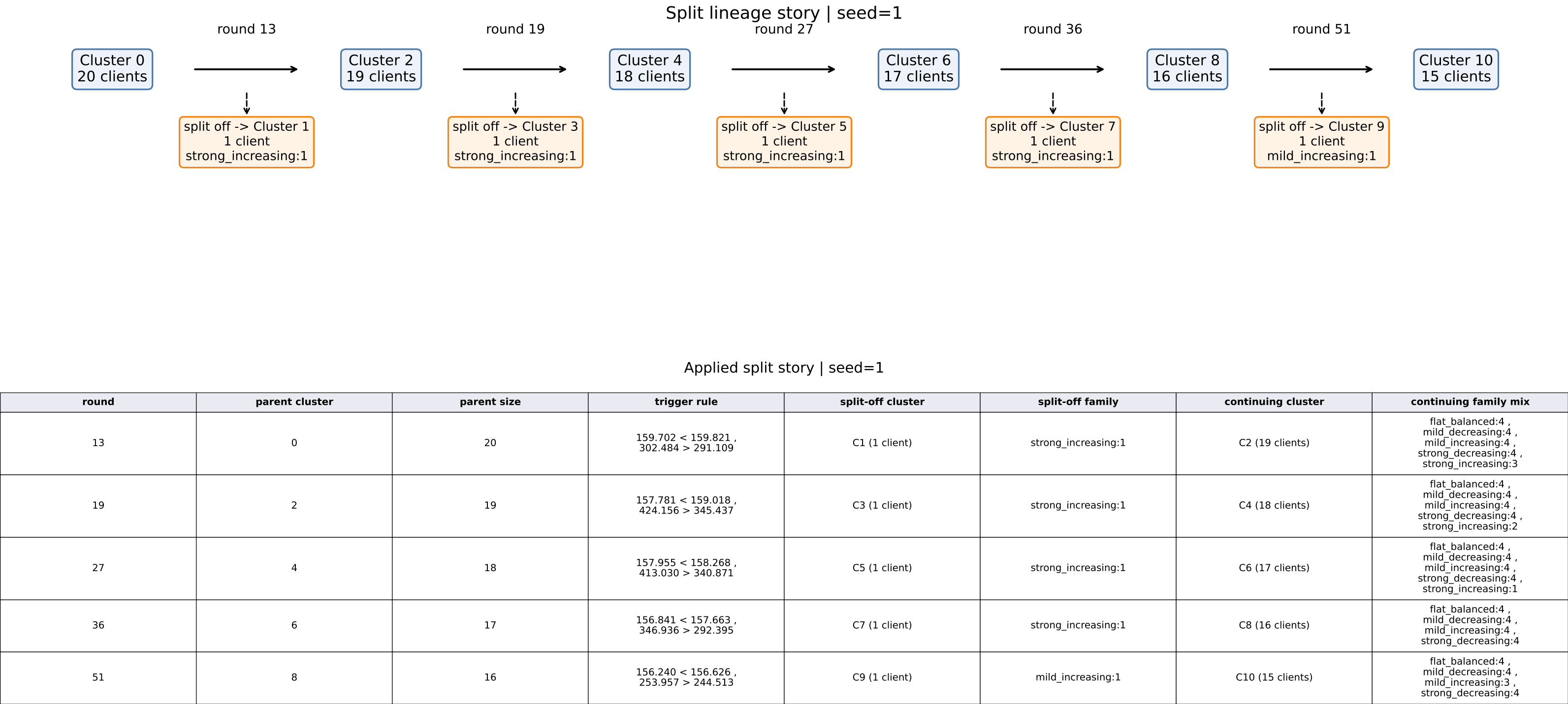


Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	13, 19, 27, 36, 51	13, 19, 27, 36, 51	1, 3, 5, 7, 9, 10	1:1, 3:1, 5:1, 7:1, 9:1, 10:15

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

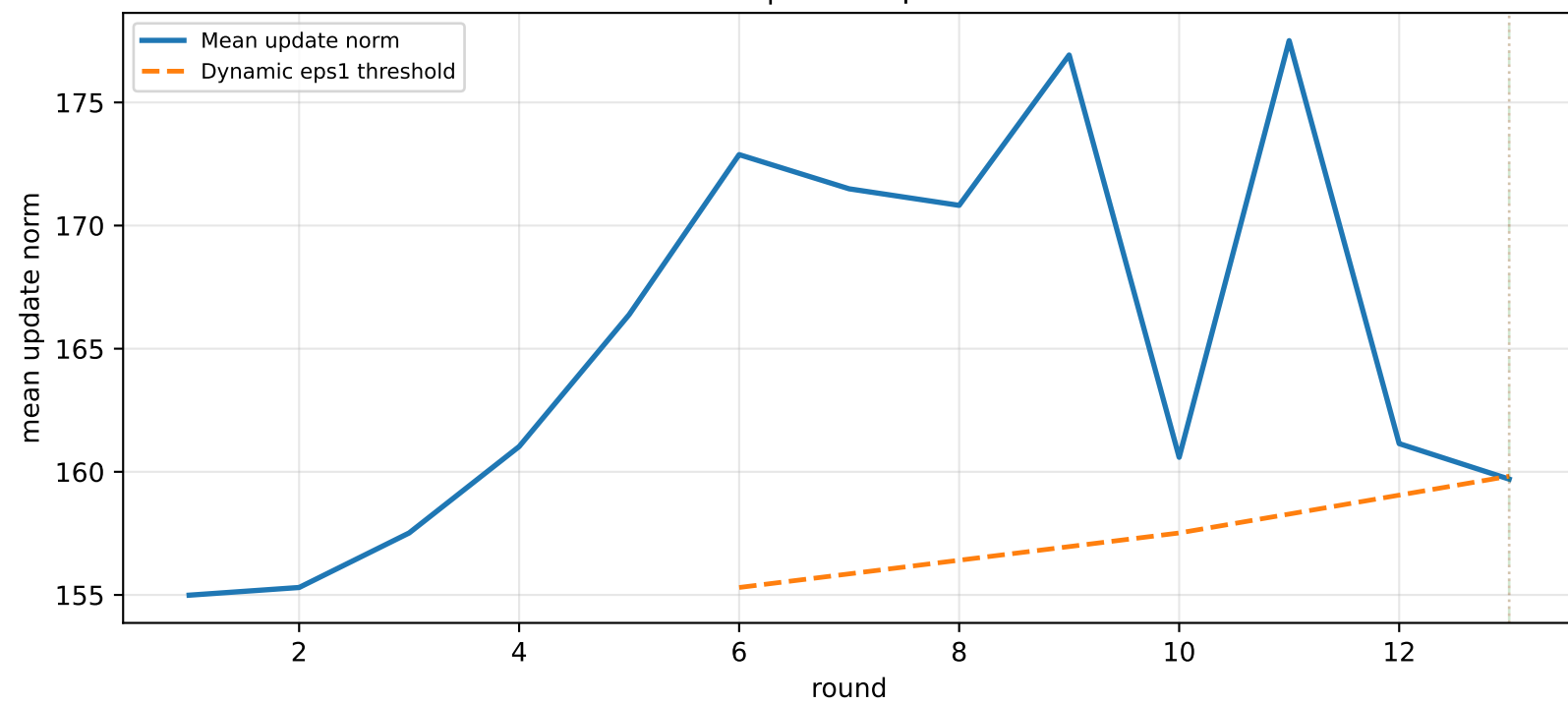
Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	68	17	strong_increasing:1
3	1	153	16	strong_increasing:1
5	1	159	19	strong_increasing:1
7	1	219	18	strong_increasing:1
9	1	196	11	mild_increasing:1
10	15	178, 96, 17, 18, 171, 169, 91, 11, 192, 33, 111, 2, 162, 82, 3	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15	flat_balanced:4 , mild_decreasing:4 , mild_increasing:3 , strong_decreasing:4

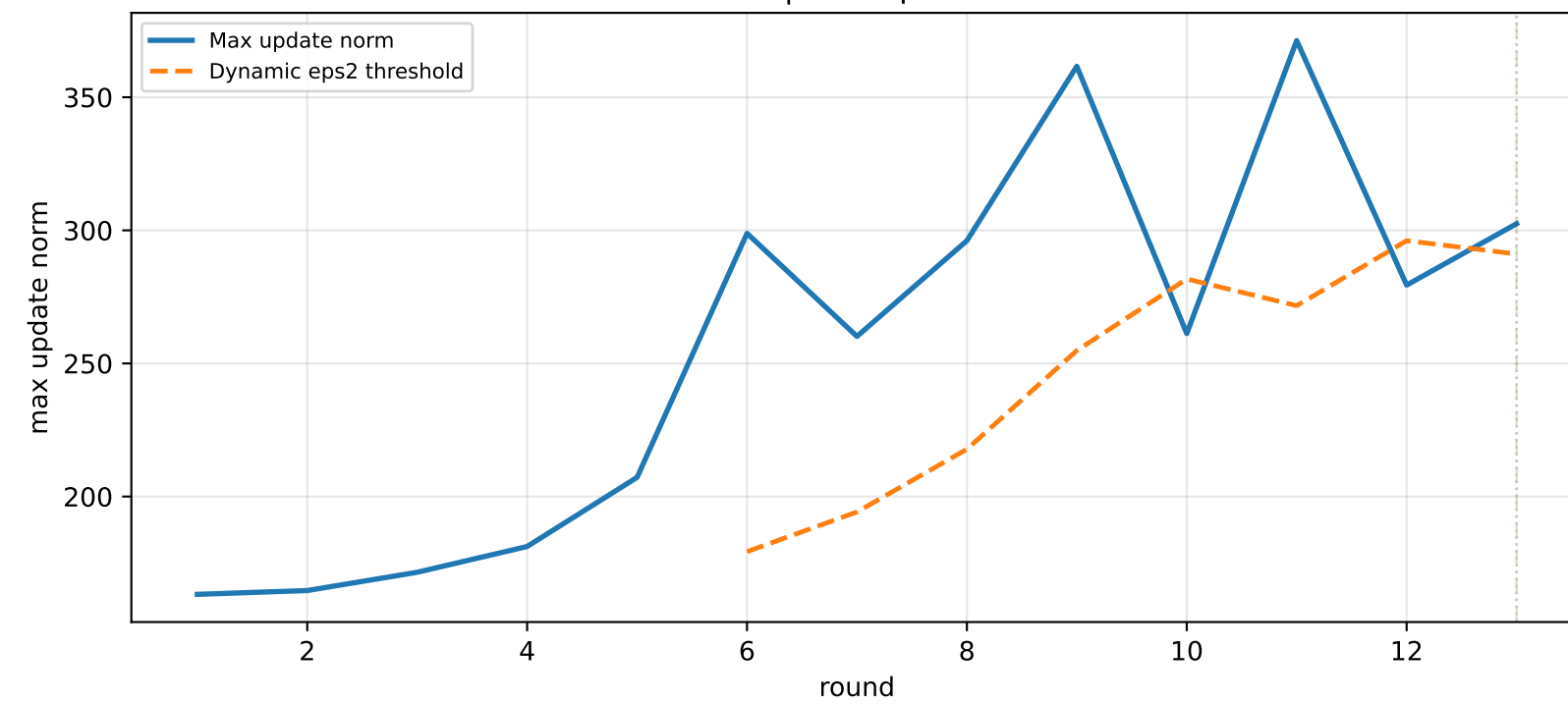
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

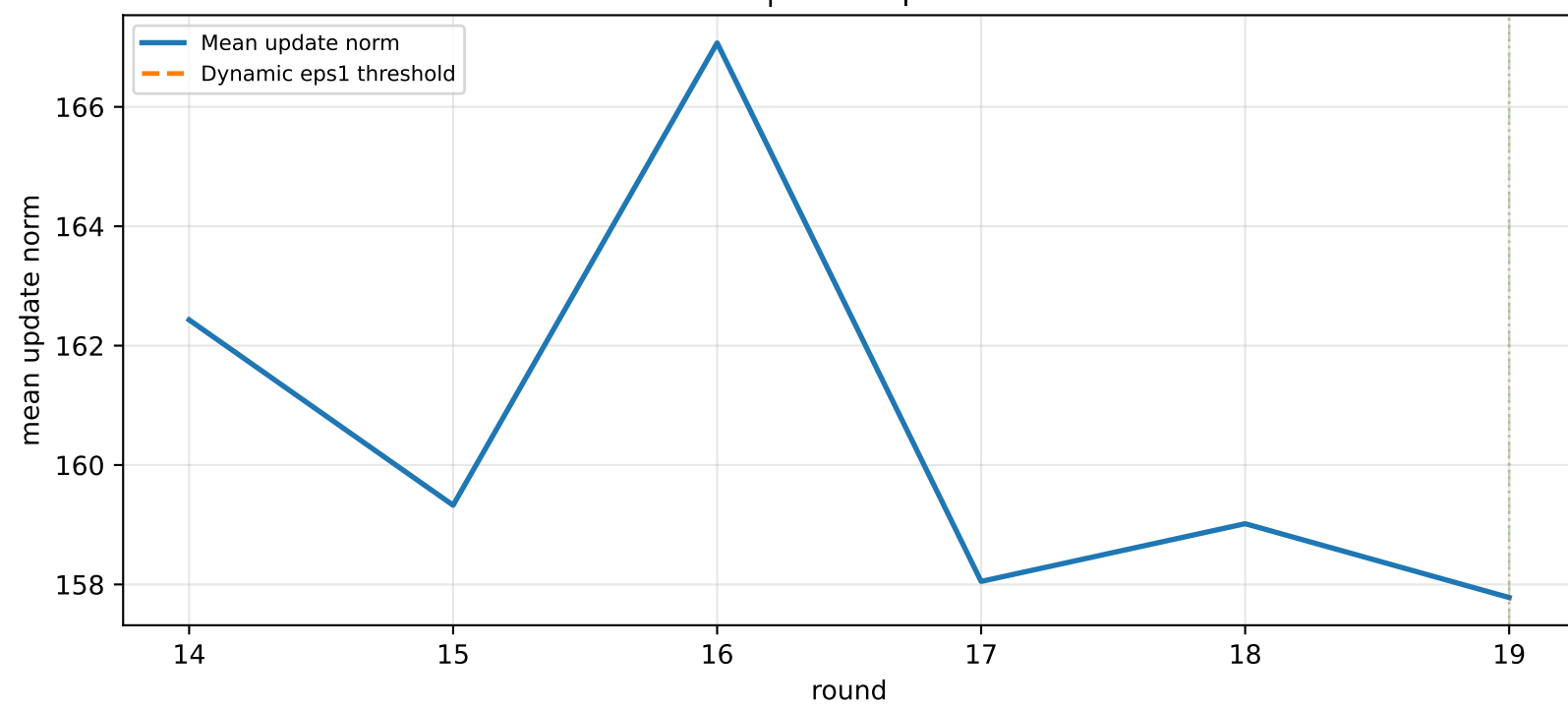
Cluster 0 | Mean update norm



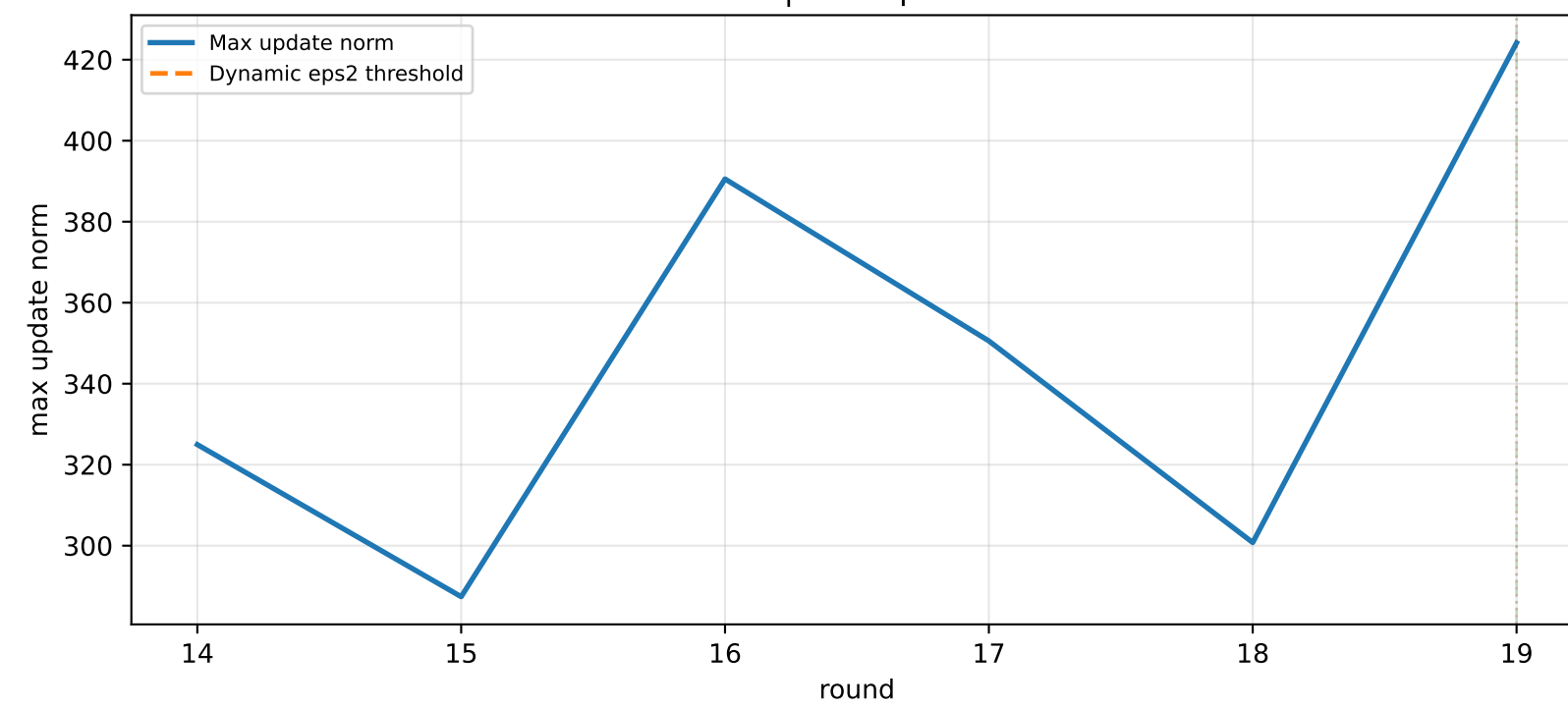
Cluster 0 | Max update norm



Cluster 2 | Mean update norm

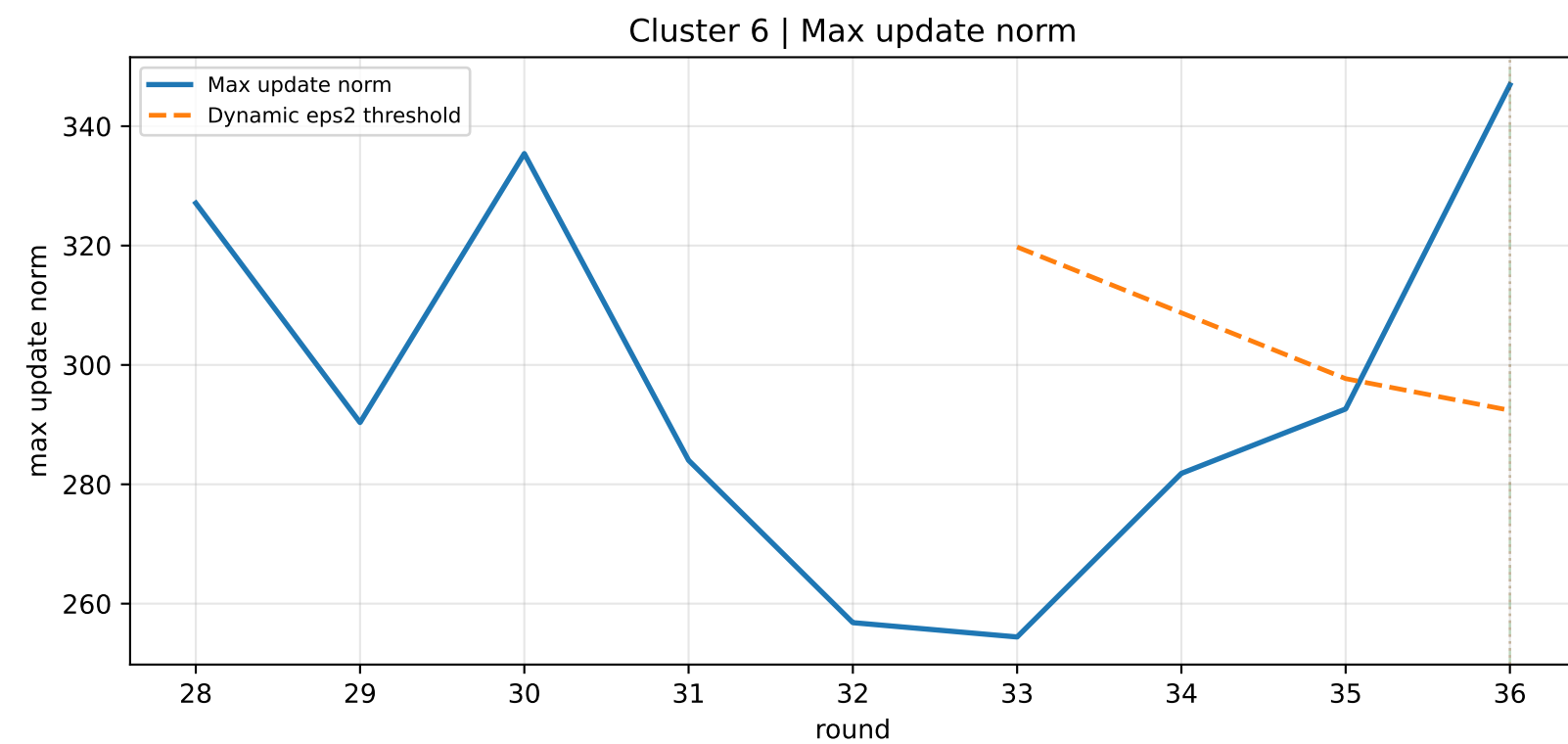
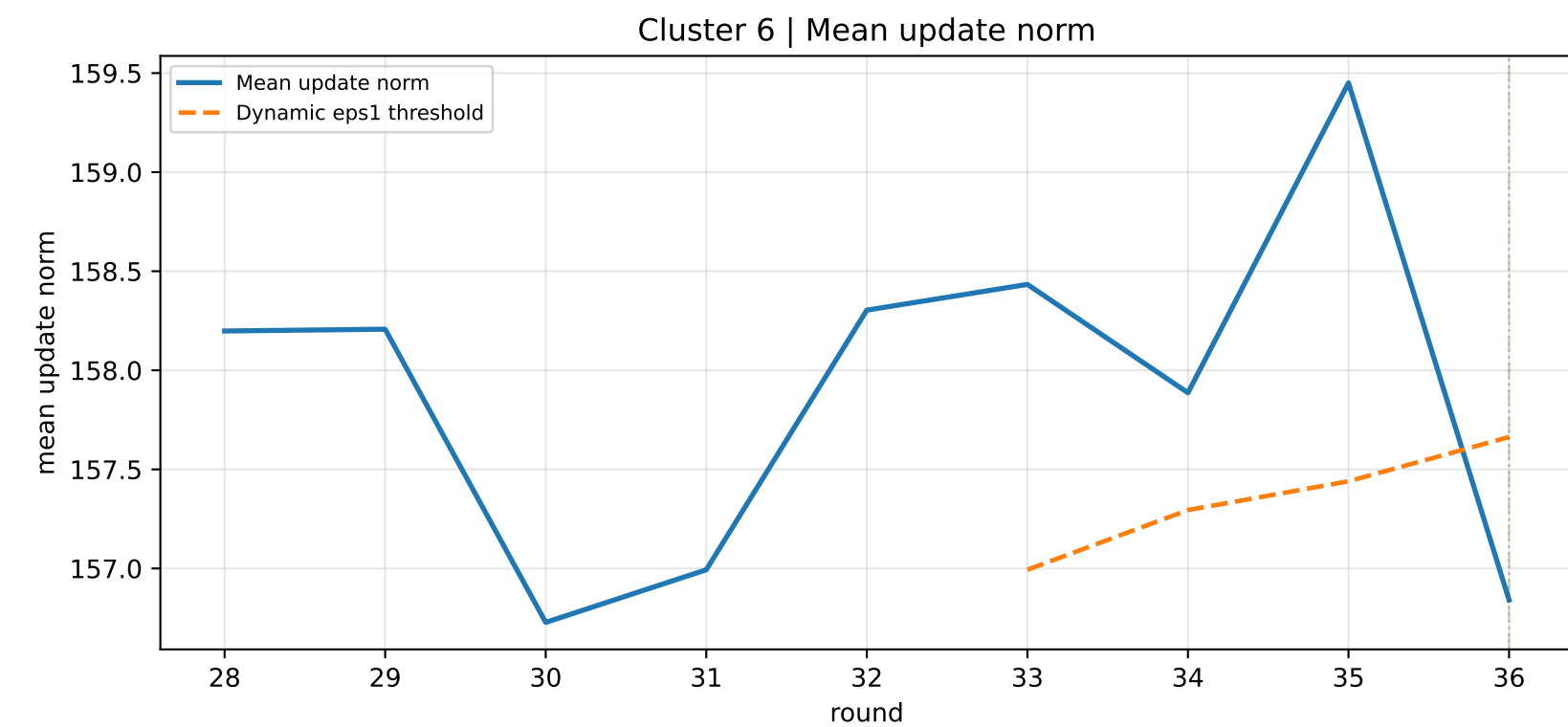
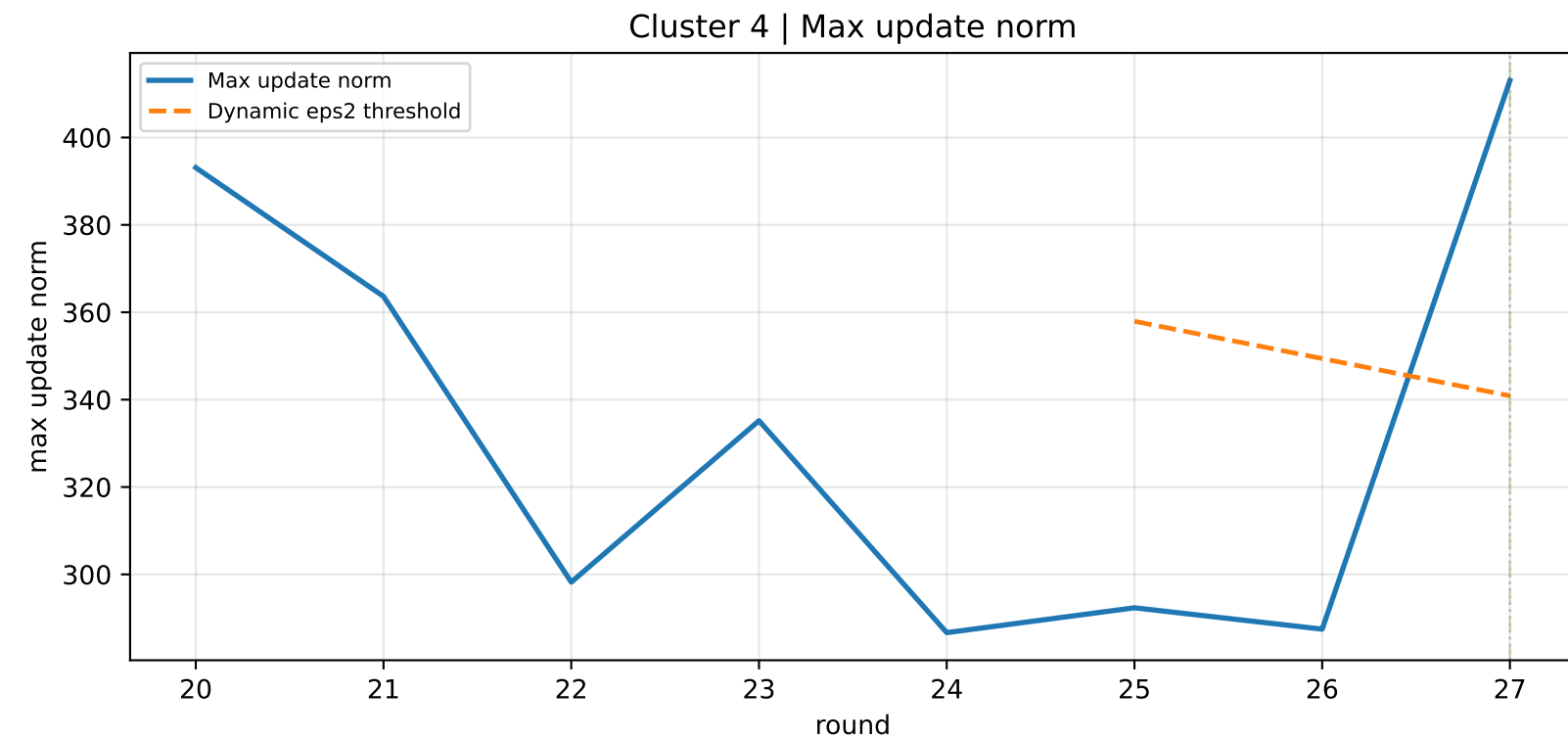
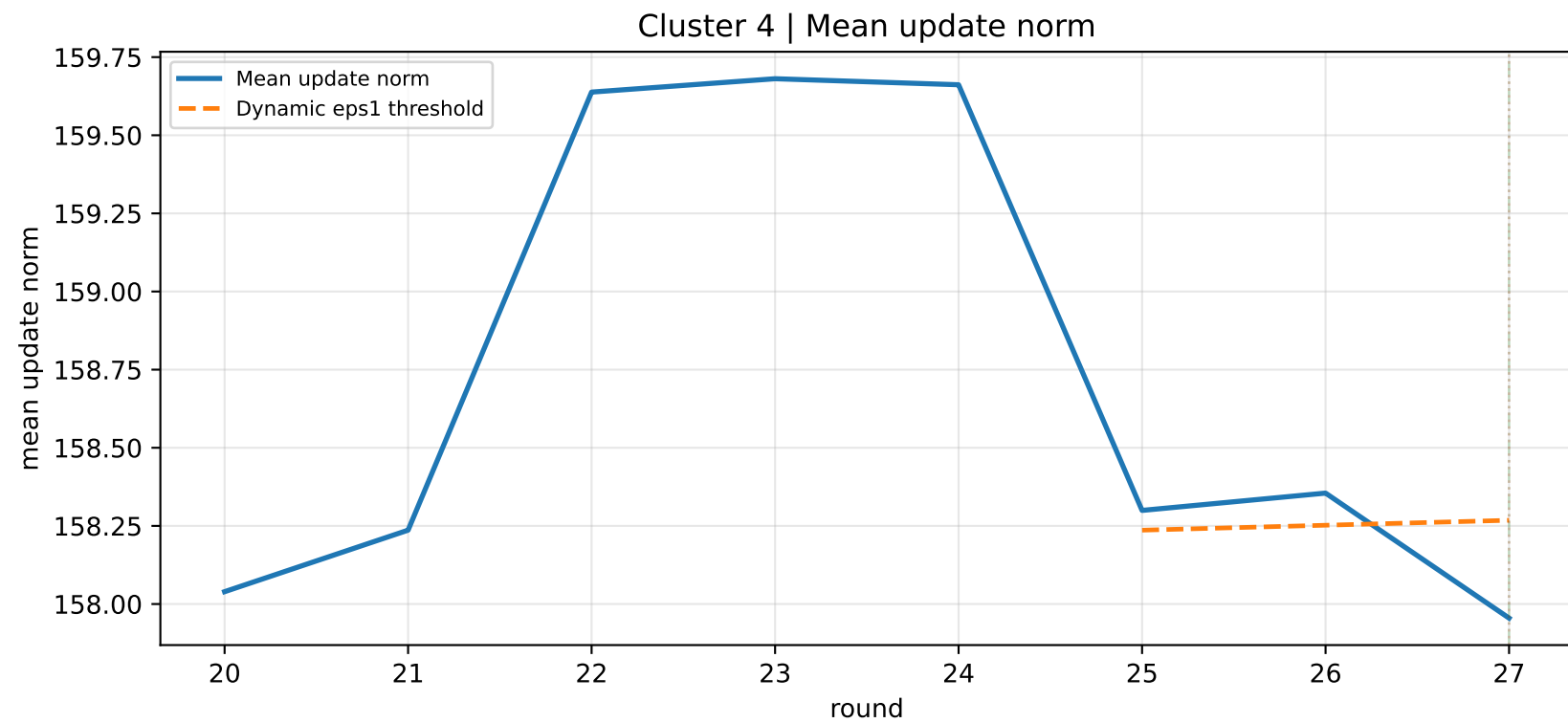


Cluster 2 | Max update norm



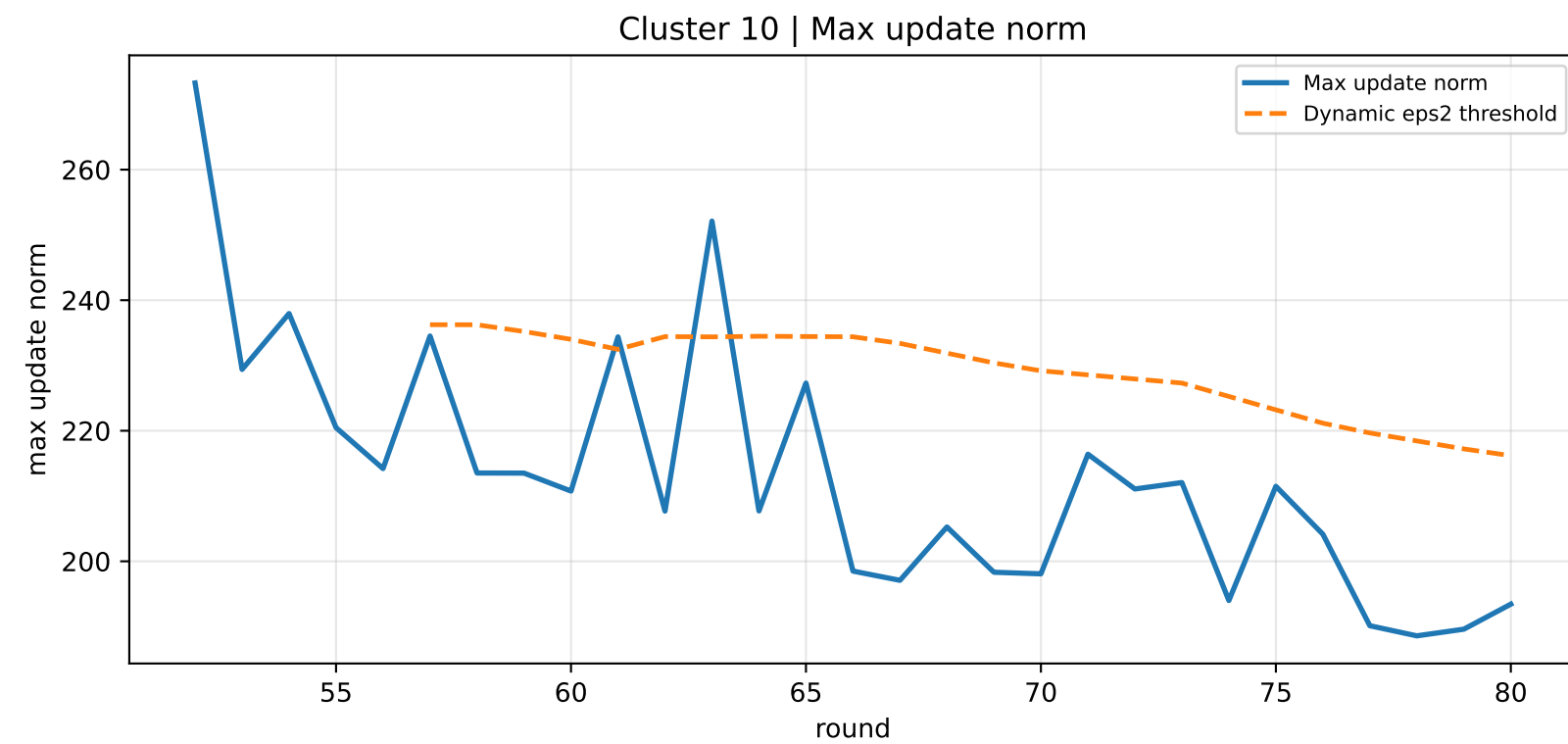
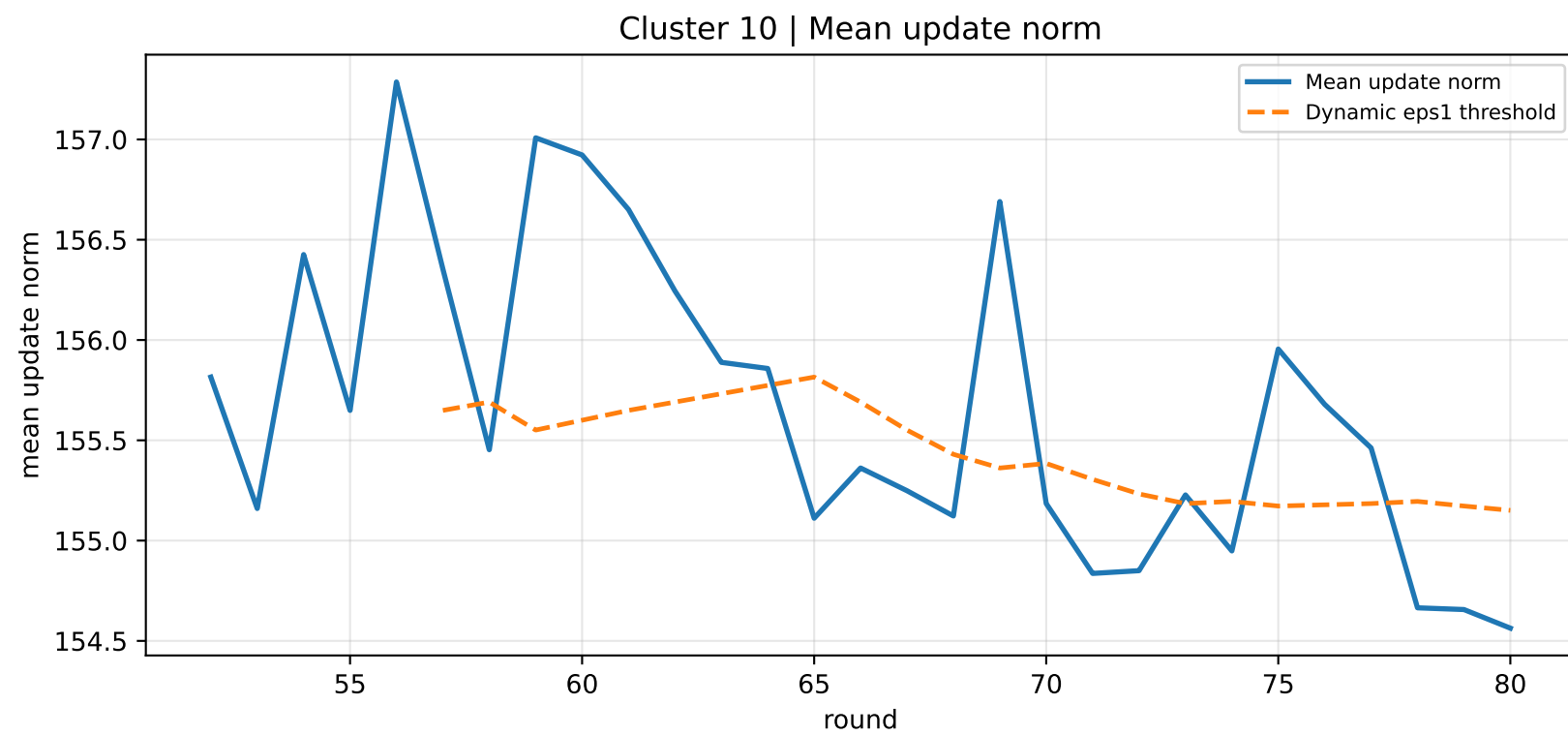
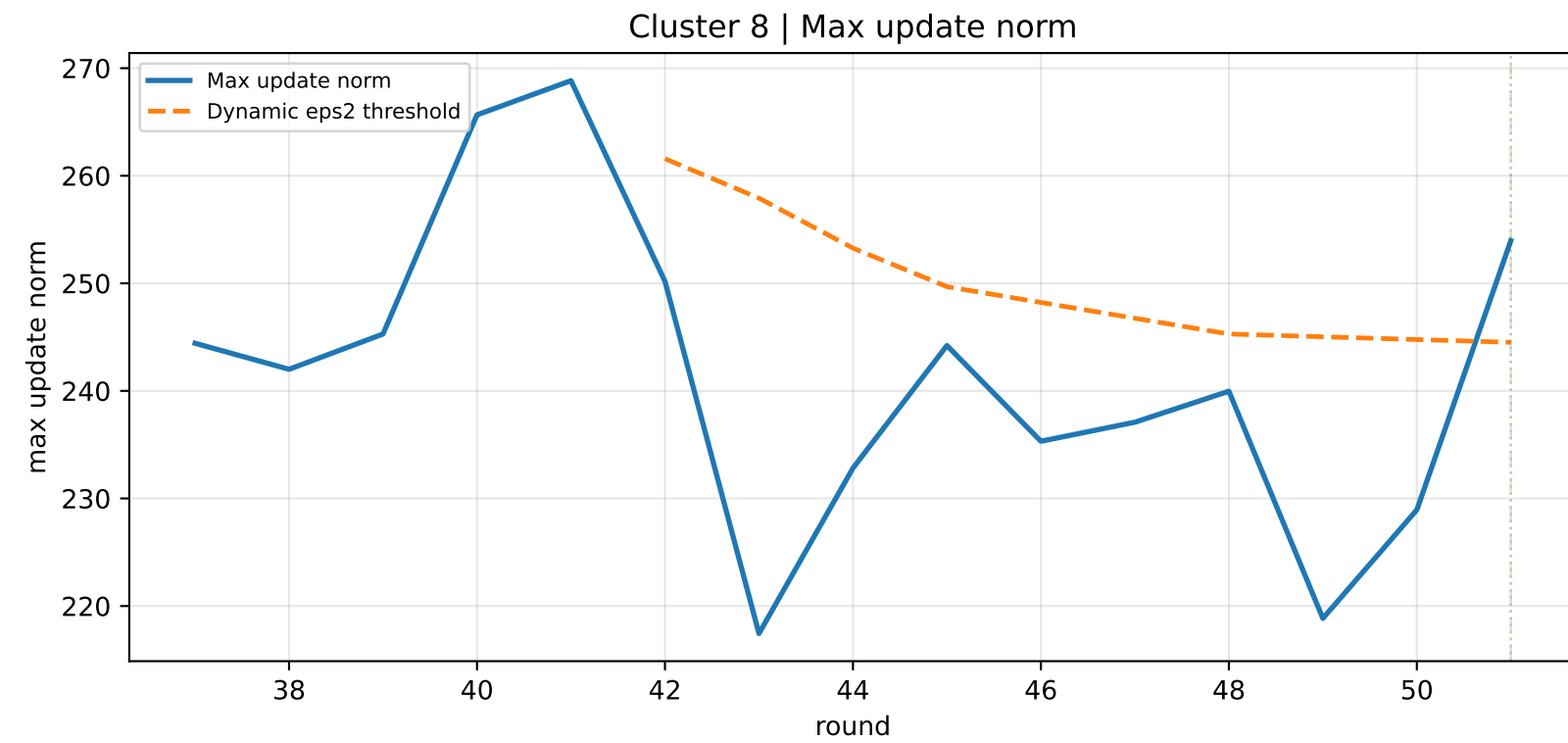
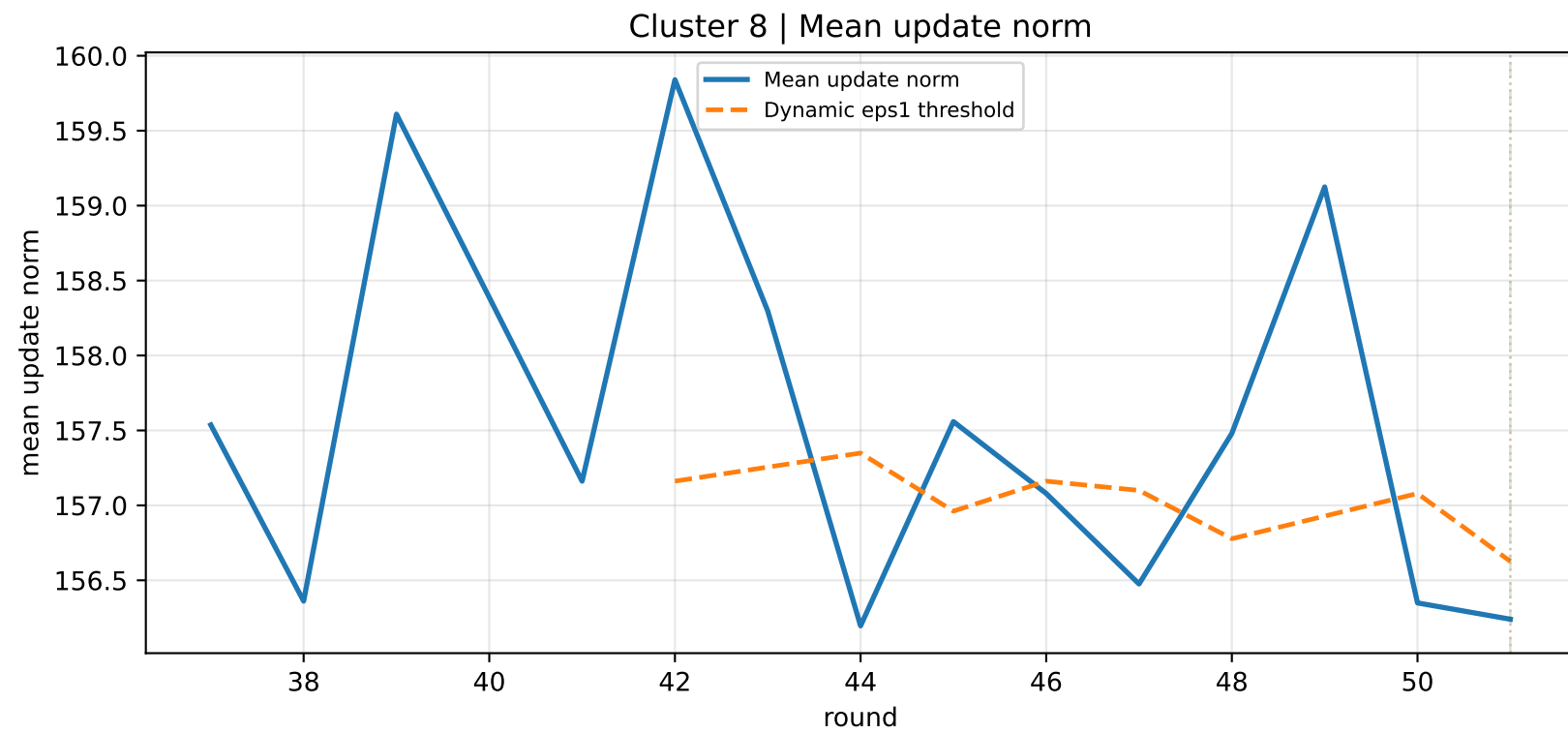
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (8-10)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

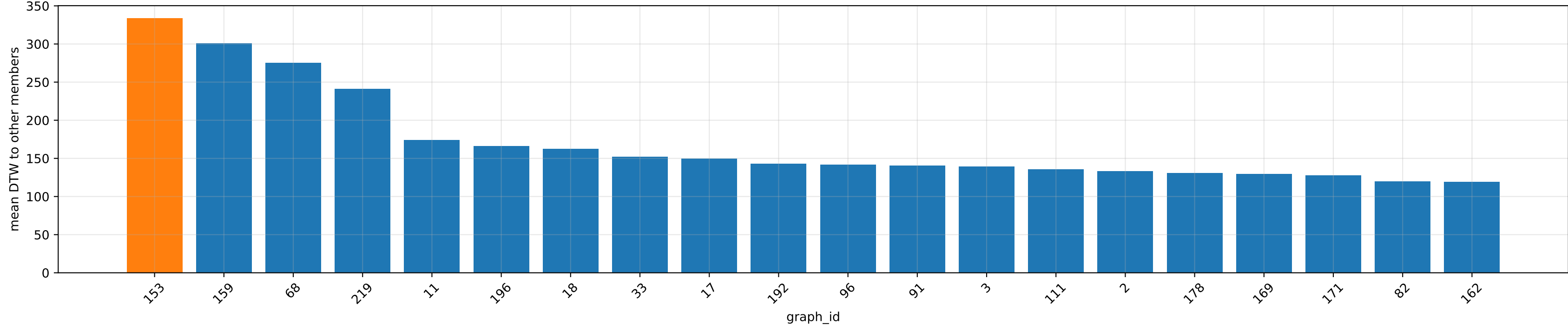
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	13	0	1 vs 19	153	153	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
333.578	nan	152.723

Per-member mean DTW



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

First applied split pairwise DTW metadata | seed=2

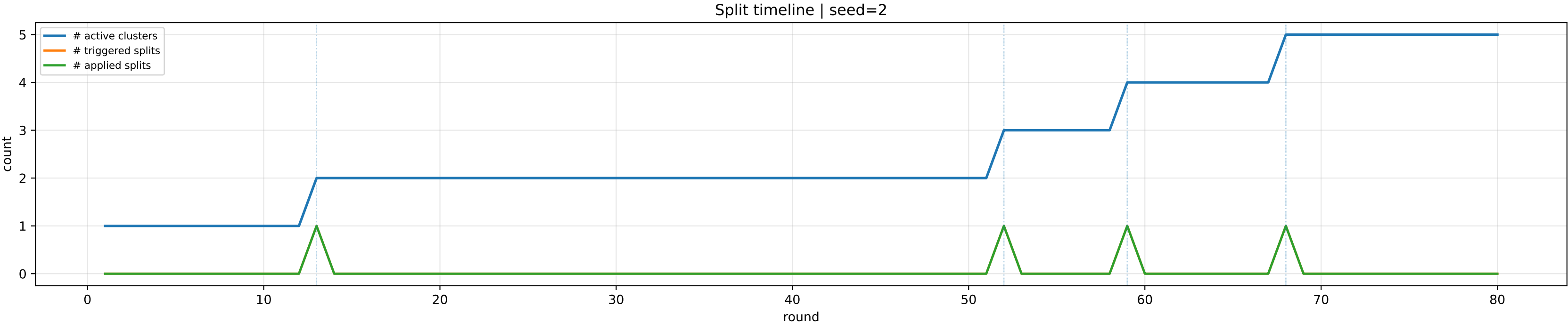
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	13	0	20	153.0	0.066	333.578	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 469.2 at (11, 153)

graph_id	178	96	17	18	171	169	91	11	192	33	111	196	2	162	82	3	153	68	219	159
178	0.0	28.1	49.1	54.2	38.9	39.7	31.4	69.7	149.4	94.8	92.9	190.0	58.9	59.1	63.8	85.2	399.6	330.5	296.5	356.2
96	28.1	0.0	57.0	43.2	53.2	38.3	23.5	45.0	174.9	102.7	118.4	215.5	61.9	83.3	89.3	77.3	425.1	356.0	322.0	381.7
17	49.1	57.0	0.0	58.5	90.1	88.6	79.7	66.0	156.7	69.8	100.1	197.3	106.8	93.7	88.7	134.3	406.9	337.9	303.8	363.6
18	54.2	43.2	58.5	0.0	66.3	69.9	56.2	27.6	203.6	116.8	147.1	244.2	75.6	115.9	118.0	95.9	453.8	384.7	350.7	410.4
171	38.9	53.2	90.1	66.3	0.0	22.4	43.1	82.1	106.3	138.6	99.9	171.0	55.6	40.2	45.7	41.3	387.1	318.1	284.0	343.8
169	39.7	38.3	88.6	69.9	22.4	0.0	26.0	74.4	134.5	134.5	90.7	184.3	57.2	47.4	52.7	45.0	393.8	324.8	290.8	350.5
91	31.4	23.5	79.7	56.2	43.1	26.0	0.0	55.2	168.6	126.2	112.0	209.2	56.0	79.4	83.0	56.9	418.7	349.7	315.6	375.4
11	69.7	45.0	66.0	27.6	82.1	74.4	55.2	0.0	219.1	132.5	162.6	259.7	90.0	131.3	133.5	91.4	469.2	400.2	366.2	425.9
192	149.4	174.9	156.7	203.6	106.3	134.5	168.6	219.1	0.0	96.5	63.8	90.9	106.1	75.6	73.1	116.9	250.2	179.3	127.1	224.1
33	94.8	102.7	69.8	116.8	138.6	134.5	126.2	132.5	96.5	0.0	43.7	167.2	120.8	127.2	126.9	170.7	337.1	268.1	197.4	316.4
111	92.9	118.4	100.1	147.1	99.9	90.7	112.0	162.6	63.8	43.7	0.0	133.0	111.1	83.3	83.4	126.9	306.7	237.7	172.5	285.9
196	190.0	215.5	197.3	244.2	171.0	184.3	209.2	259.7	90.9	167.2	133.0	0.0	141.9	85.3	81.9	132.5	209.6	178.9	111.9	151.6
2	58.9	61.9	106.8	75.6	55.6	57.2	56.0	90.0	106.1	120.8	111.1	141.9	0.0	81.9	82.8	67.0	377.6	294.7	274.5	316.2
162	59.1	83.3	93.7	115.9	40.2	47.4	79.4	131.3	75.6	127.2	83.3	85.3	81.9	0.0	14.4	56.7	337.9	245.5	237.1	274.0
82	63.8	89.3	88.7	118.0	45.7	52.7	83.0	133.5	73.1	126.9	83.4	81.9	82.8	14.4	0.0	61.1	335.7	240.9	237.4	268.2
3	85.2	77.3	134.3	95.9	41.3	45.0	56.9	91.4	116.9	170.7	126.9	132.5	67.0	56.7	61.1	0.0	381.8	296.0	278.7	327.9
153	399.6	425.1	406.9	453.8	387.1	393.8	418.7	469.2	250.2	337.1	306.7	209.6	377.6	337.9	335.7	381.8	0.0	156.7	110.5	180.1
68	330.5	356.0	337.9	384.7	318.1	324.8	349.7	400.2	179.3	268.1	237.7	178.9	294.7	245.5	240.9	296.0	156.7	0.0	135.4	191.0
219	296.5	322.0	303.8	350.7	284.0	290.8	315.6	366.2	127.1	197.4	172.5	111.9	274.5	237.1	237.4	278.7	110.5	135.4	0.0	169.9
159	356.2	381.7	363.6	410.4	343.8	350.5	375.4	425.9	224.1	316.4	285.9	151.6	316.2	274.0	268.2	327.9	180.1	191.0	169.9	0.0

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

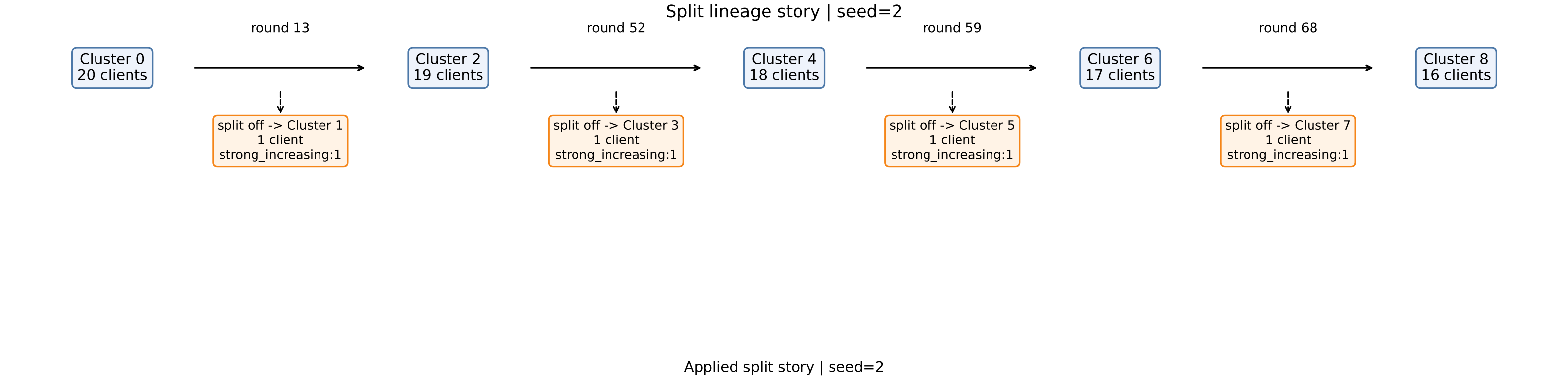


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	13, 52, 59, 68	13, 52, 59, 68	1, 3, 5, 7, 8	1:1, 3:1, 5:1, 7:1, 8:16

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087



round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
13	0	20	159.213 < 159.947 , 292.810 > 259.496	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
52	2	19	156.003 < 156.643 , 310.954 > 300.823	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
59	4	18	155.640 < 155.771 , 263.145 > 240.515	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1
68	6	17	155.550 < 156.085 , 223.751 > 222.427	C7 (1 client)	strong_increasing:1	C8 (16 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4

Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

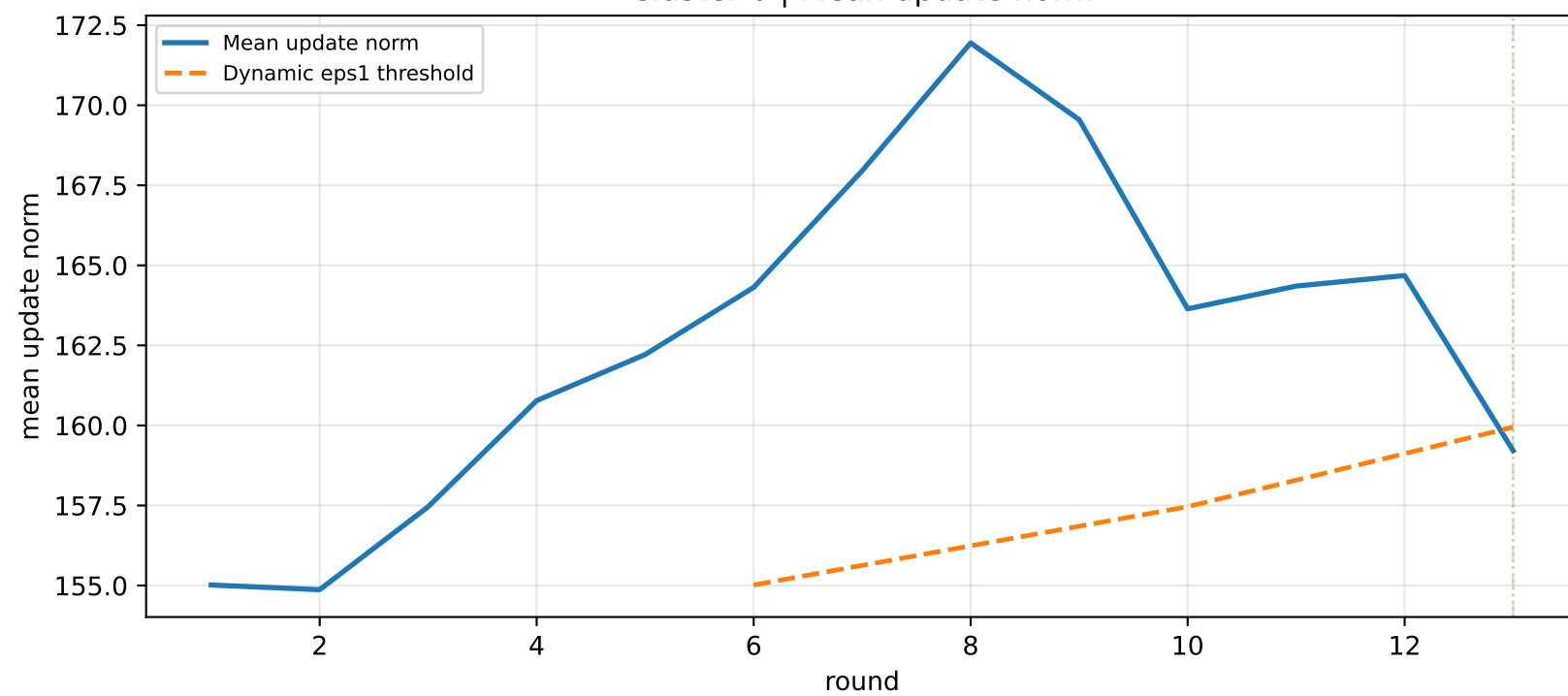
Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	153	16	strong_increasing:1
3	1	159	19	strong_increasing:1
5	1	68	17	strong_increasing:1
7	1	219	18	strong_increasing:1
8	16	178, 96, 17, 18, 171, 169, 91, 11, 192, 33, 111, 196, 2, 162, 82, 3	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4

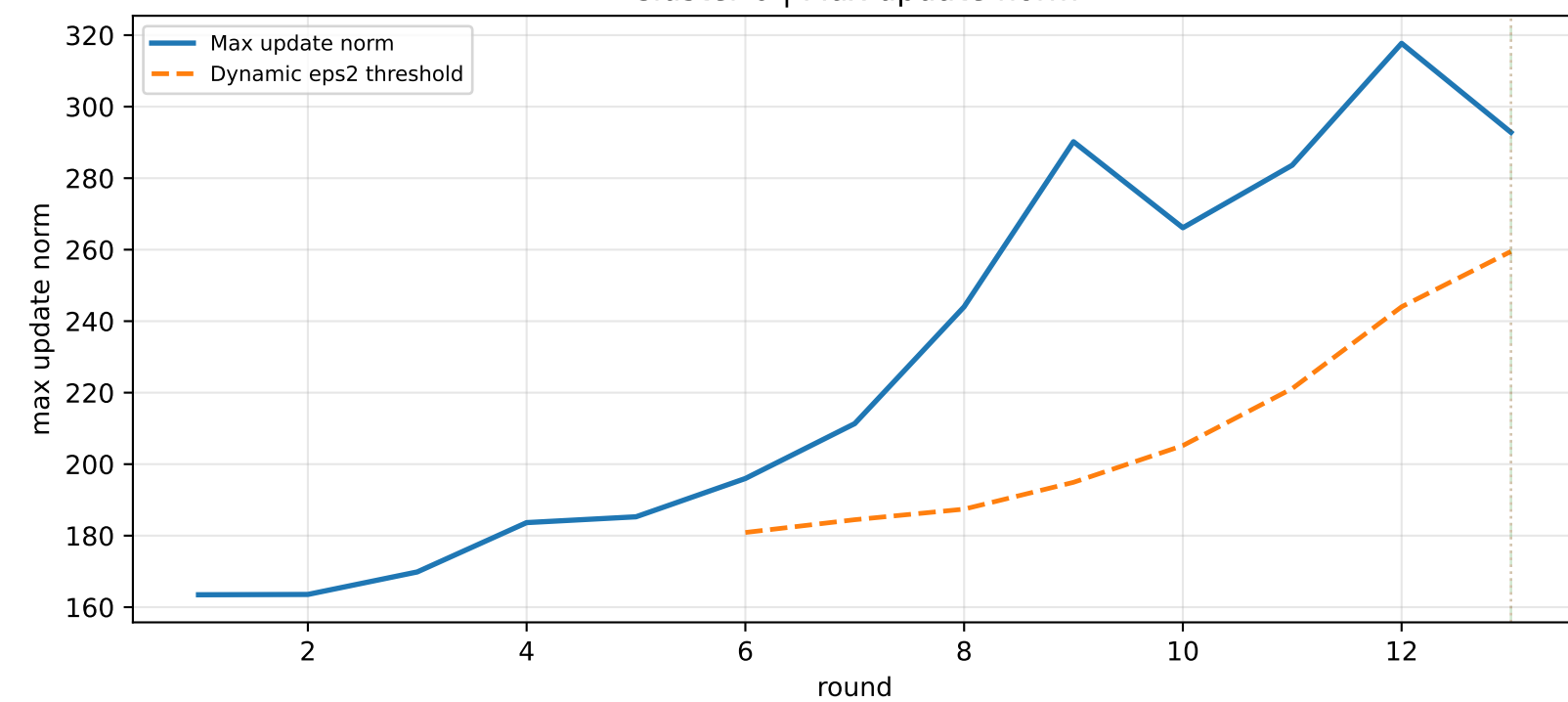
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

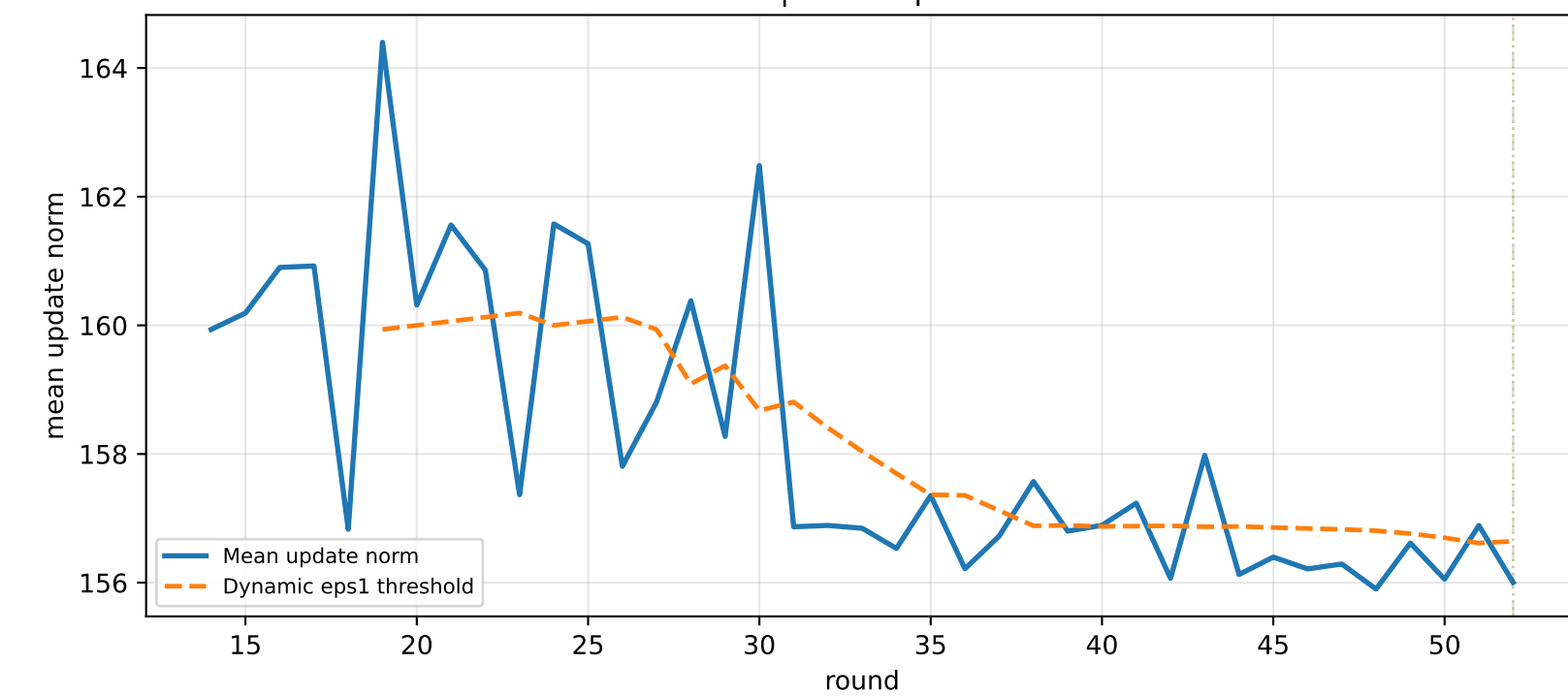
Cluster 0 | Mean update norm



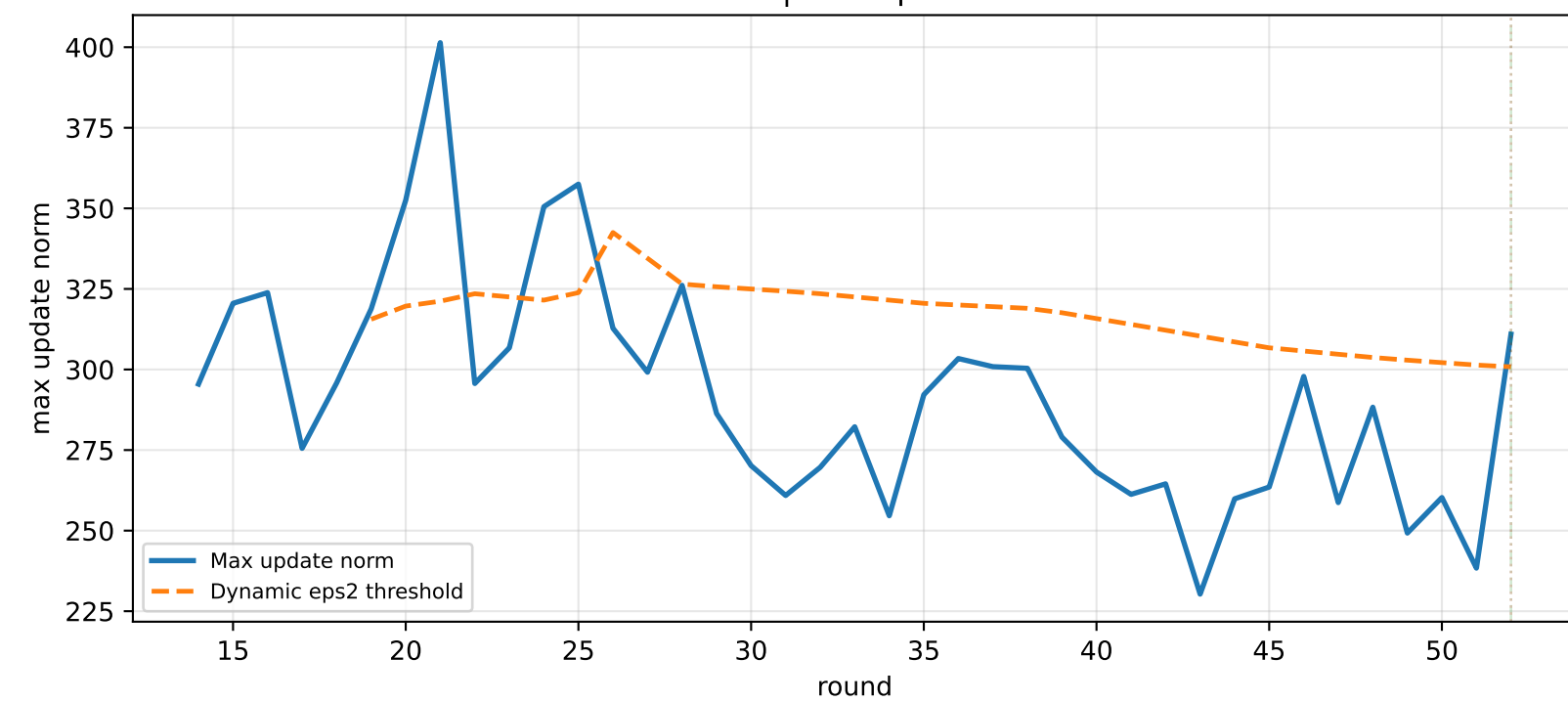
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



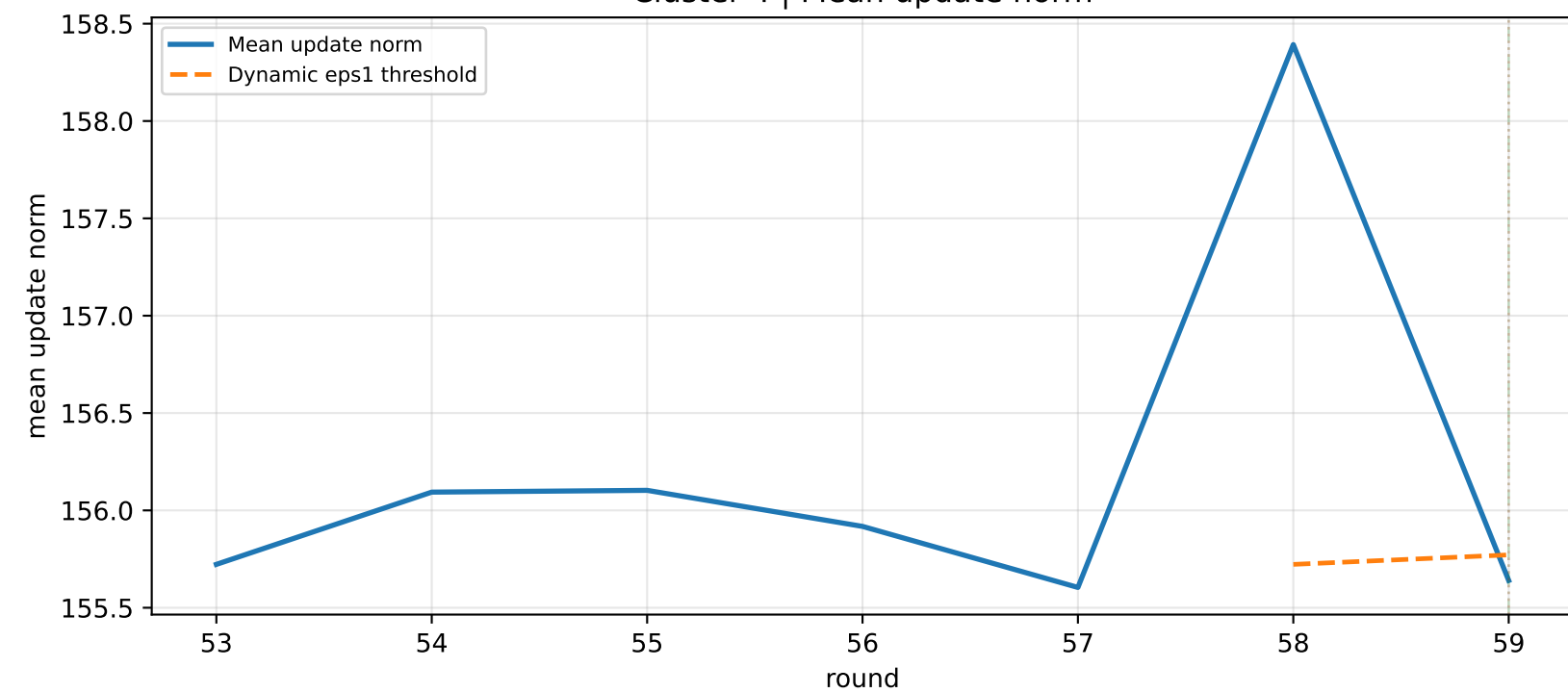
Cluster 2 | Max update norm



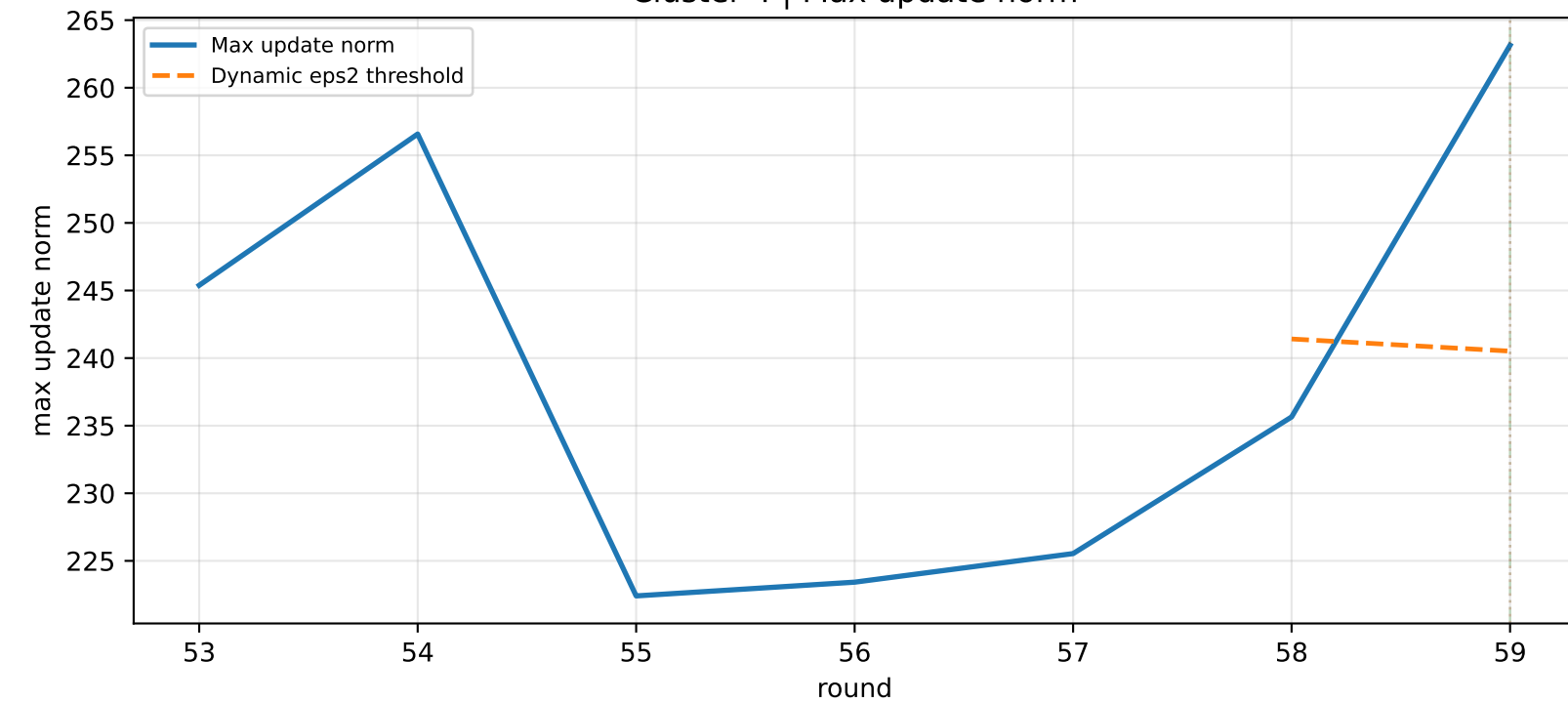
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

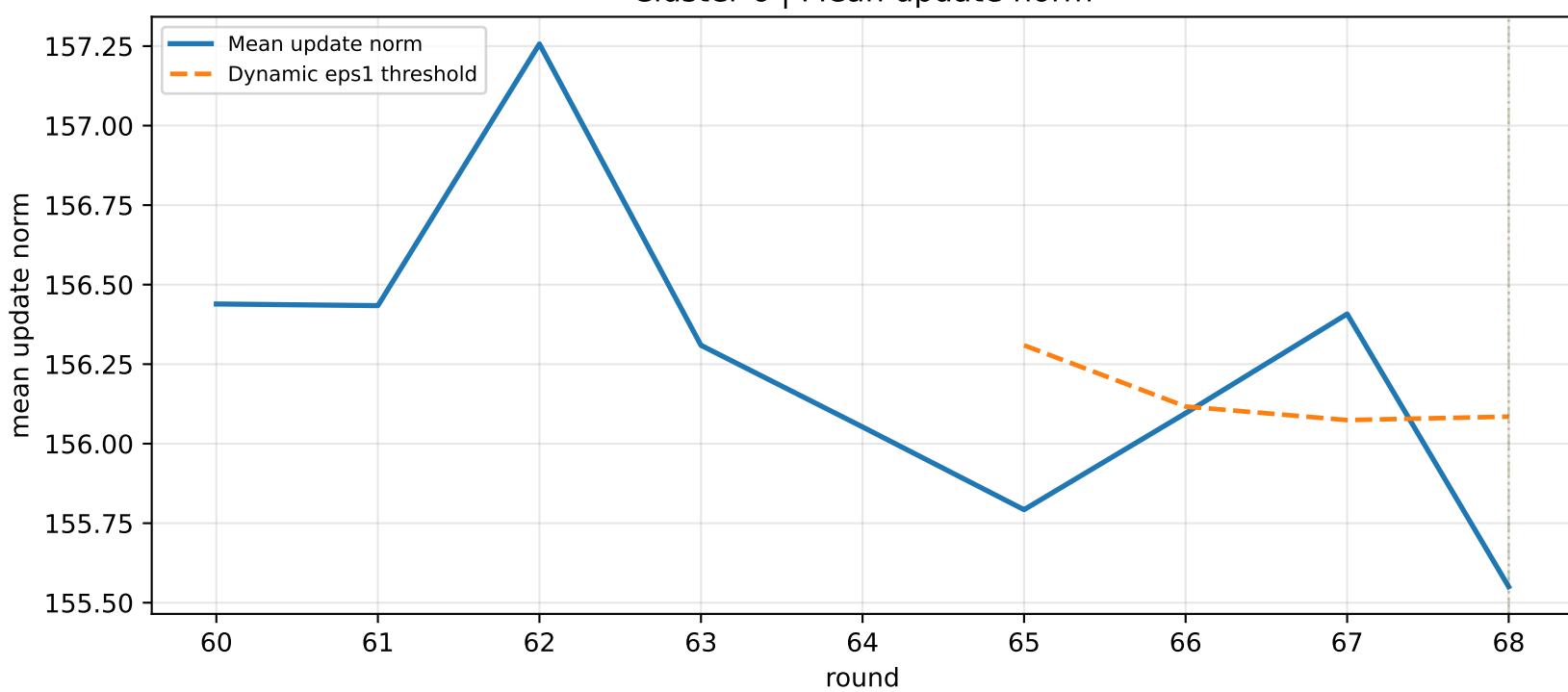
Cluster 4 | Mean update norm



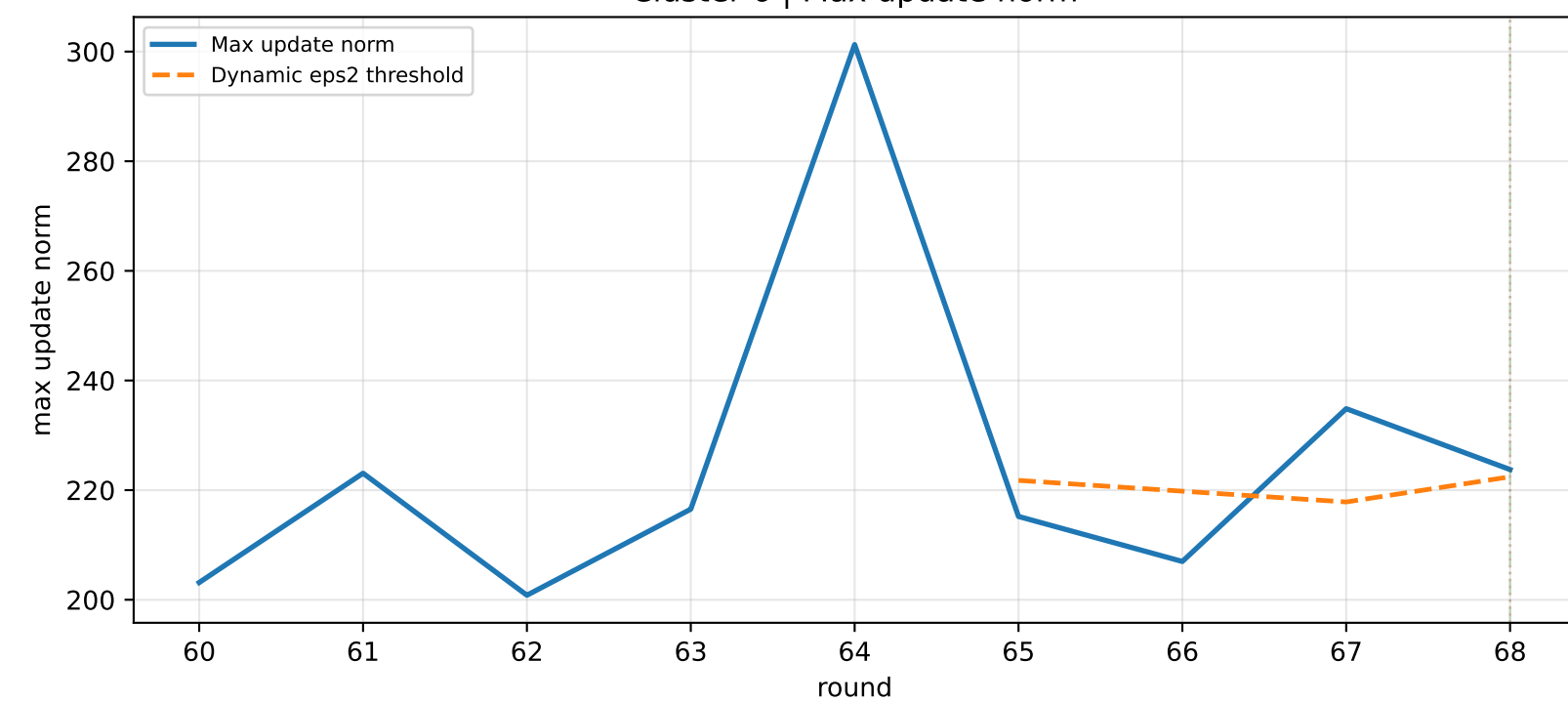
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



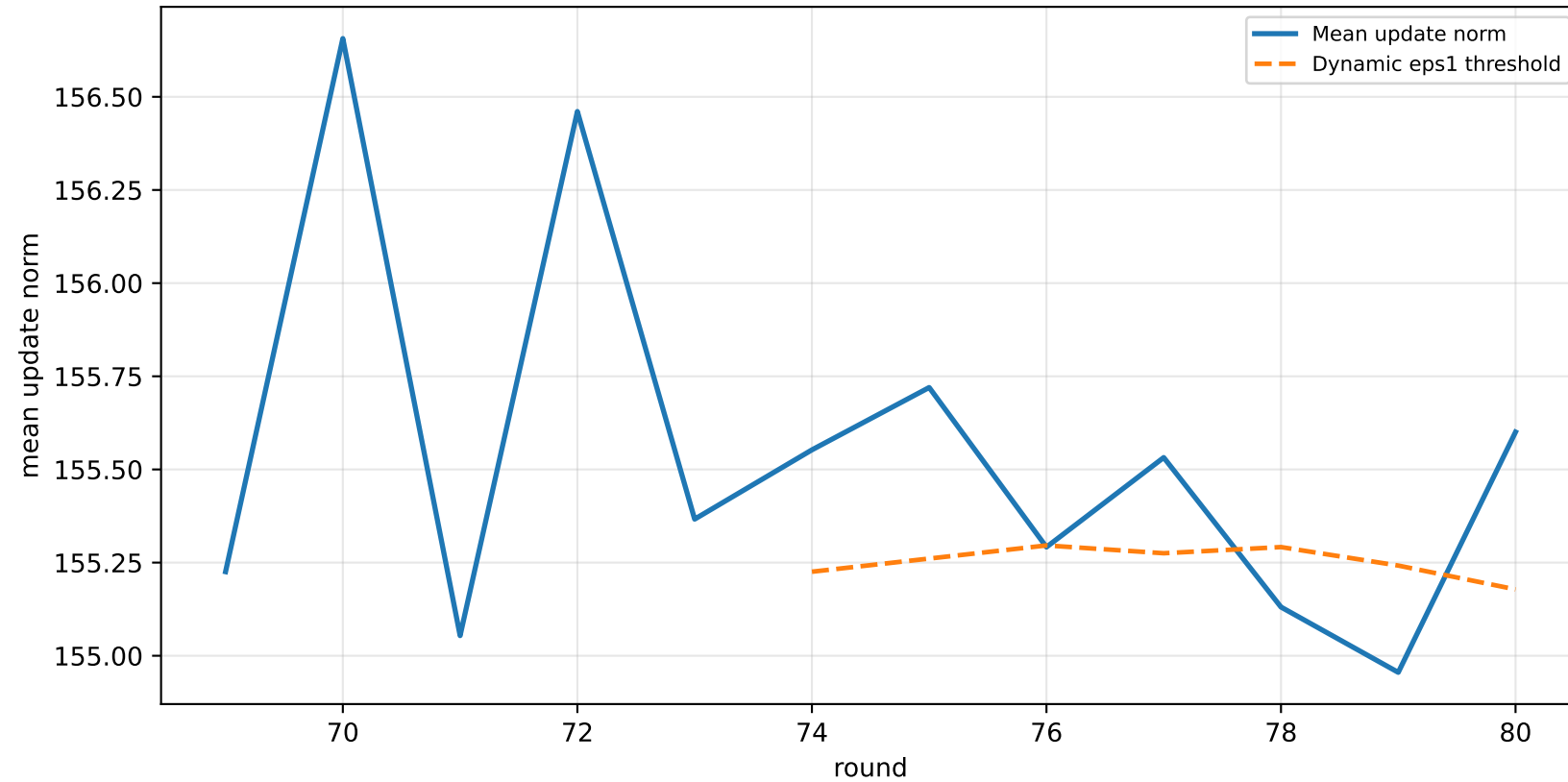
Cluster 6 | Max update norm



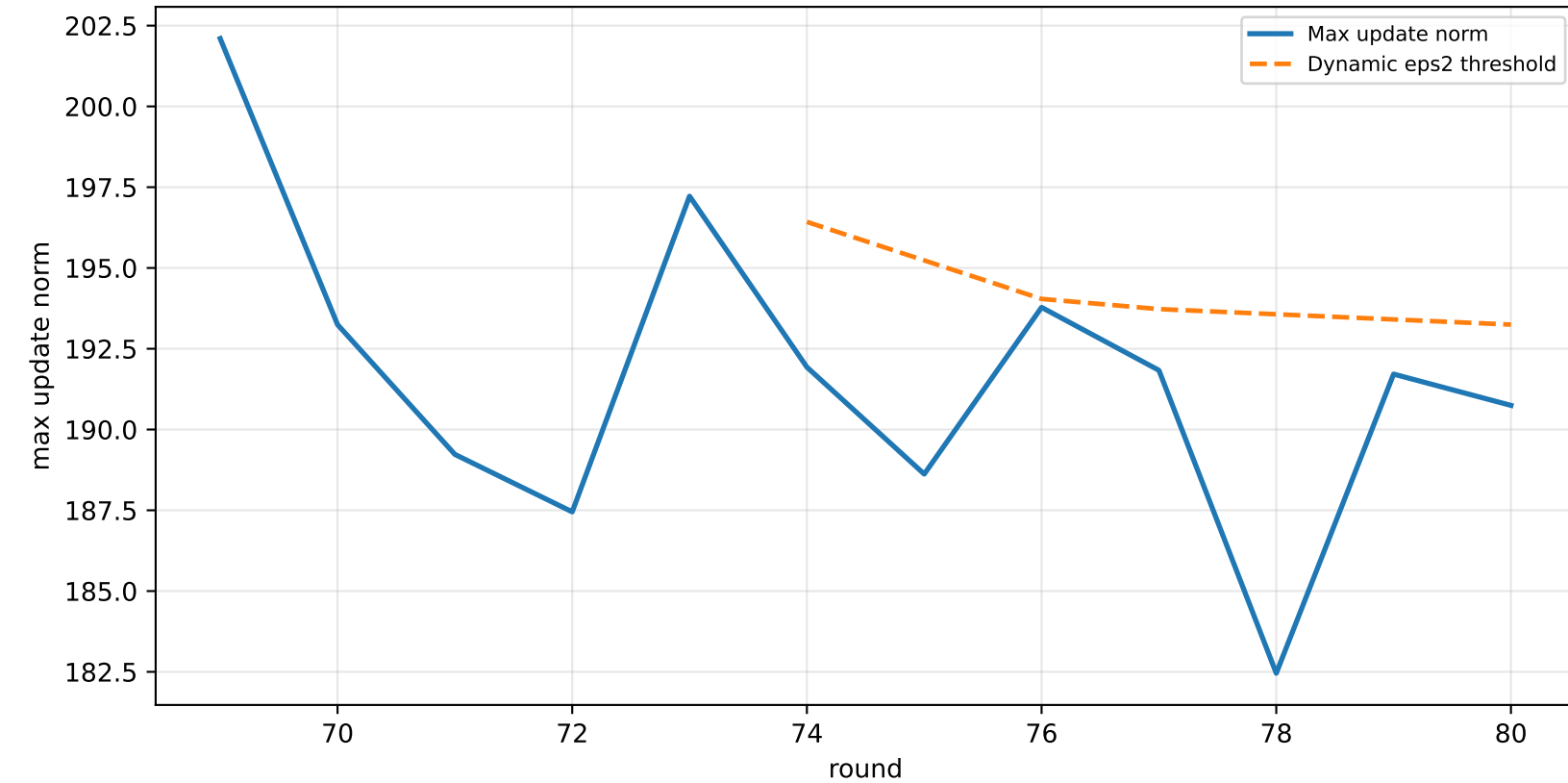
Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (8-8)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Cluster 8 | Mean update norm



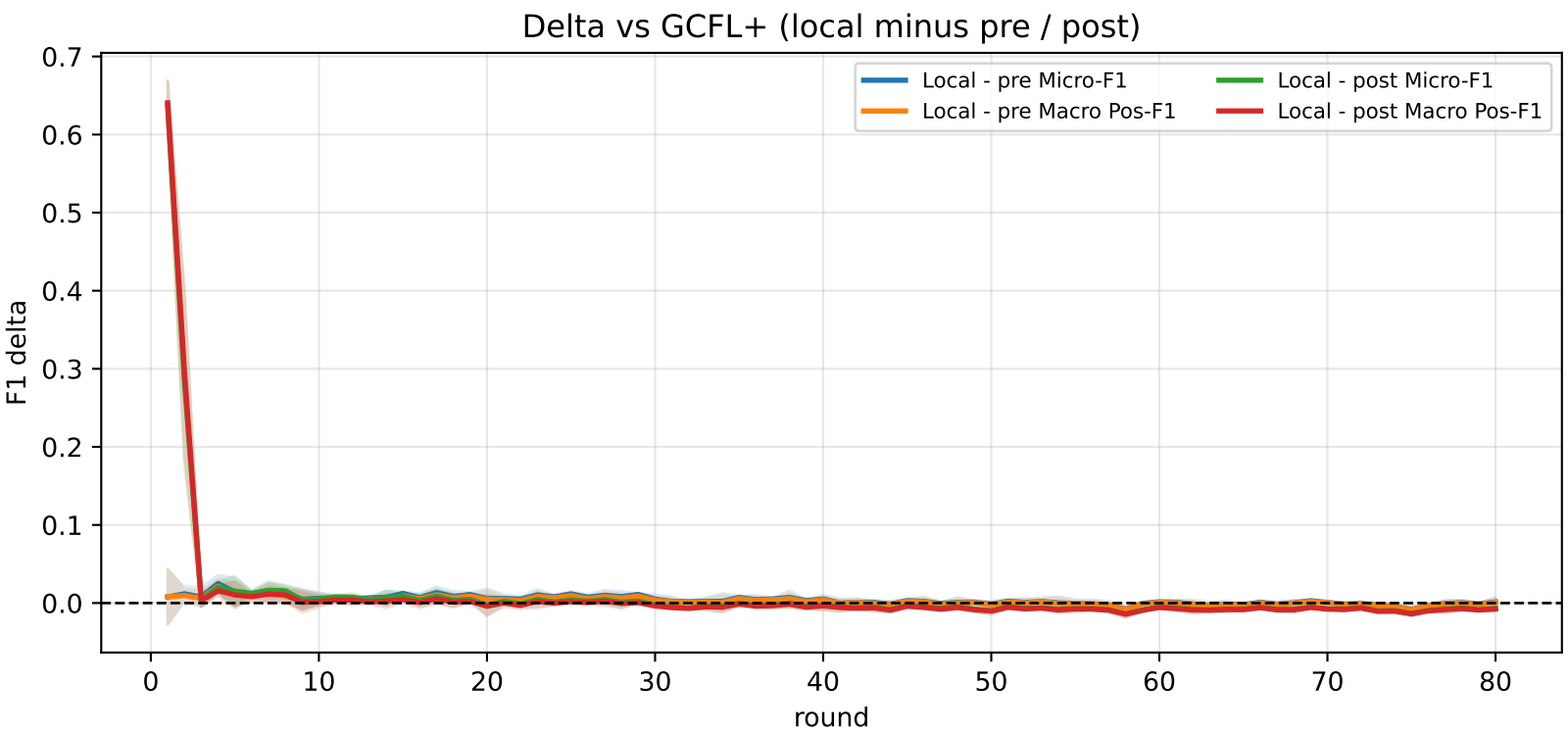
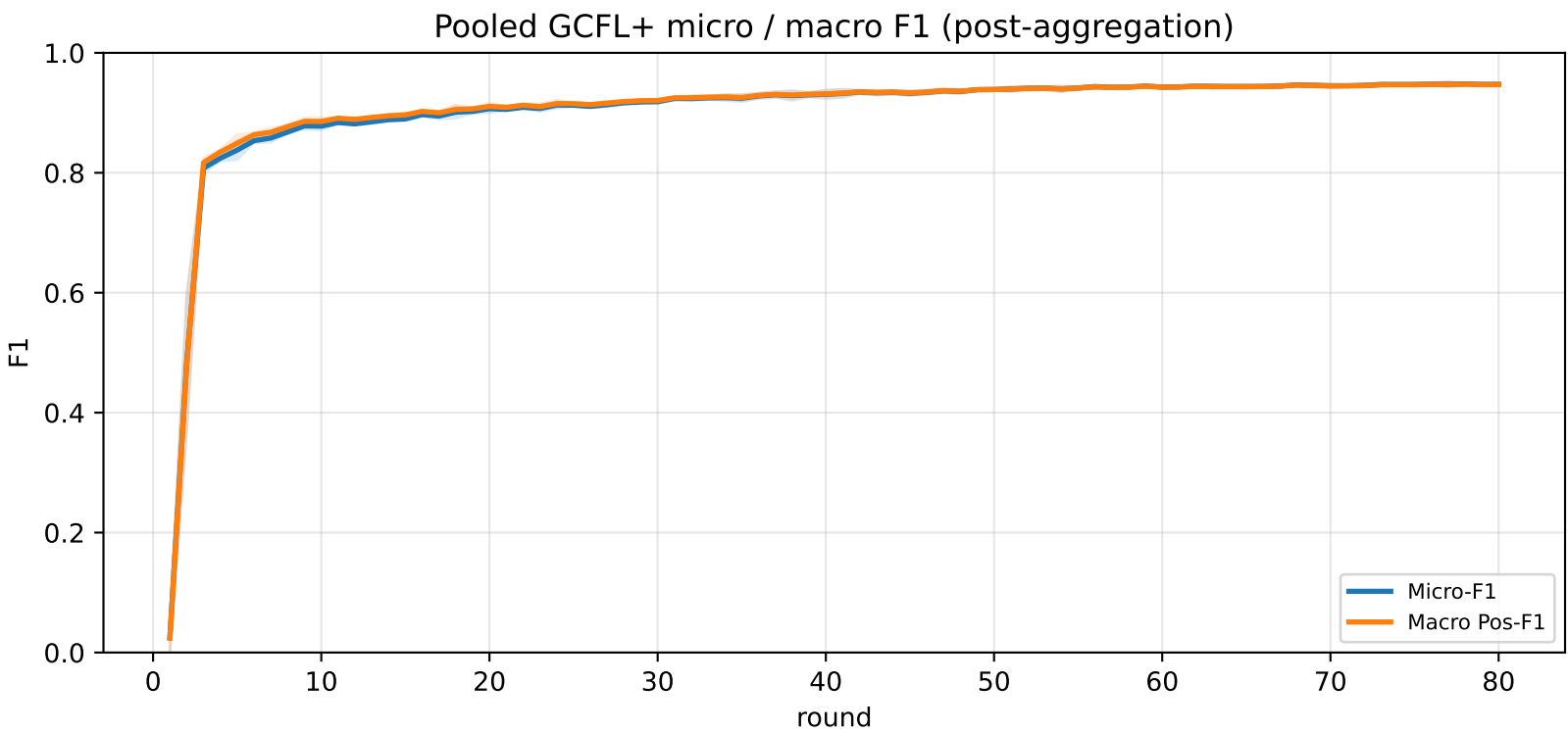
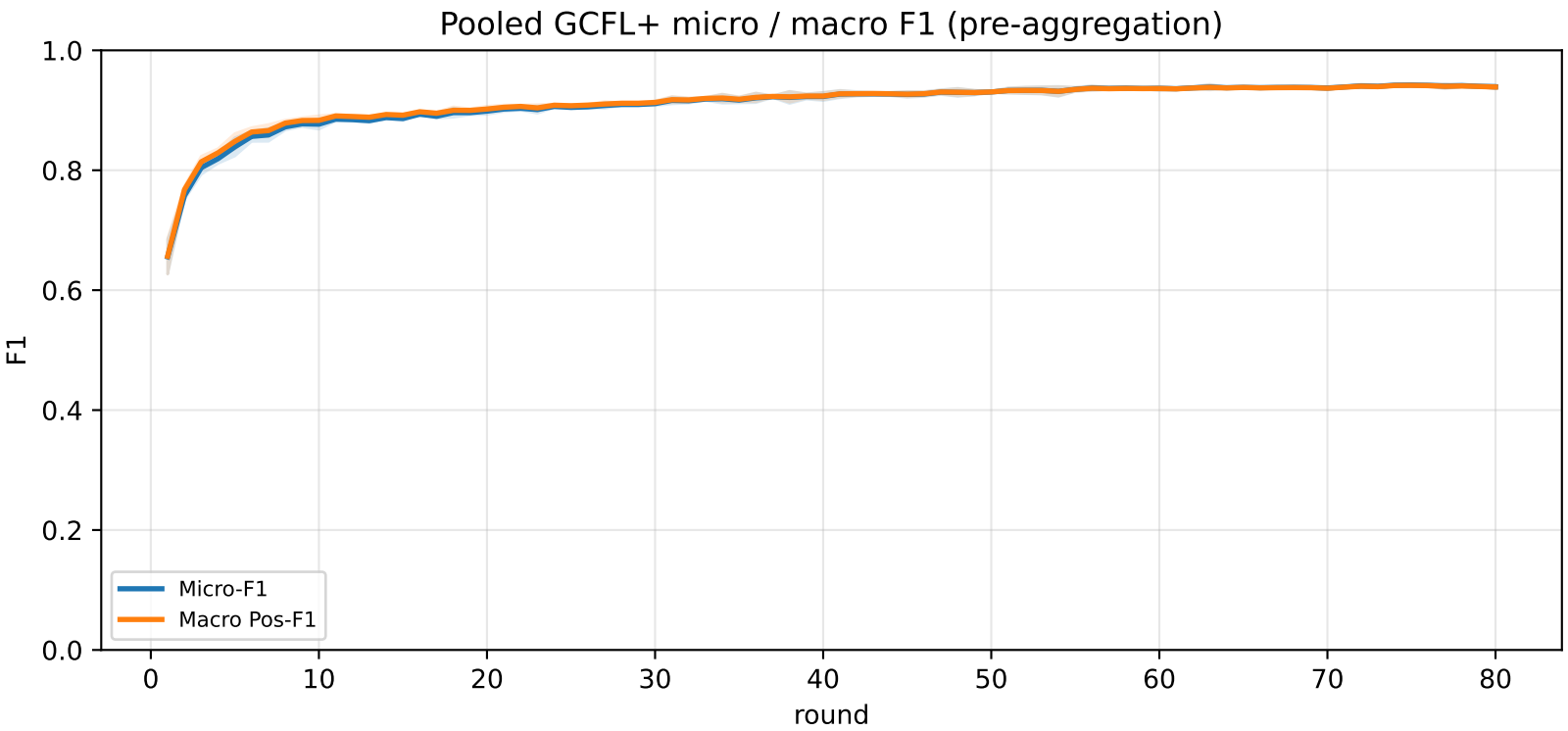
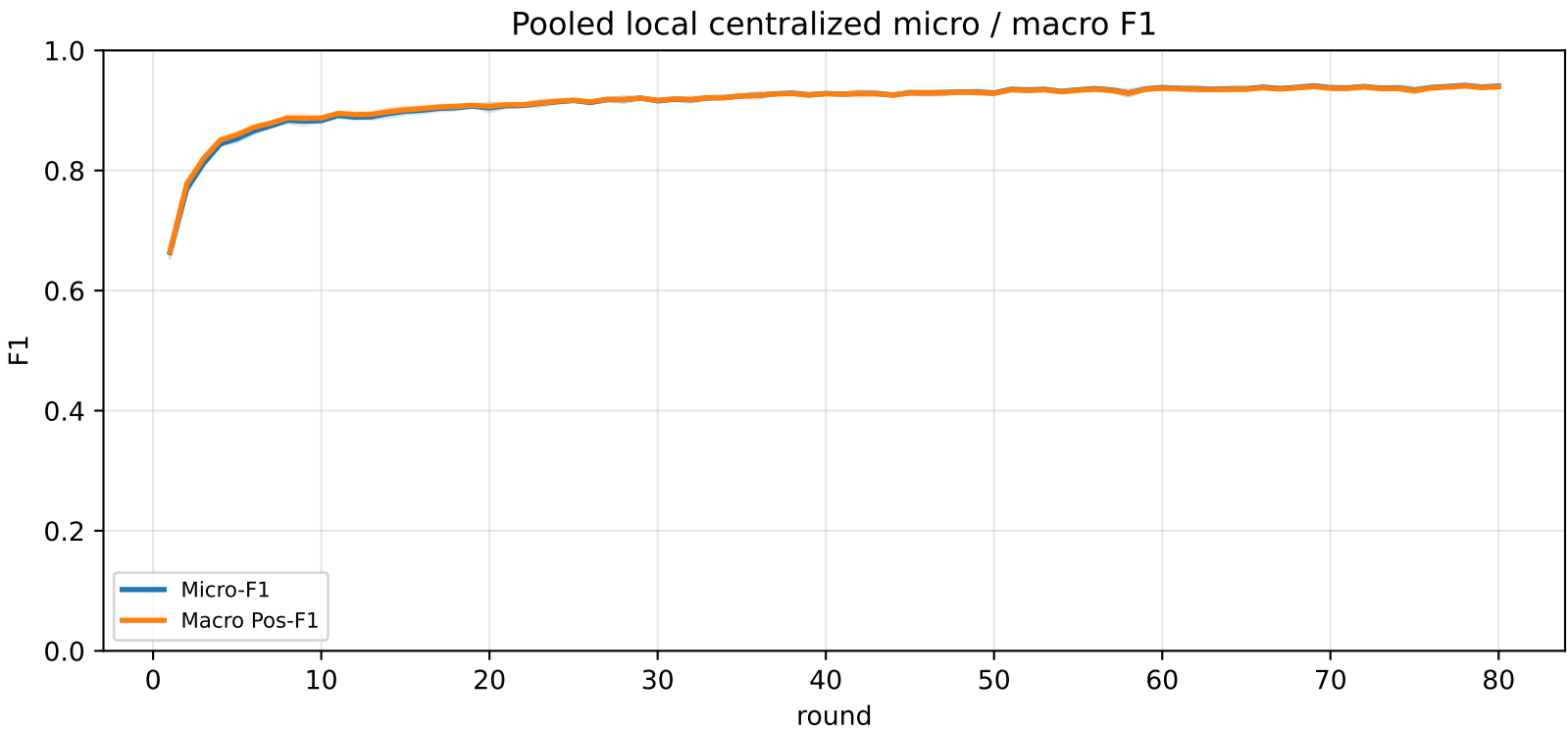
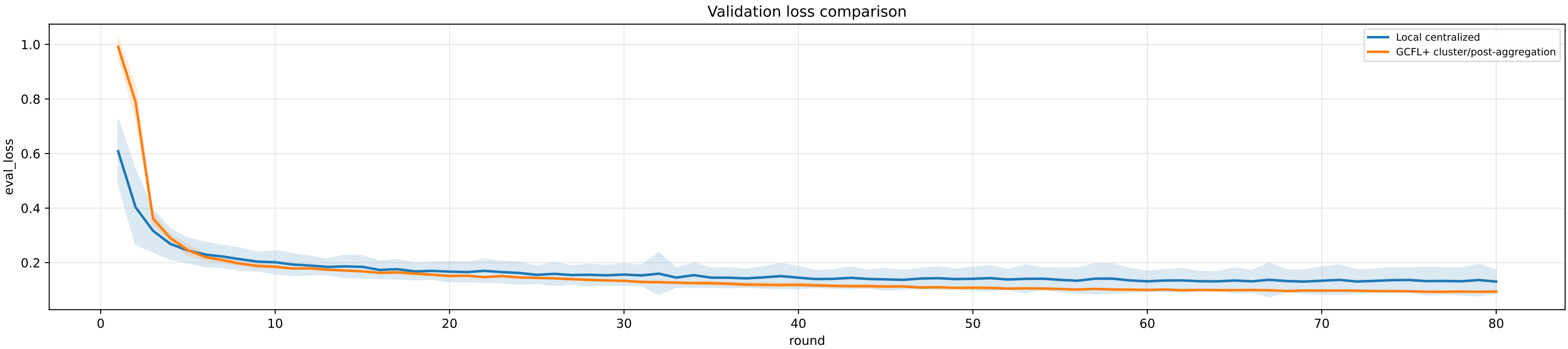
Cluster 8 | Max update norm



Heterogeneity level: High | GCFL+ diagnostics | aggregated / pooled over 20 clients | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.8% ± 0.1%	92.8% ± 0.1%	99.3% ± 0.1%	98.5% ± 0.1%	96.1% ± 0.1%	91.4% ± 0.2%	84.2% ± 0.3%
GCFL+	94.7% ± 0.1%	93.9% ± 0.1%	99.3% ± 0.0%	98.5% ± 0.0%	96.7% ± 0.1%	92.9% ± 0.0%	86.7% ± 0.3%



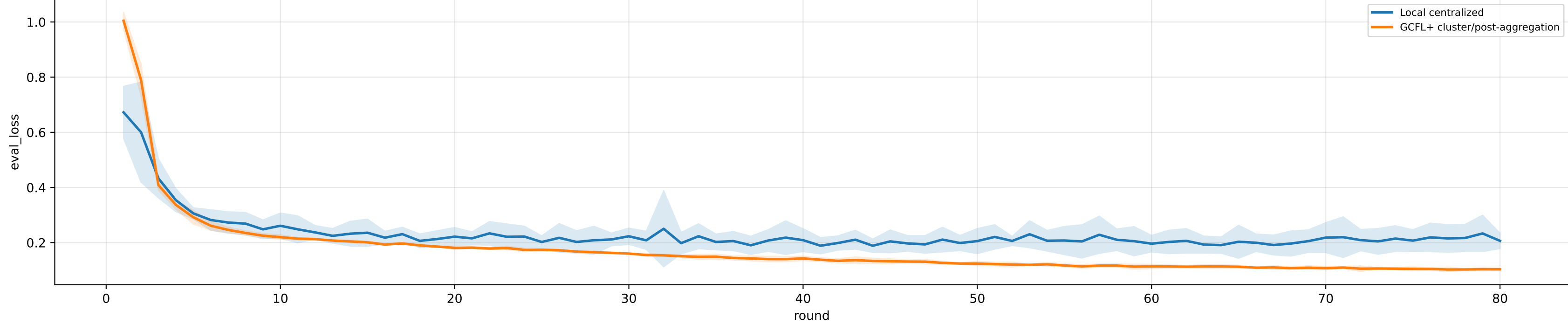
Heterogeneity level: High | GCFL+ diagnostics | family=strong_decreasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_decreasing | cycle2=44.0%±0.3% | cycle3=34.3%±0.7% | cycle4=22.7%±0.7% | cycle5=13.9%±0.4% | cycle6=7.8%±0.6%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

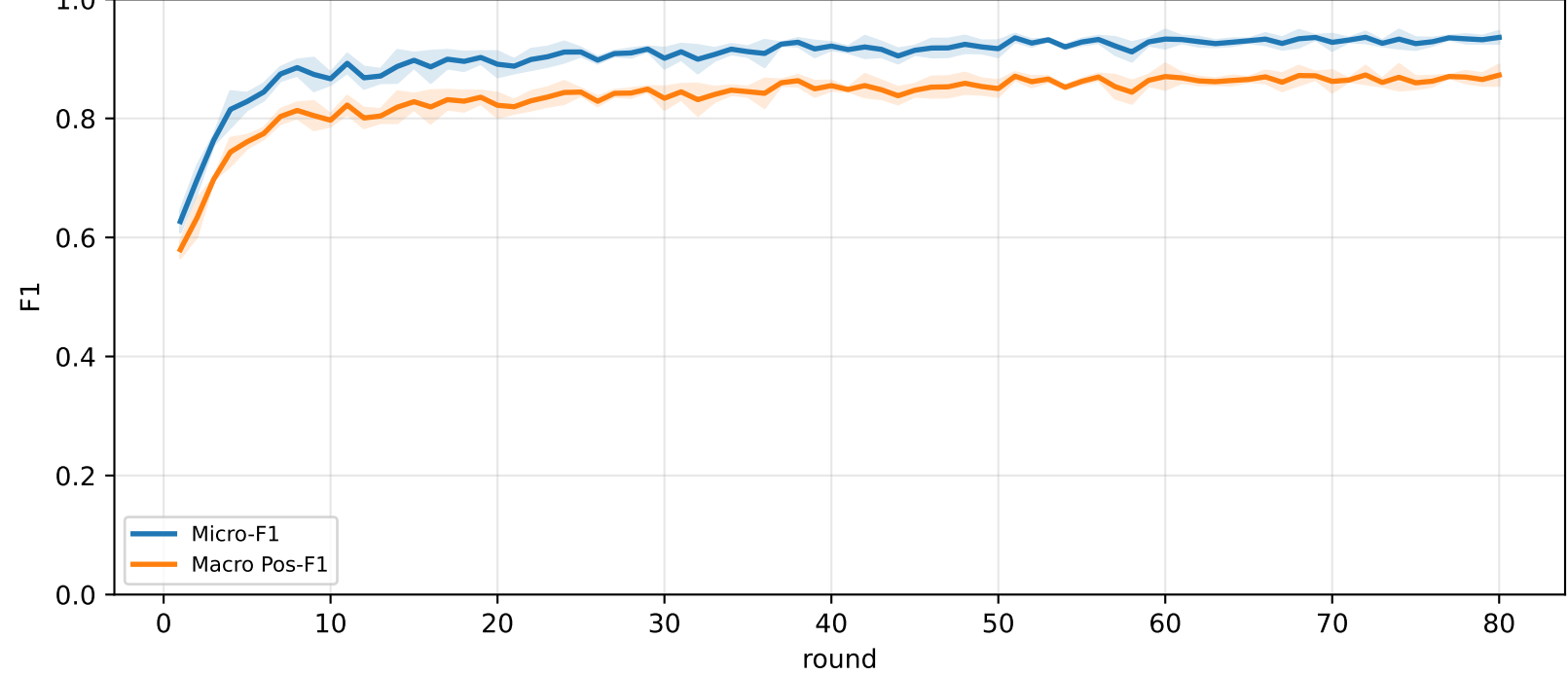
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.1% ± 0.6%	85.7% ± 0.7%	99.6% ± 0.1%	98.9% ± 0.1%	95.0% ± 0.4%	79.3% ± 1.4%	55.6% ± 2.3%
GCFL+	94.5% ± 0.3%	89.2% ± 0.4%	99.8% ± 0.0%	99.3% ± 0.1%	97.0% ± 0.2%	85.6% ± 0.3%	64.3% ± 1.5%

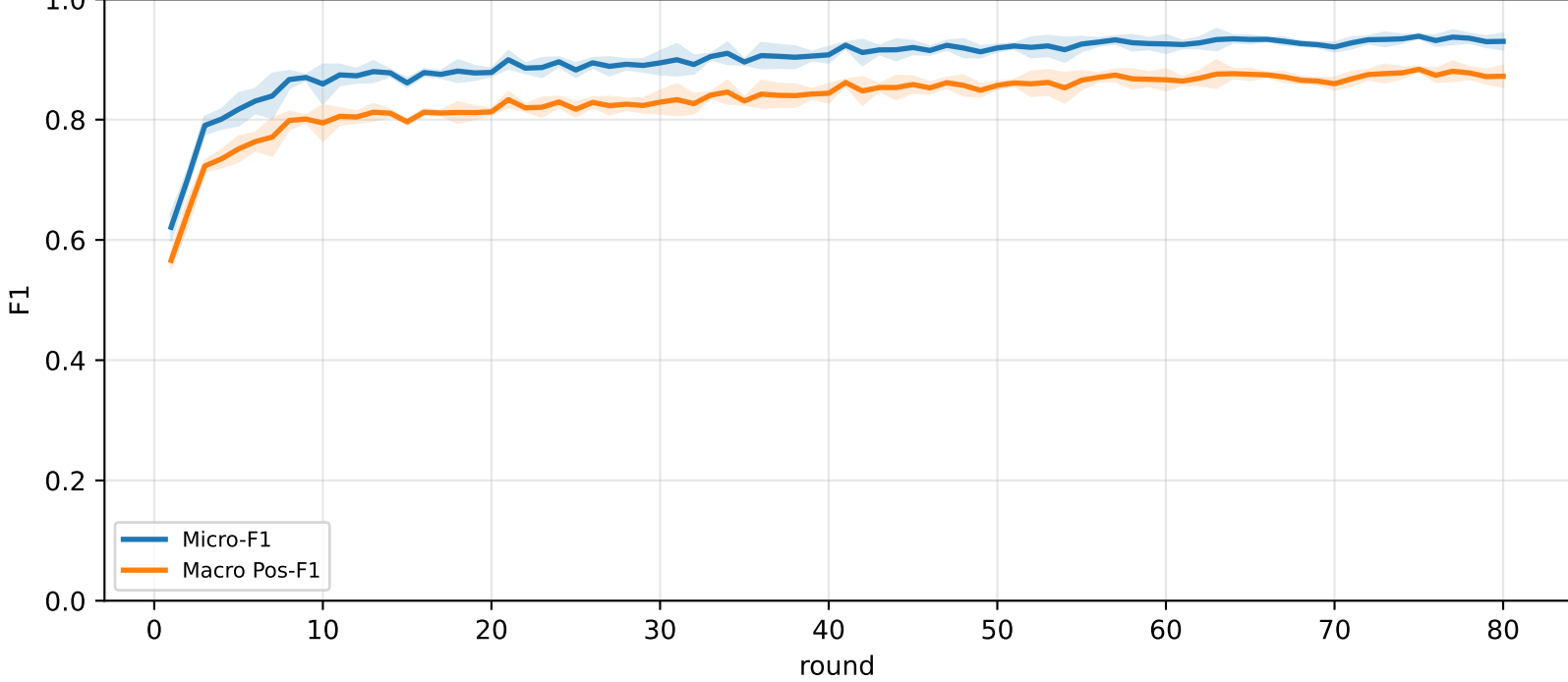
Validation loss comparison



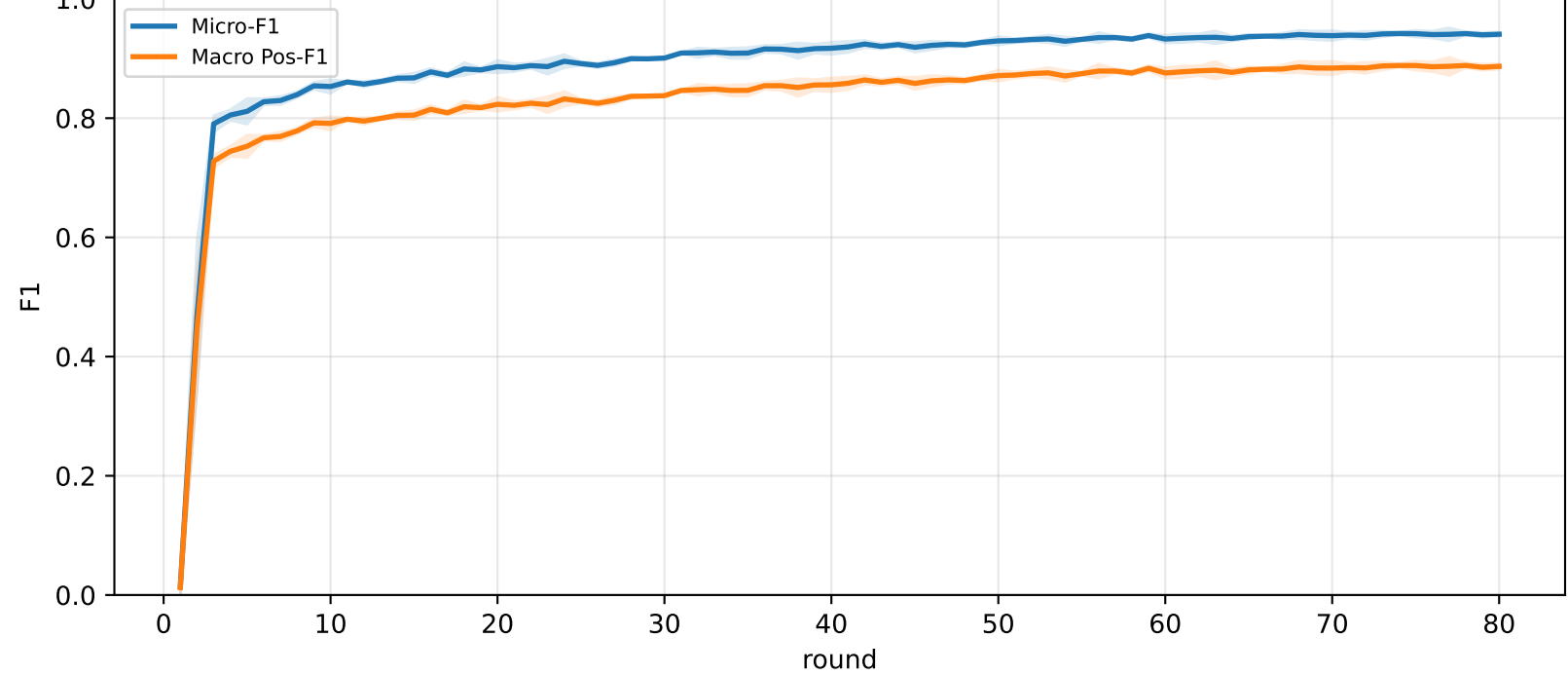
Pooled local centralized micro / macro F1



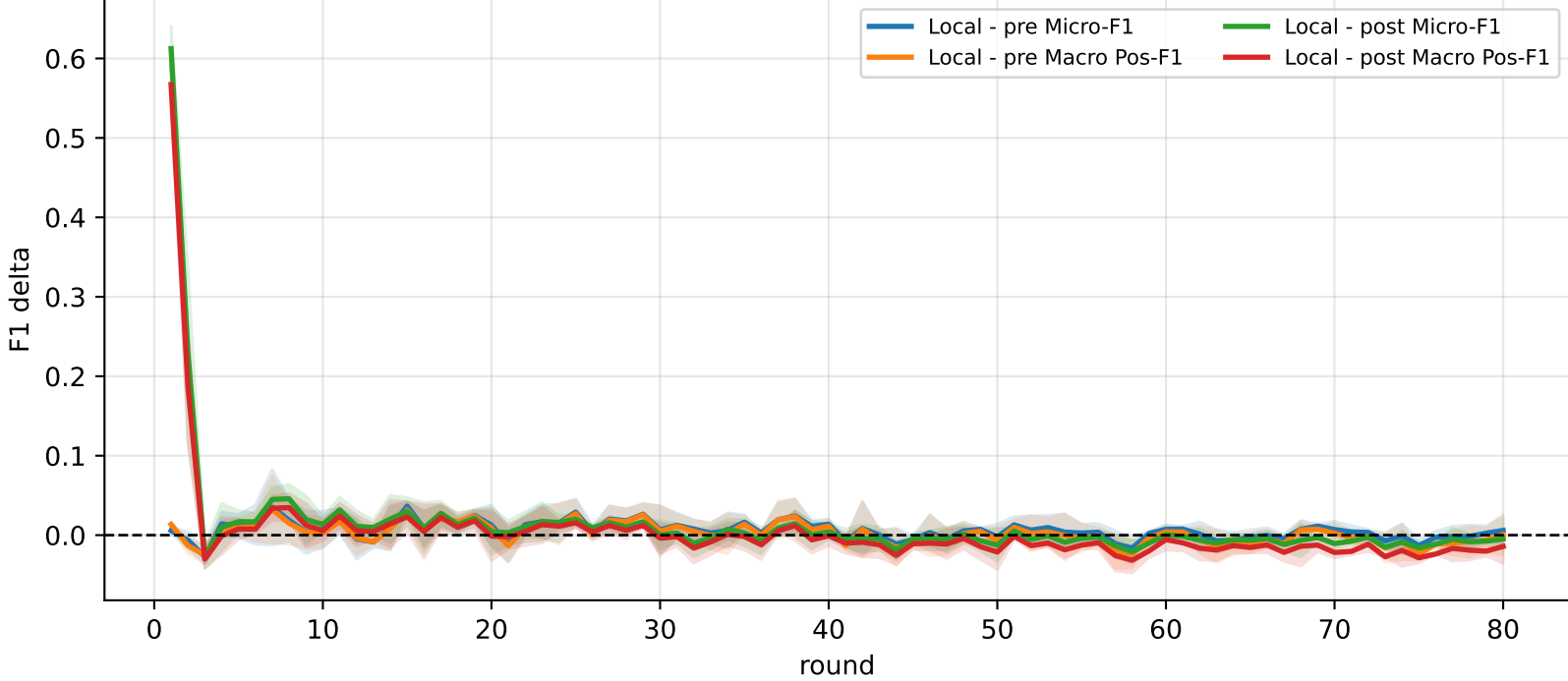
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



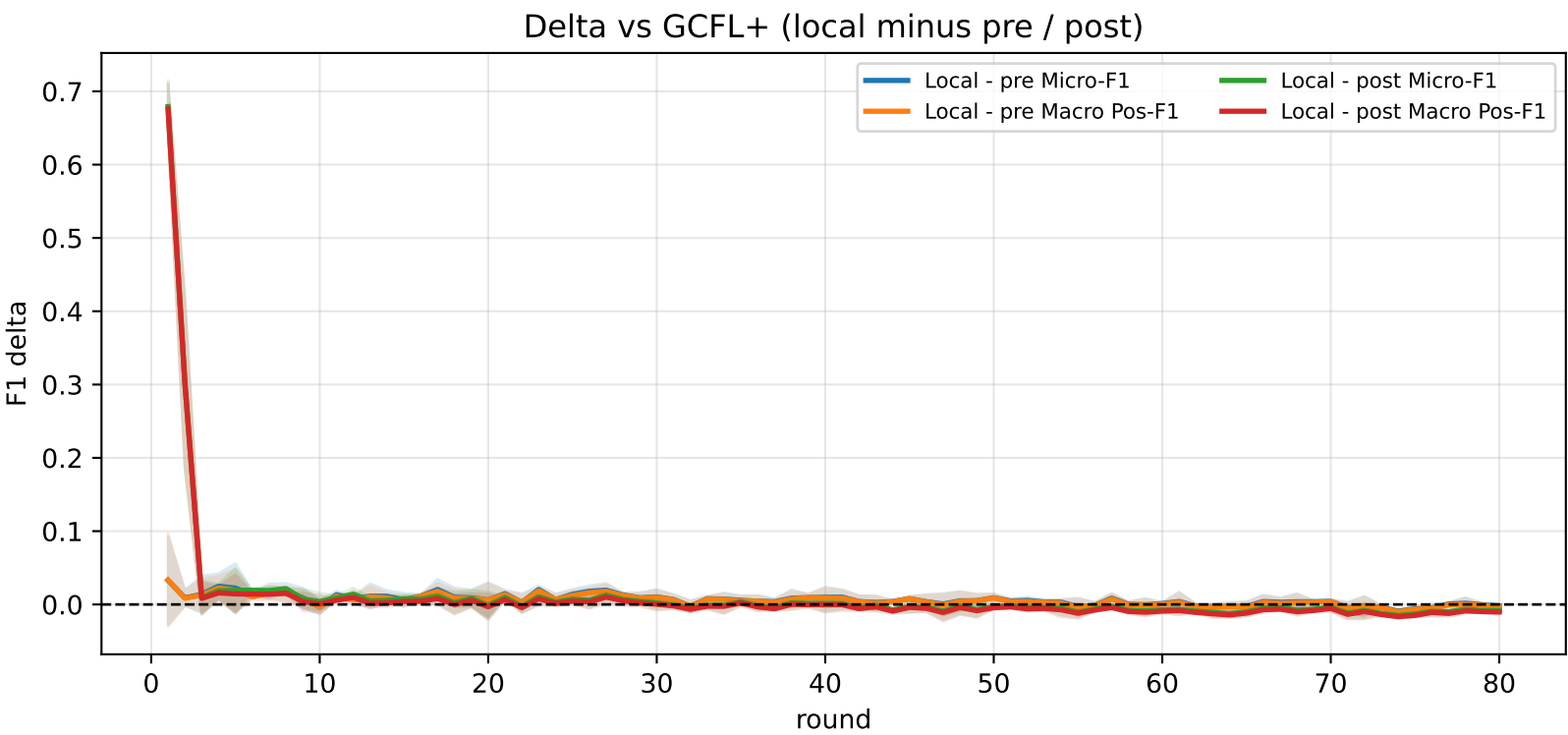
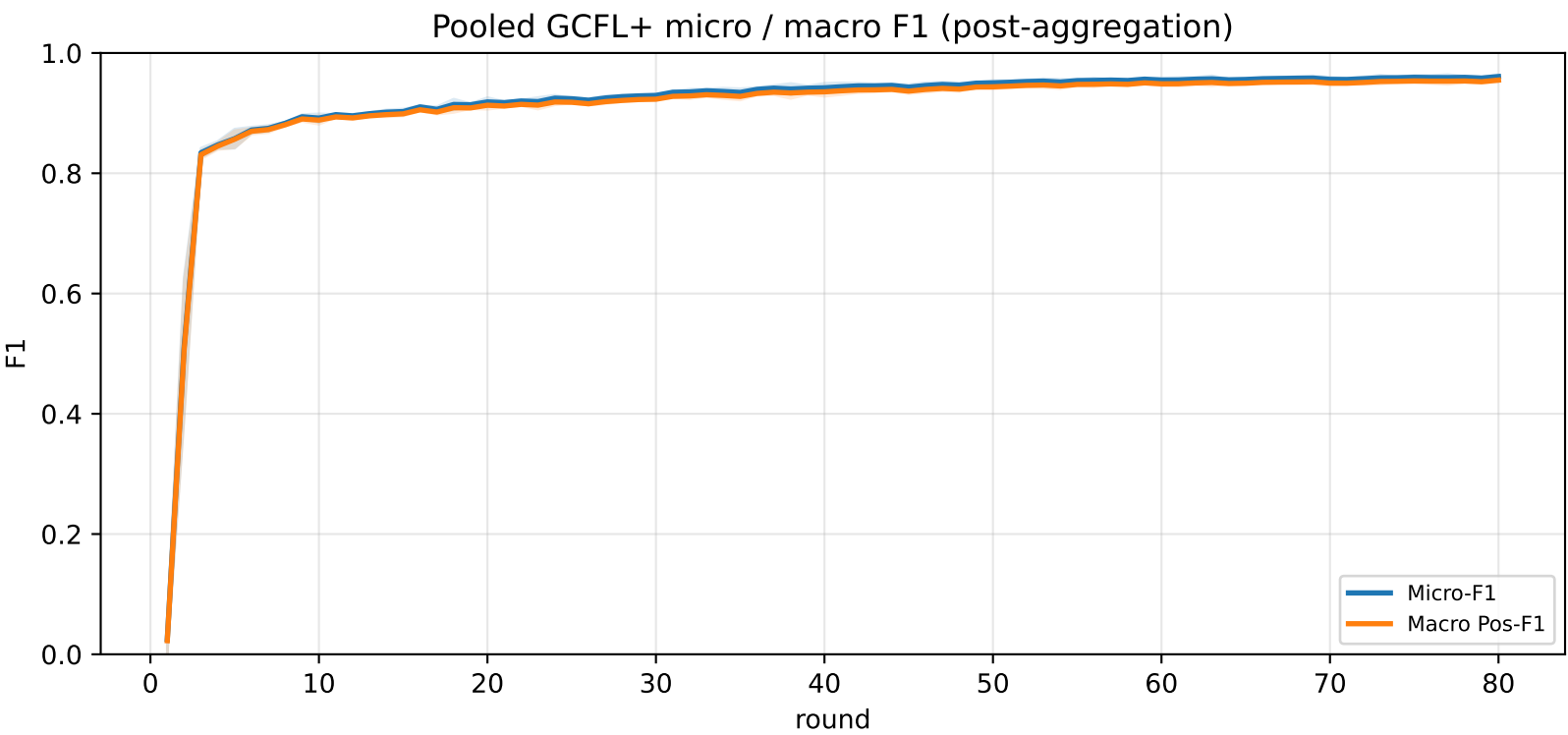
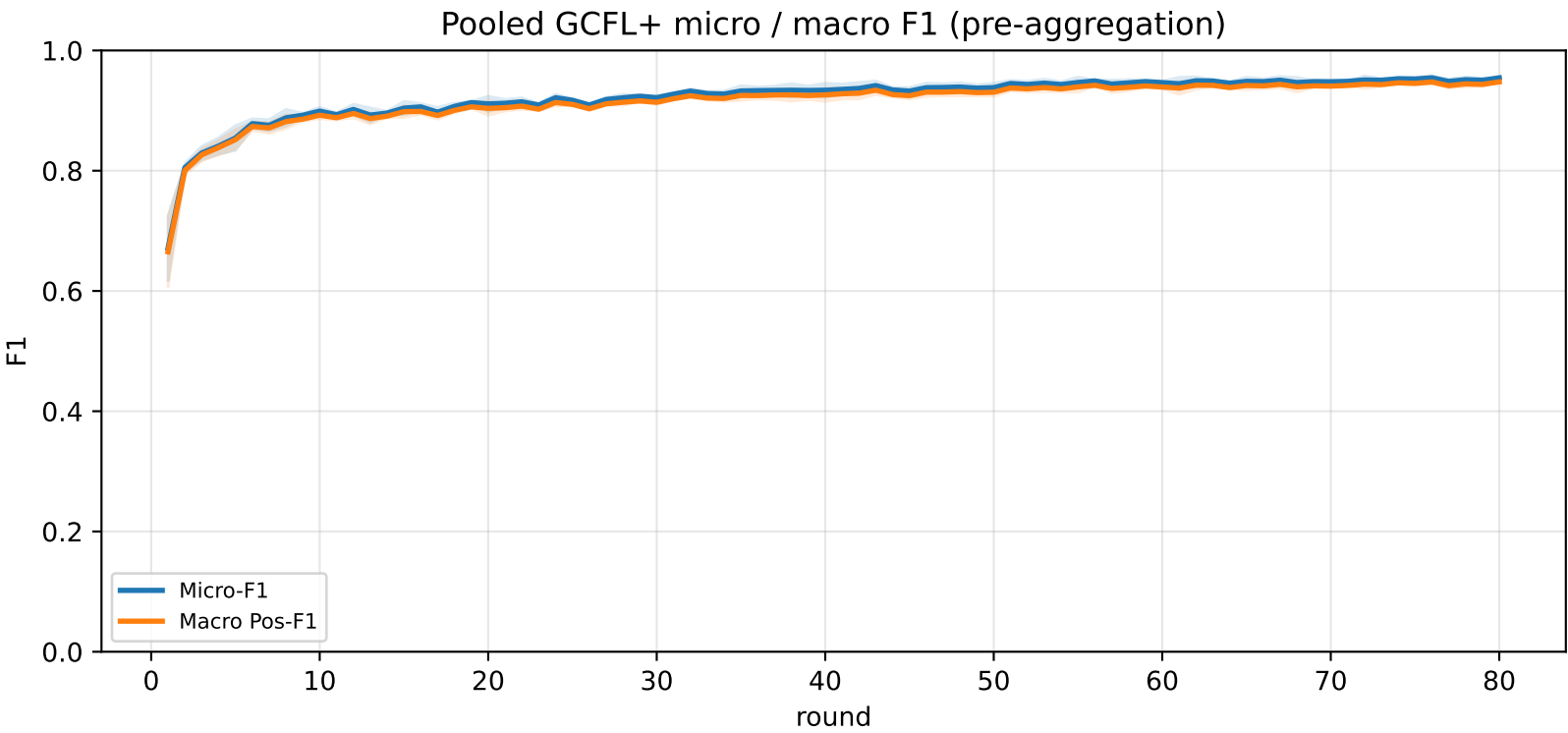
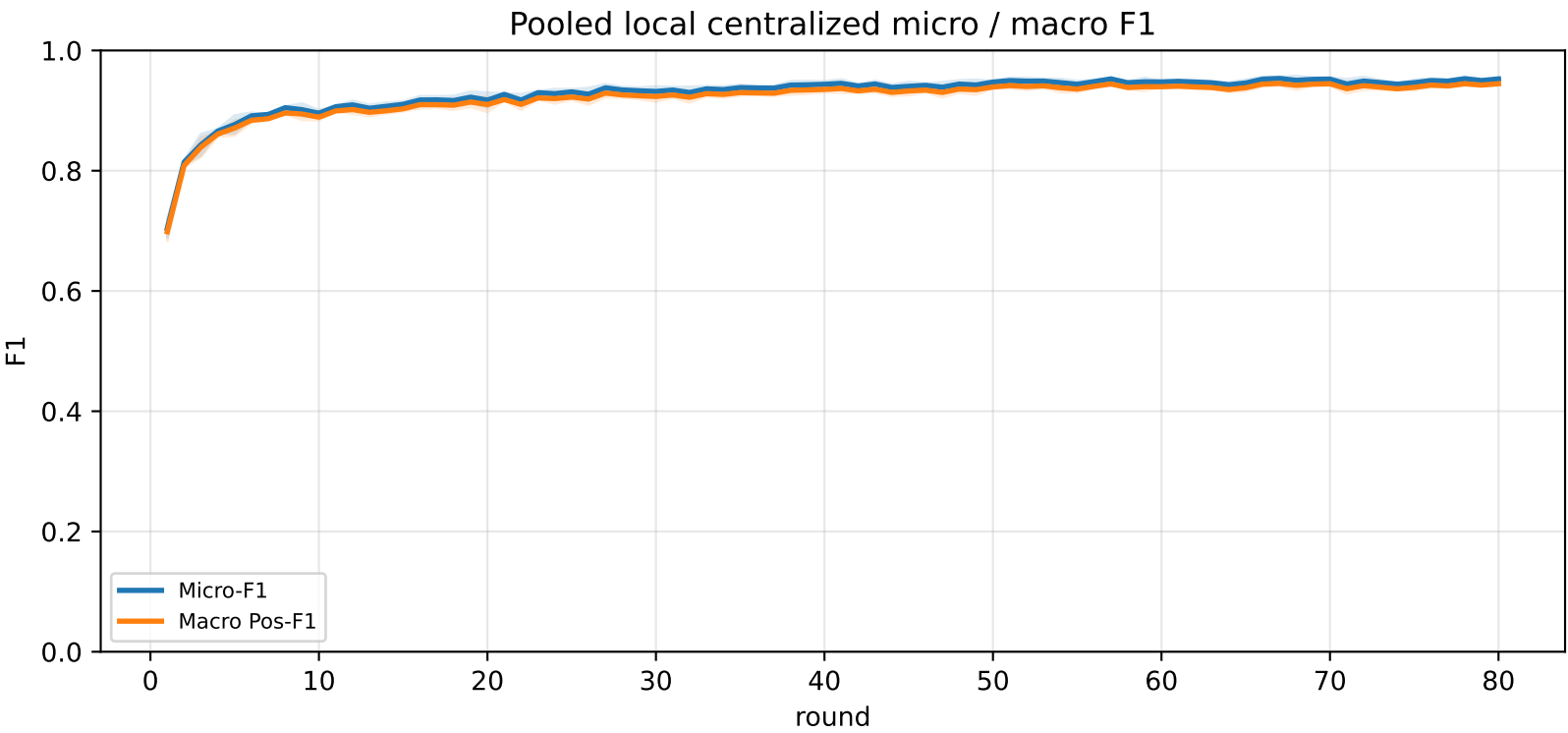
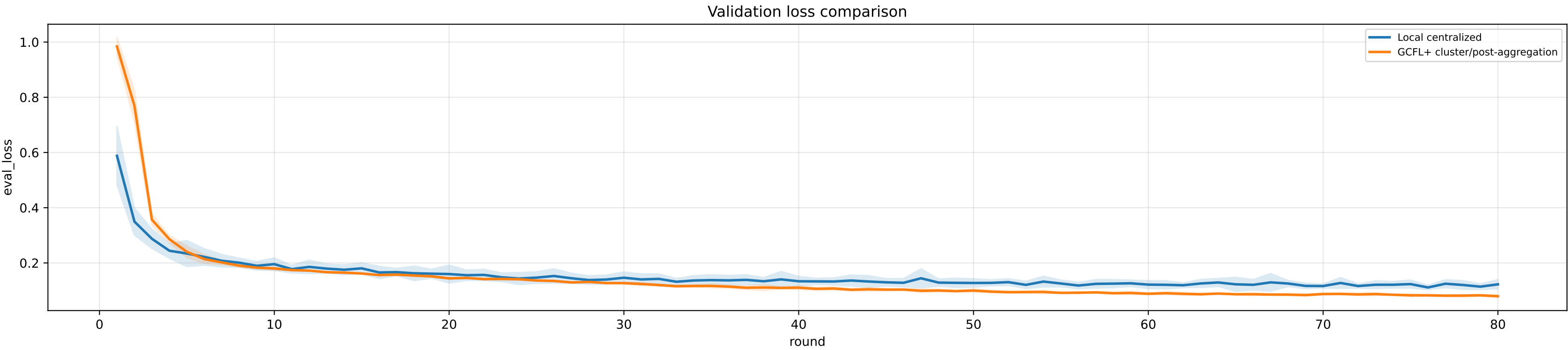
Delta vs GCFL+ (local minus pre / post)



Heterogeneity level: High | GCFL+ diagnostics | family=mild_decreasing |
n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_decreasing | cycle2=32.5%±0.7% | cycle3=30.7%±1.0% | cycle4=26.8%±1.1% | cycle5=24.5%±1.0% | cycle6=19.2%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

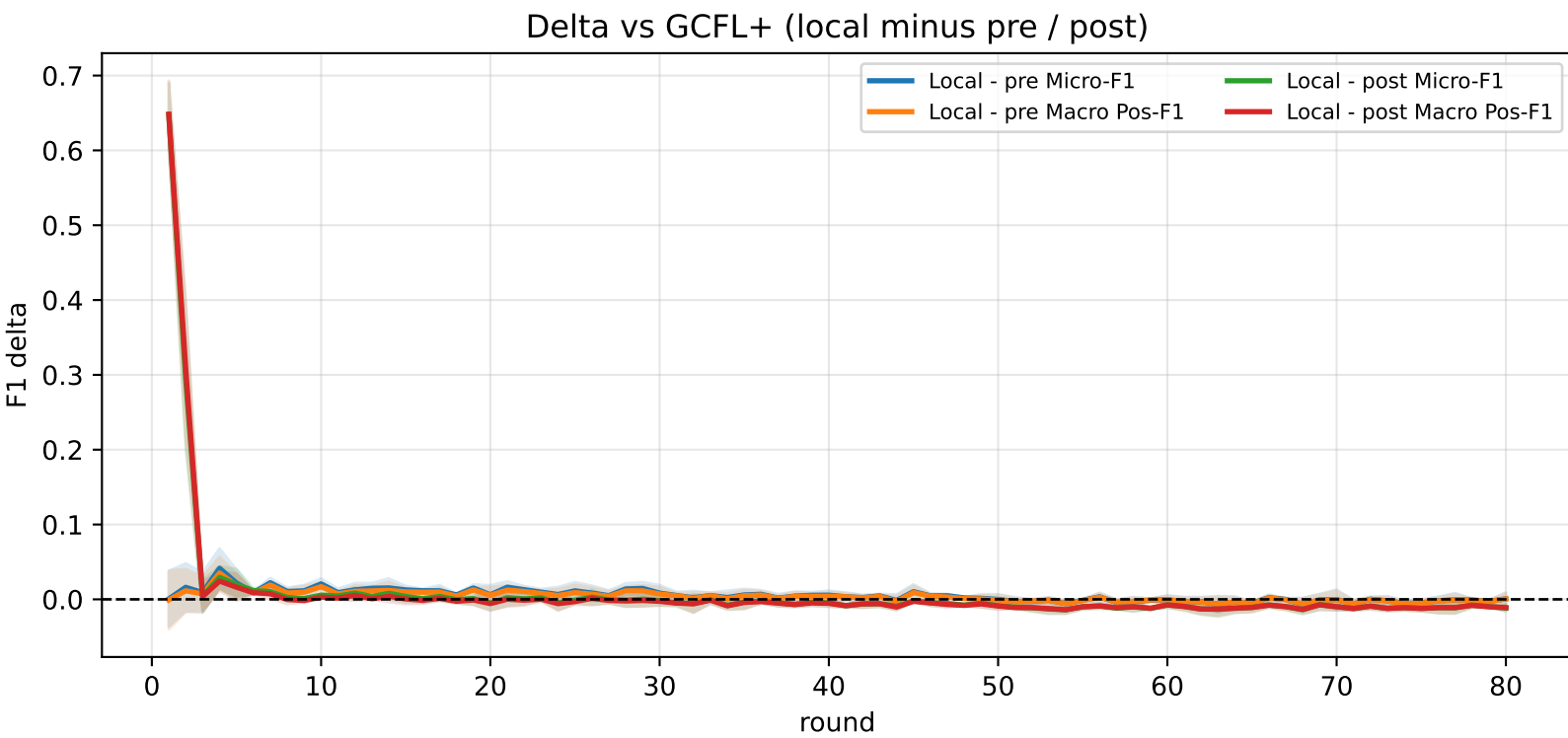
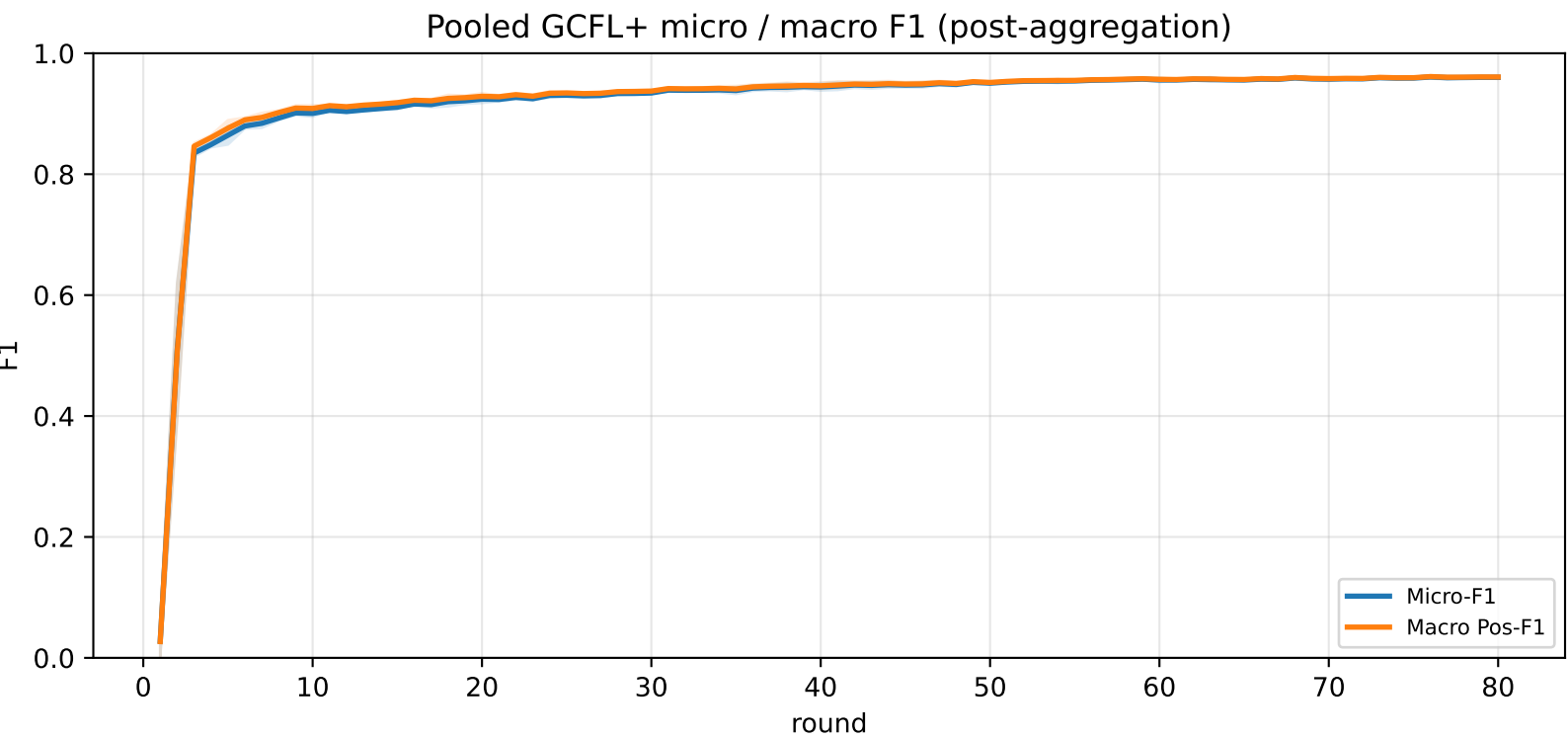
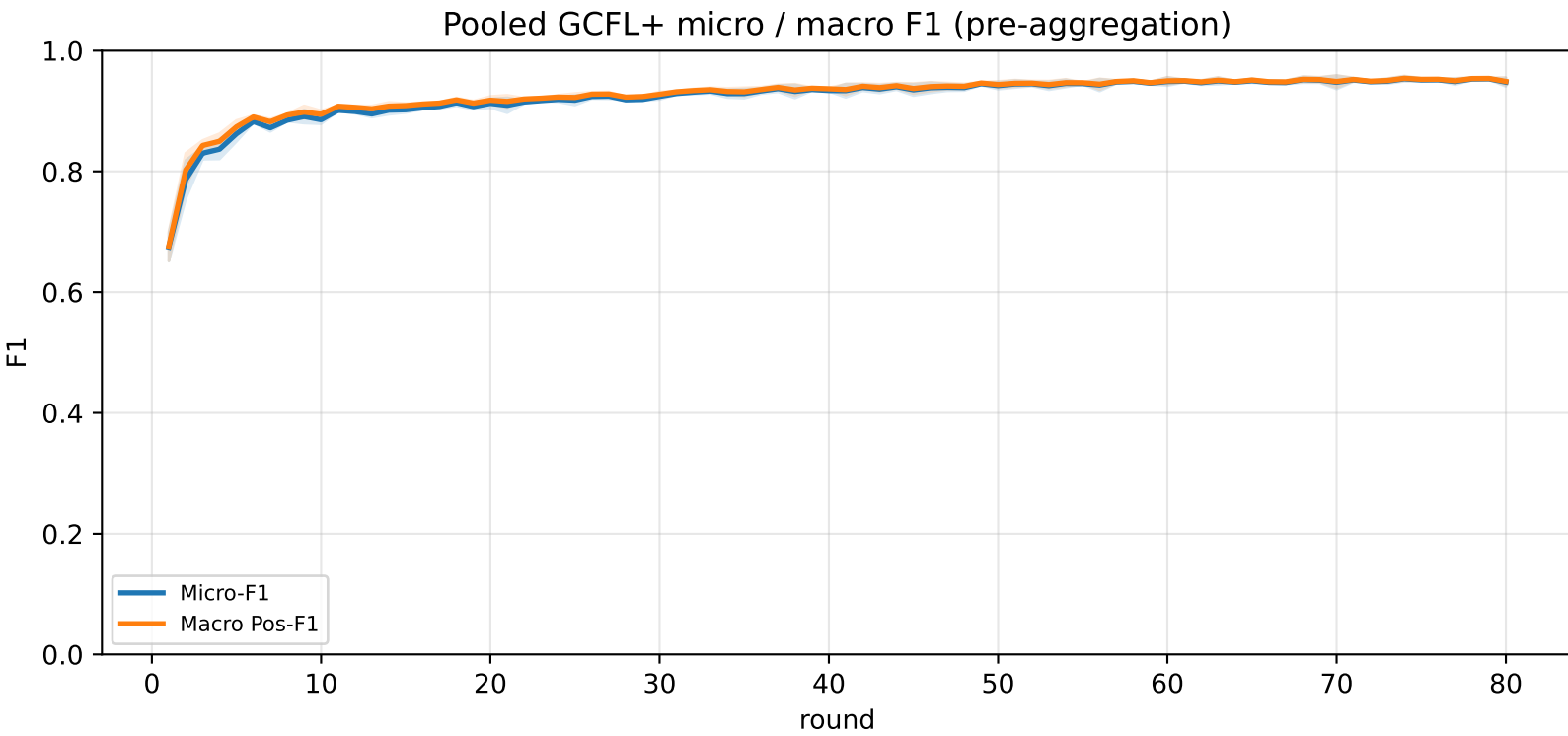
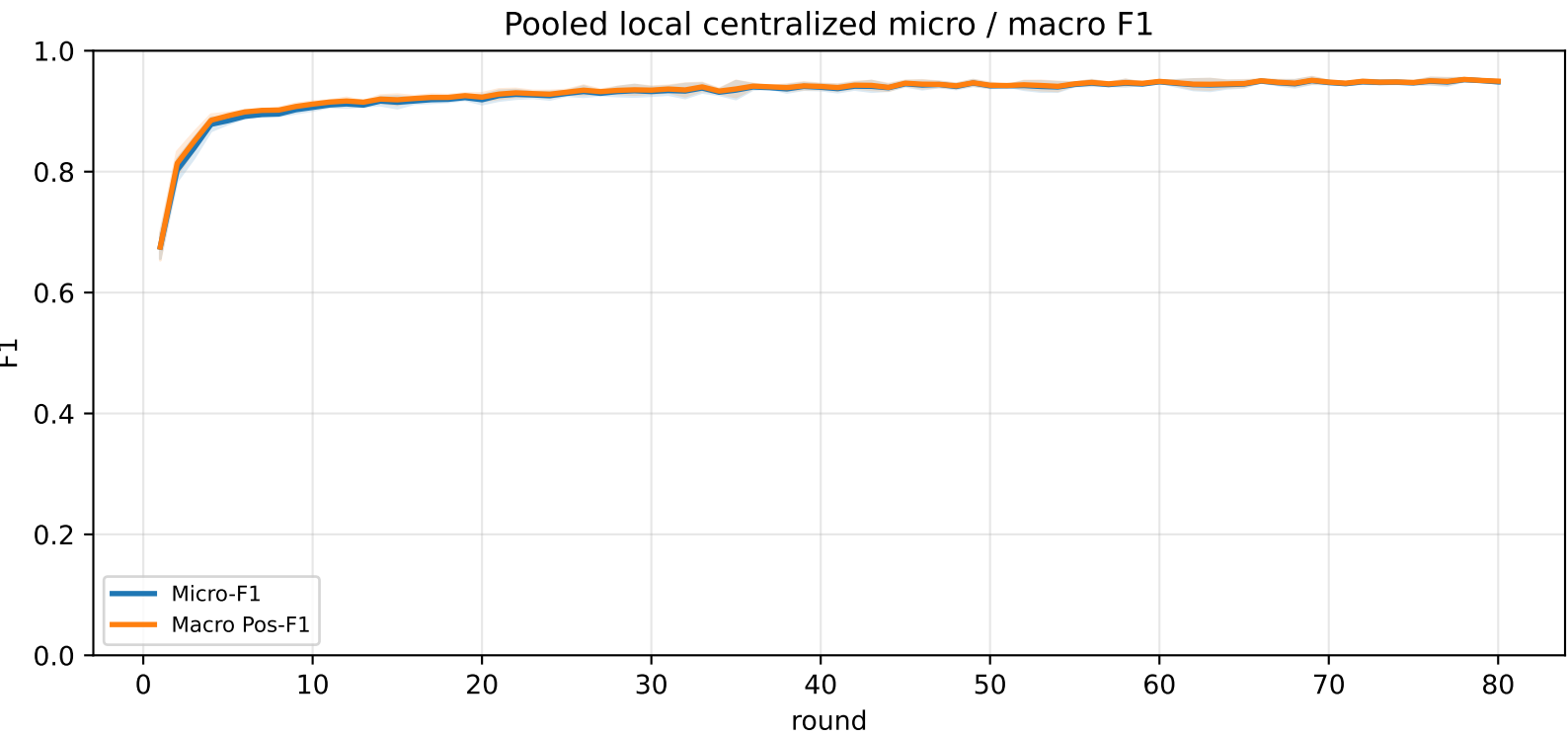
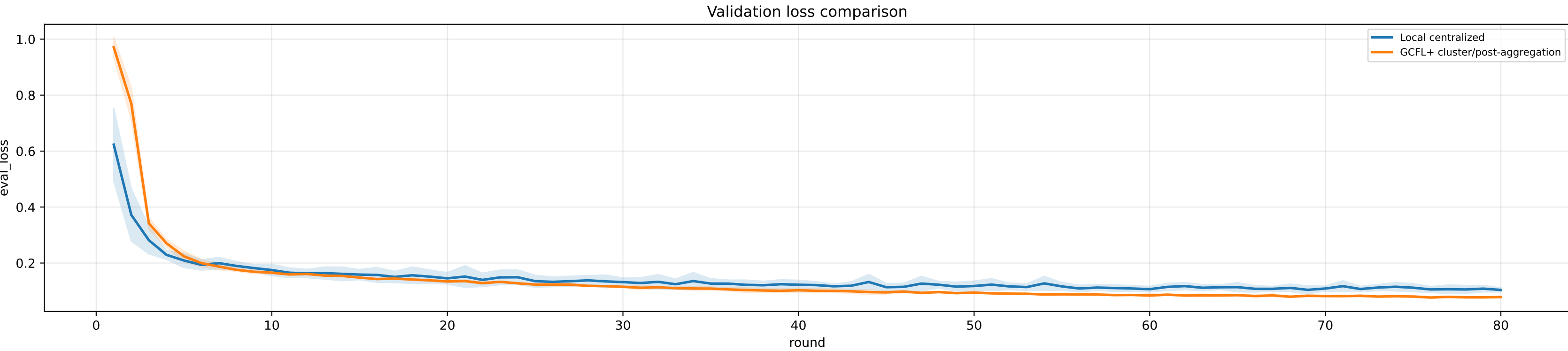
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.2% ± 0.2%	94.4% ± 0.2%	99.5% ± 0.1%	99.1% ± 0.1%	97.3% ± 0.1%	92.9% ± 0.1%	83.2% ± 0.7%
GCFL+	96.0% ± 0.1%	95.3% ± 0.2%	99.6% ± 0.1%	99.1% ± 0.1%	98.1% ± 0.1%	94.3% ± 0.1%	85.5% ± 0.7%



Heterogeneity level: High | GCFL+ diagnostics | family=flat_balanced | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=flat_balanced | cycle2=25.1%±0.7% | cycle3=26.6%±0.8% | cycle4=27.5%±0.8% | cycle5=27.8%±1.2% | cycle6=25.6%±0.3%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

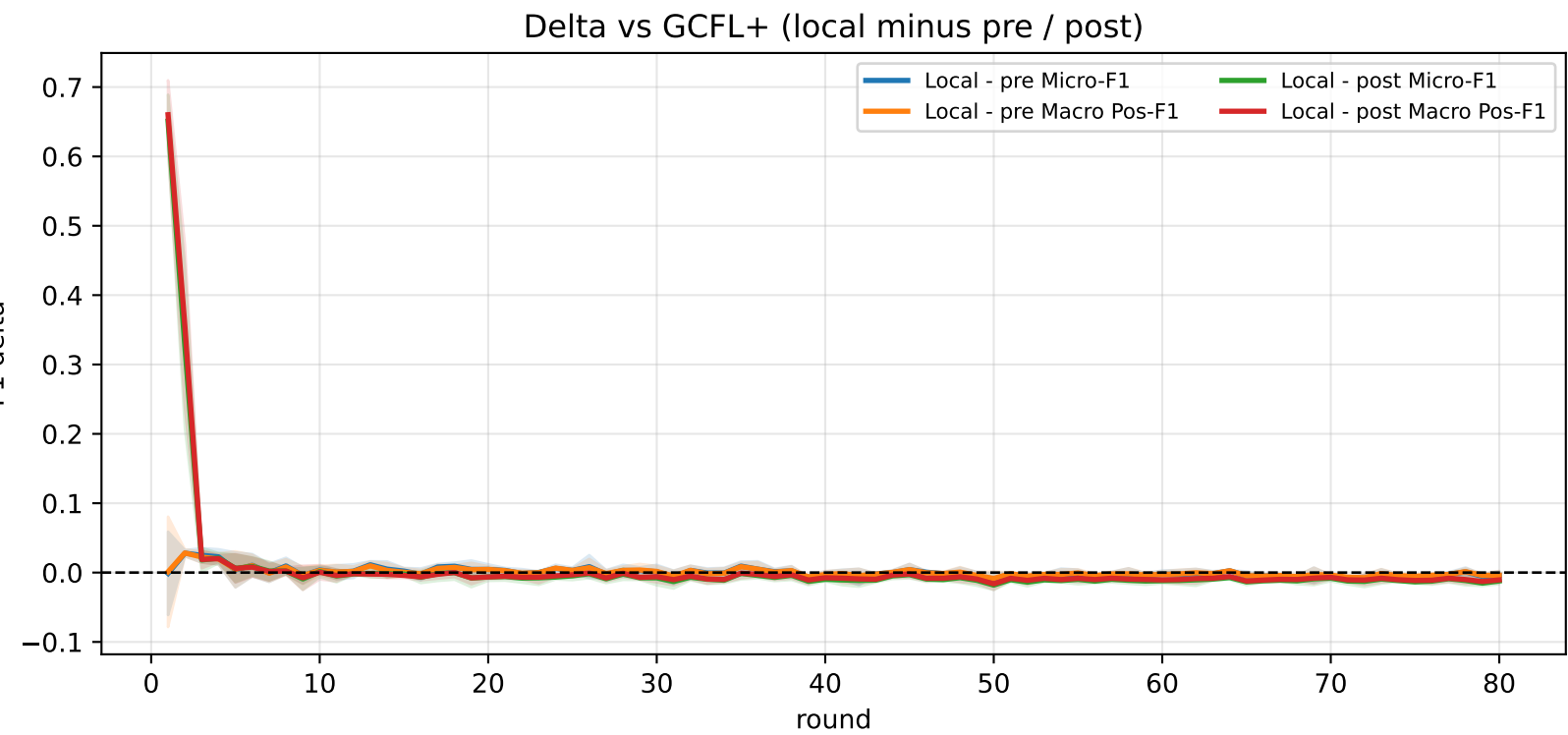
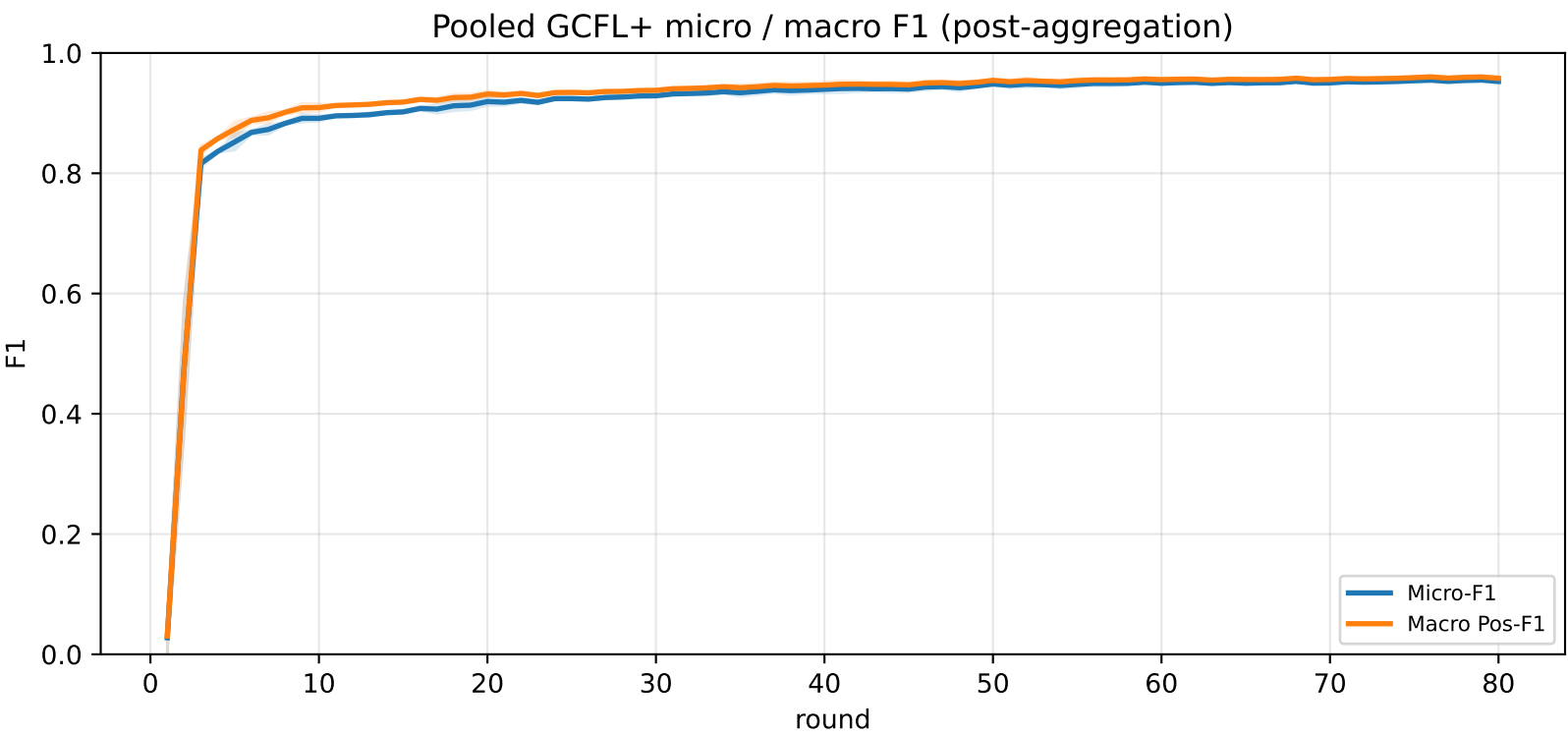
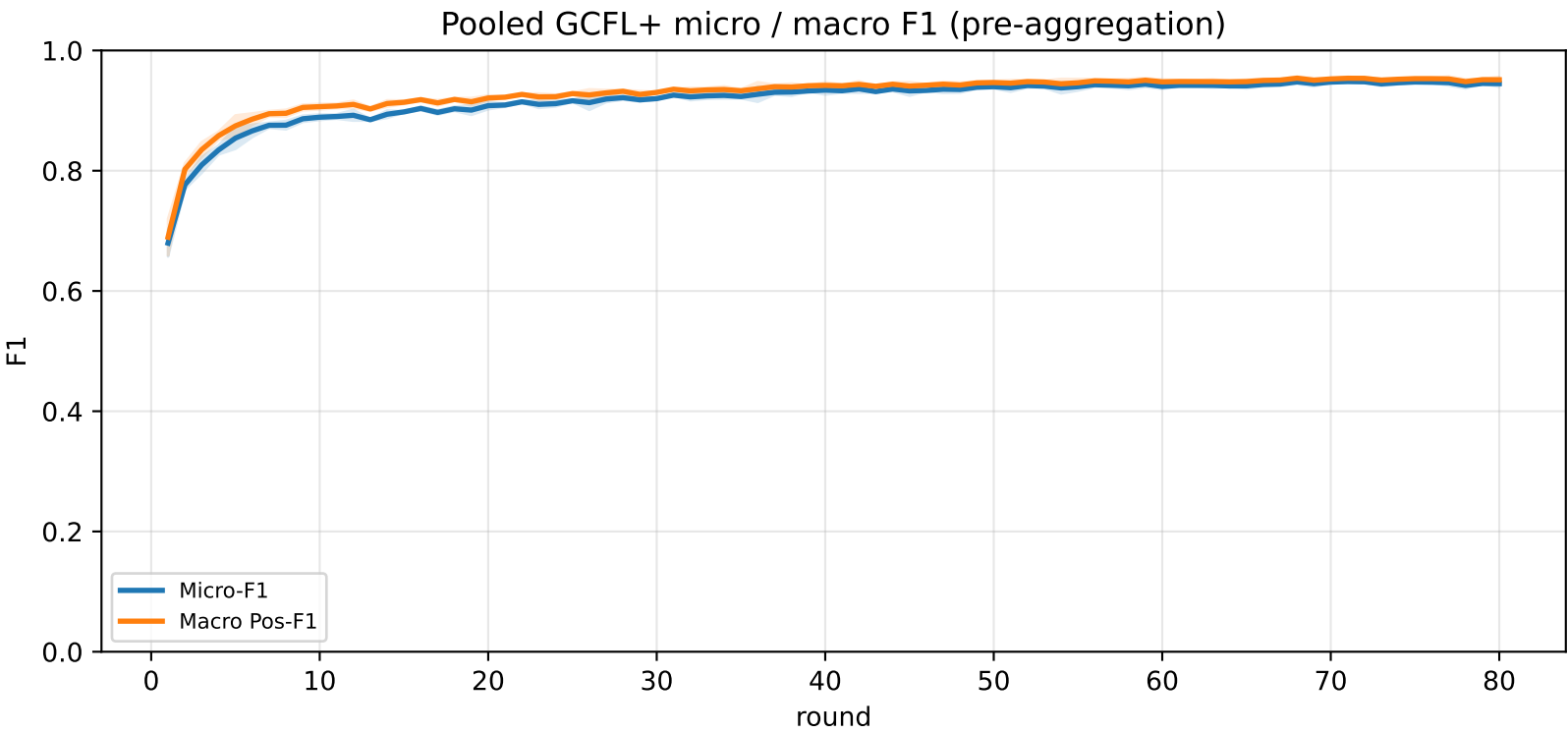
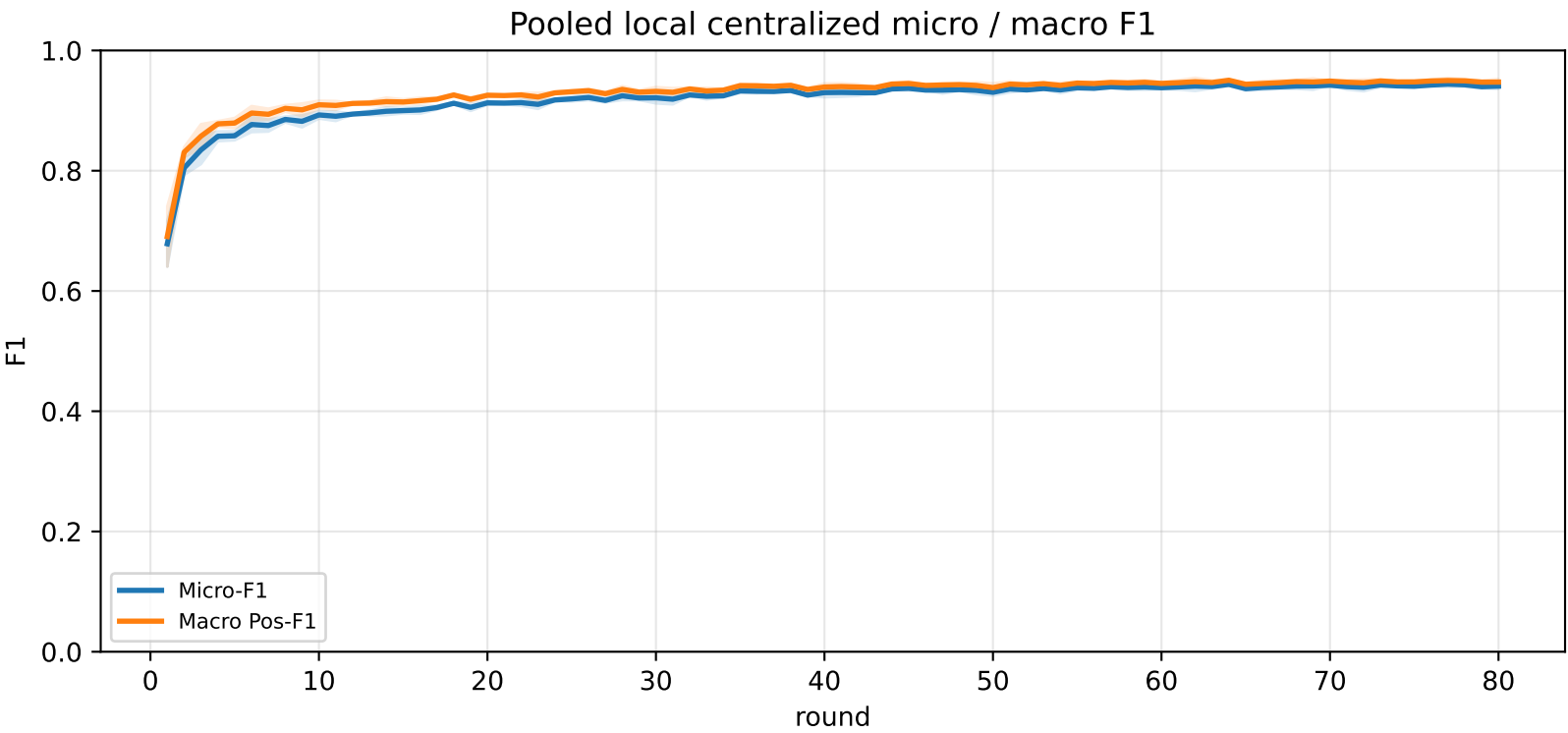
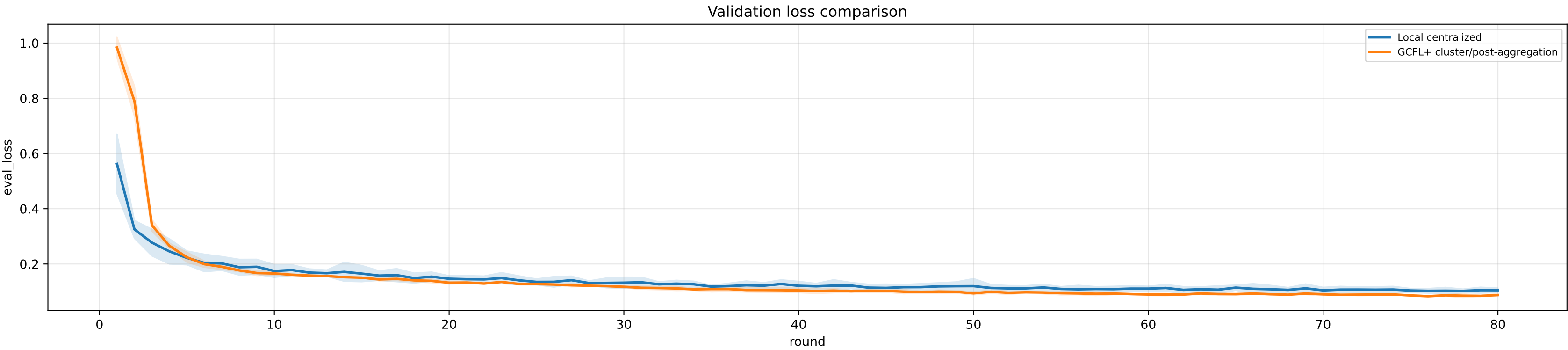
Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	95.0% ± 0.1%	95.2% ± 0.1%	99.1% ± 0.1%	98.8% ± 0.2%	97.1% ± 0.1%	94.2% ± 0.2%	86.6% ± 0.5%
GCFL+	96.0% ± 0.1%	96.2% ± 0.1%	99.3% ± 0.2%	98.7% ± 0.1%	97.4% ± 0.1%	95.3% ± 0.2%	90.1% ± 0.5%



Heterogeneity level: High | GCFL+ diagnostics | family=mild_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=mild_increasing | cycle2=16.7%±0.8% | cycle3=21.4%±0.7% | cycle4=25.6%±1.2% | cycle5=28.8%±0.7% | cycle6=30.0%±0.4%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.4% ± 0.1%	95.1% ± 0.1%	98.7% ± 0.2%	97.8% ± 0.2%	96.2% ± 0.2%	93.8% ± 0.4%	89.2% ± 0.5%
GCFL+	95.4% ± 0.2%	95.9% ± 0.2%	98.7% ± 0.1%	98.1% ± 0.2%	96.5% ± 0.2%	94.3% ± 0.4%	91.9% ± 0.3%



Heterogeneity level: High | GCFL+ diagnostics | family=strong_increasing | n_clients=4 | model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

family=strong_increasing | cycle2=9.9%±1.2% | cycle3=14.1%±0.8% | cycle4=19.4%±1.7% | cycle5=25.4%±1.6% | cycle6=31.3%±1.0%
heterogeneity=high | task_jsd_mean=0.028 | task_jsd_median=0.018 | task_jsd_max=0.120 | family_centroid_jsd_mean=0.033
per-task pos-rate std | cycle2_std=0.120 | cycle3_std=0.071 | cycle4_std=0.032 | cycle5_std=0.054 | cycle6_std=0.087

Best logged test performance (GCFL+ vs Local centralized)							
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.3% ± 0.2%	93.6% ± 0.2%	98.2% ± 0.1%	96.3% ± 0.1%	94.2% ± 0.0%	91.2% ± 0.2%	88.2% ± 0.7%
GCFL+	91.7% ± 0.4%	93.1% ± 0.3%	97.9% ± 0.1%	95.8% ± 0.3%	93.9% ± 0.4%	91.1% ± 0.5%	86.8% ± 0.5%

